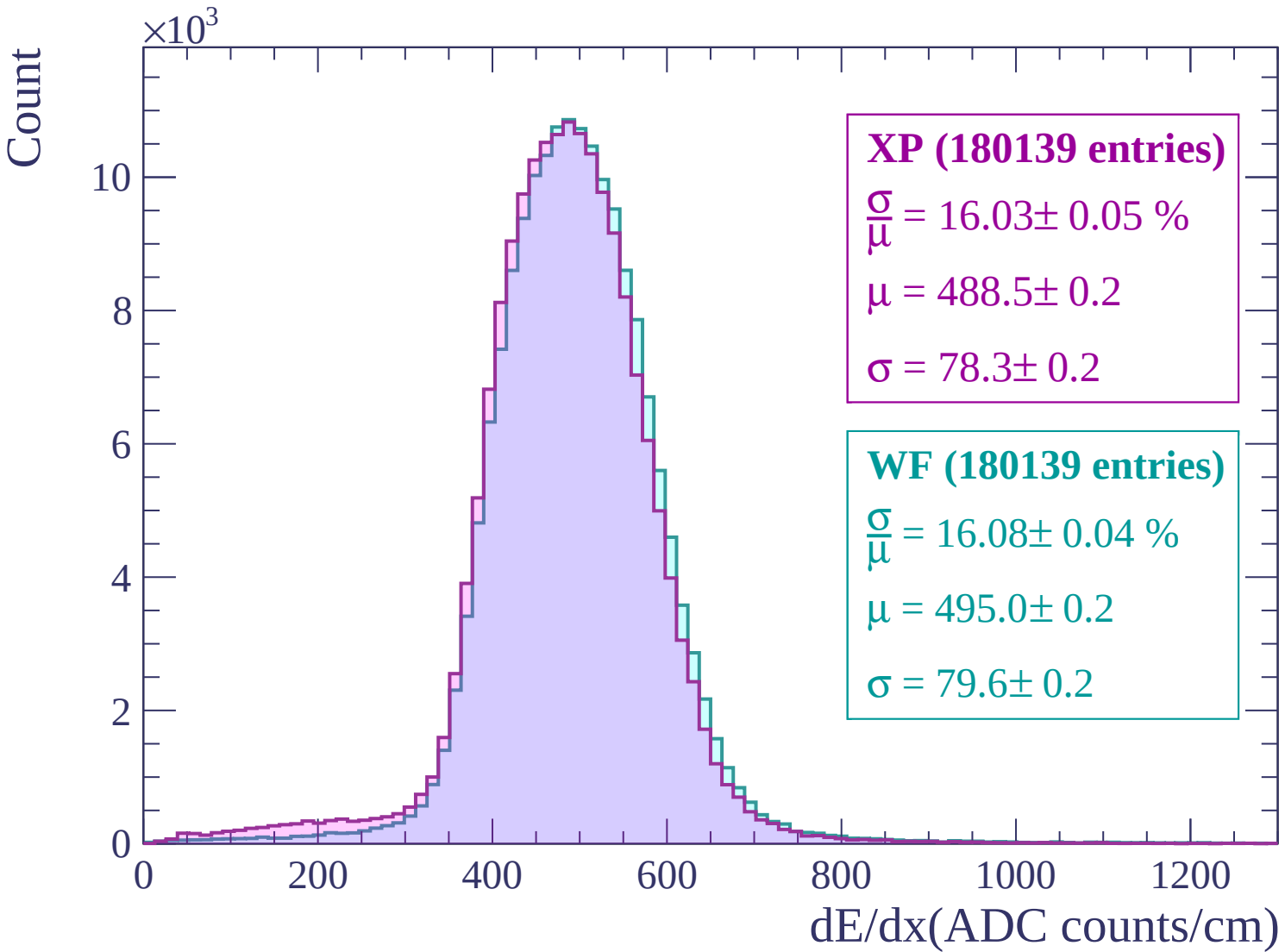
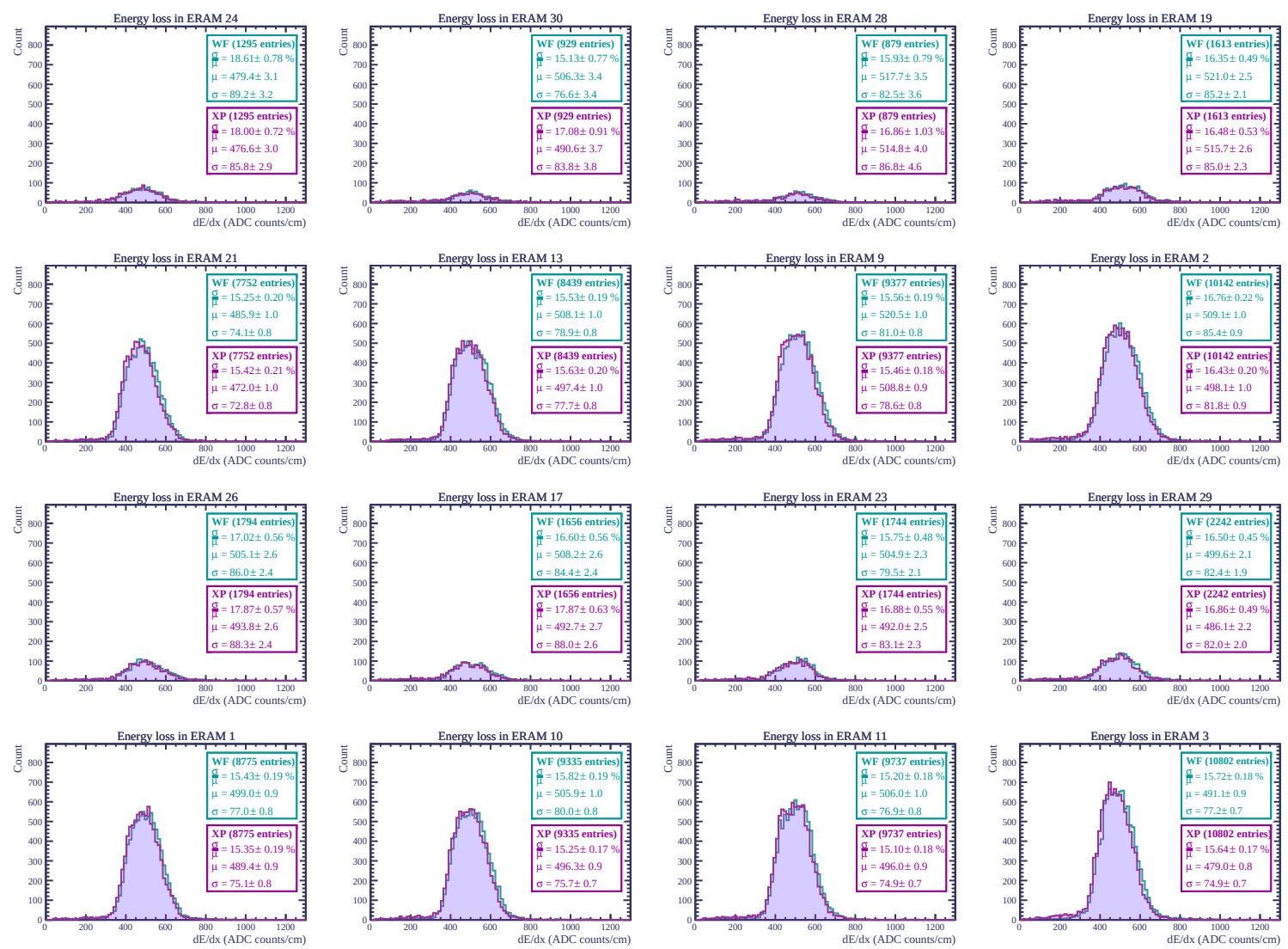
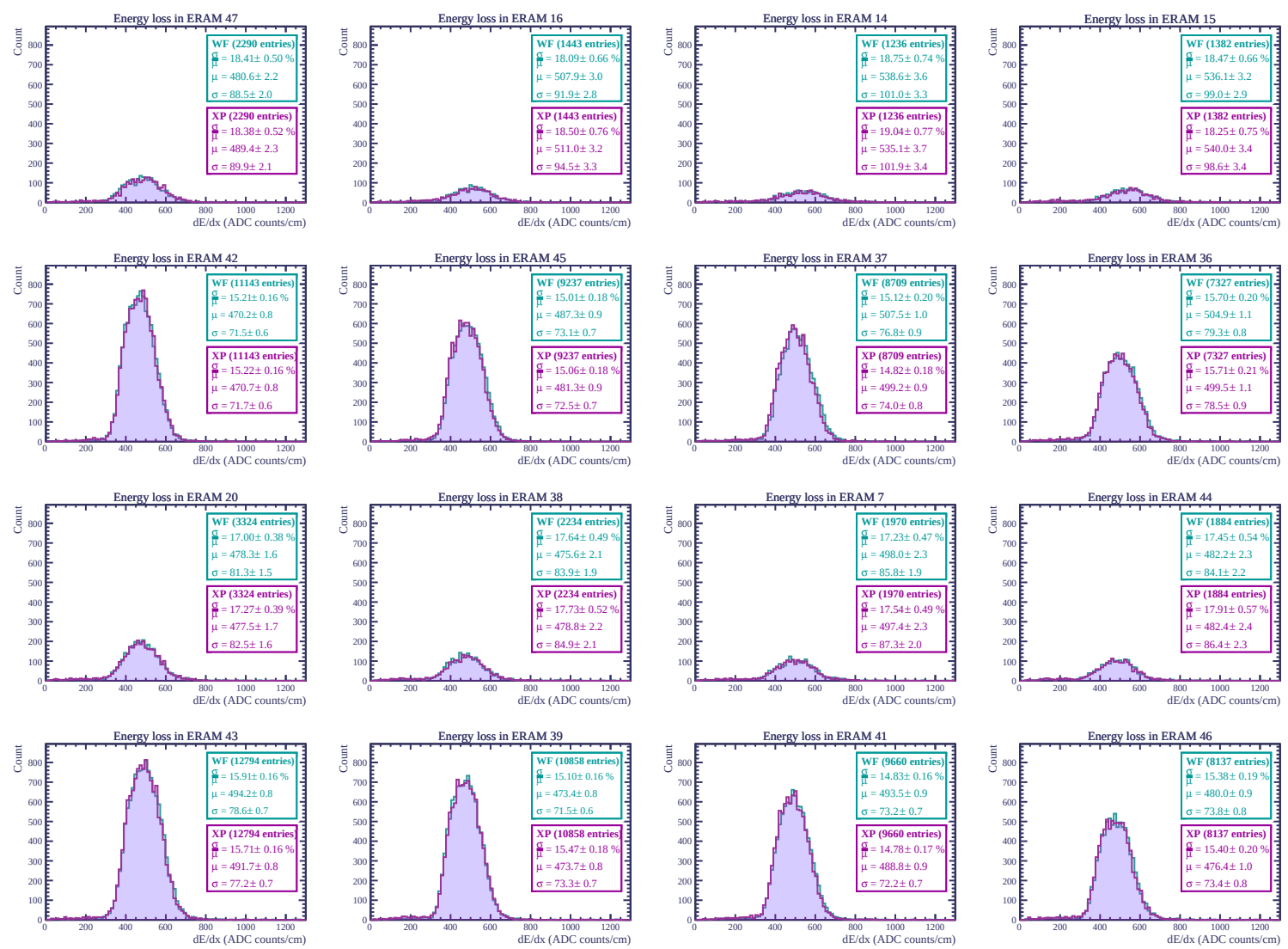
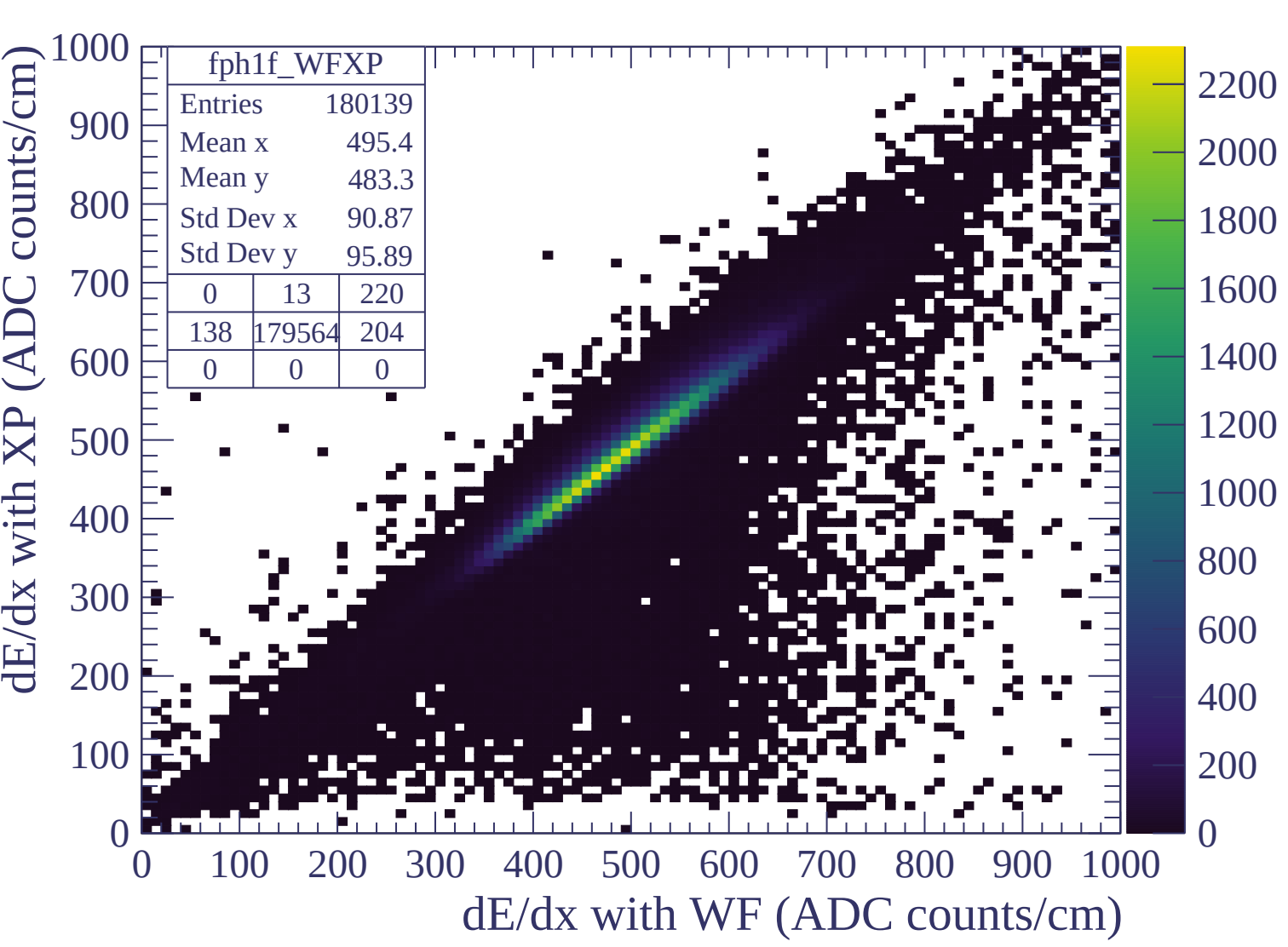
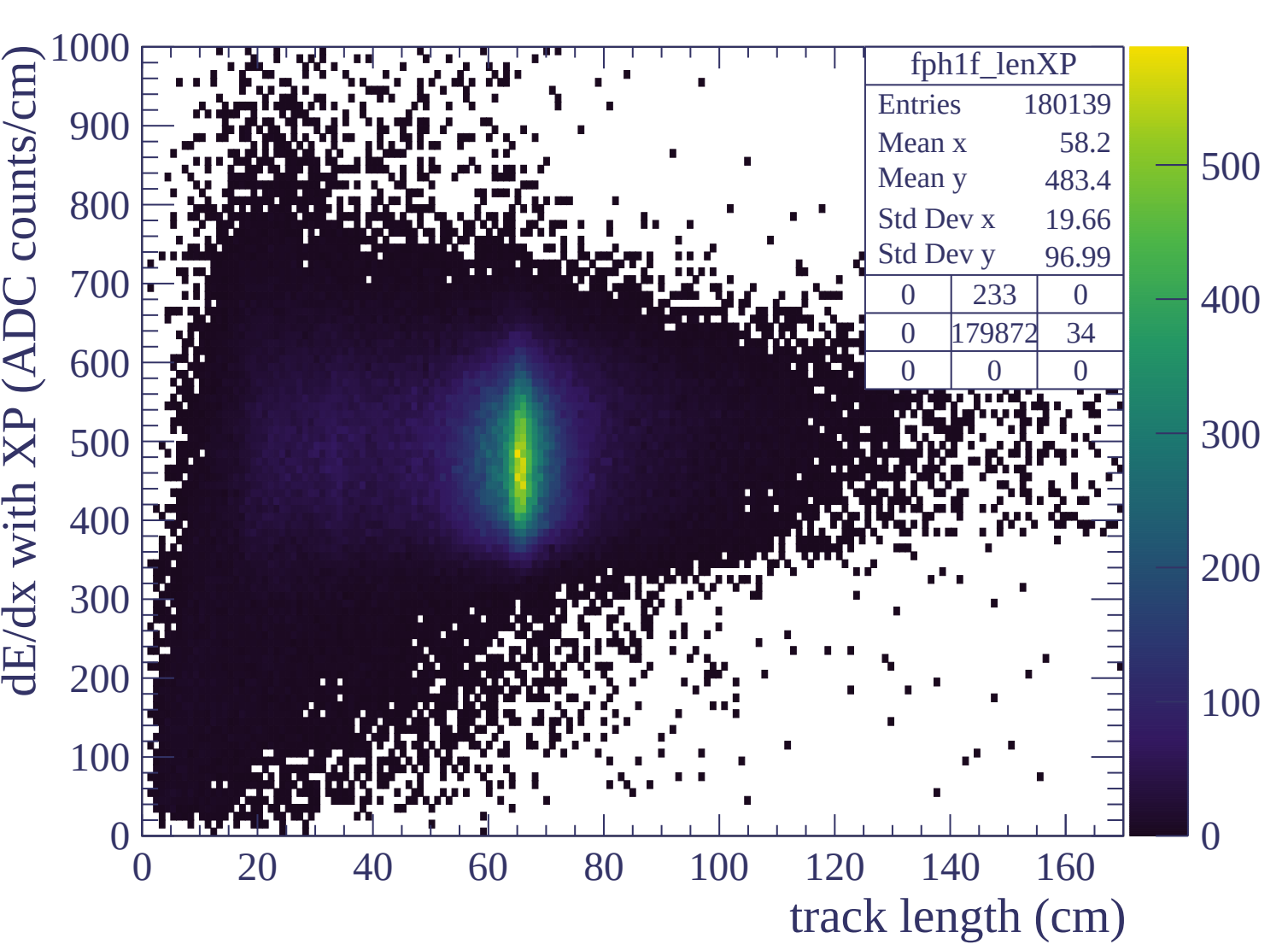


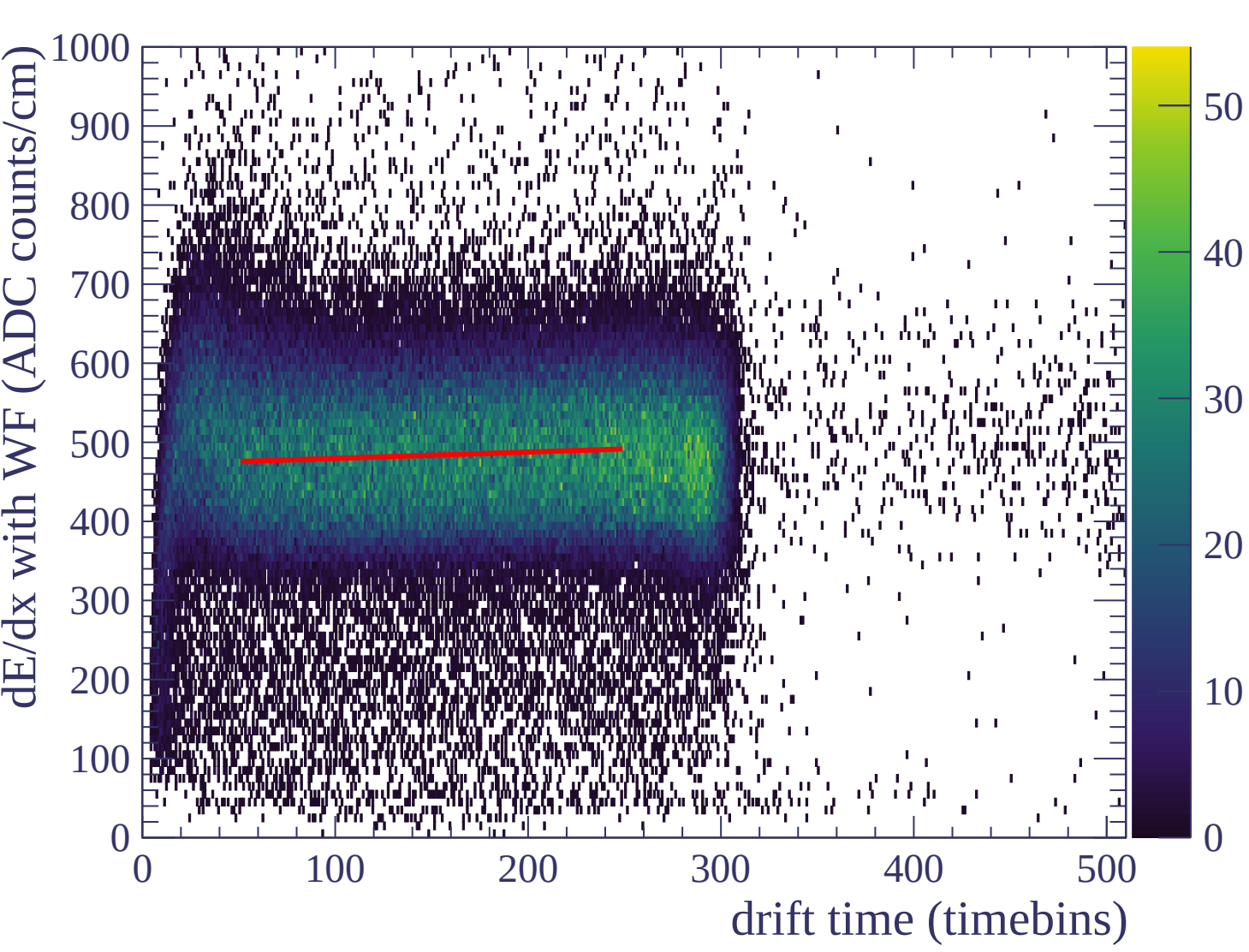


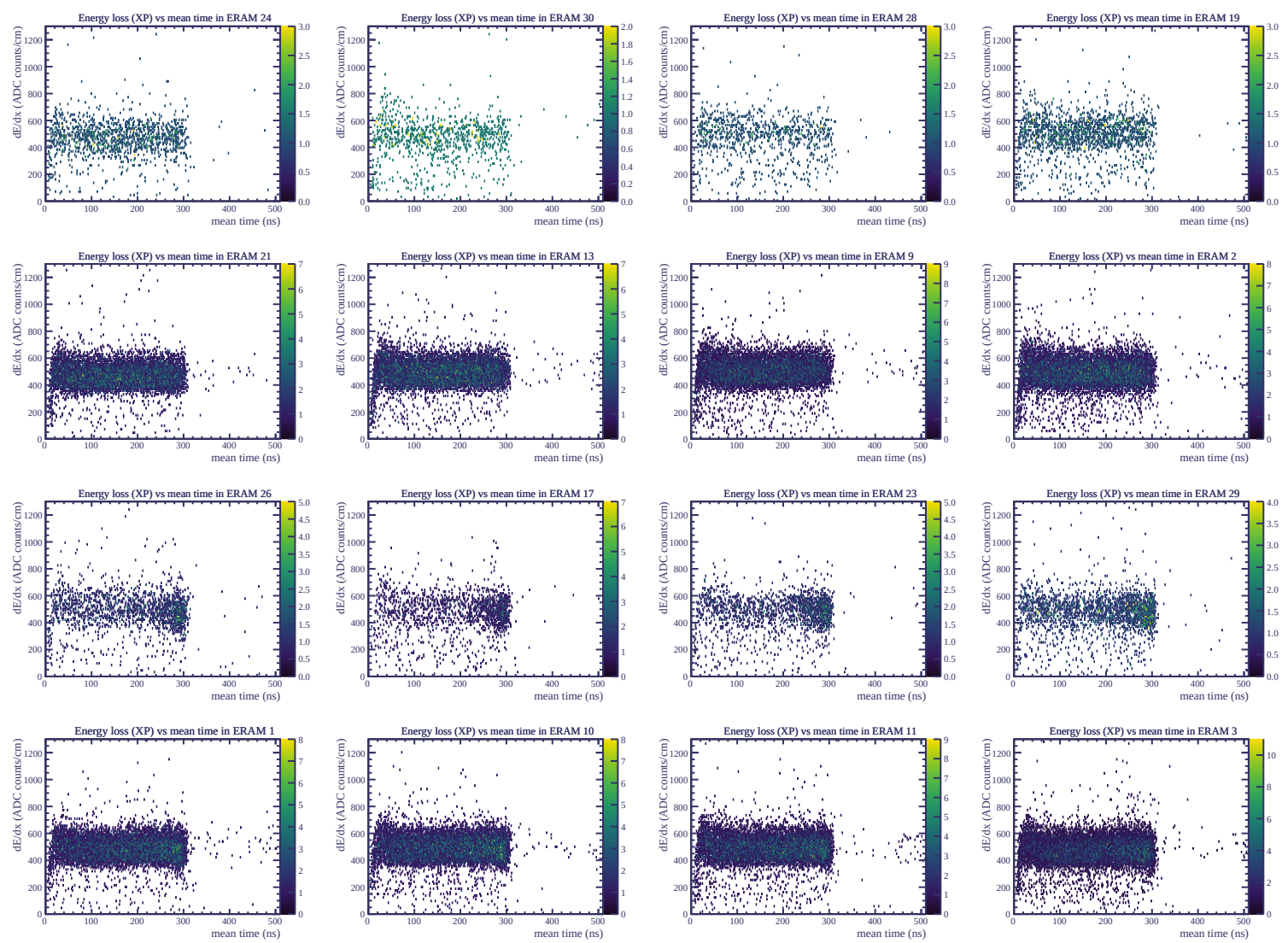


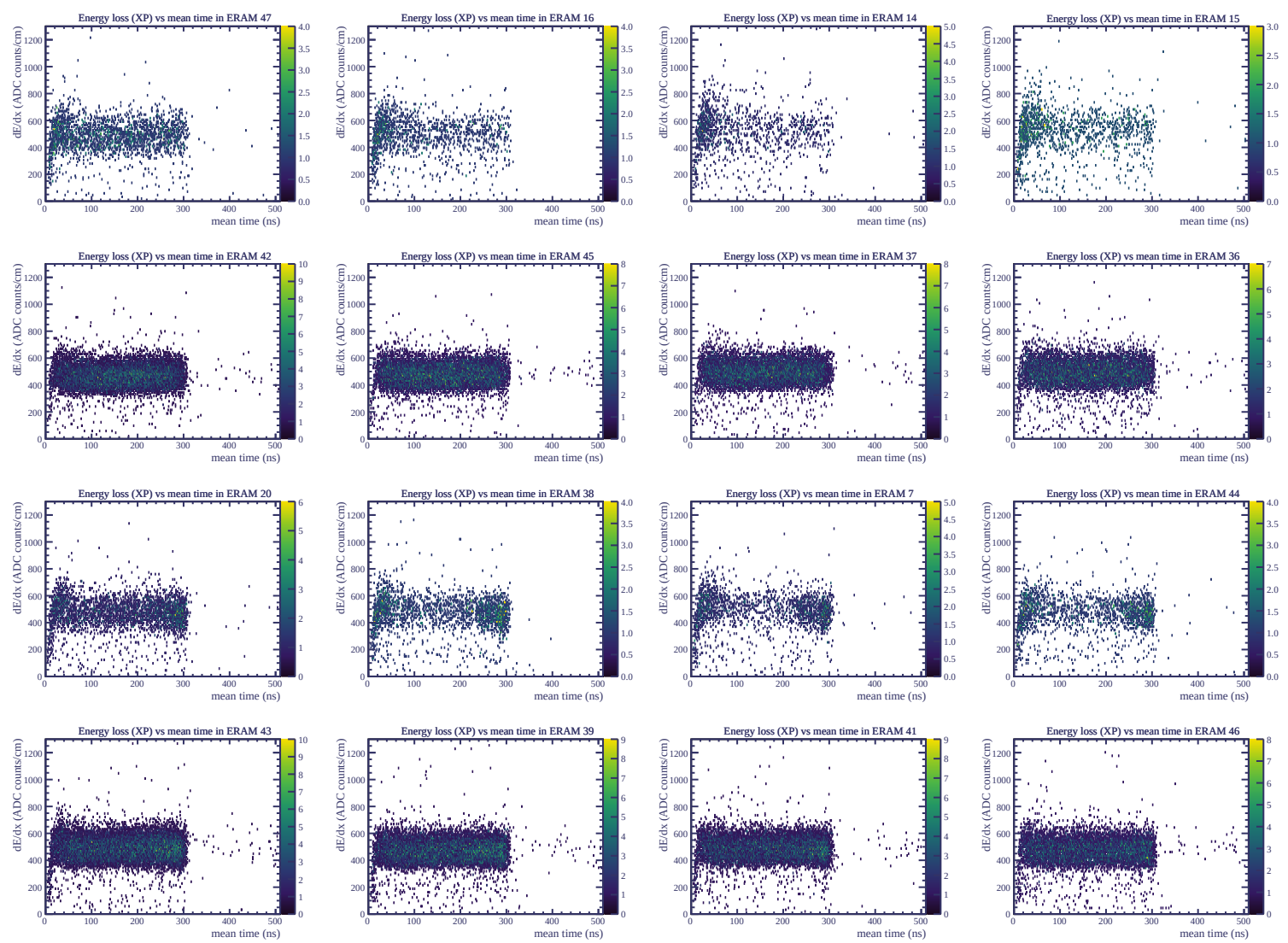




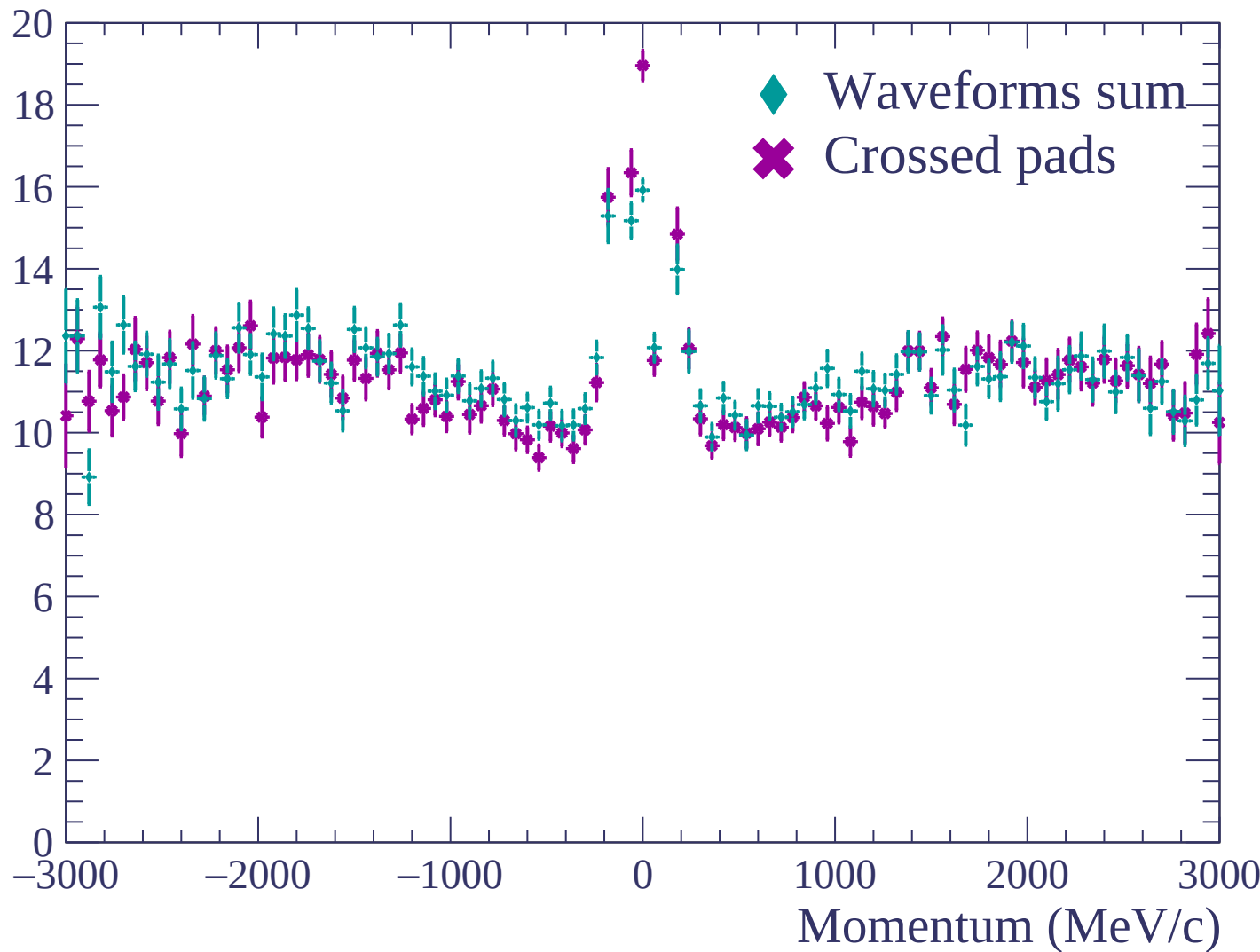


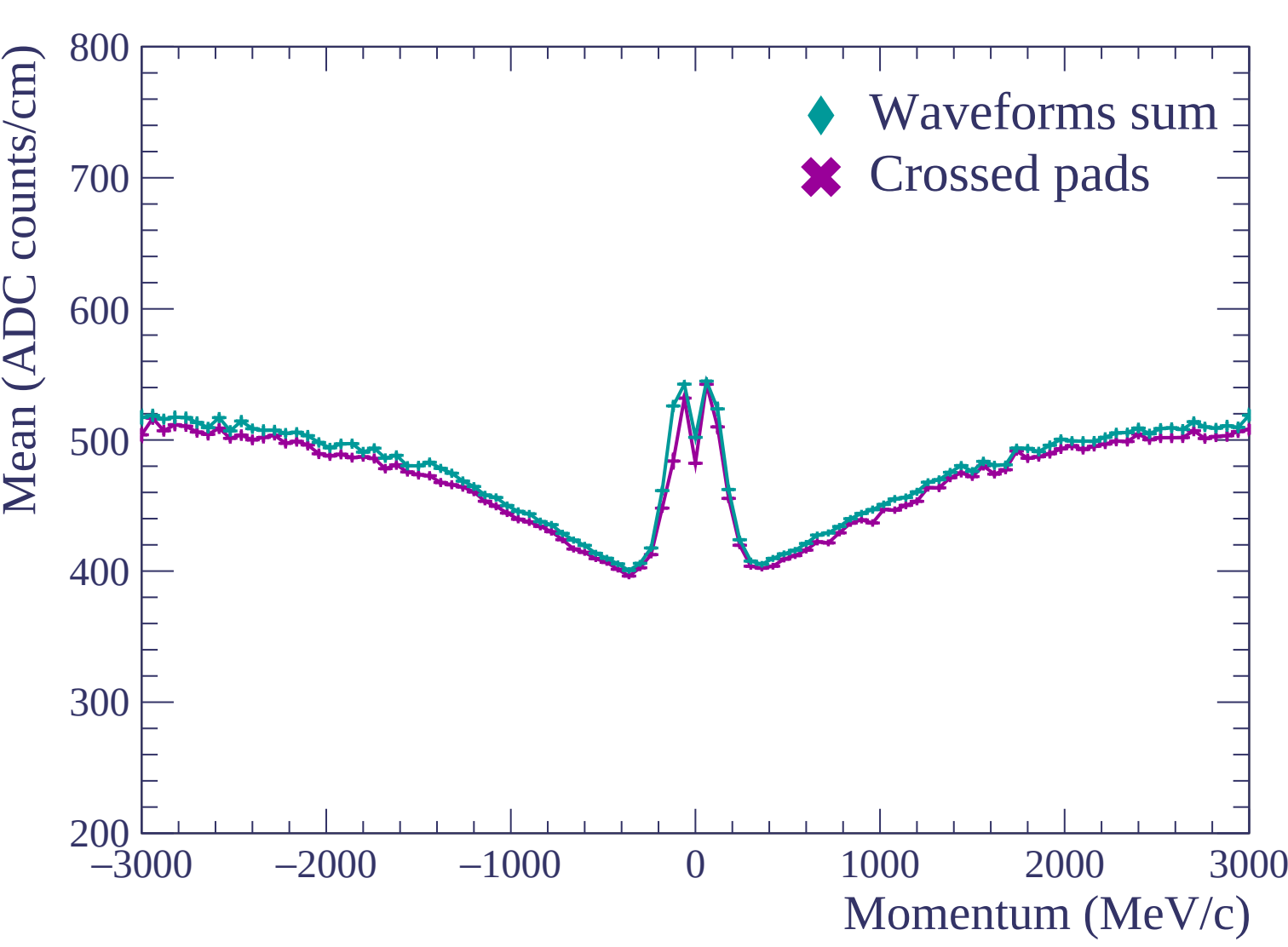


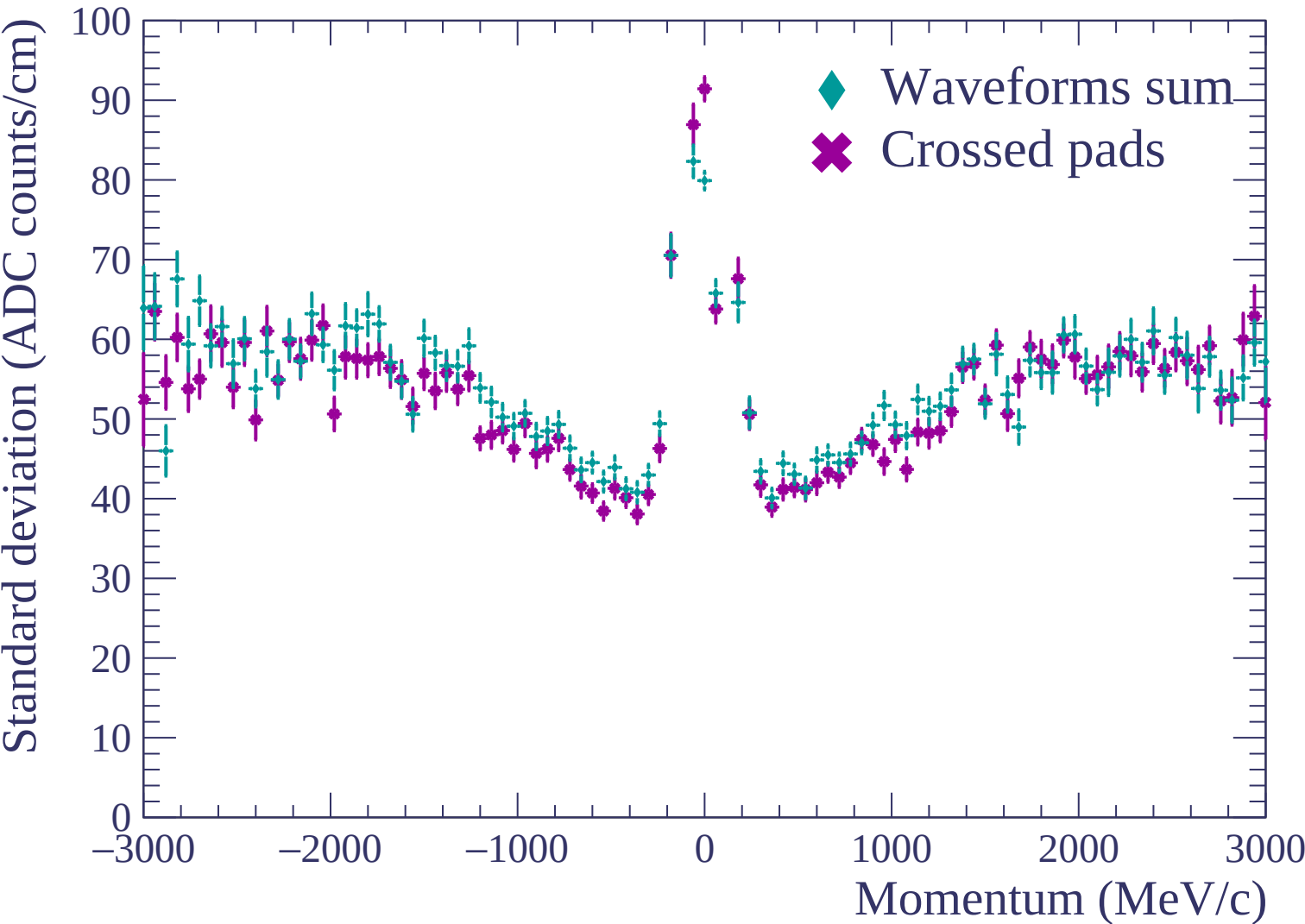


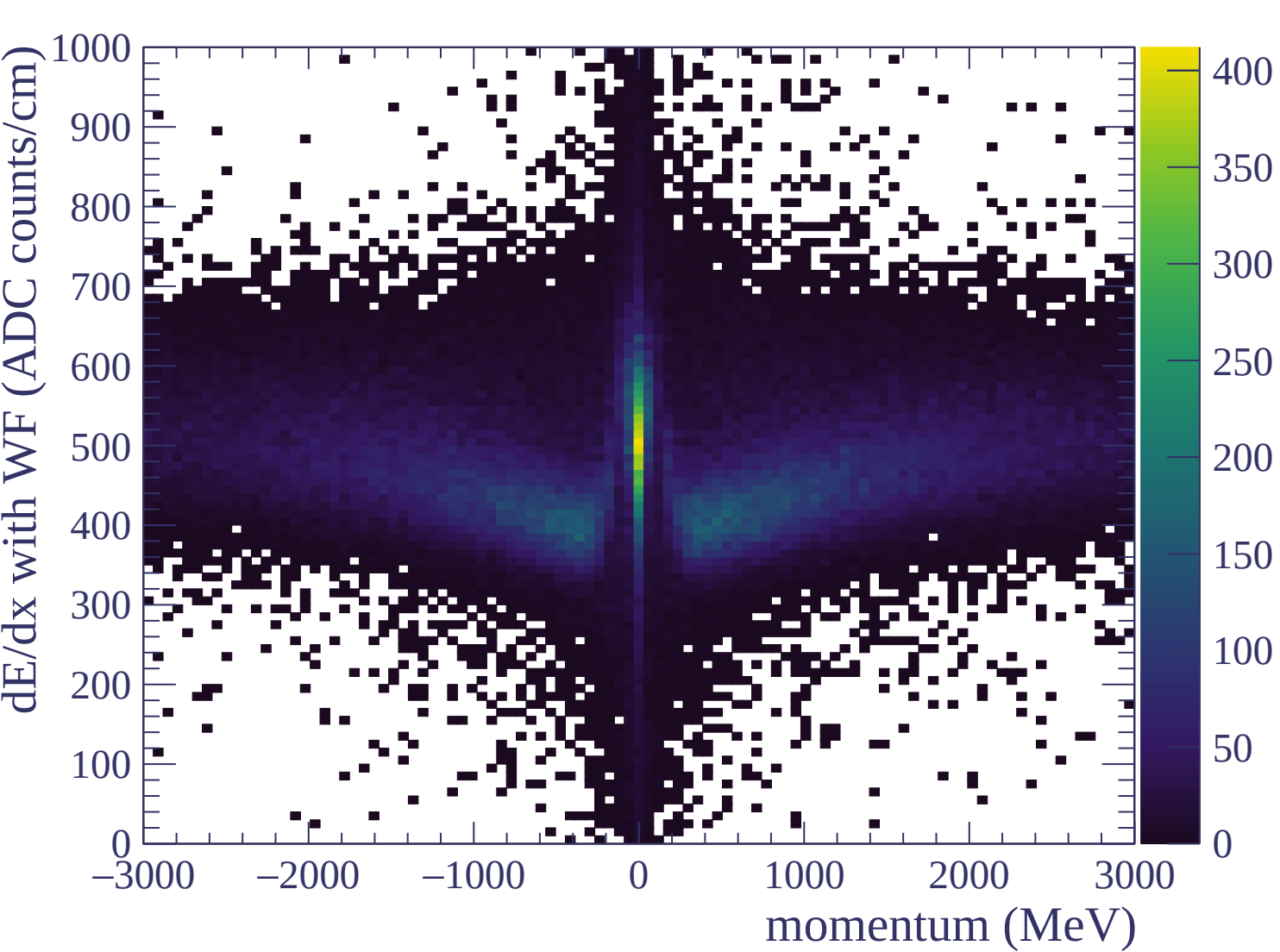


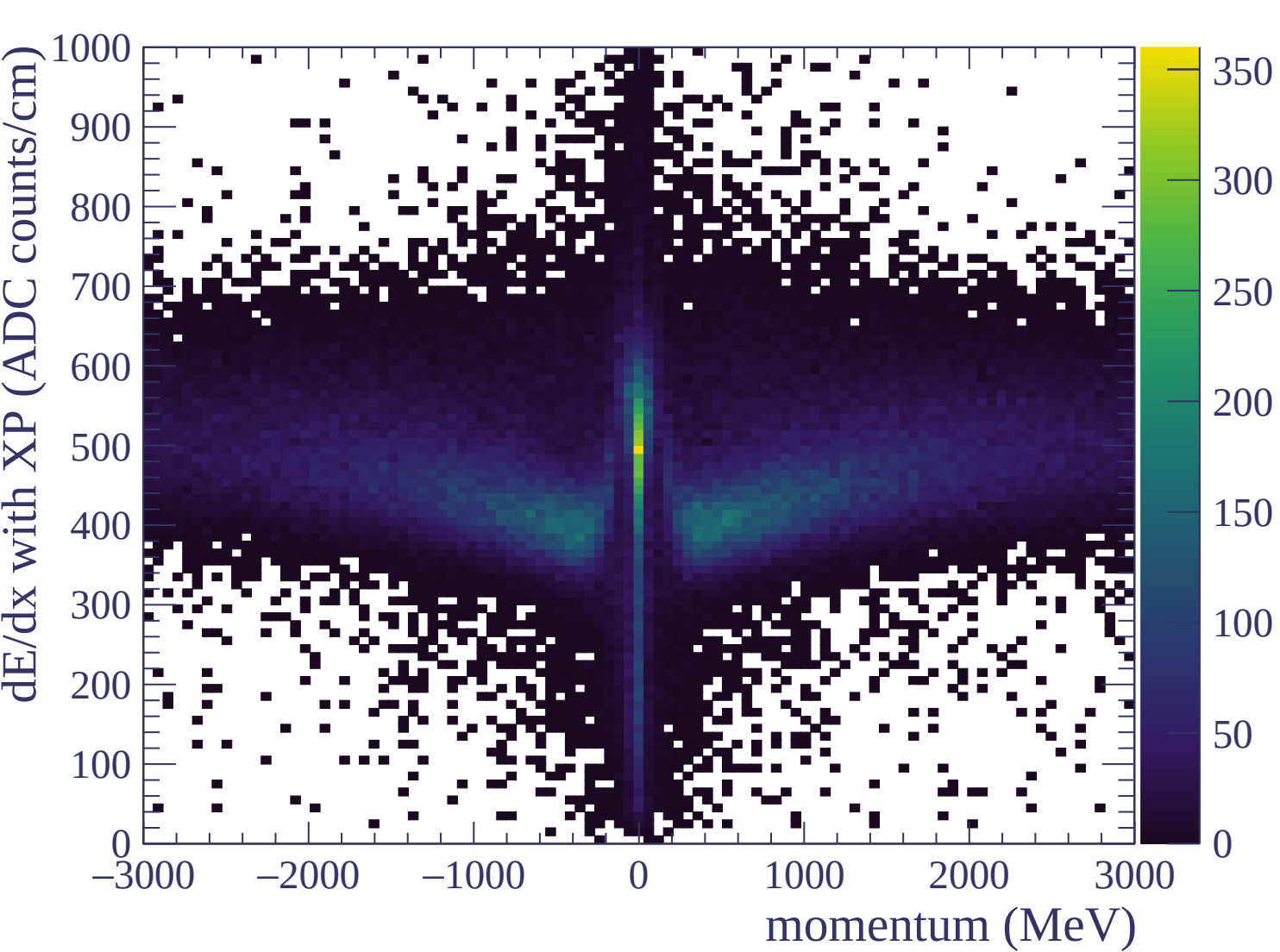
Resolution (%)

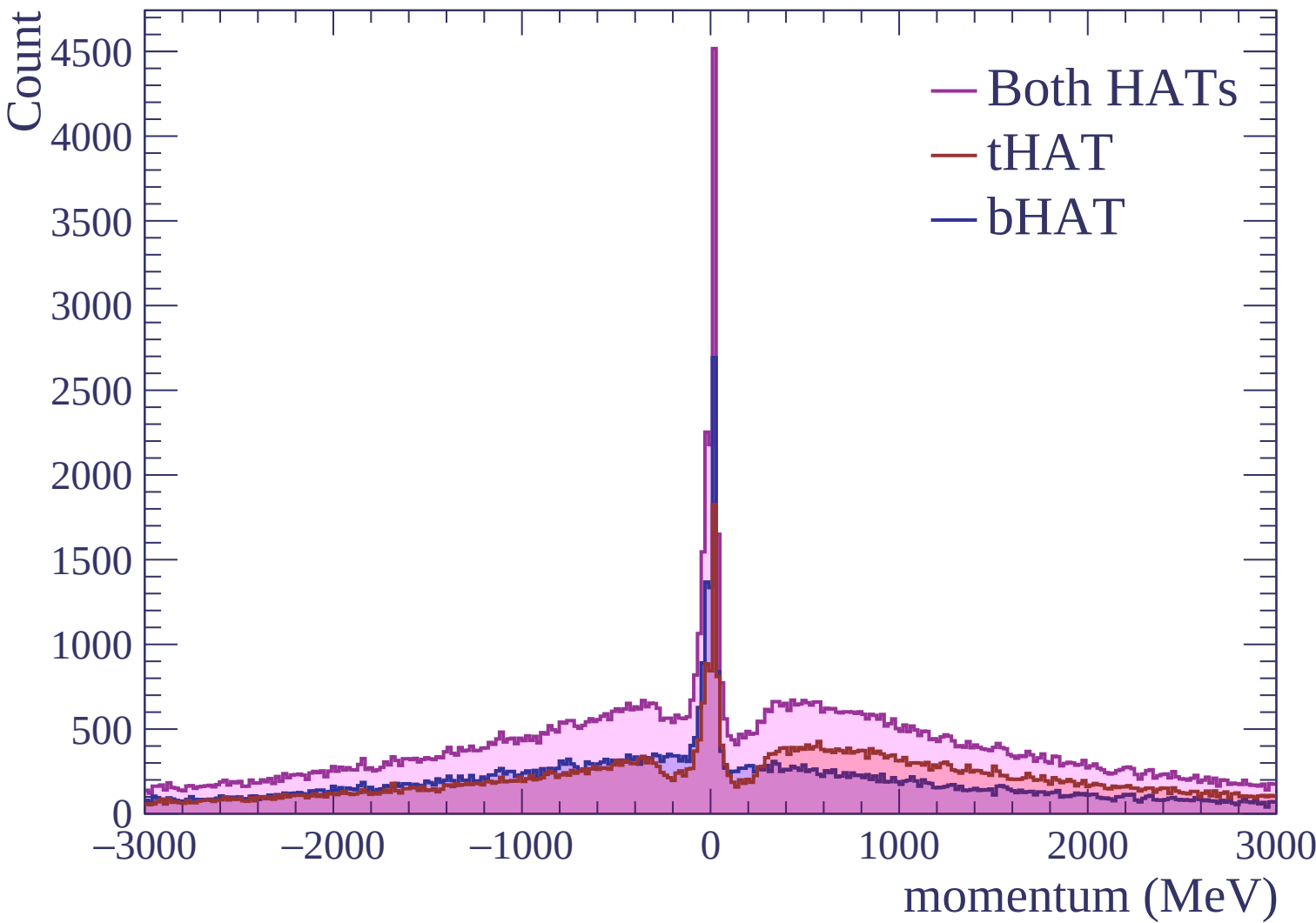


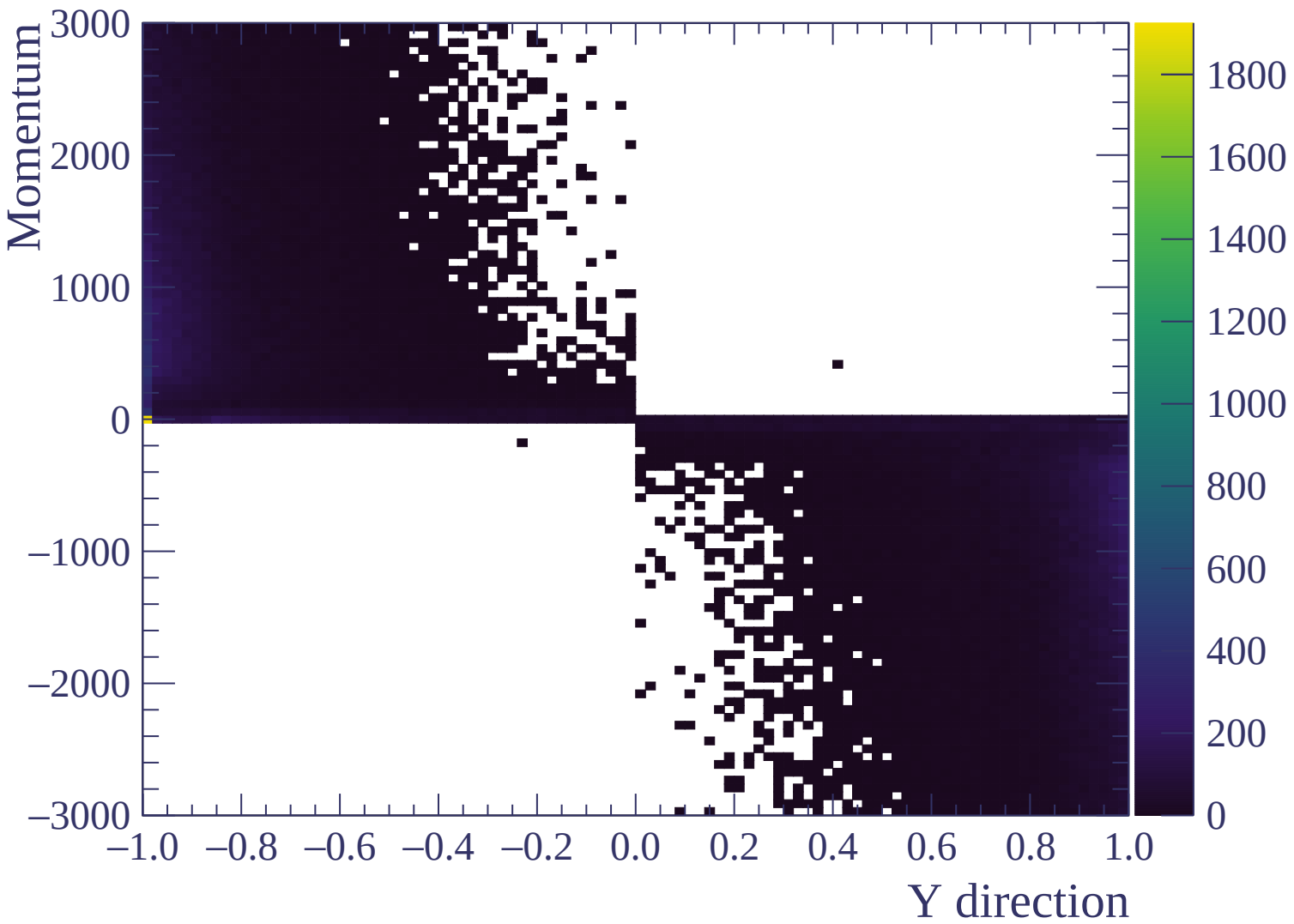


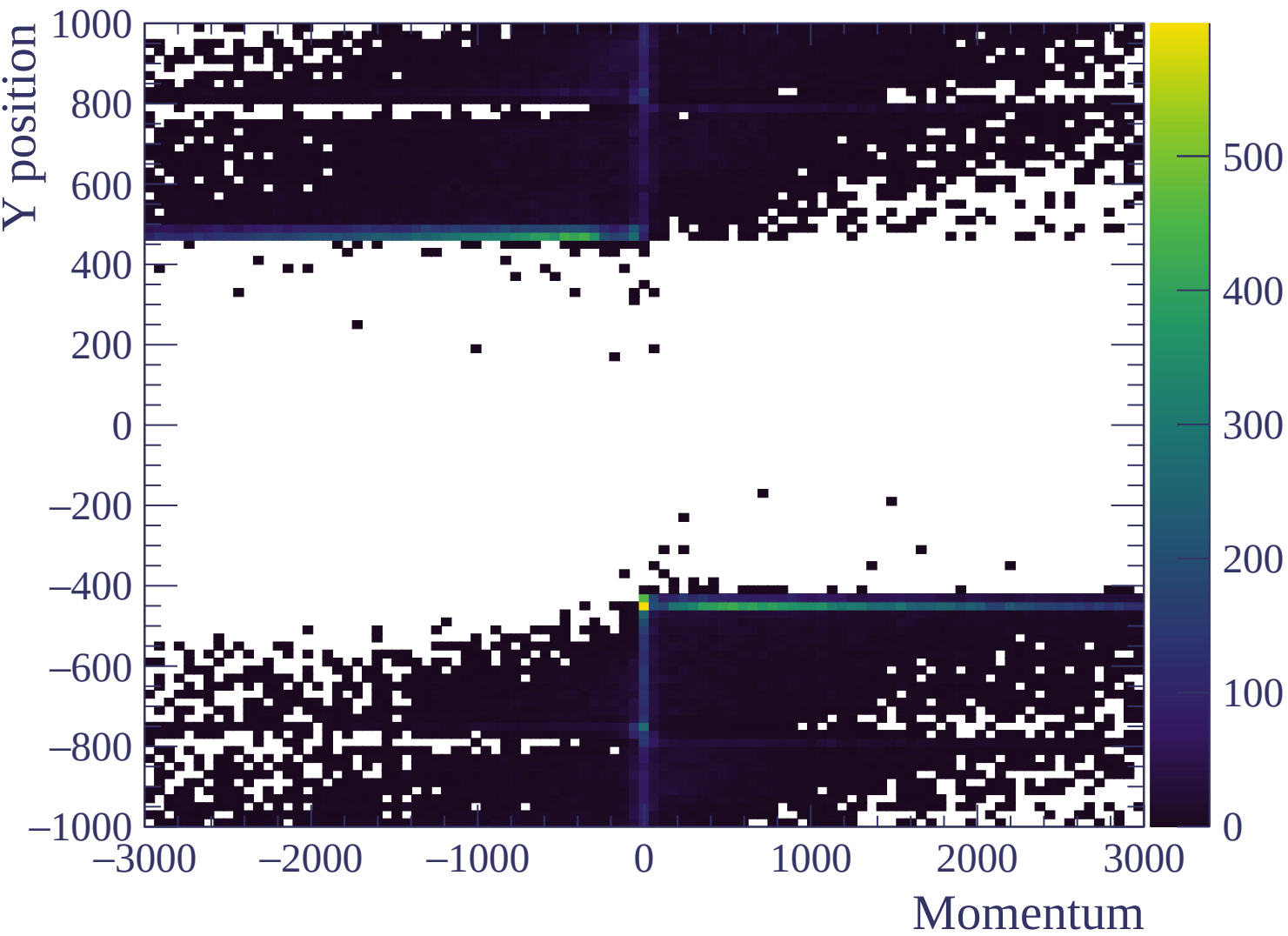


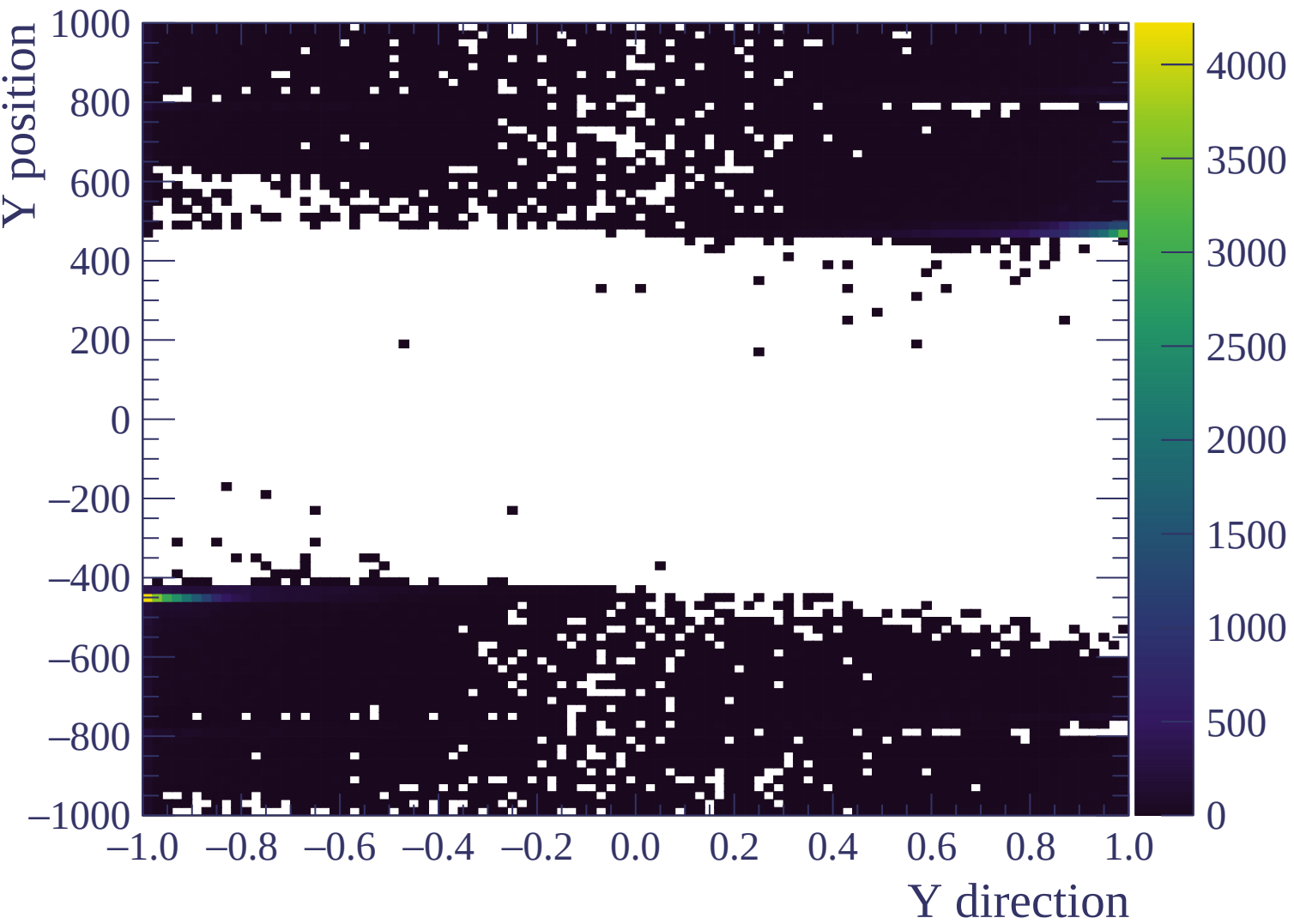






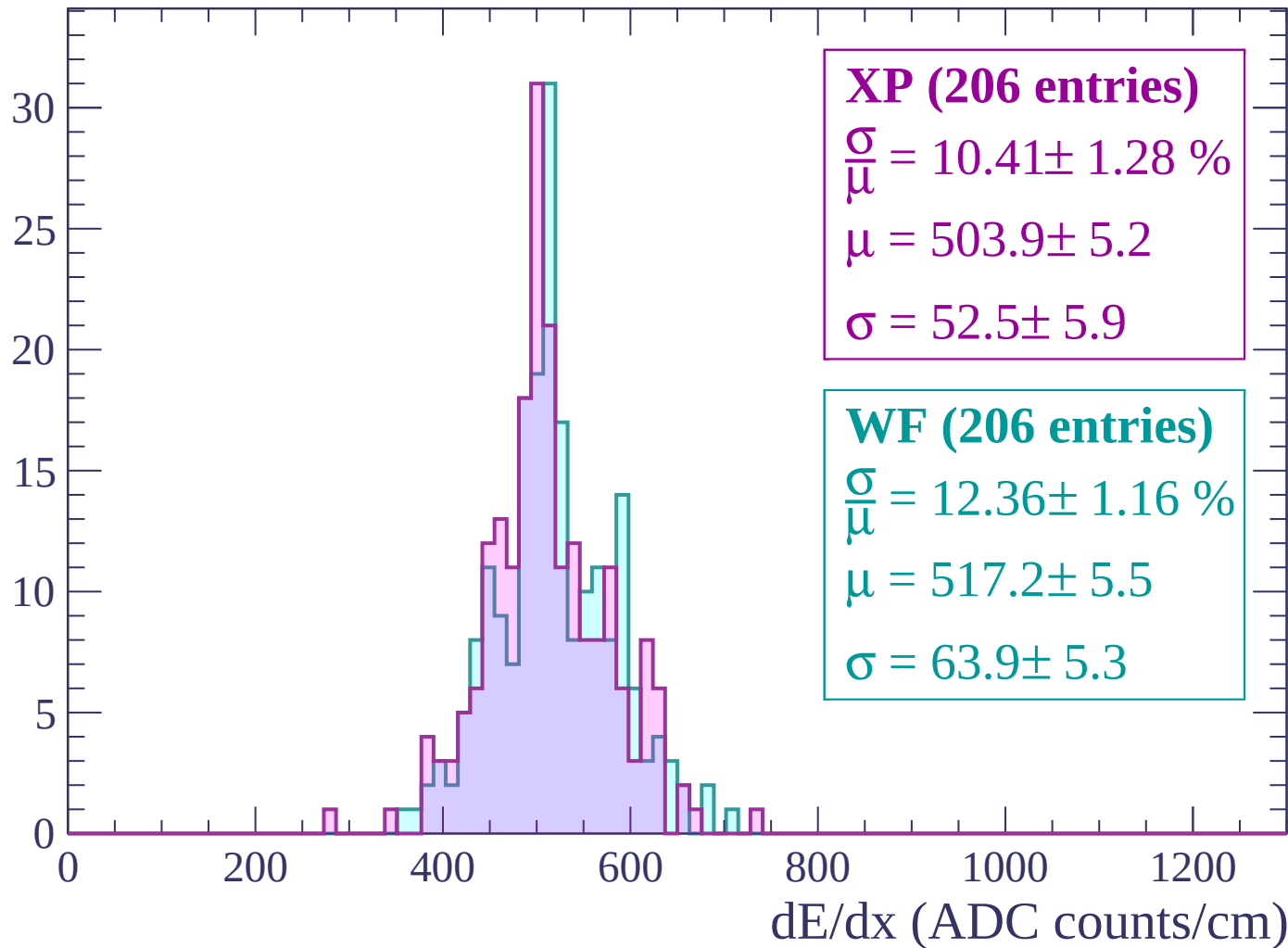






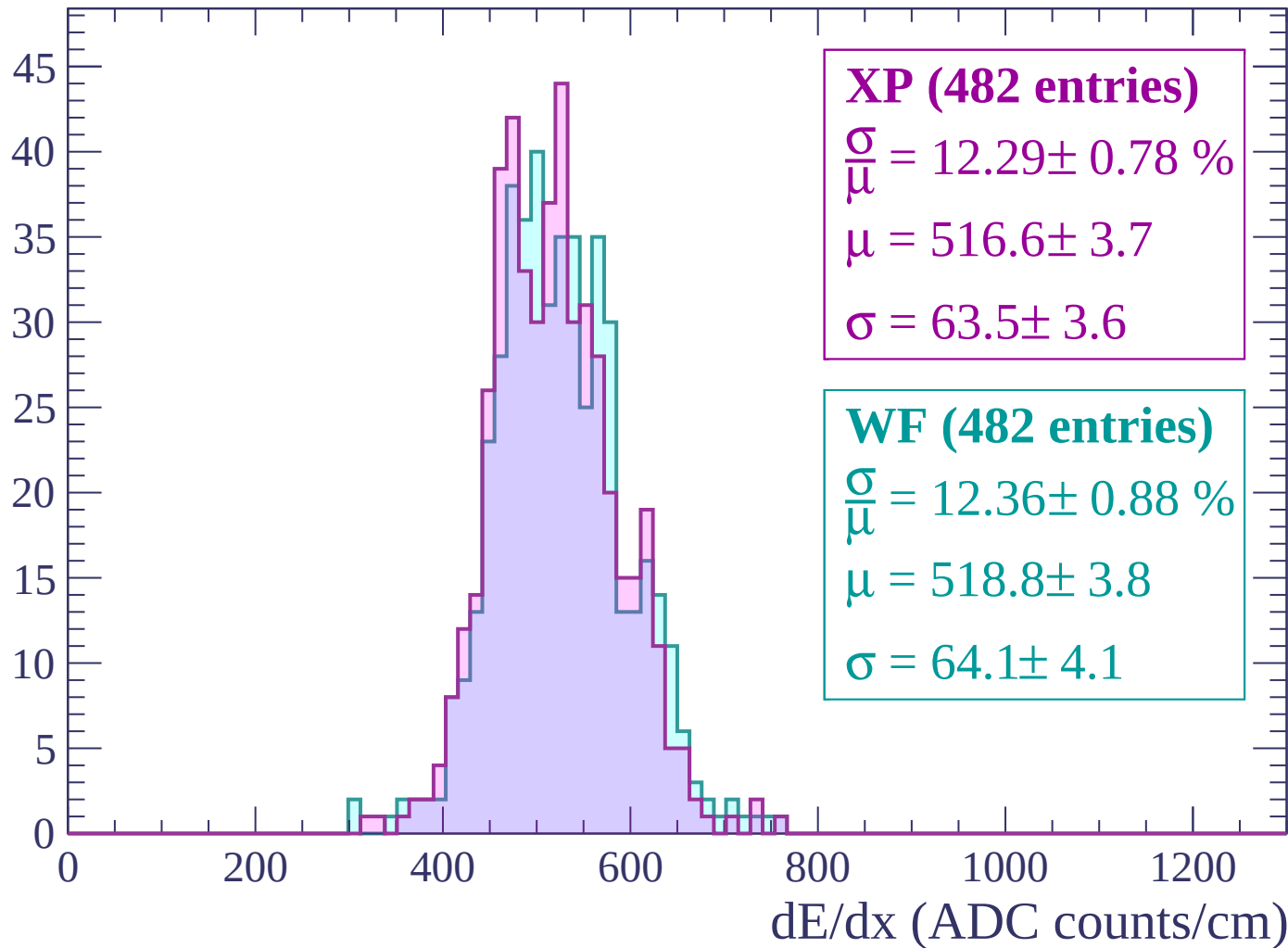
Energy loss | -3000 < p < -2940

Count



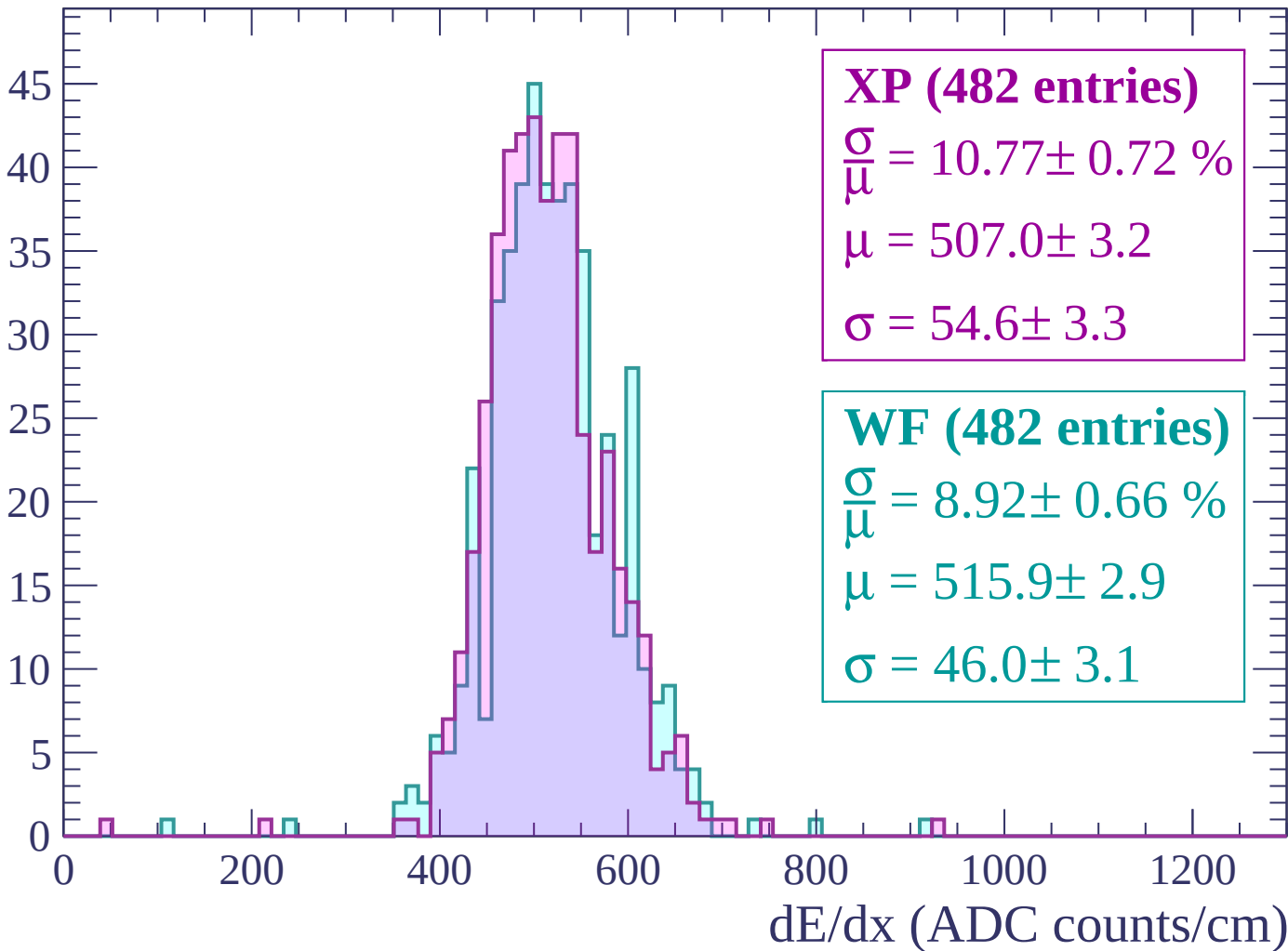
Energy loss | $-2940 < p < -2880$

Count



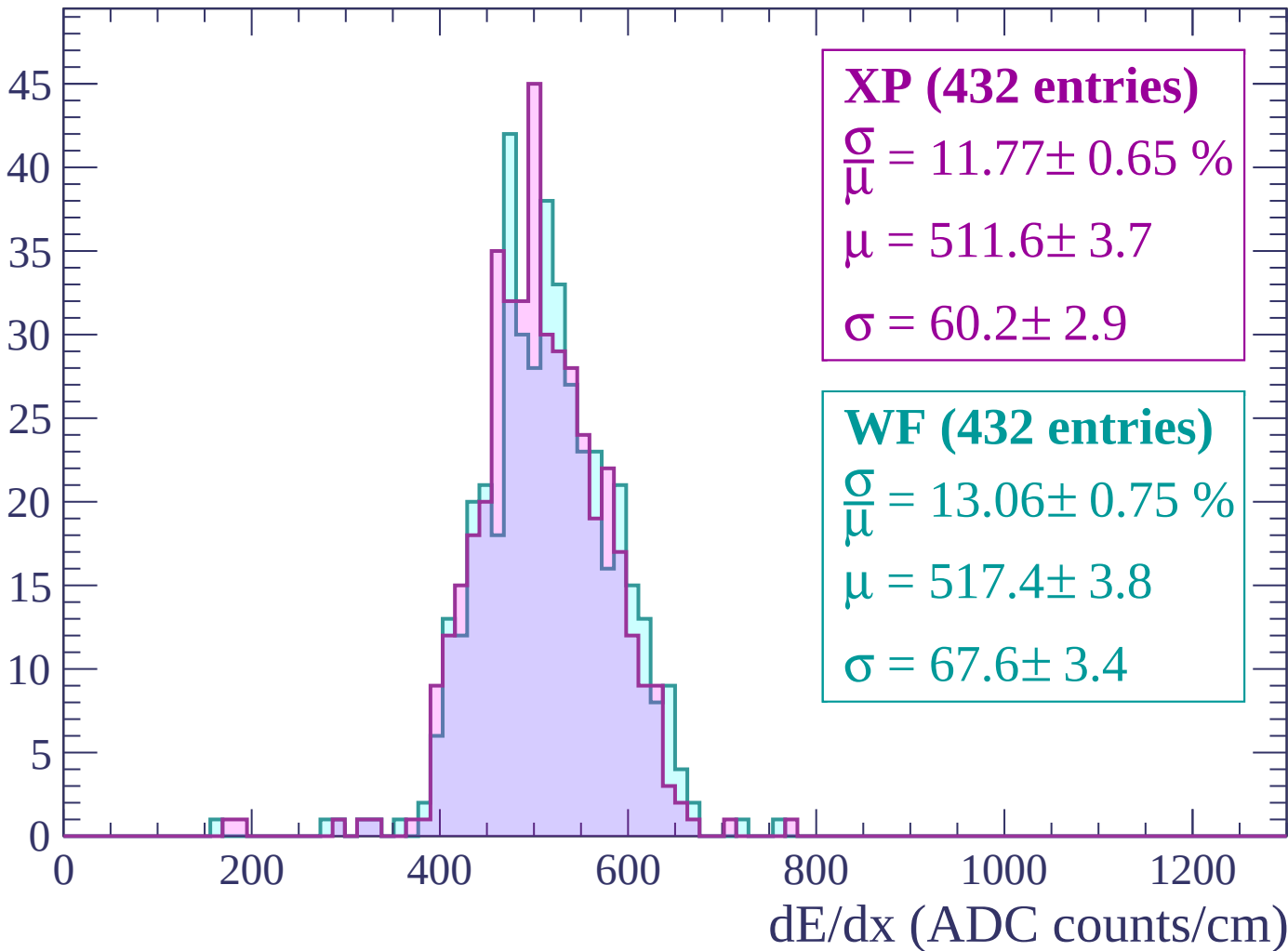
Energy loss | $-2880 < p < -2820$

Count



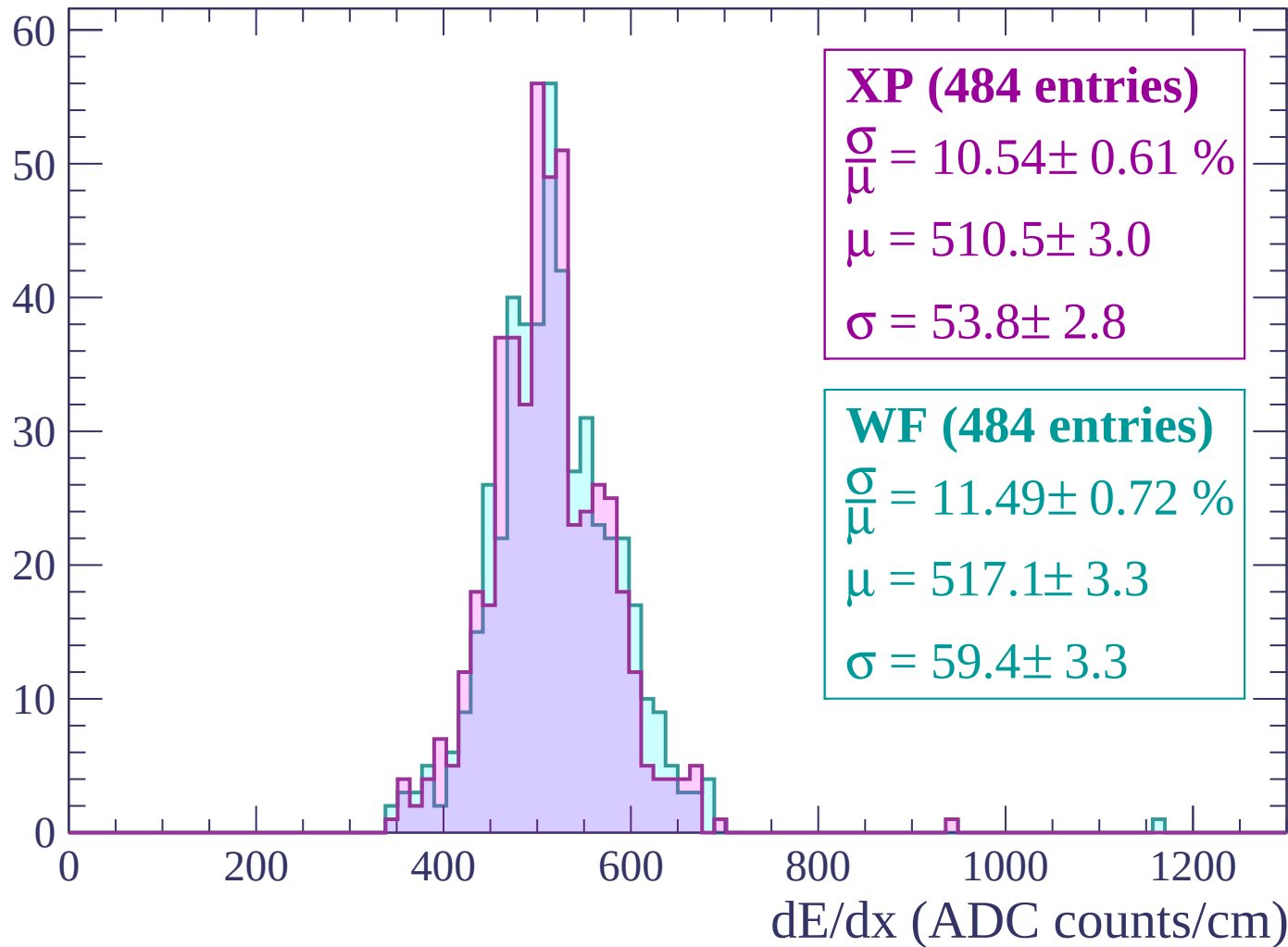
Energy loss | $-2820 < p < -2760$

Count



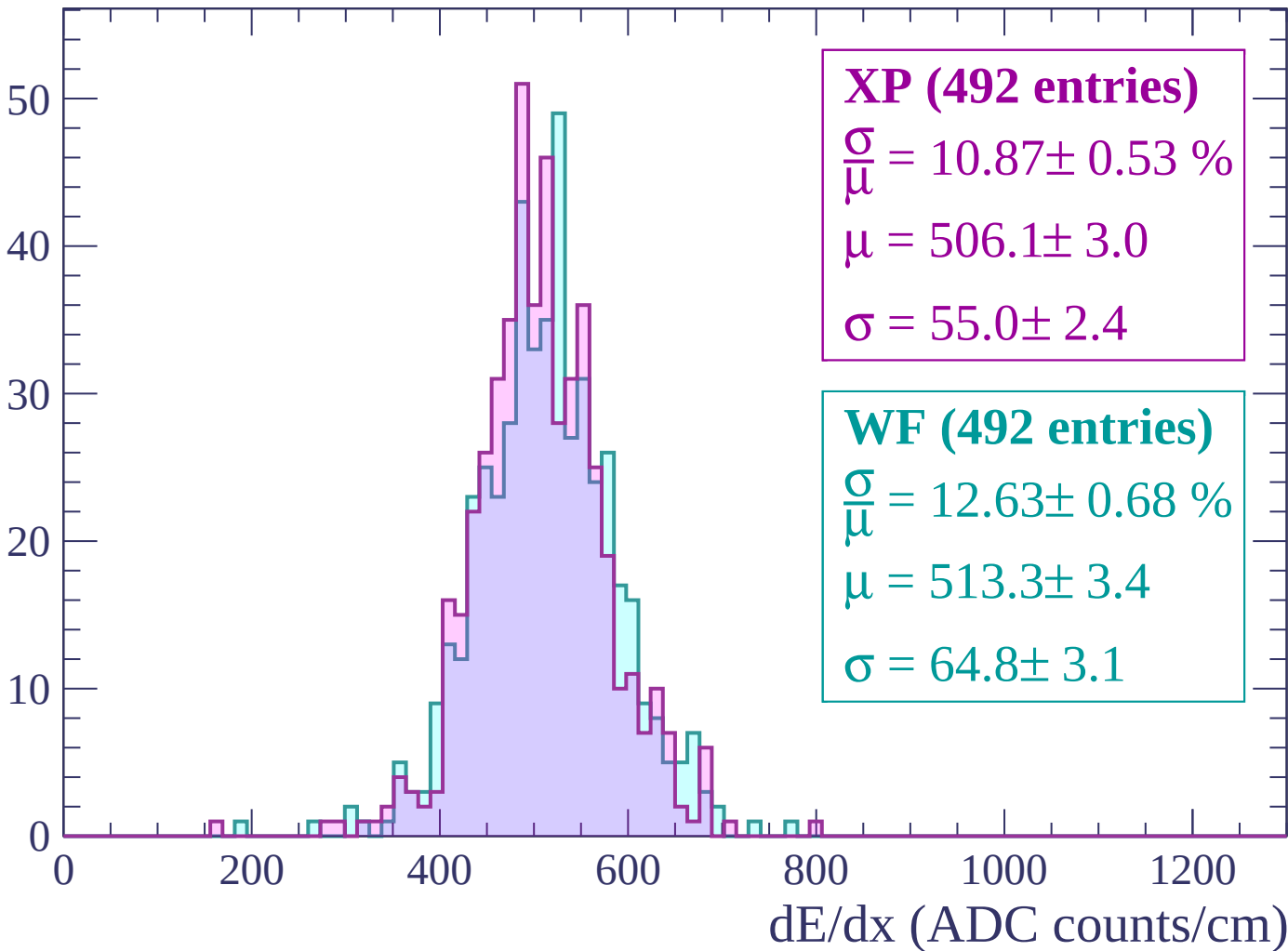
Energy loss | $-2760 < p < -2700$

Count



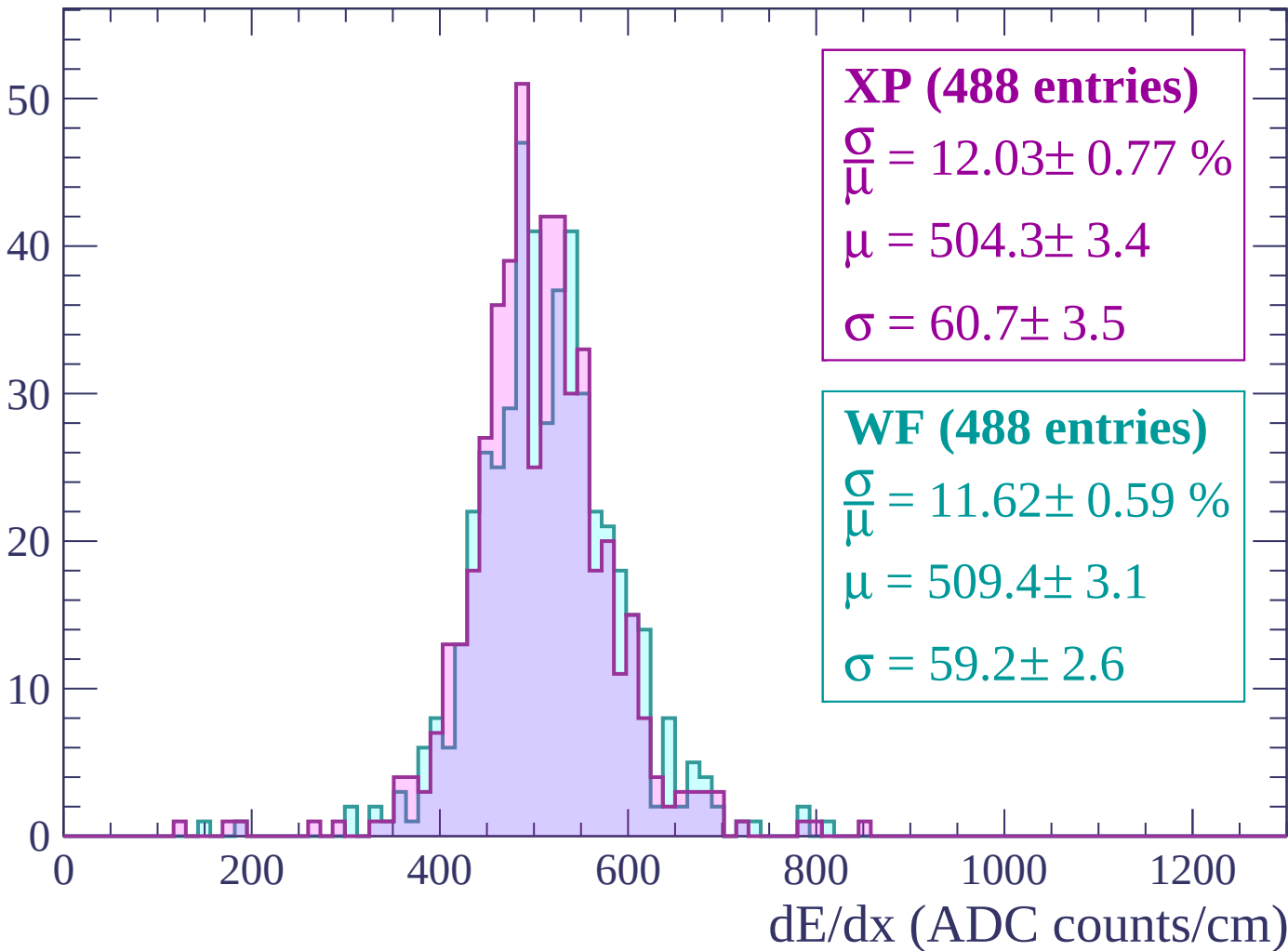
Energy loss | $-2700 < p < -2640$

Count



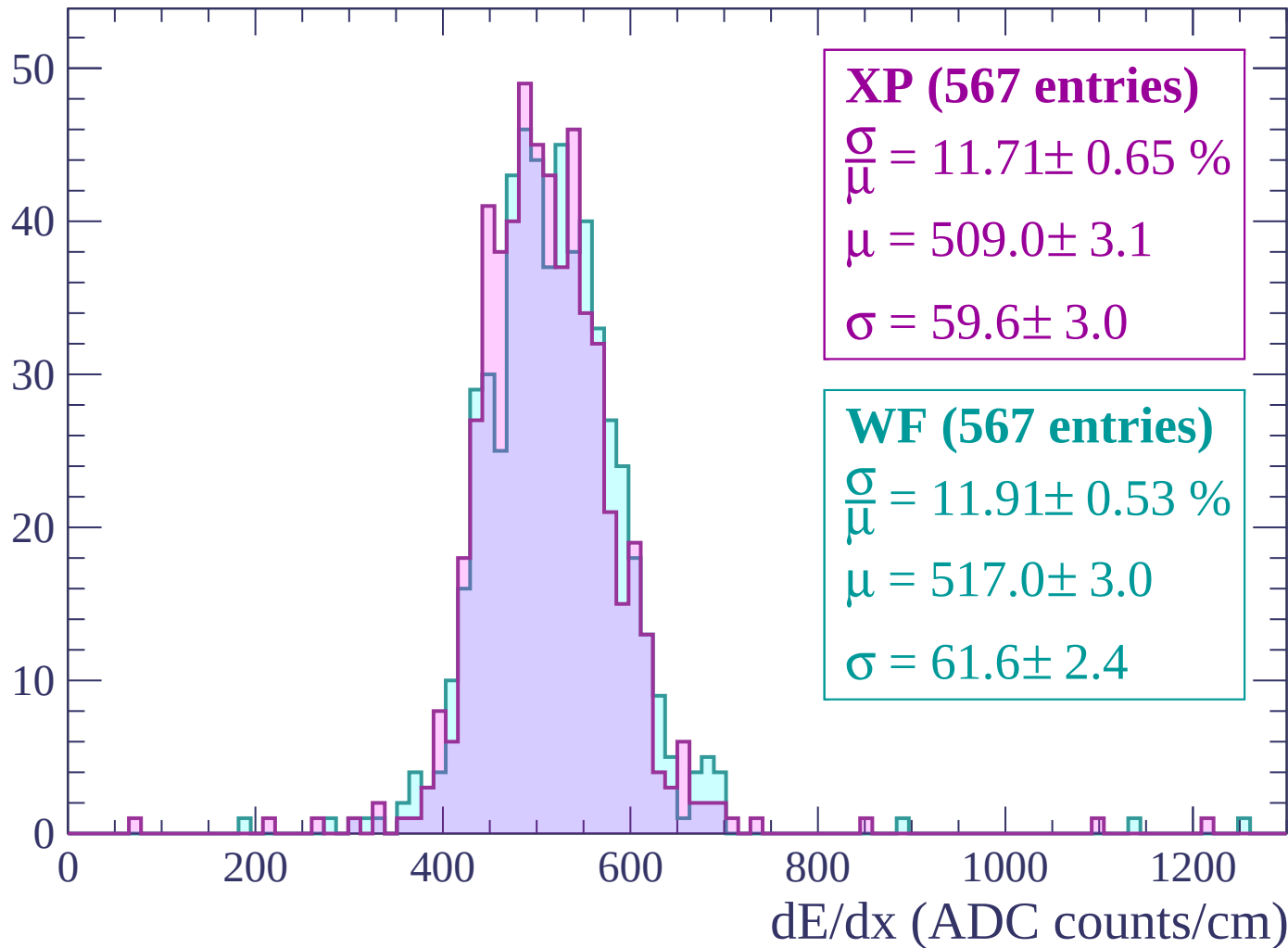
Energy loss | $-2640 < p < -2580$

Count



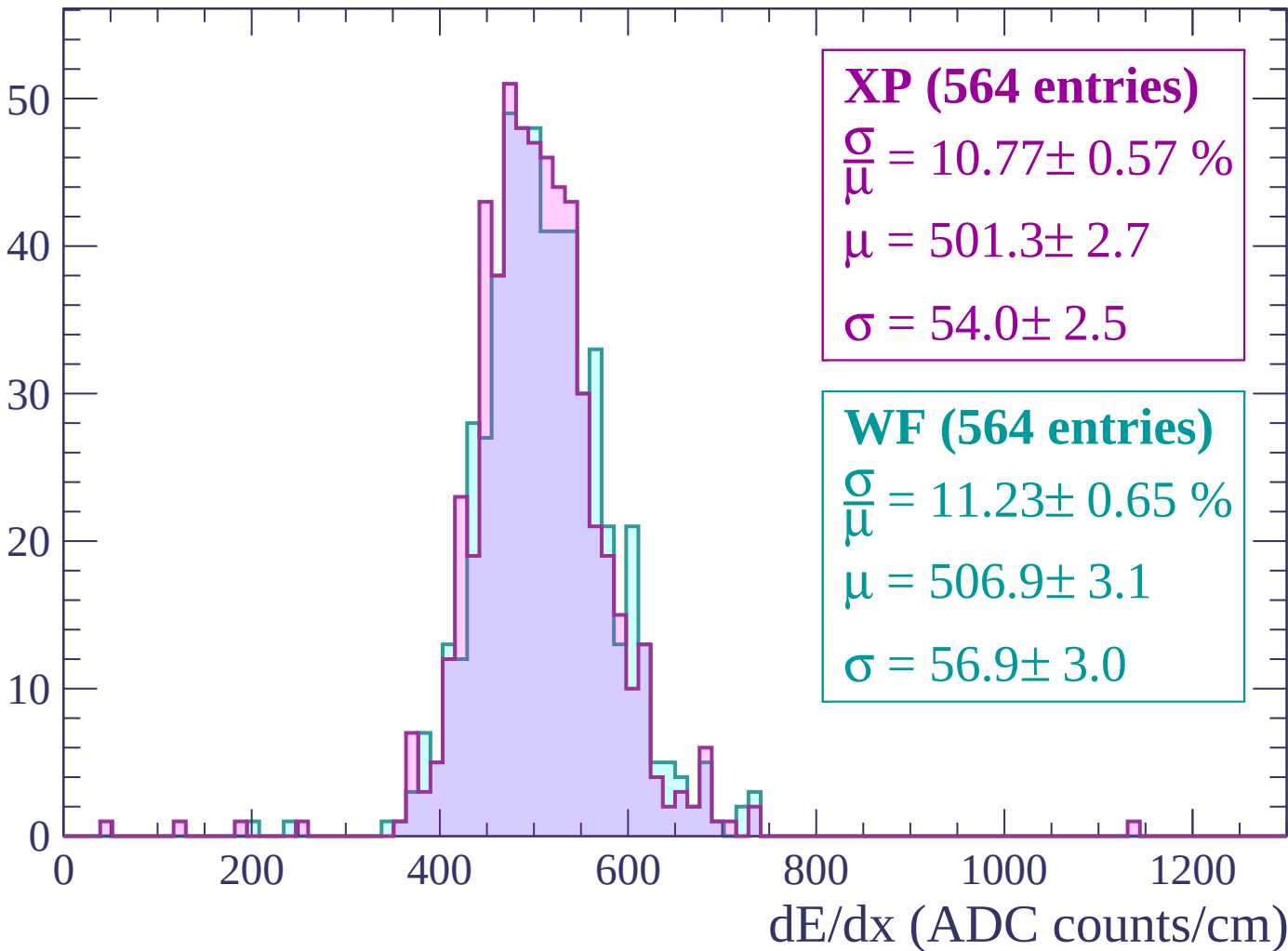
Energy loss | -2580 < p < -2520

Count



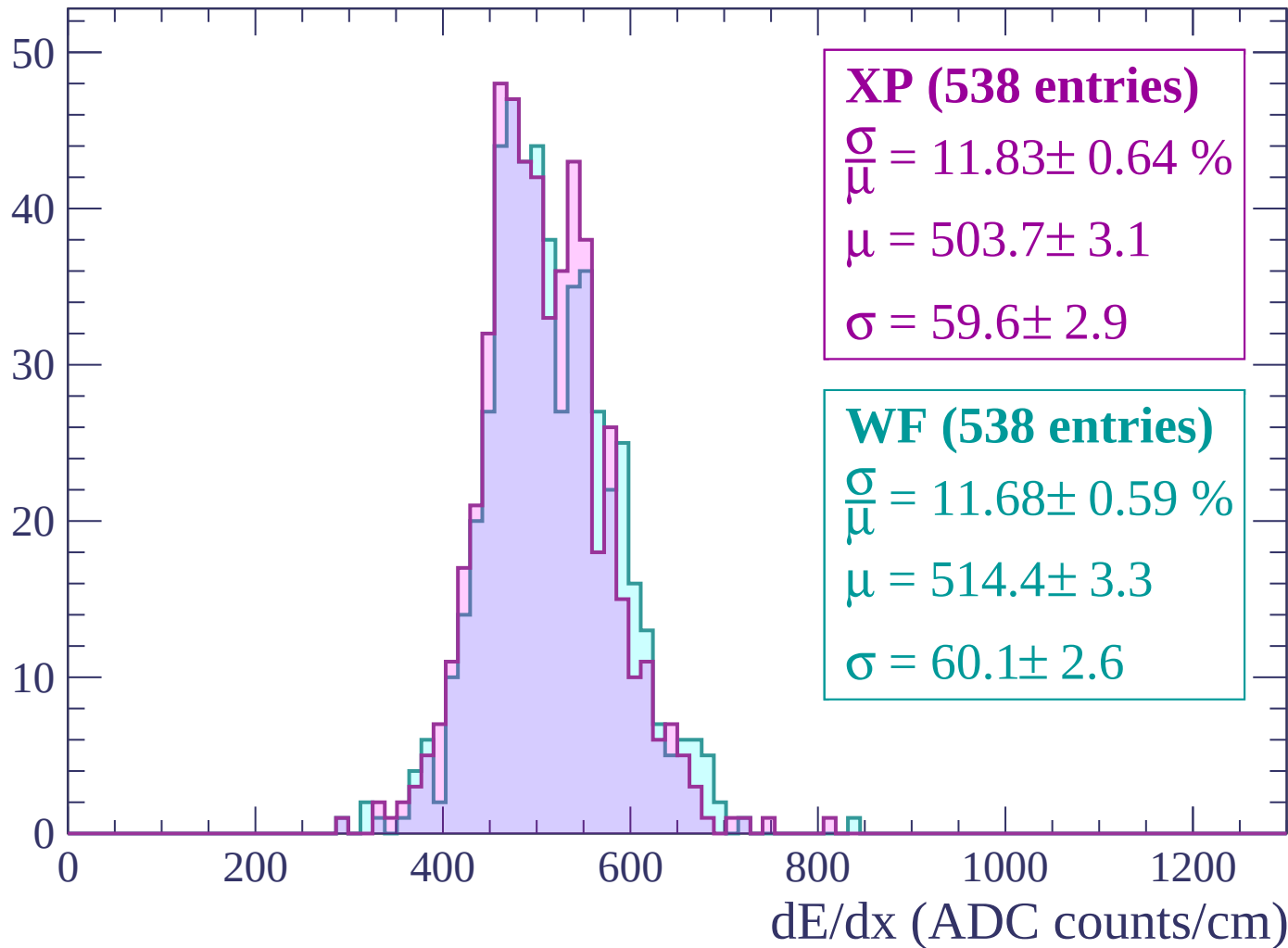
Energy loss | $-2520 < p < -2460$

Count



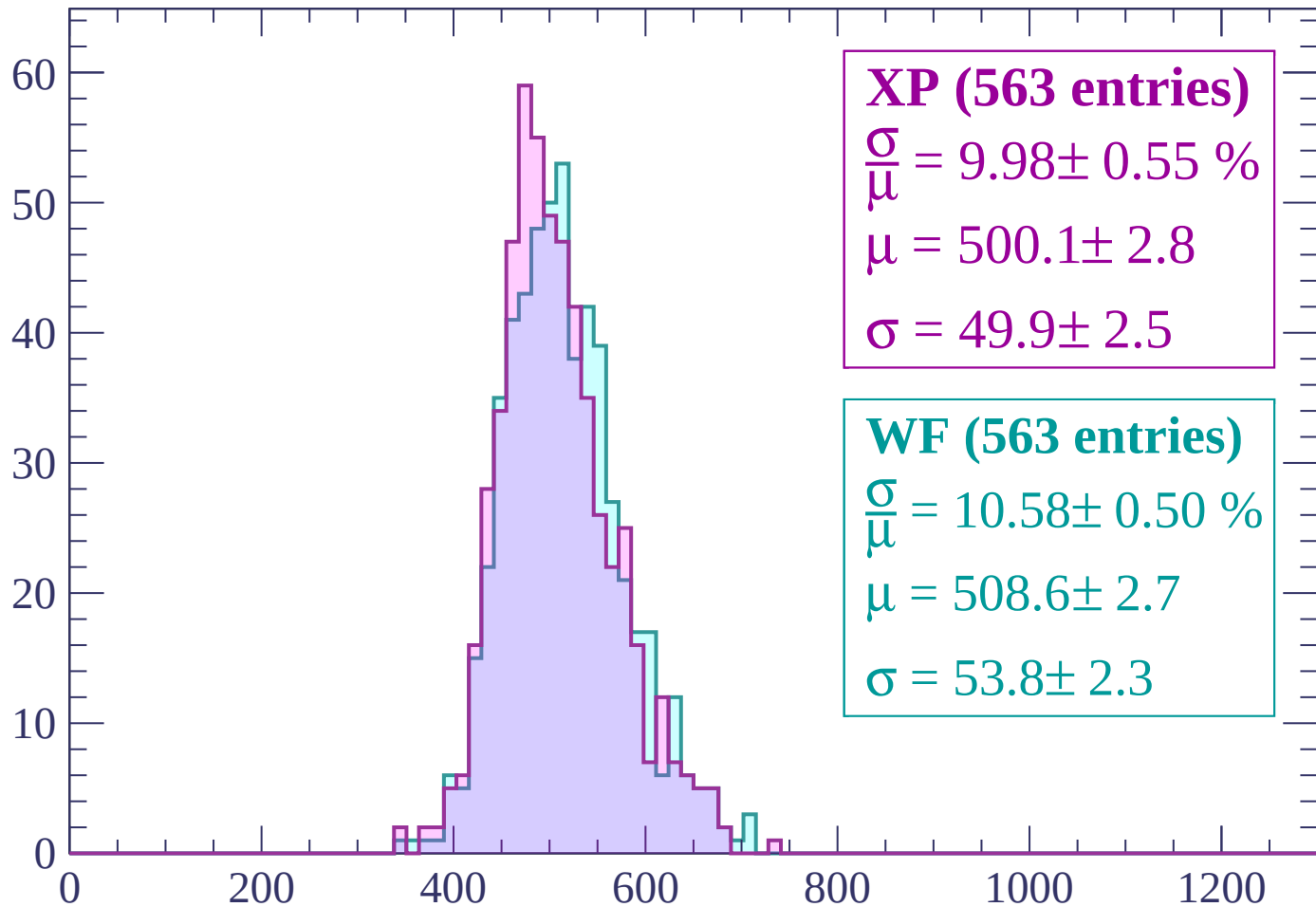
Energy loss | $-2460 < p < -2400$

Count



Energy loss | $-2400 < p < -2340$

Count



XP (563 entries)

$$\frac{\sigma}{\mu} = 9.98 \pm 0.55 \%$$

$$\mu = 500.1 \pm 2.8$$

$$\sigma = 49.9 \pm 2.5$$

WF (563 entries)

$$\frac{\sigma}{\mu} = 10.58 \pm 0.50 \%$$

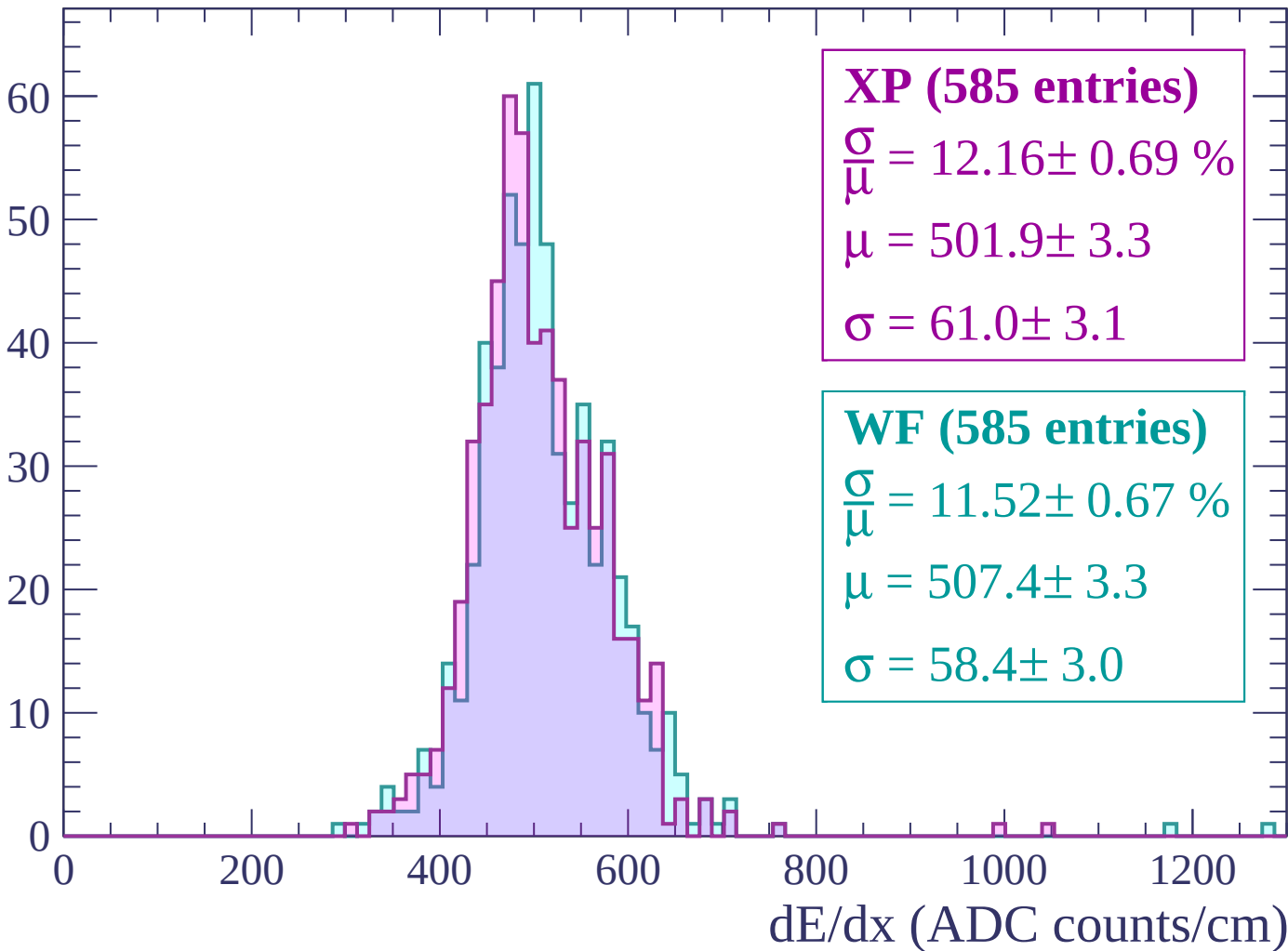
$$\mu = 508.6 \pm 2.7$$

$$\sigma = 53.8 \pm 2.3$$

dE/dx (ADC counts/cm)

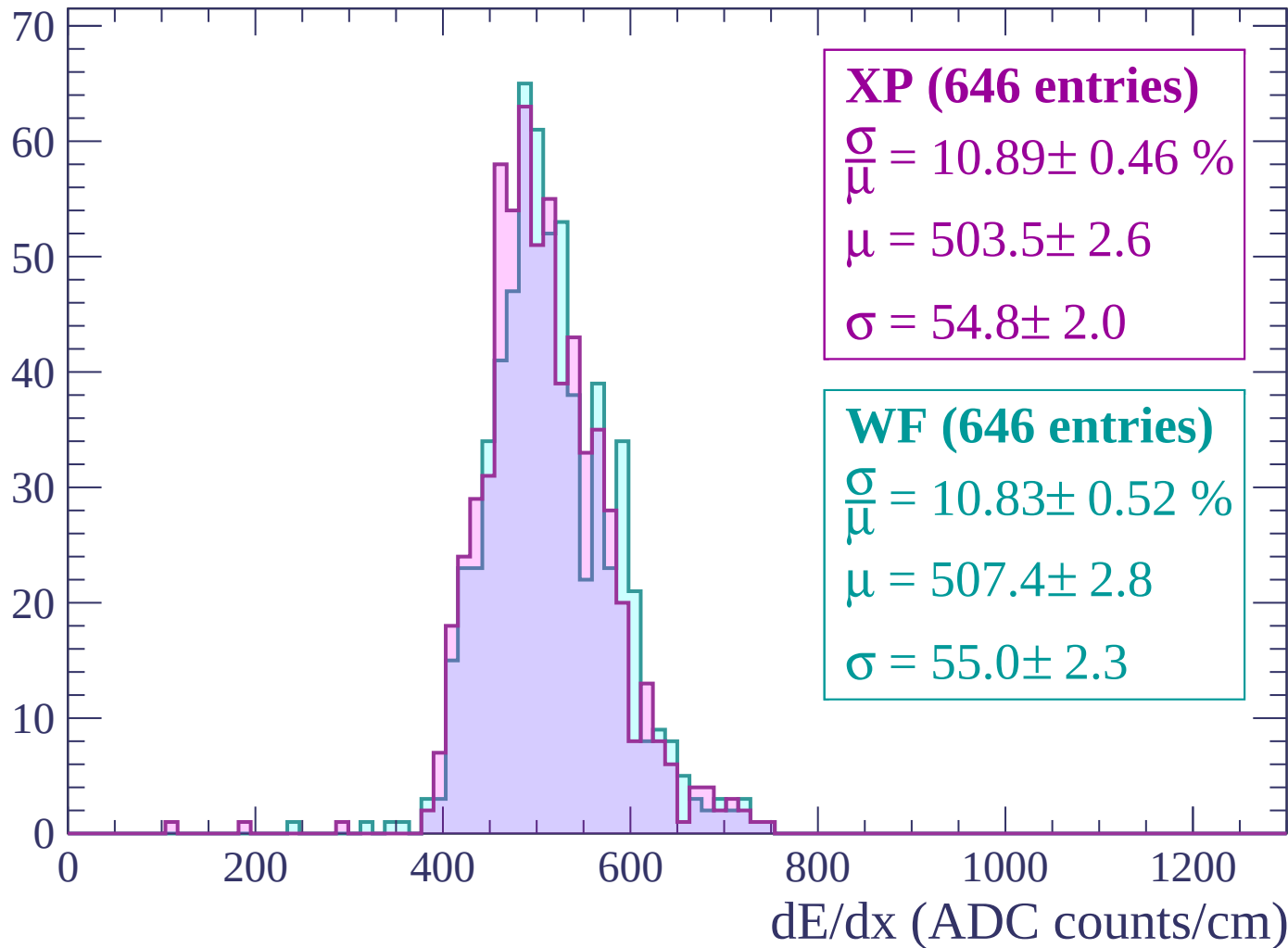
Energy loss | $-2340 < p < -2280$

Count



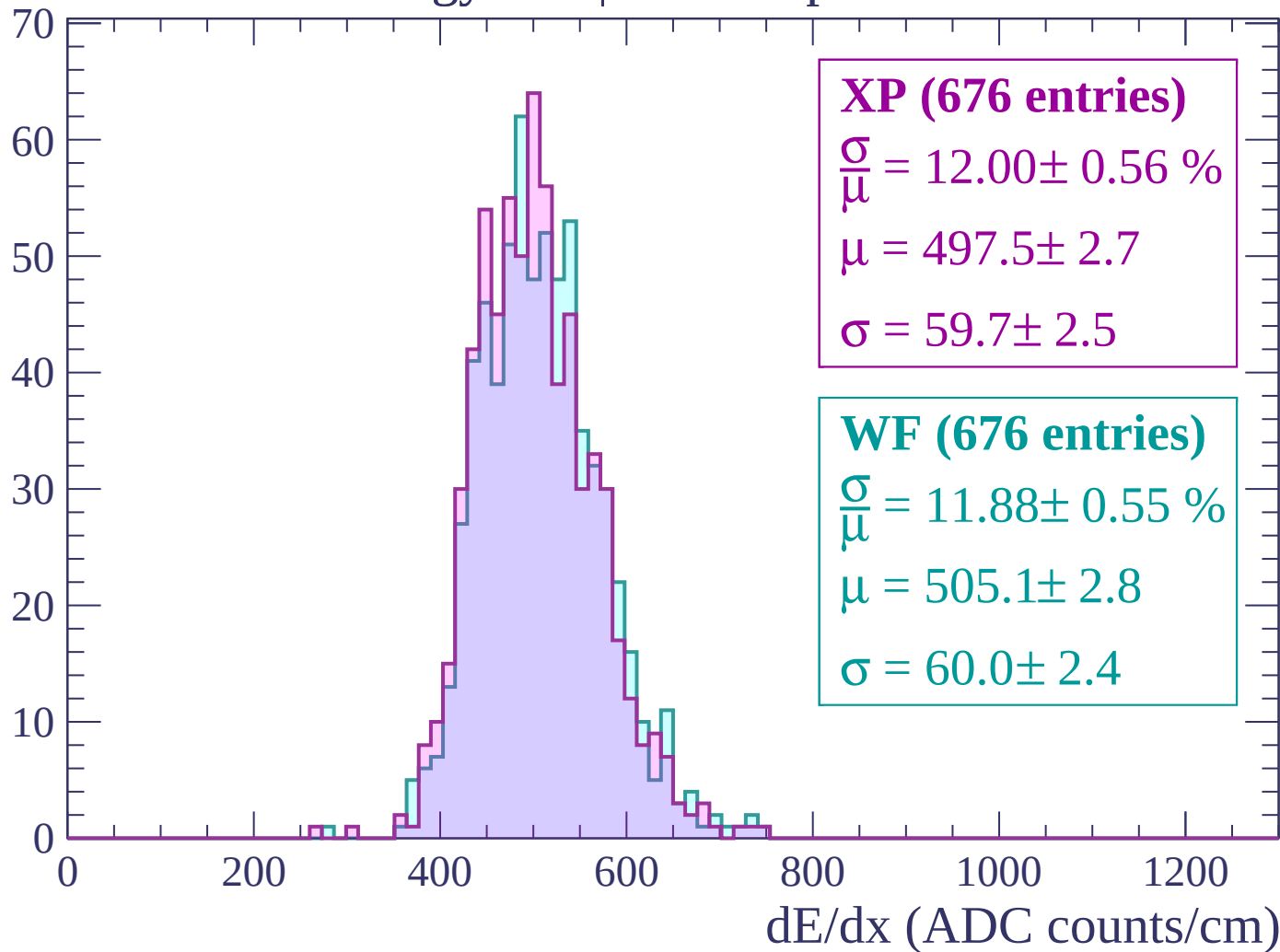
Energy loss | $-2280 < p < -2220$

Count



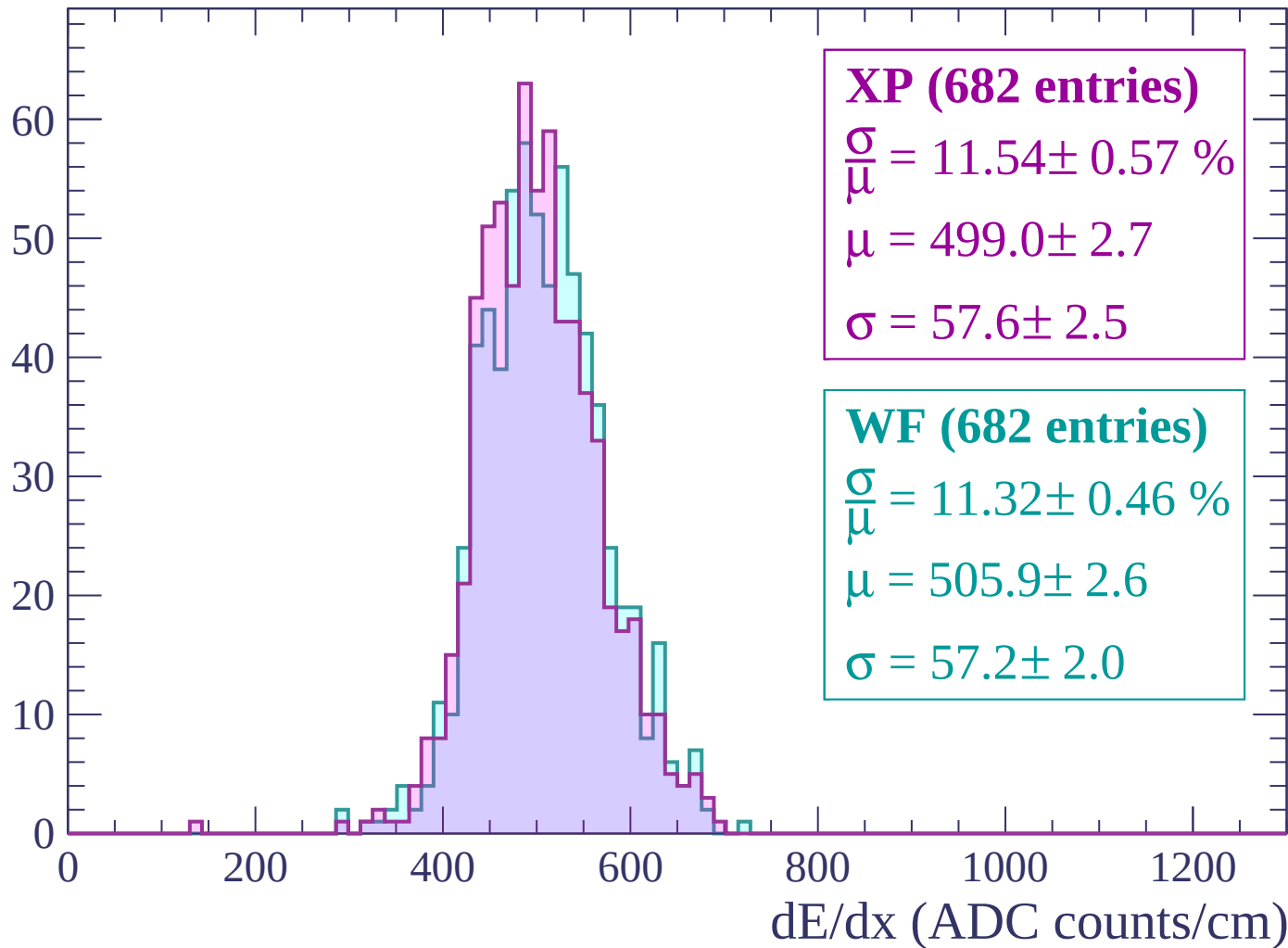
Energy loss | -2220 < p < -2160

Count



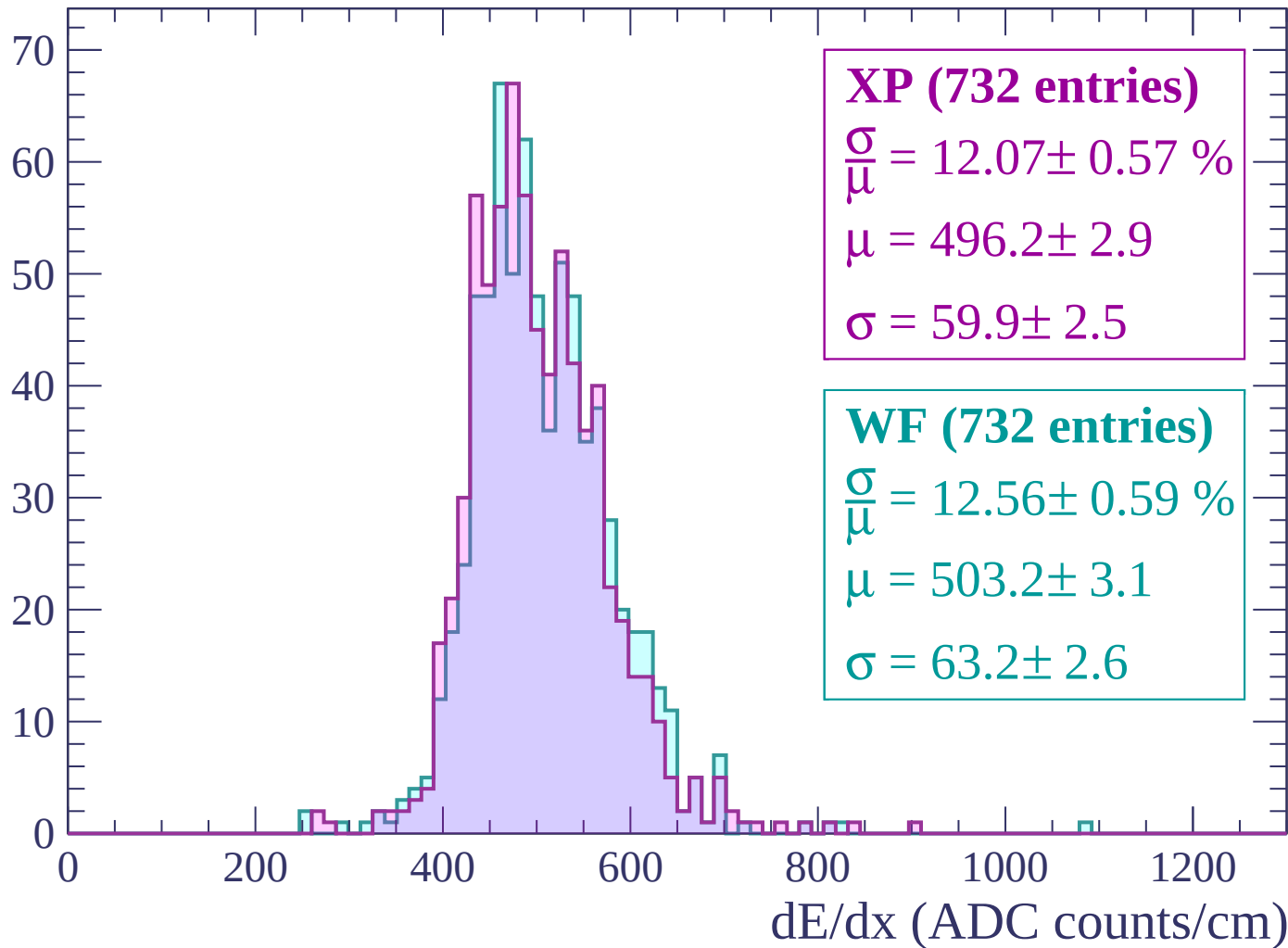
Energy loss | -2160 < p < -2100

Count



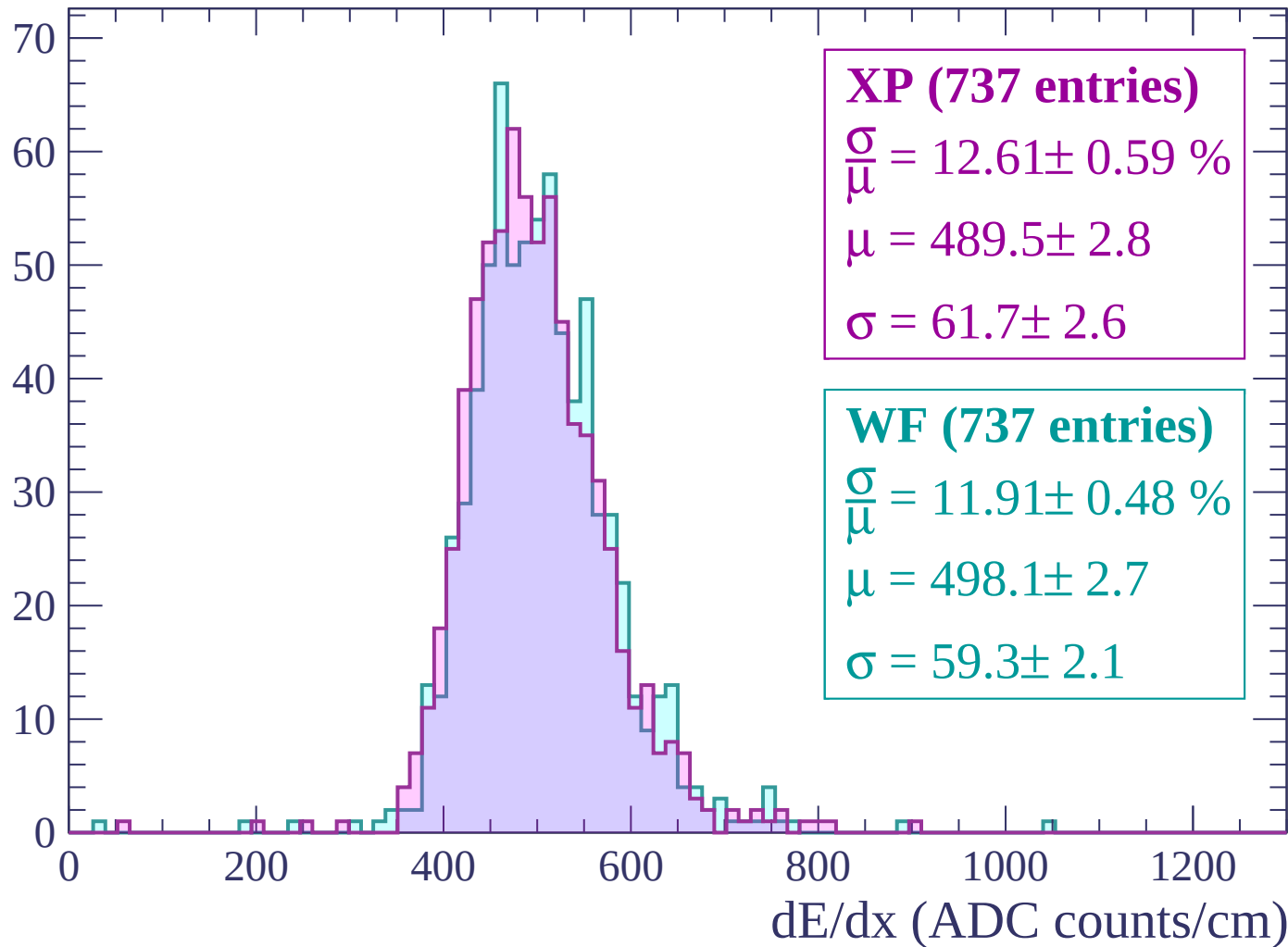
Energy loss | -2100 < p < -2040

Count



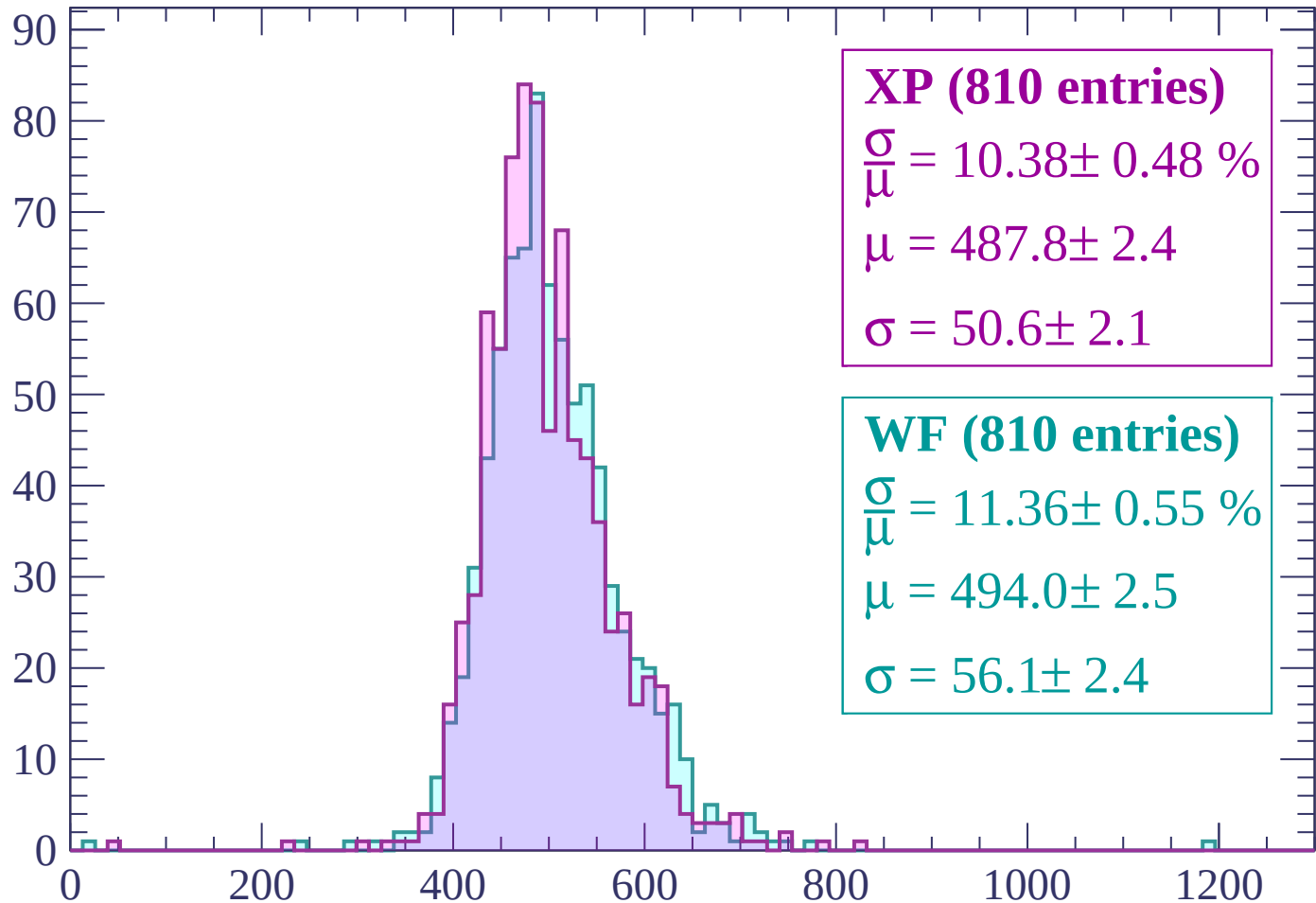
Energy loss | -2040 < p < -1980

Count



Energy loss | -1980 < p < -1920

Count



XP (810 entries)

$$\frac{\sigma}{\mu} = 10.38 \pm 0.48 \%$$

$$\mu = 487.8 \pm 2.4$$

$$\sigma = 50.6 \pm 2.1$$

WF (810 entries)

$$\frac{\sigma}{\mu} = 11.36 \pm 0.55 \%$$

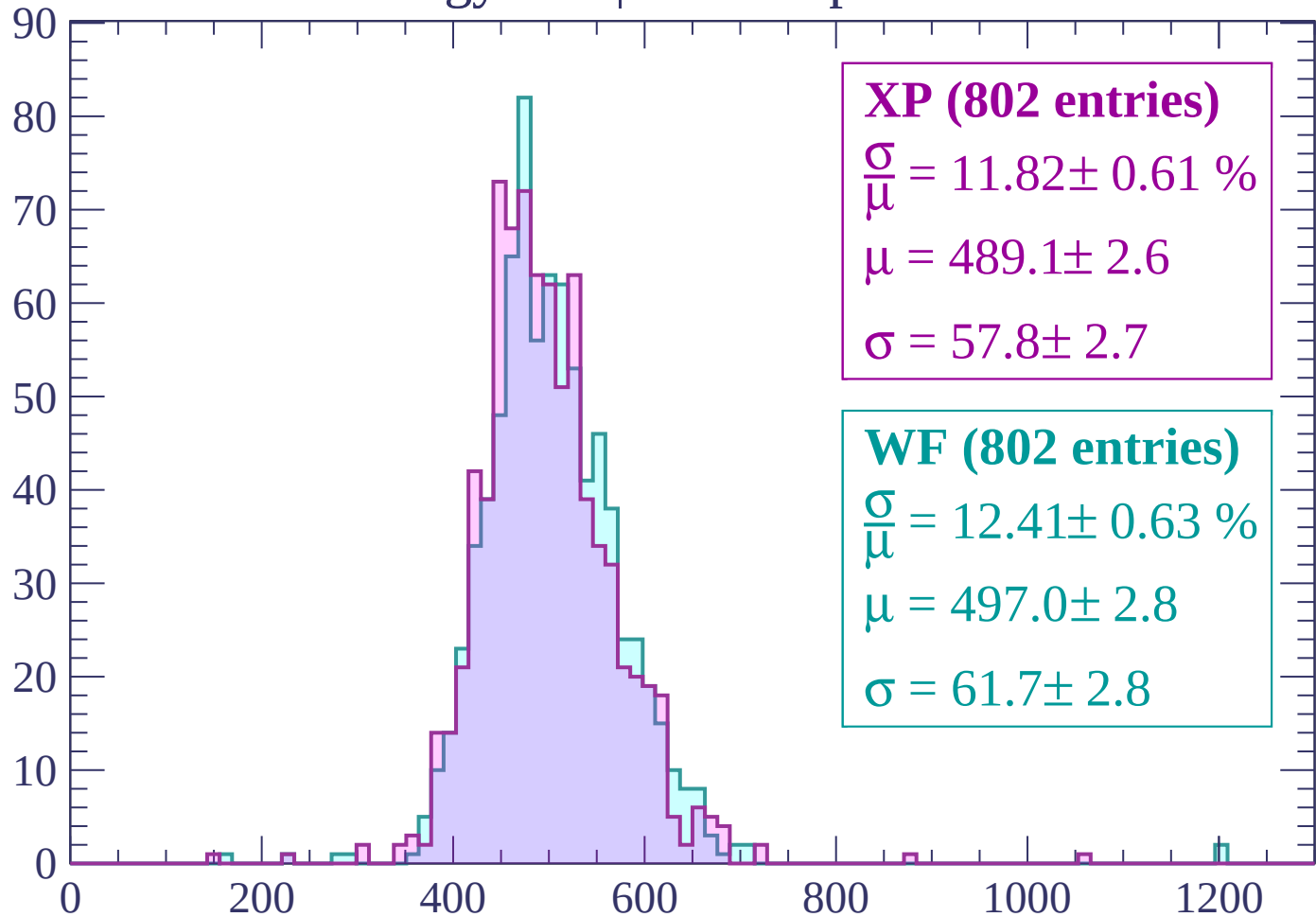
$$\mu = 494.0 \pm 2.5$$

$$\sigma = 56.1 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1860

Count



XP (802 entries)

$$\frac{\sigma}{\mu} = 11.82 \pm 0.61 \%$$

$$\mu = 489.1 \pm 2.6$$

$$\sigma = 57.8 \pm 2.7$$

WF (802 entries)

$$\frac{\sigma}{\mu} = 12.41 \pm 0.63 \%$$

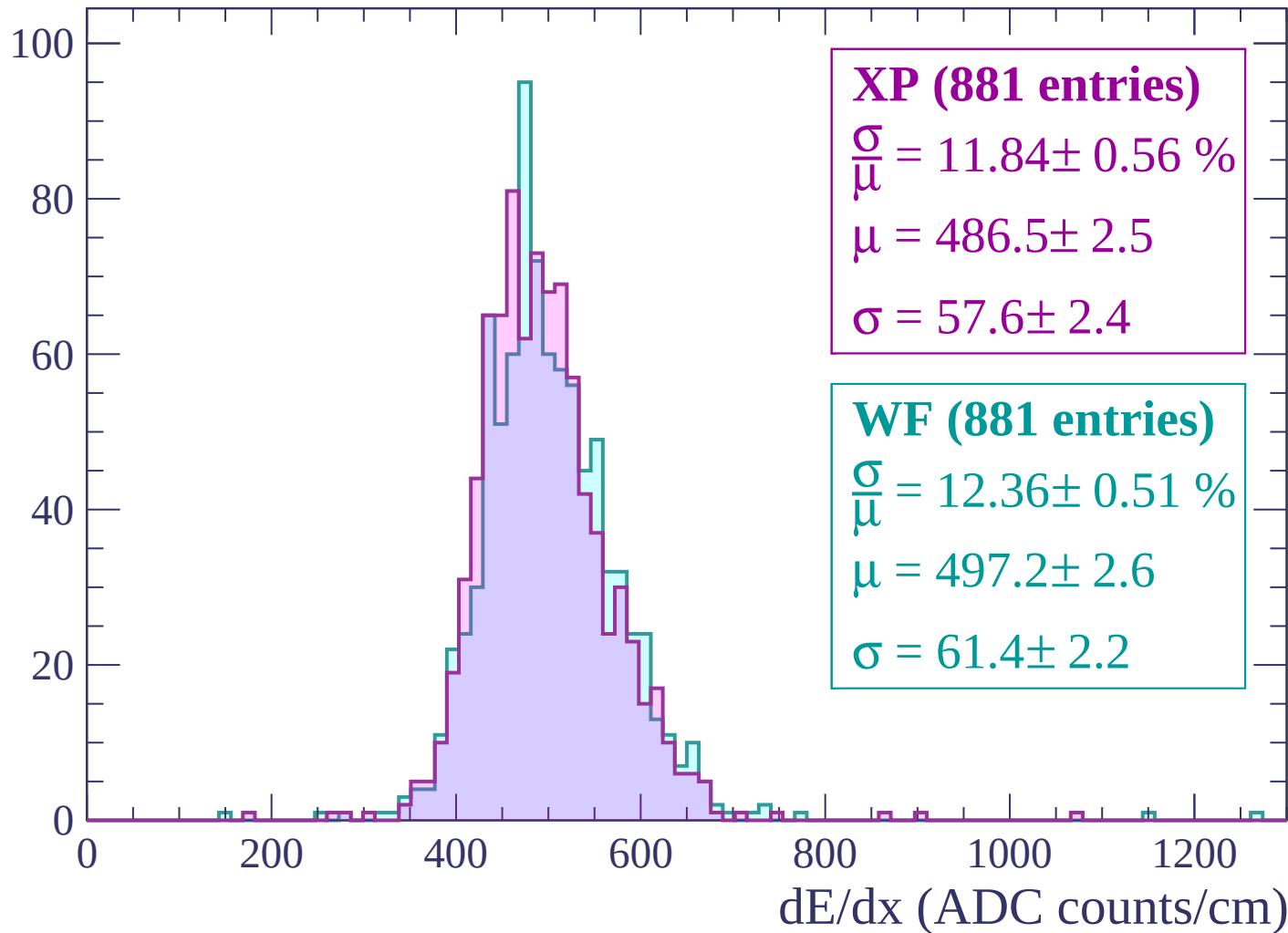
$$\mu = 497.0 \pm 2.8$$

$$\sigma = 61.7 \pm 2.8$$

dE/dx (ADC counts/cm)

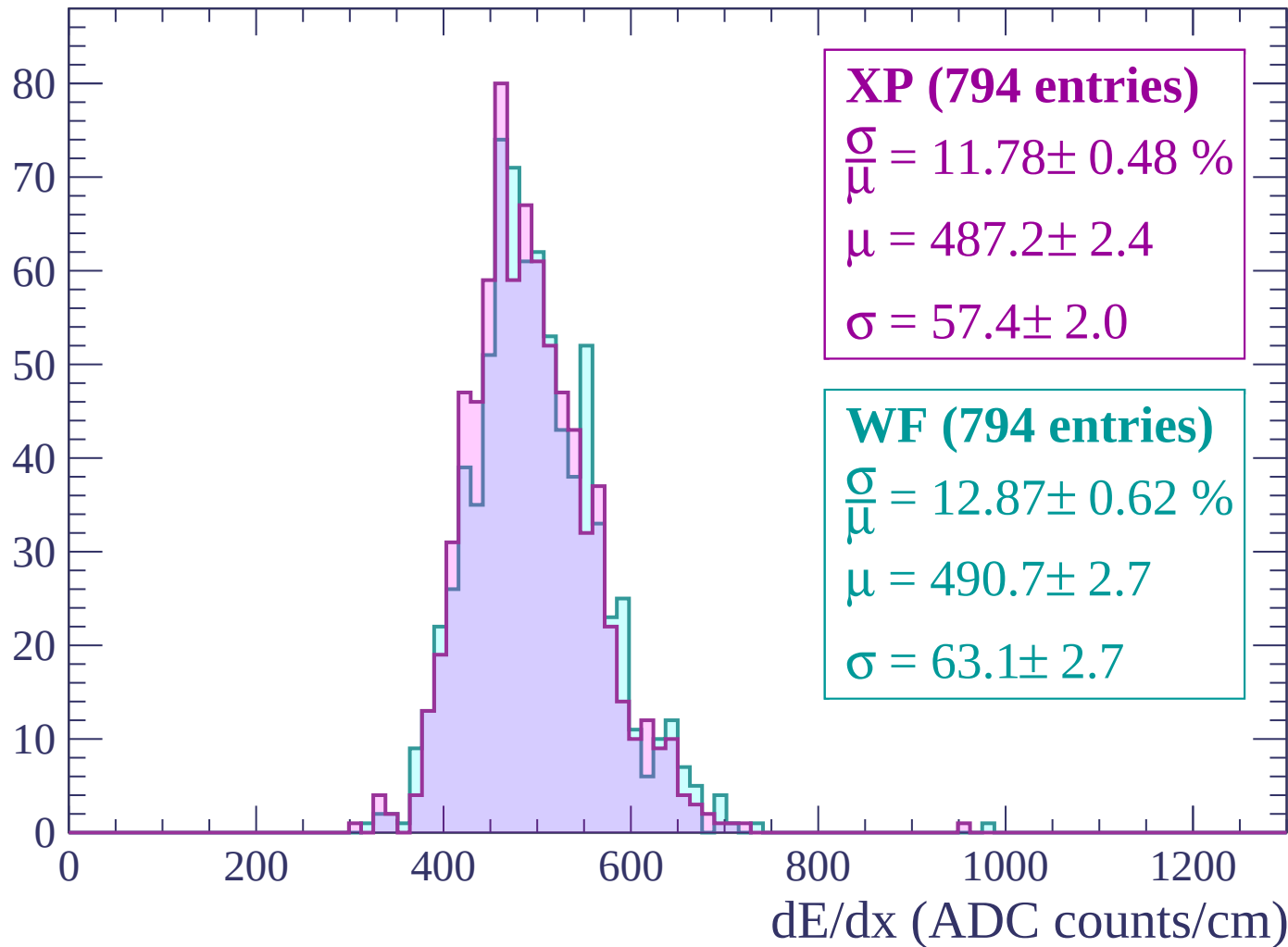
Energy loss | $-1860 < p < -1800$

Count



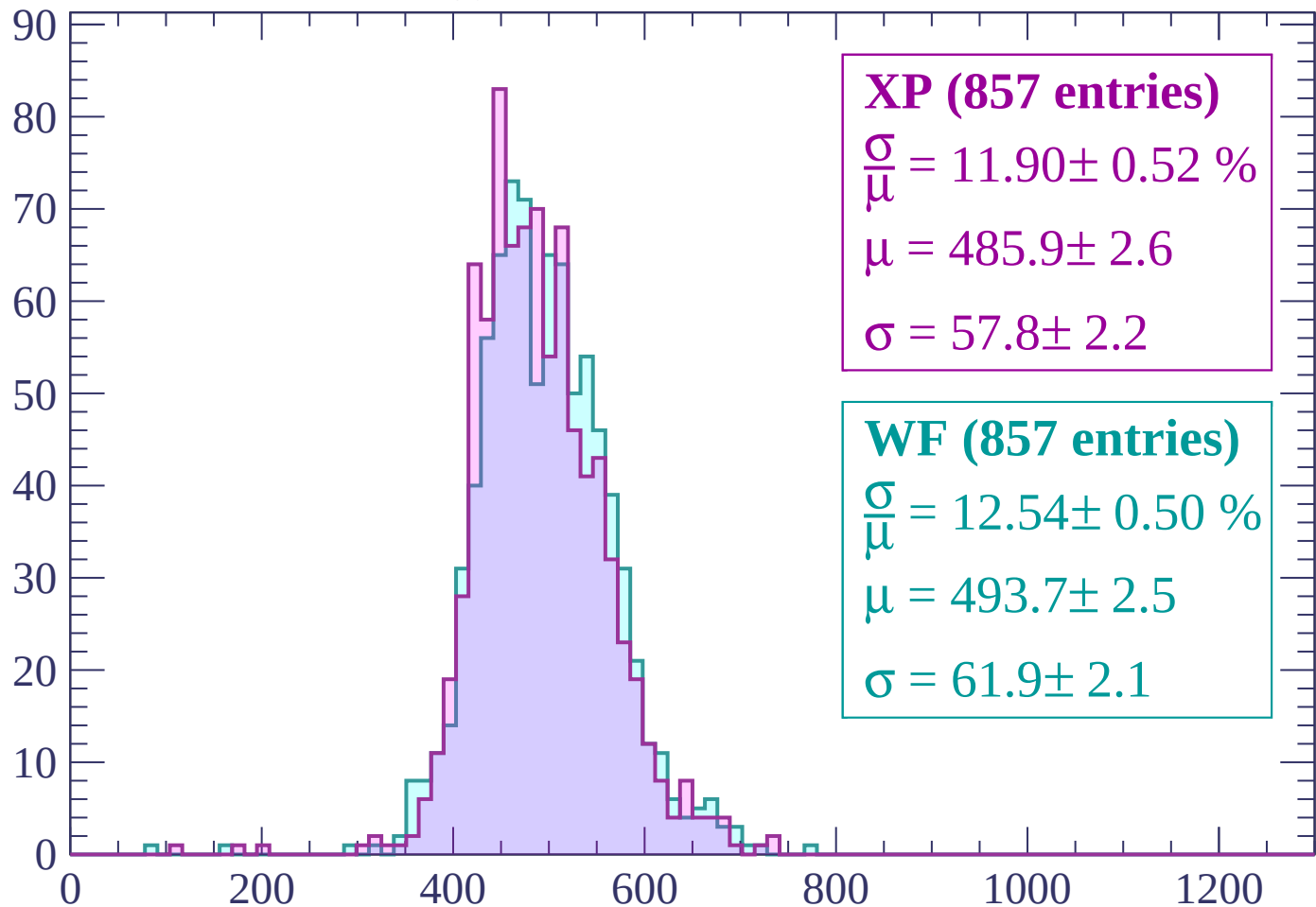
Energy loss | $-1800 < p < -1740$

Count



Energy loss | $-1740 < p < -1680$

Count



XP (857 entries)

$$\frac{\sigma}{\mu} = 11.90 \pm 0.52 \%$$

$$\mu = 485.9 \pm 2.6$$

$$\sigma = 57.8 \pm 2.2$$

WF (857 entries)

$$\frac{\sigma}{\mu} = 12.54 \pm 0.50 \%$$

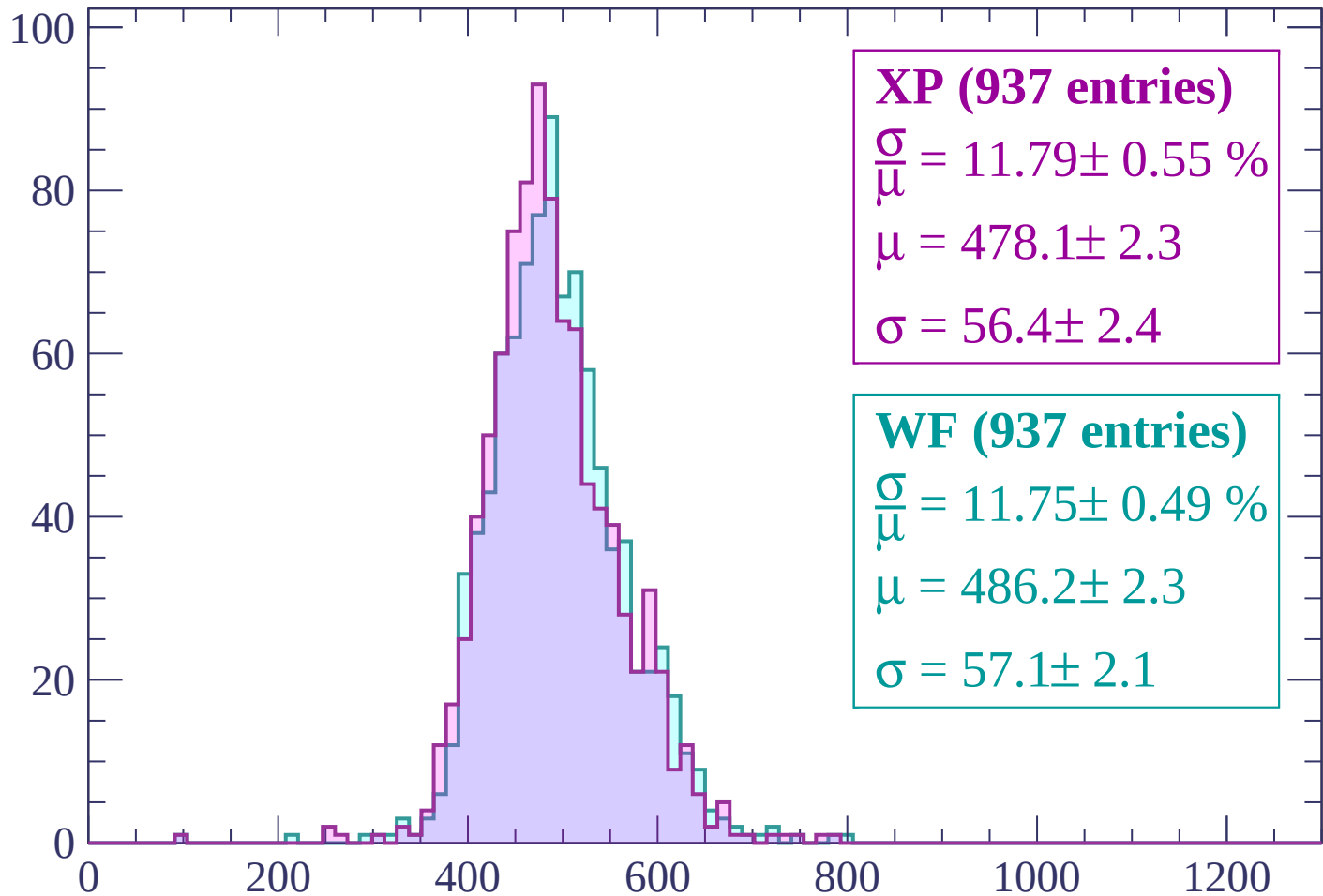
$$\mu = 493.7 \pm 2.5$$

$$\sigma = 61.9 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1620$

Count



XP (937 entries)

$\frac{\sigma}{\mu} = 11.79 \pm 0.55 \%$

$\mu = 478.1 \pm 2.3$

$\sigma = 56.4 \pm 2.4$

WF (937 entries)

$\frac{\sigma}{\mu} = 11.75 \pm 0.49 \%$

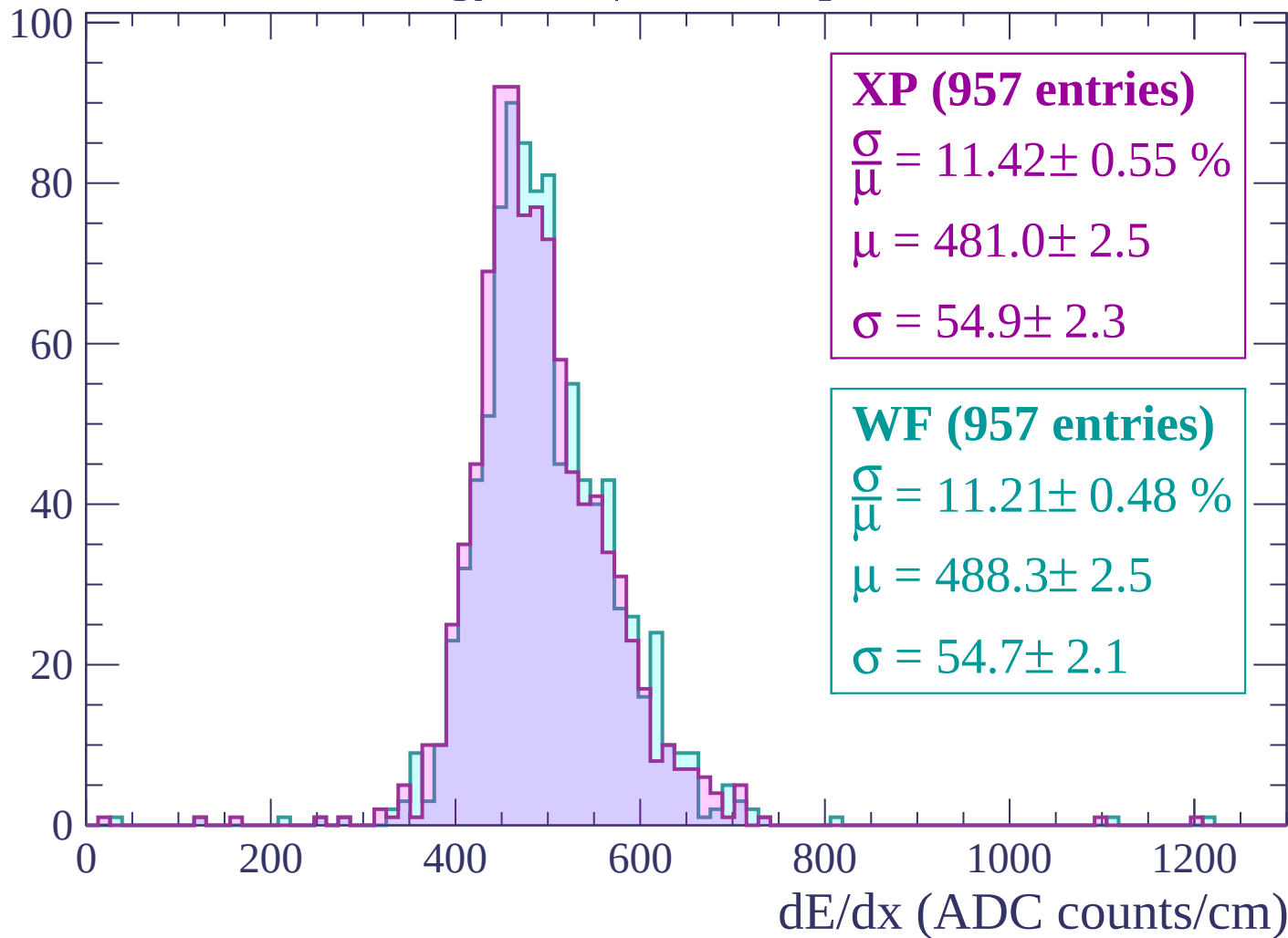
$\mu = 486.2 \pm 2.3$

$\sigma = 57.1 \pm 2.1$

dE/dx (ADC counts/cm)

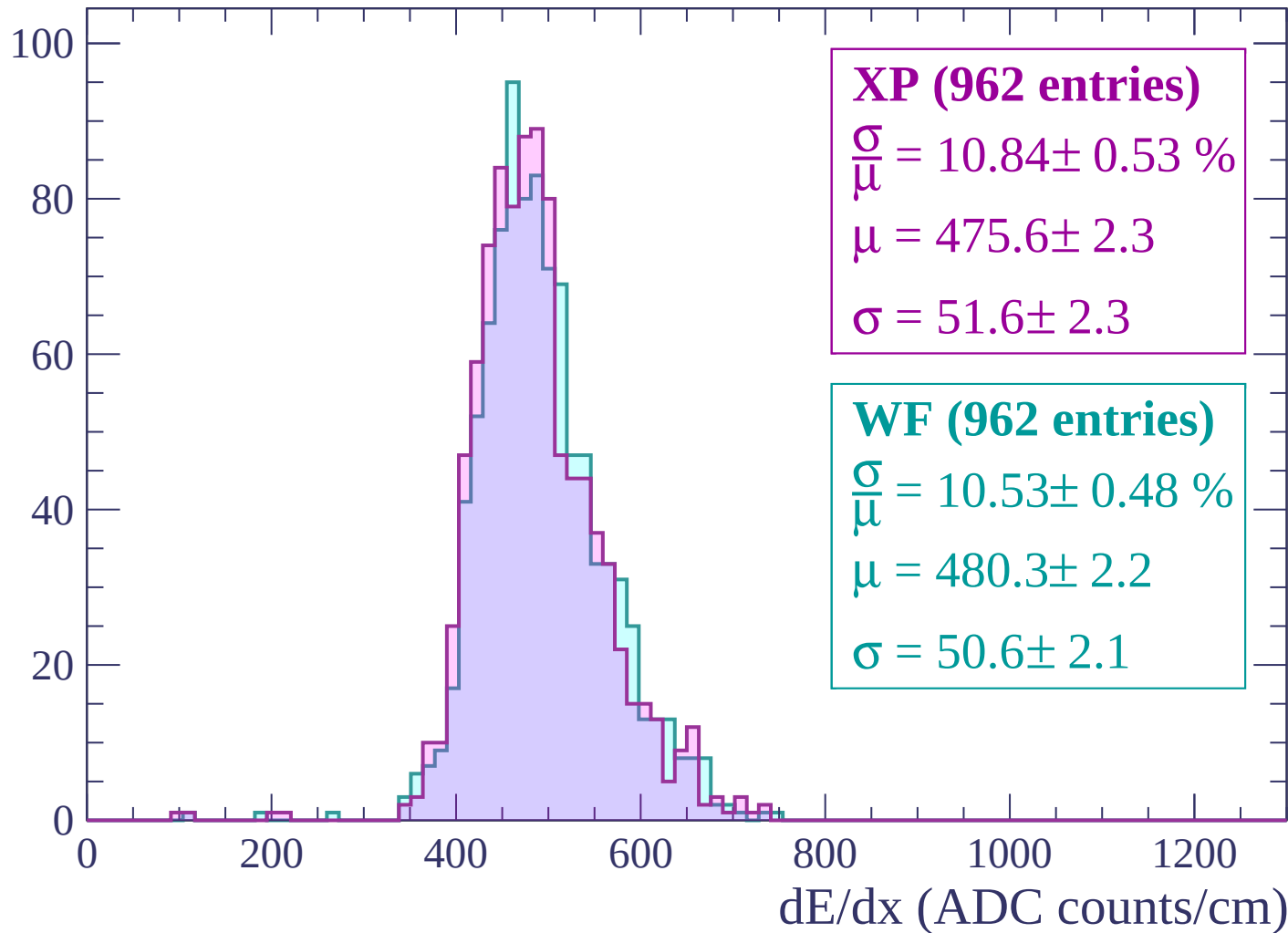
Energy loss | -1620 < p < -1560

Count



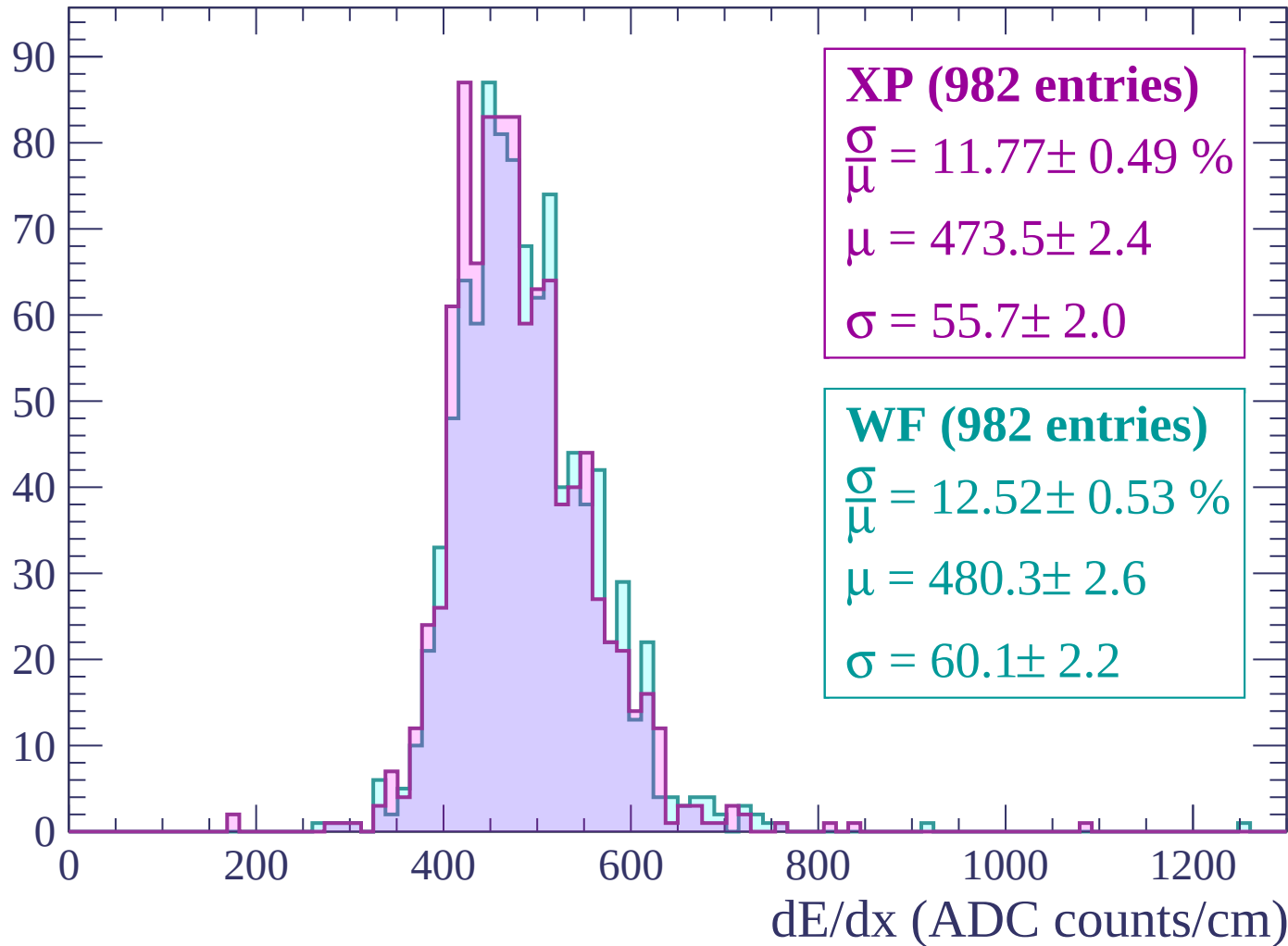
Energy loss | $-1560 < p < -1500$

Count



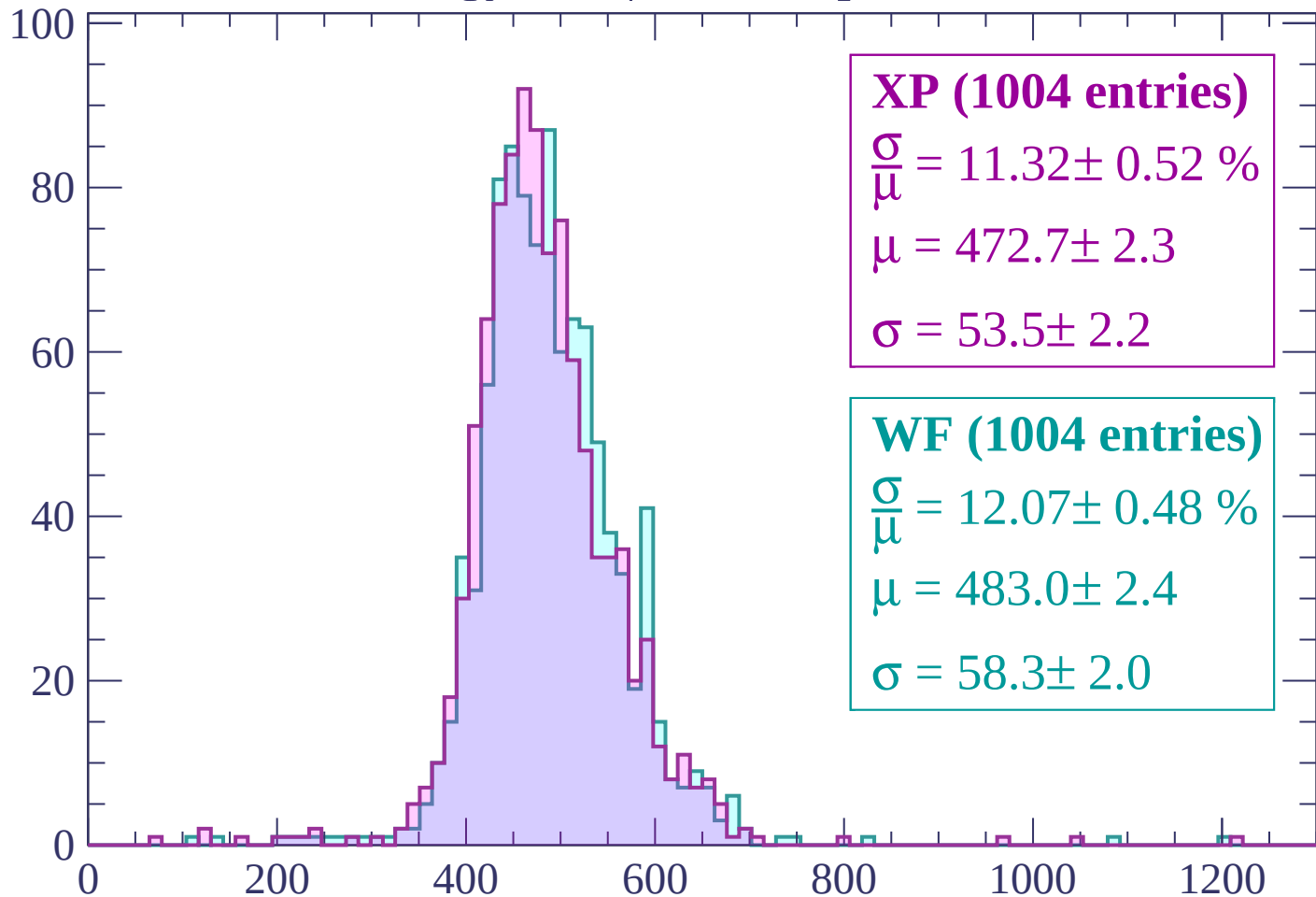
Energy loss | $-1500 < p < -1440$

Count



Energy loss | $-1440 < p < -1380$

Count



XP (1004 entries)

$$\frac{\sigma}{\mu} = 11.32 \pm 0.52 \%$$

$$\mu = 472.7 \pm 2.3$$

$$\sigma = 53.5 \pm 2.2$$

WF (1004 entries)

$$\frac{\sigma}{\mu} = 12.07 \pm 0.48 \%$$

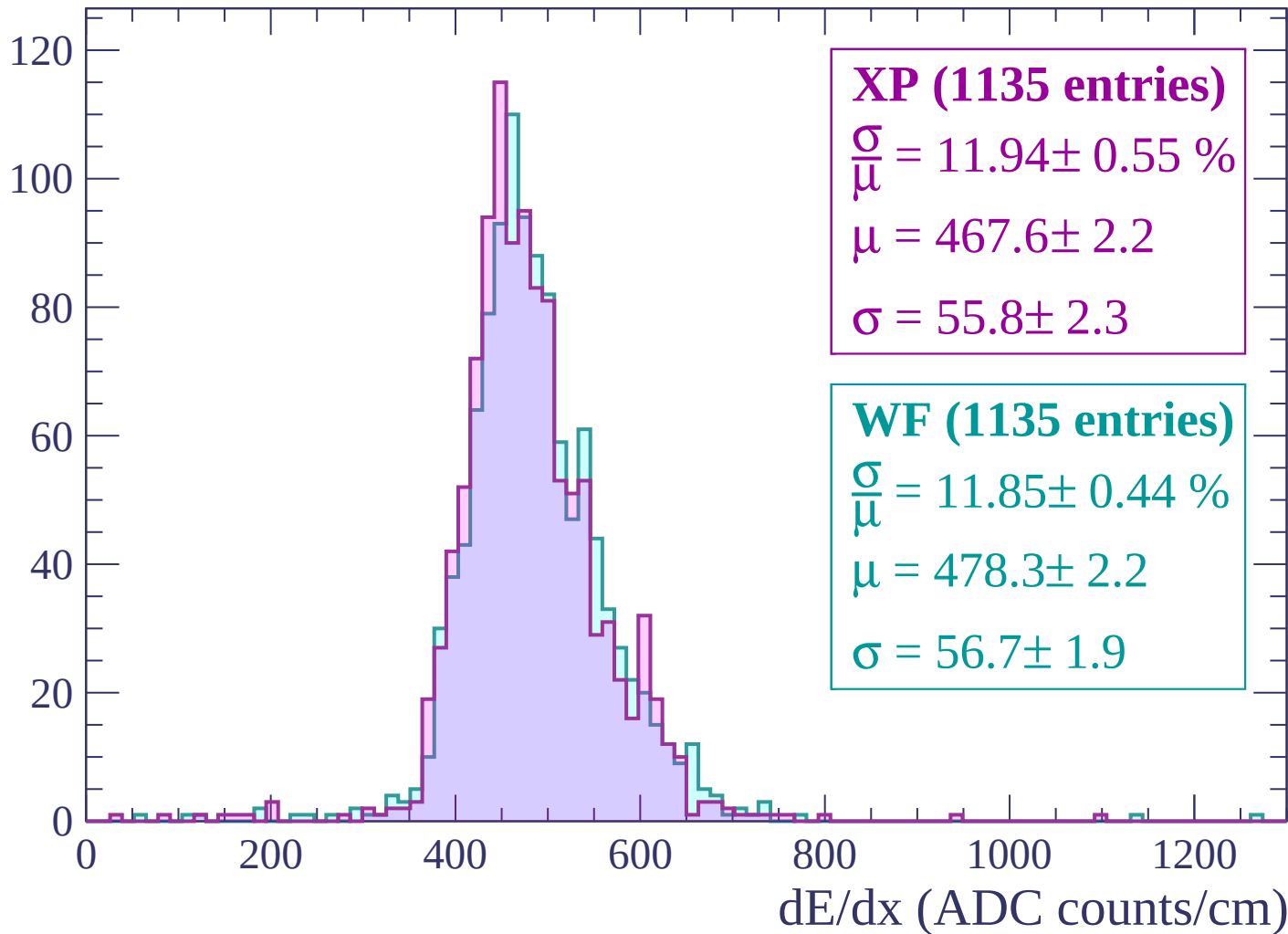
$$\mu = 483.0 \pm 2.4$$

$$\sigma = 58.3 \pm 2.0$$

dE/dx (ADC counts/cm)

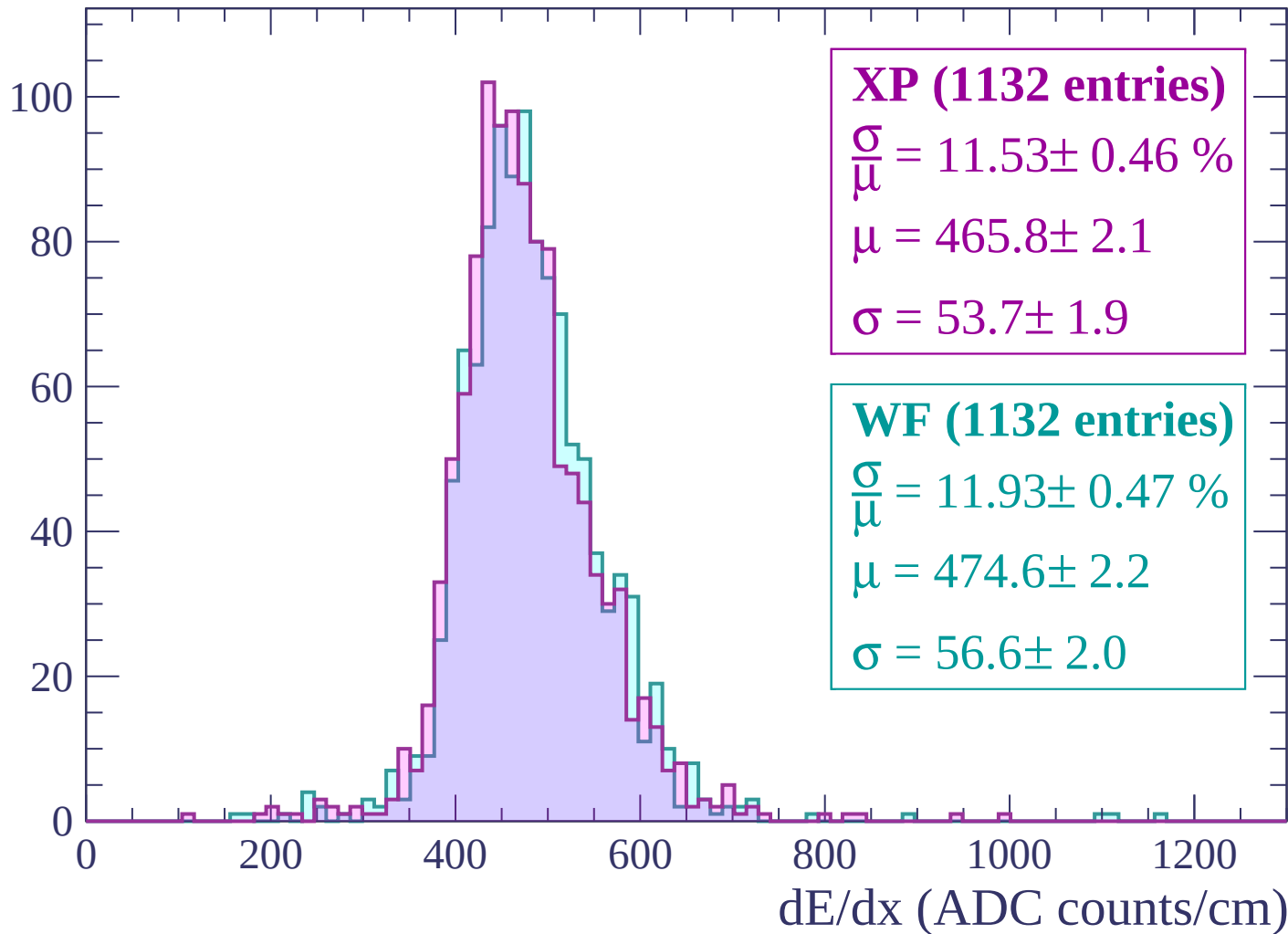
Energy loss | -1380 < p < -1320

Count



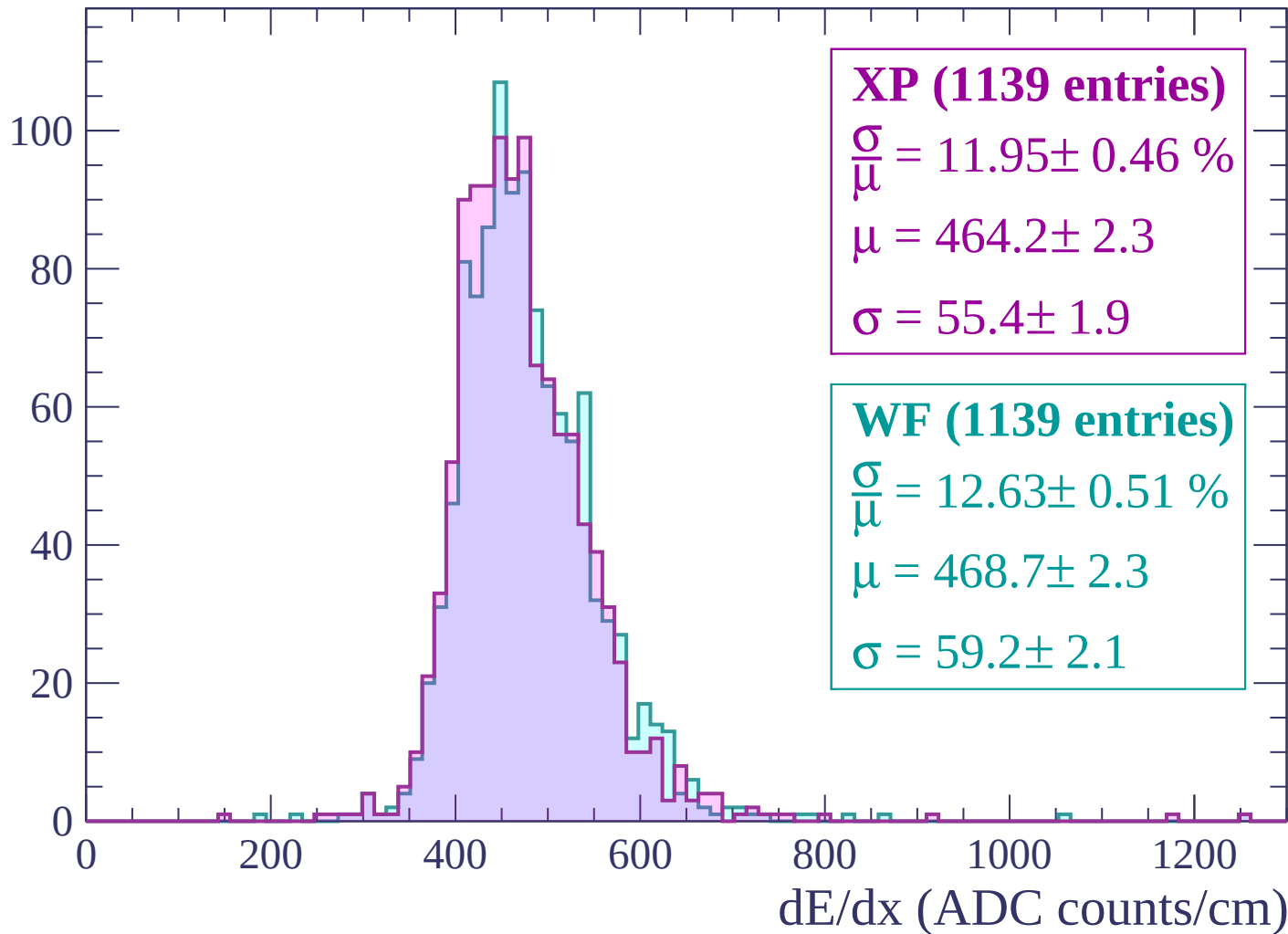
Energy loss | $-1320 < p < -1260$

Count



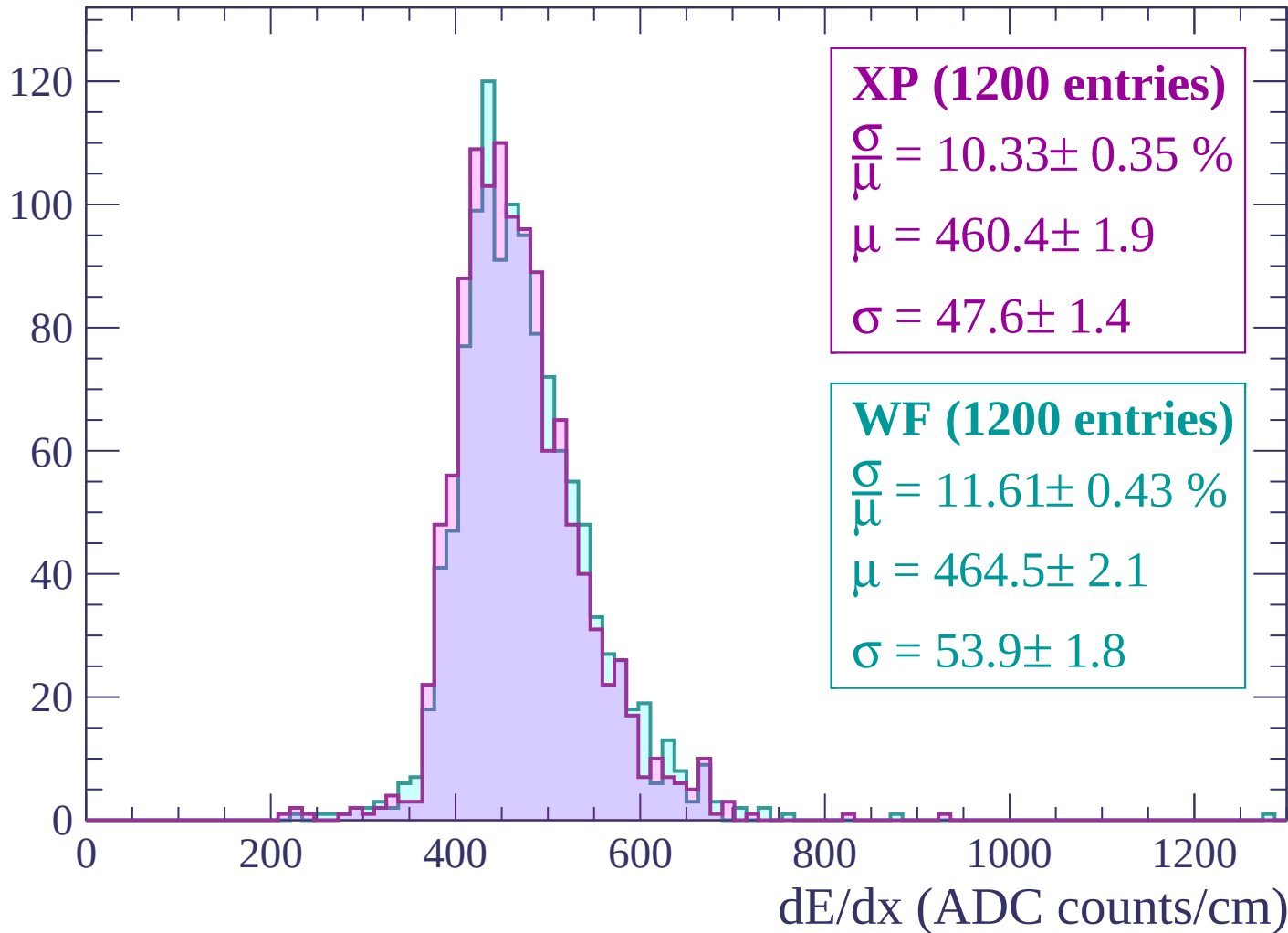
Energy loss | $-1260 < p < -1200$

Count



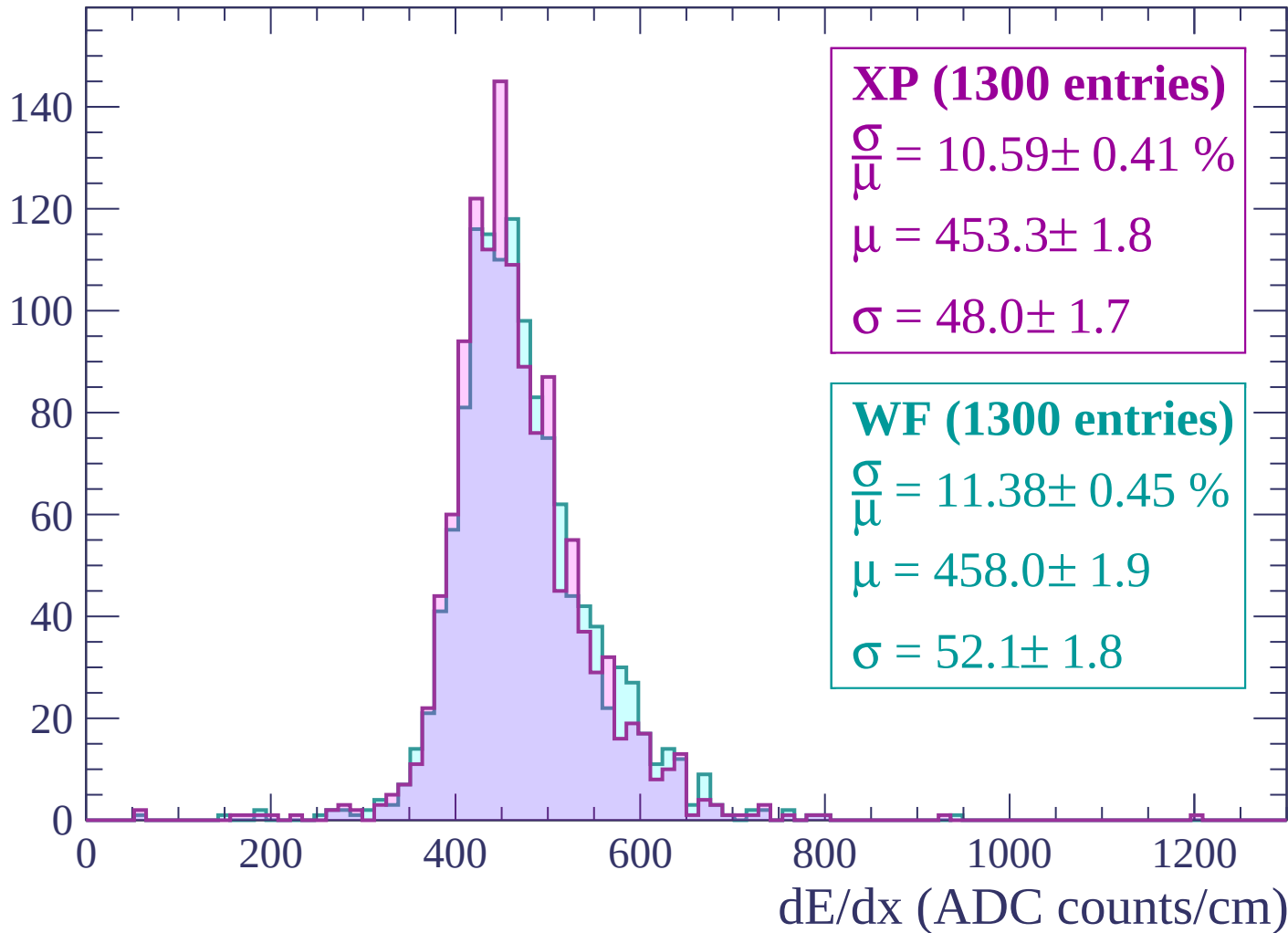
Energy loss | -1200 < p < -1140

Count



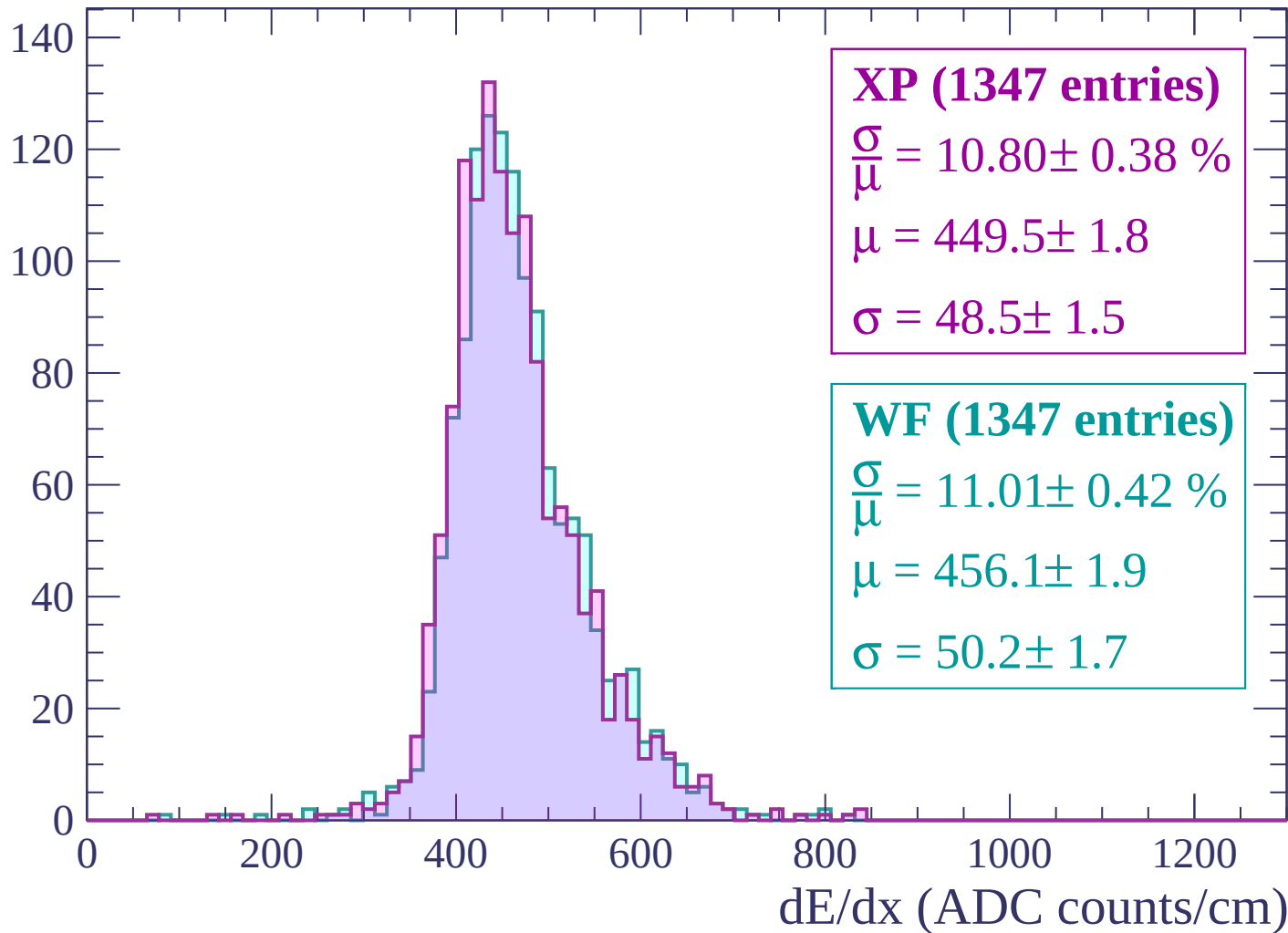
Energy loss | -1140 < p < -1080

Count



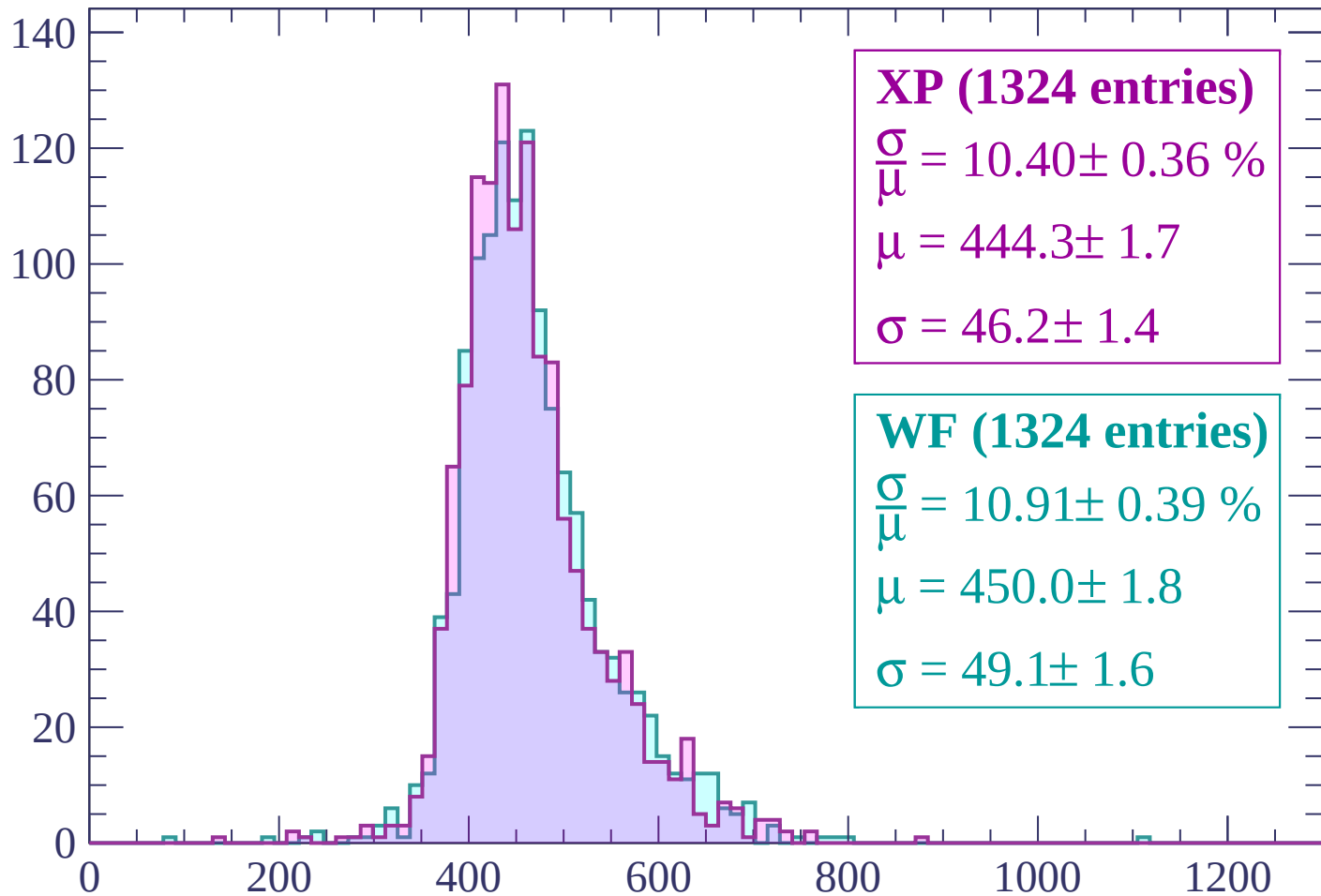
Energy loss | -1080 < p < -1020

Count



Energy loss | $-1020 < p < -960$

Count



XP (1324 entries)

$\frac{\sigma}{\mu} = 10.40 \pm 0.36 \%$

$\mu = 444.3 \pm 1.7$

$\sigma = 46.2 \pm 1.4$

WF (1324 entries)

$\frac{\sigma}{\mu} = 10.91 \pm 0.39 \%$

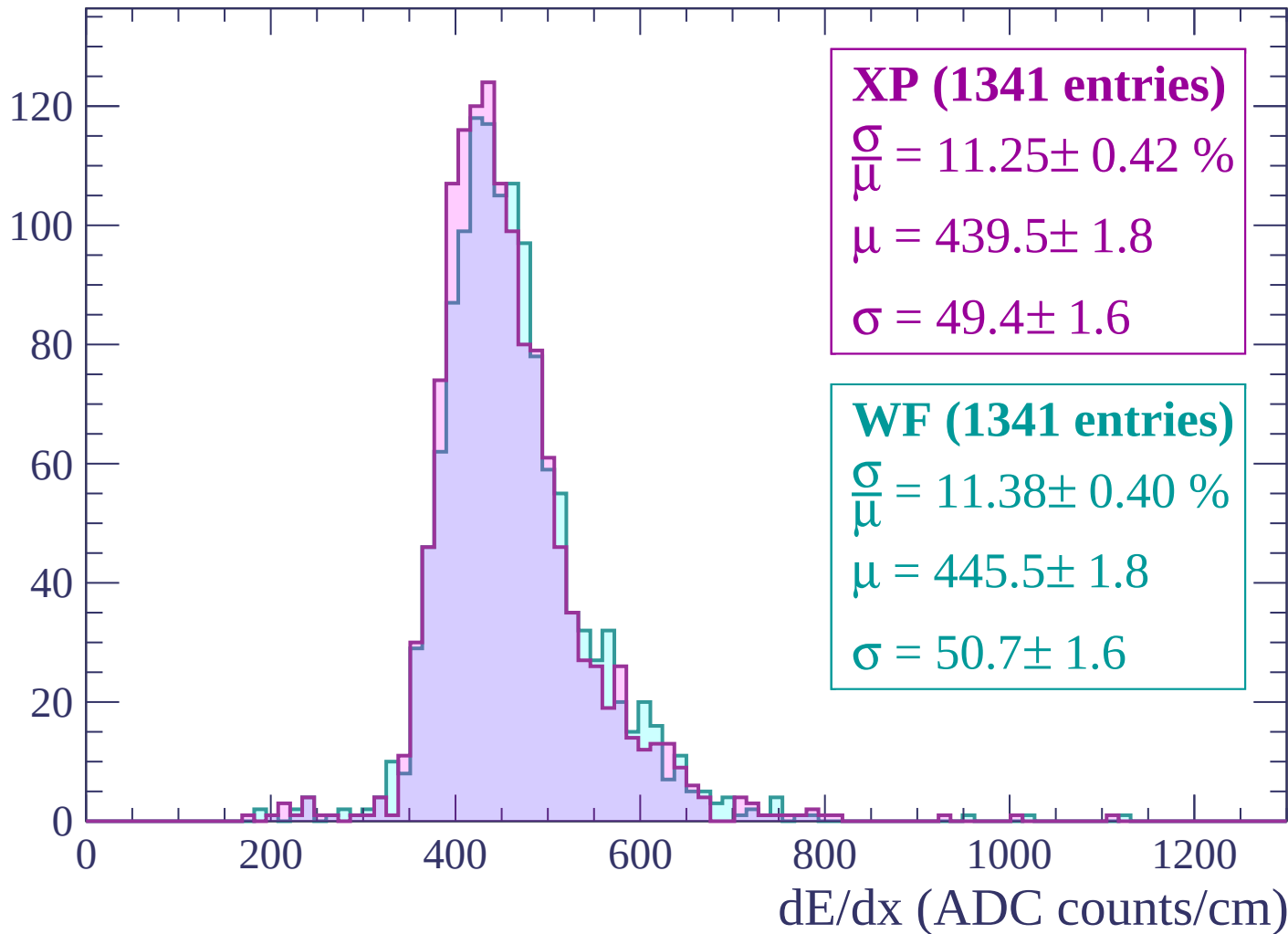
$\mu = 450.0 \pm 1.8$

$\sigma = 49.1 \pm 1.6$

dE/dx (ADC counts/cm)

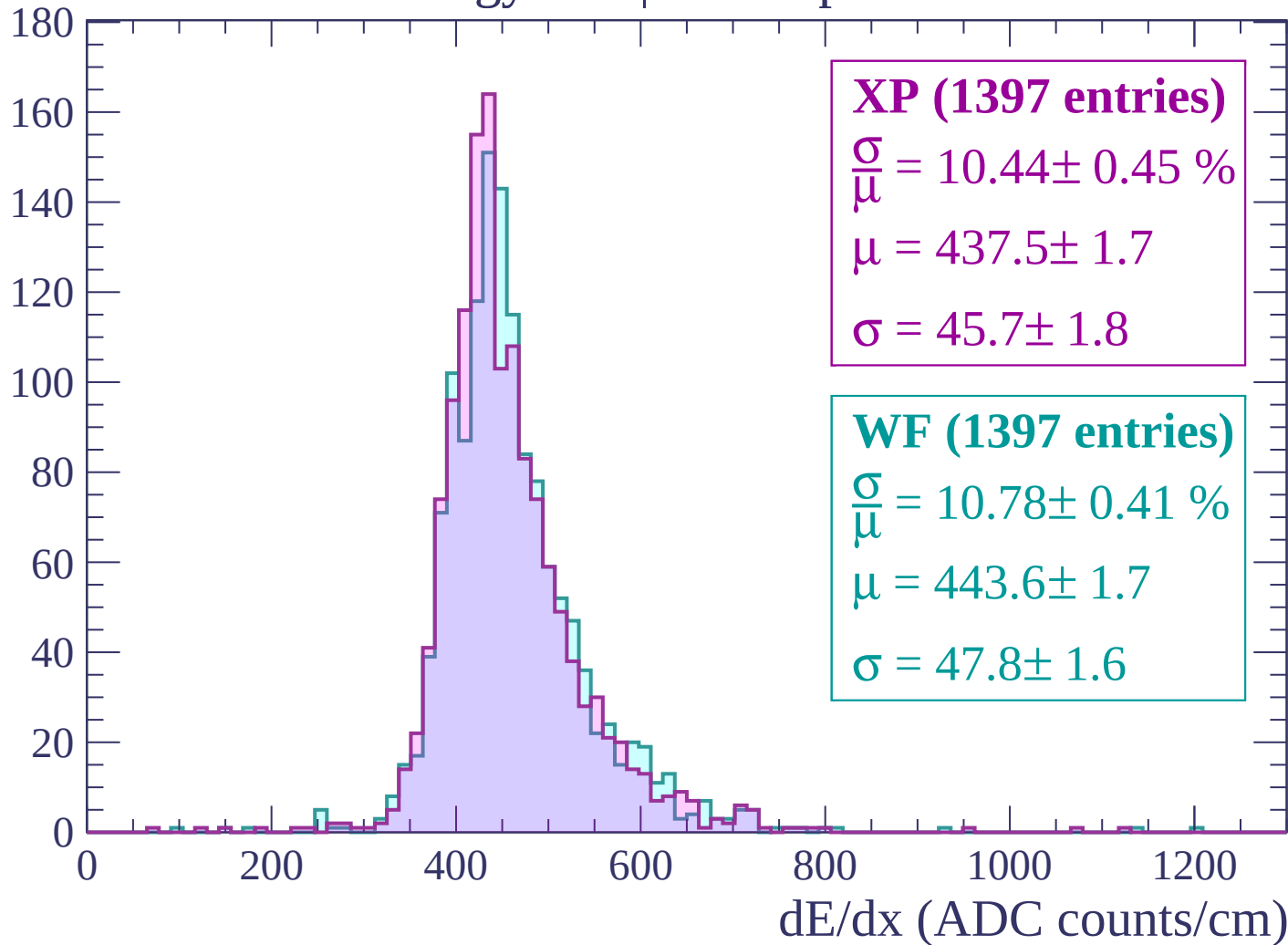
Energy loss | -960 < p < -900

Count



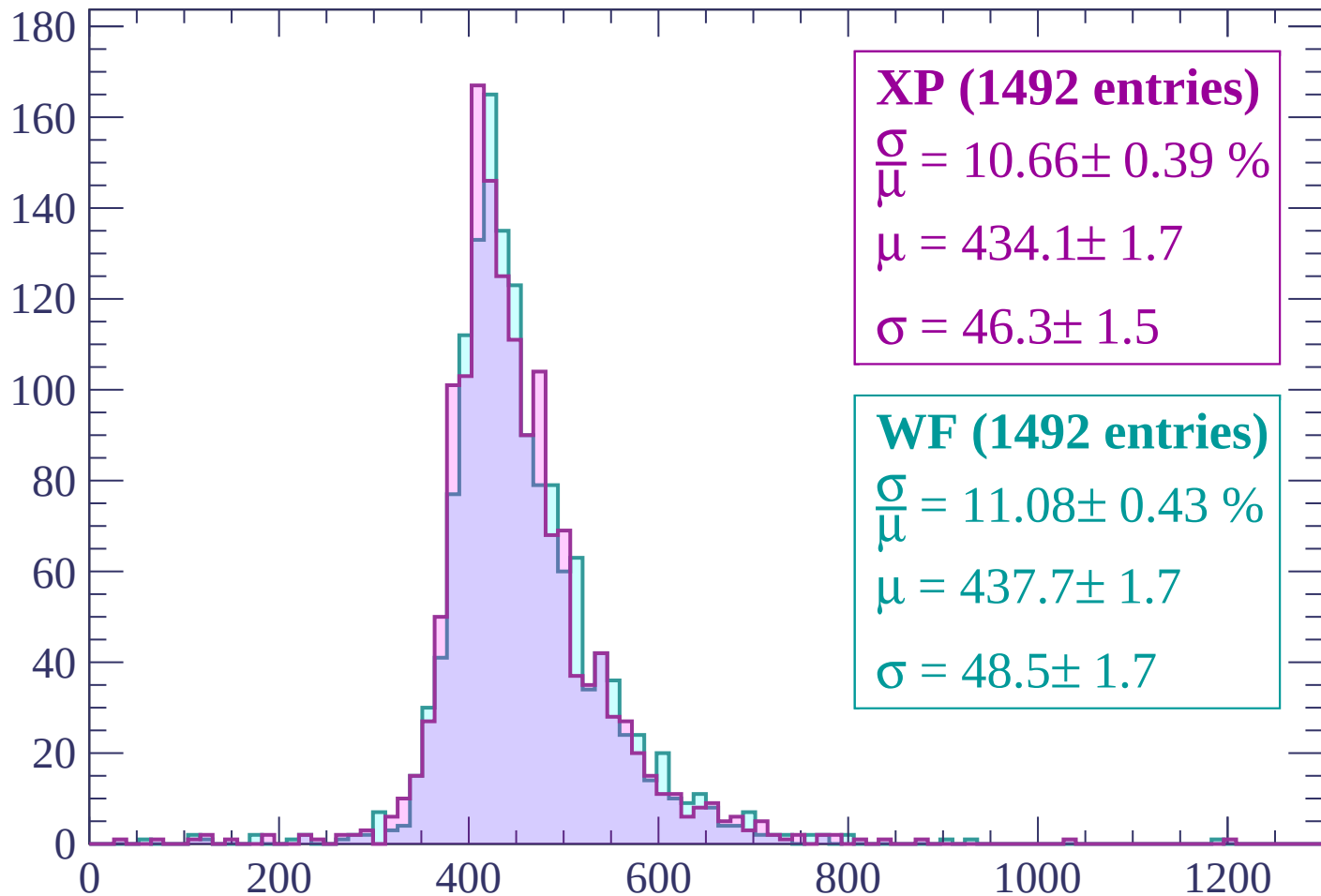
Energy loss | $-900 < p < -840$

Count



Energy loss | $-840 < p < -780$

Count



XP (1492 entries)

$$\frac{\sigma}{\mu} = 10.66 \pm 0.39 \%$$

$$\mu = 434.1 \pm 1.7$$

$$\sigma = 46.3 \pm 1.5$$

WF (1492 entries)

$$\frac{\sigma}{\mu} = 11.08 \pm 0.43 \%$$

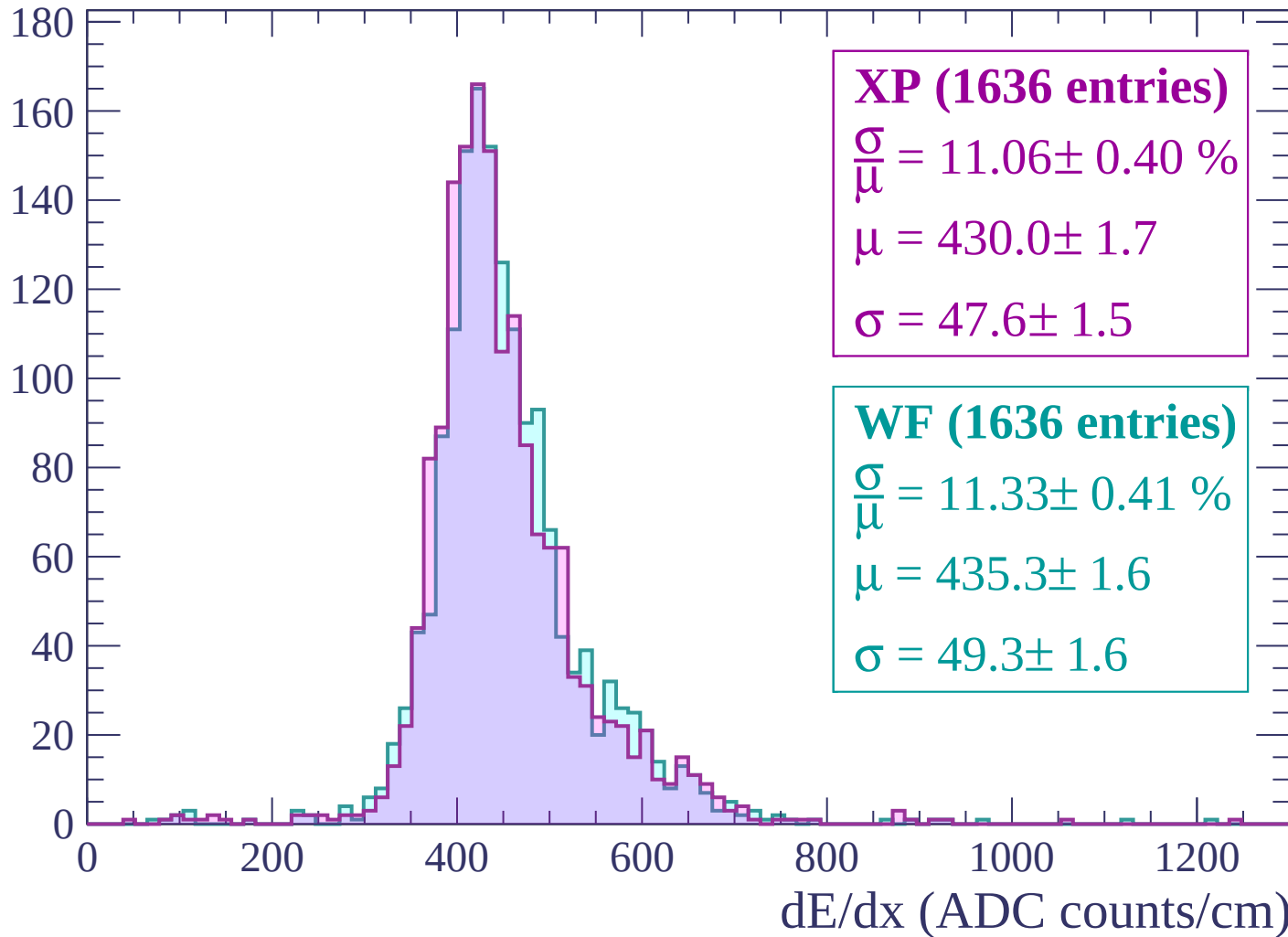
$$\mu = 437.7 \pm 1.7$$

$$\sigma = 48.5 \pm 1.7$$

dE/dx (ADC counts/cm)

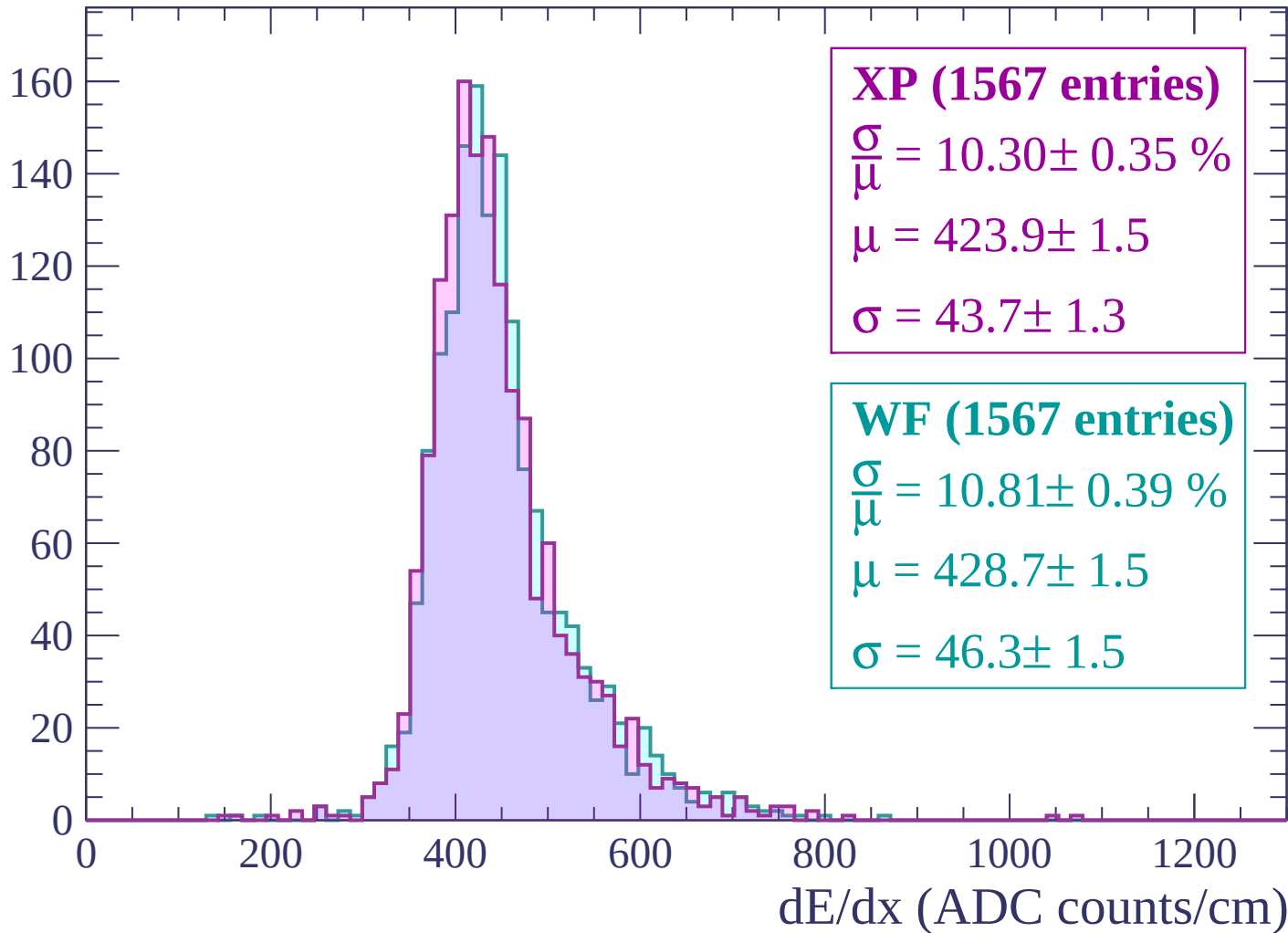
Energy loss | $-780 < p < -720$

Count



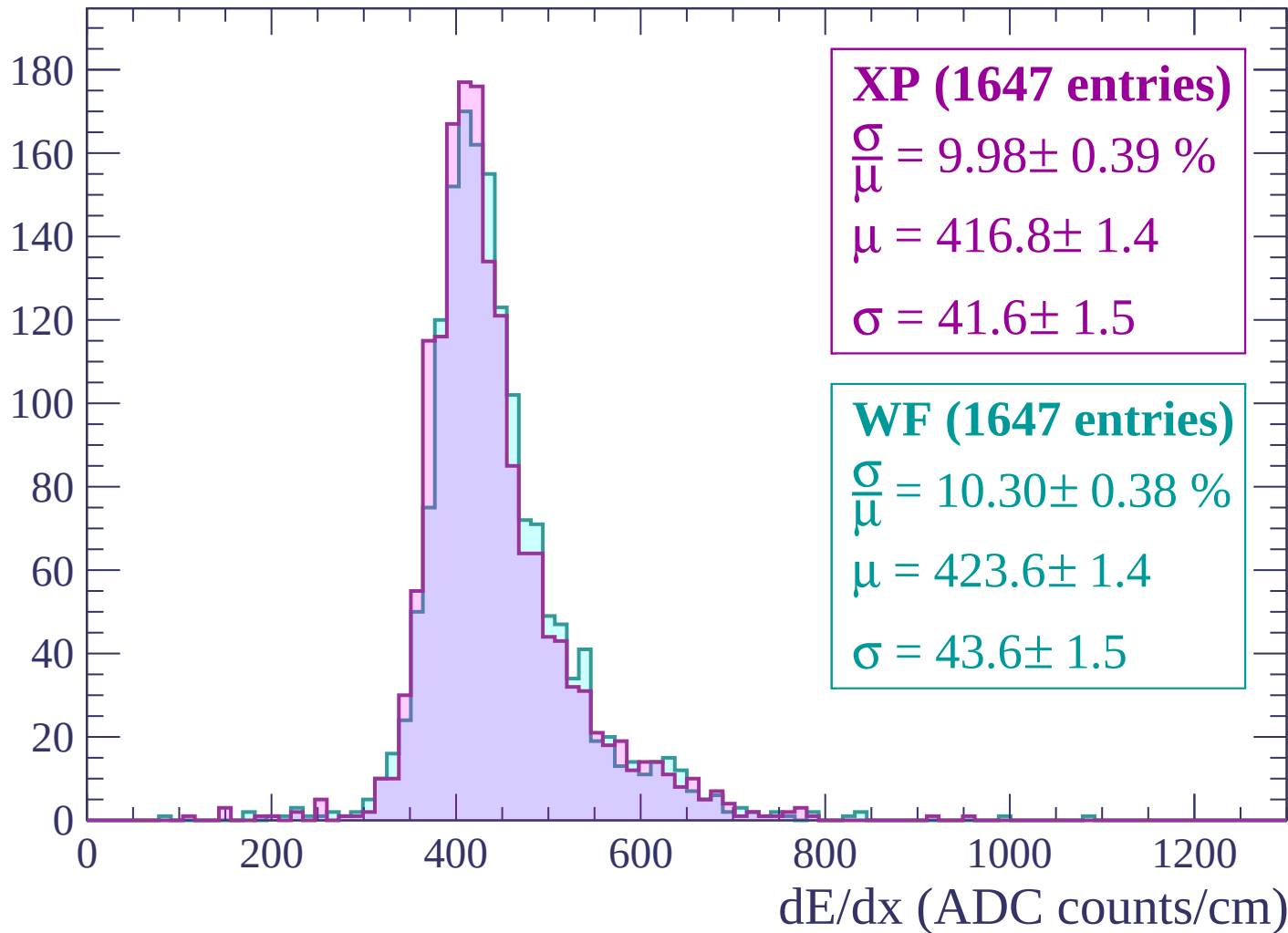
Energy loss | $-720 < p < -660$

Count



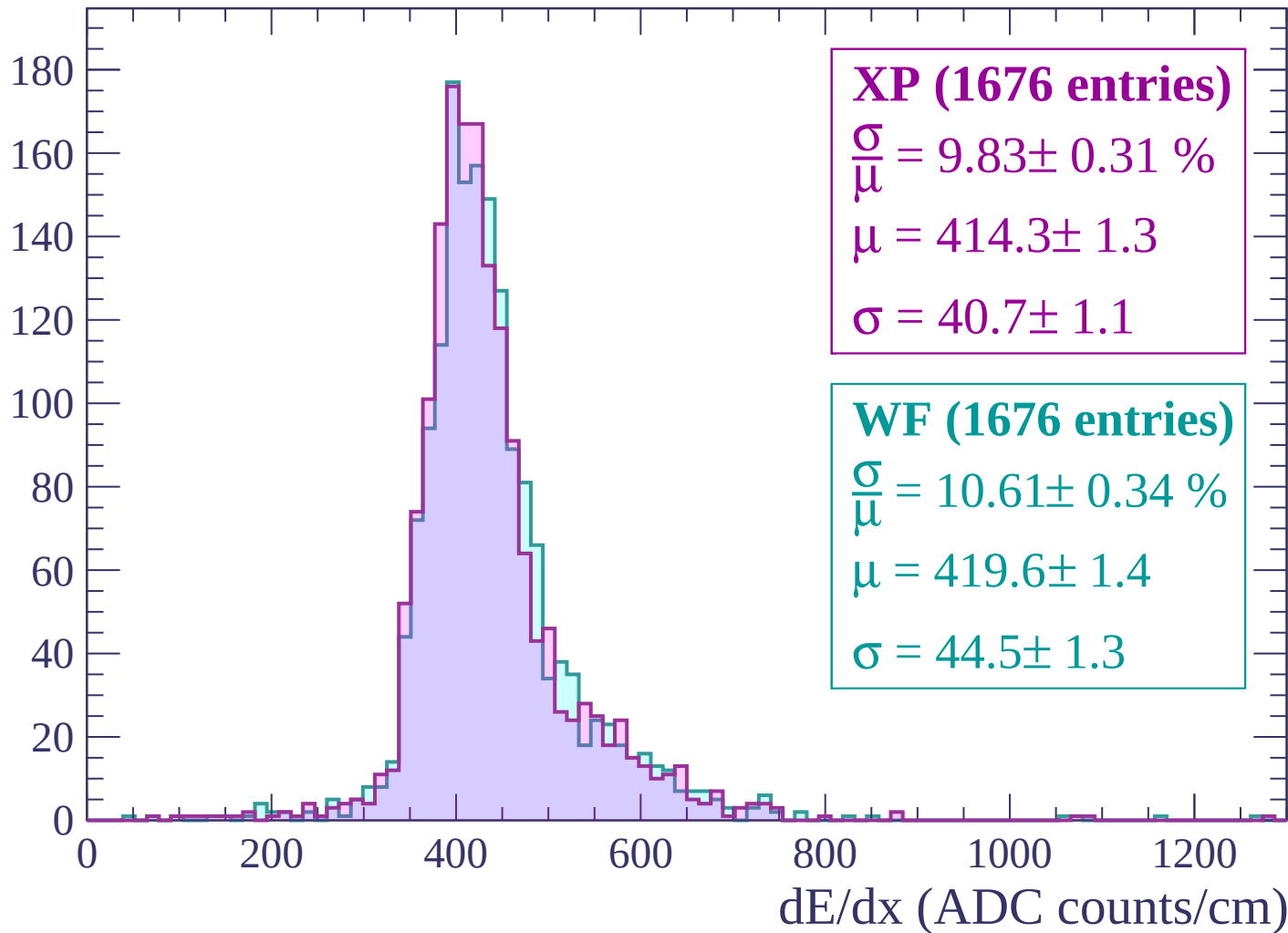
Energy loss | $-660 < p < -600$

Count



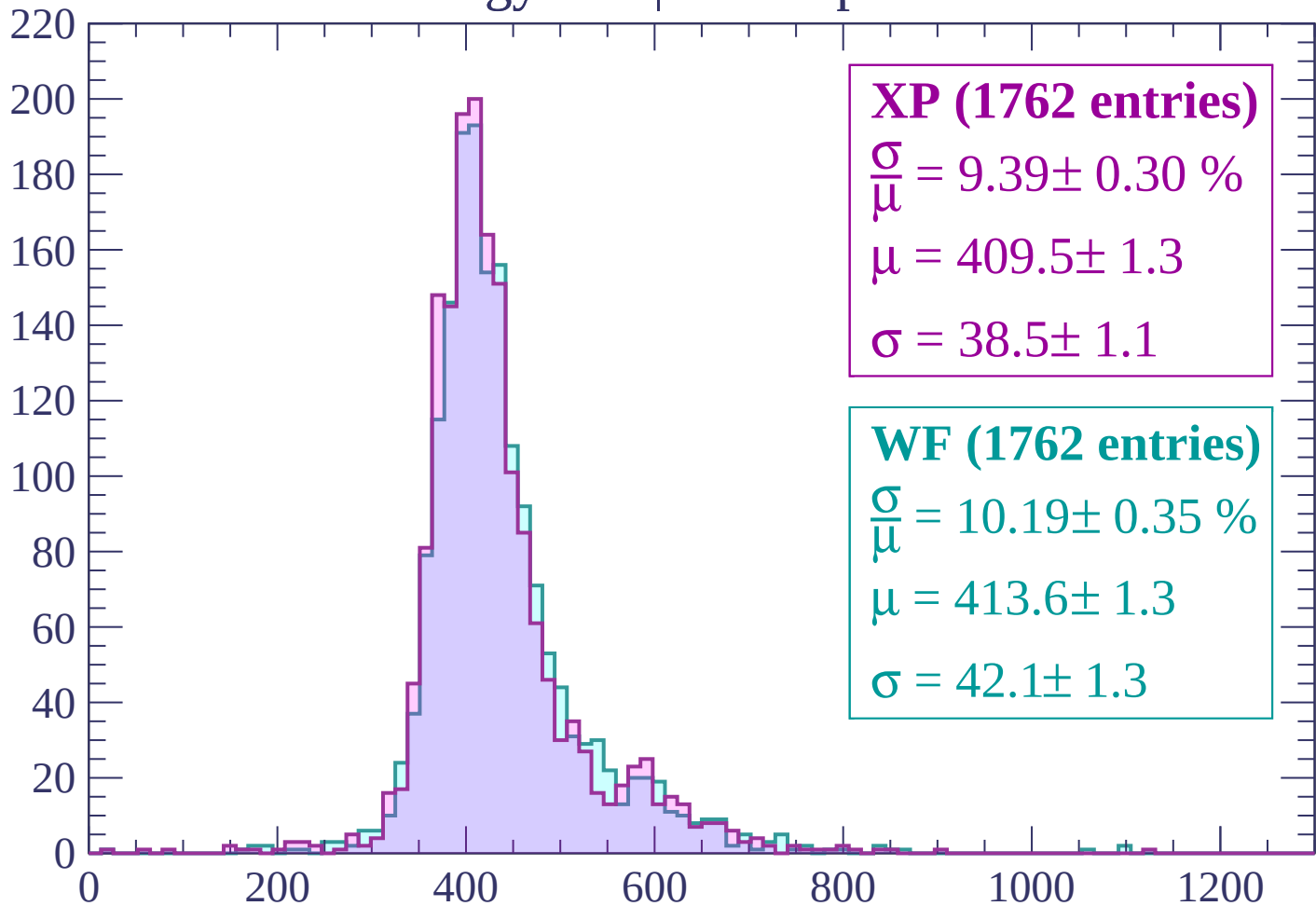
Energy loss | $-600 < p < -540$

Count



Energy loss | $-540 < p < -480$

Count



XP (1762 entries)

$$\frac{\sigma}{\mu} = 9.39 \pm 0.30 \%$$

$$\mu = 409.5 \pm 1.3$$

$$\sigma = 38.5 \pm 1.1$$

WF (1762 entries)

$$\frac{\sigma}{\mu} = 10.19 \pm 0.35 \%$$

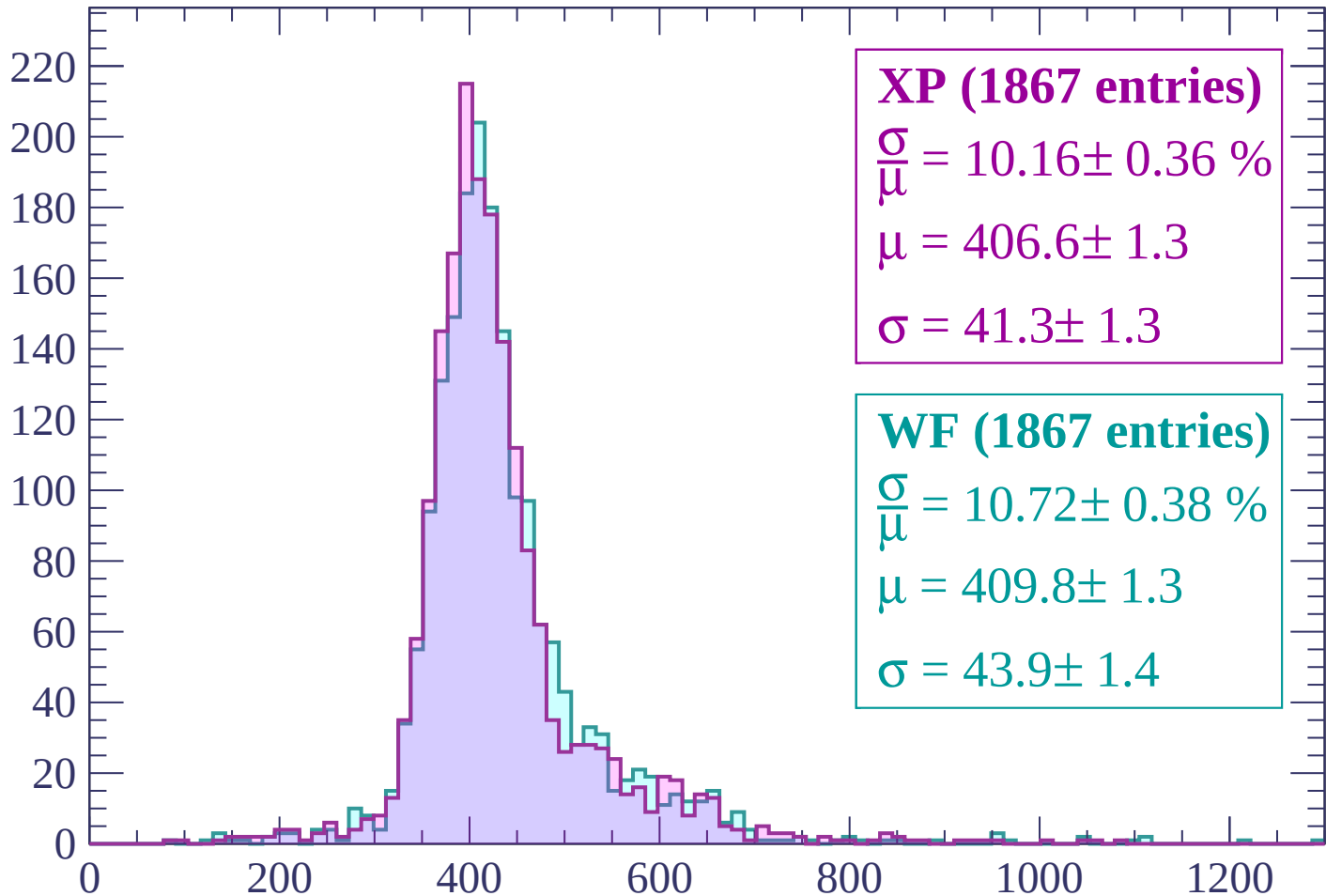
$$\mu = 413.6 \pm 1.3$$

$$\sigma = 42.1 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count



XP (1867 entries)

$$\frac{\sigma}{\mu} = 10.16 \pm 0.36 \%$$

$$\mu = 406.6 \pm 1.3$$

$$\sigma = 41.3 \pm 1.3$$

WF (1867 entries)

$$\frac{\sigma}{\mu} = 10.72 \pm 0.38 \%$$

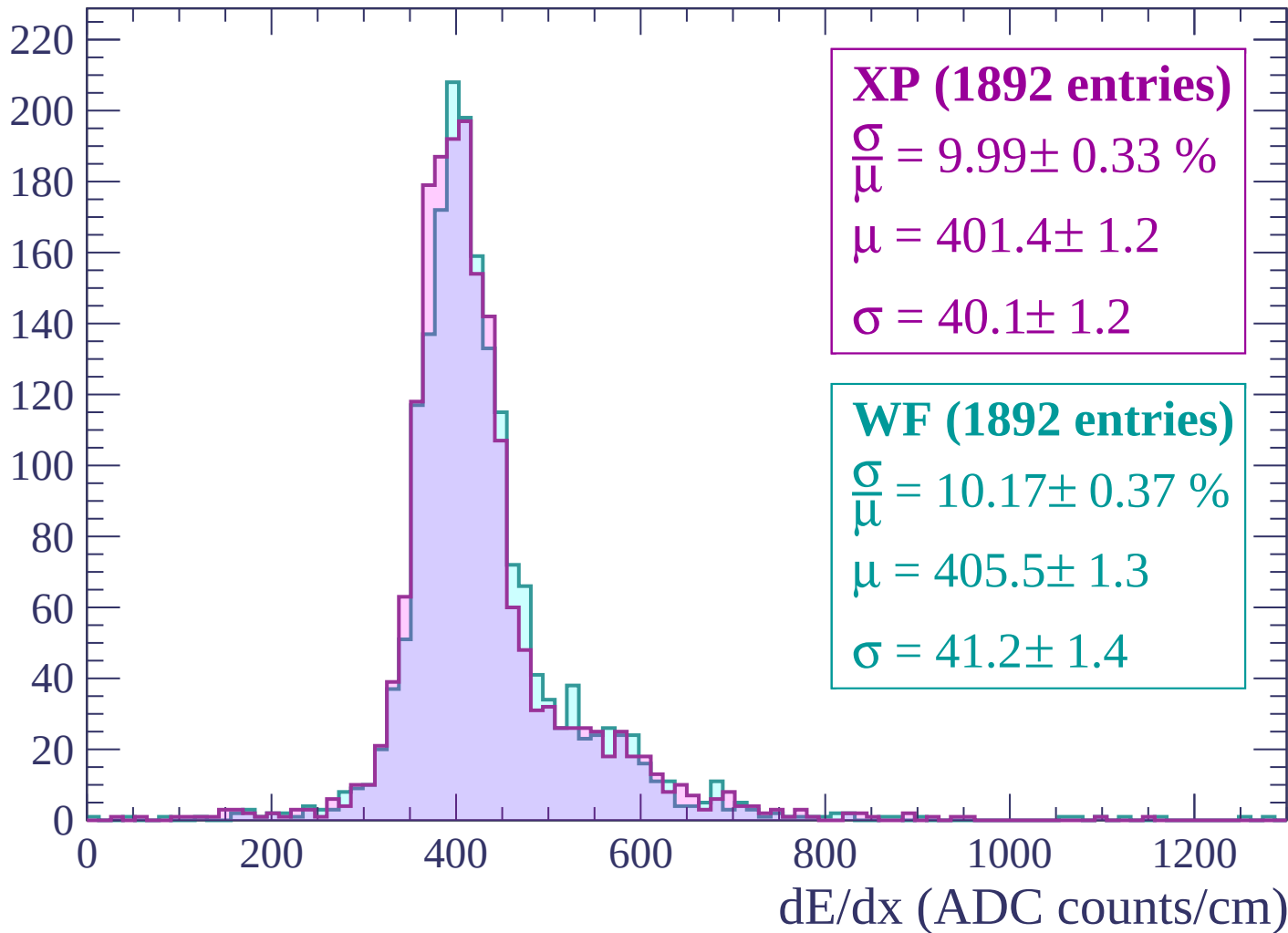
$$\mu = 409.8 \pm 1.3$$

$$\sigma = 43.9 \pm 1.4$$

dE/dx (ADC counts/cm)

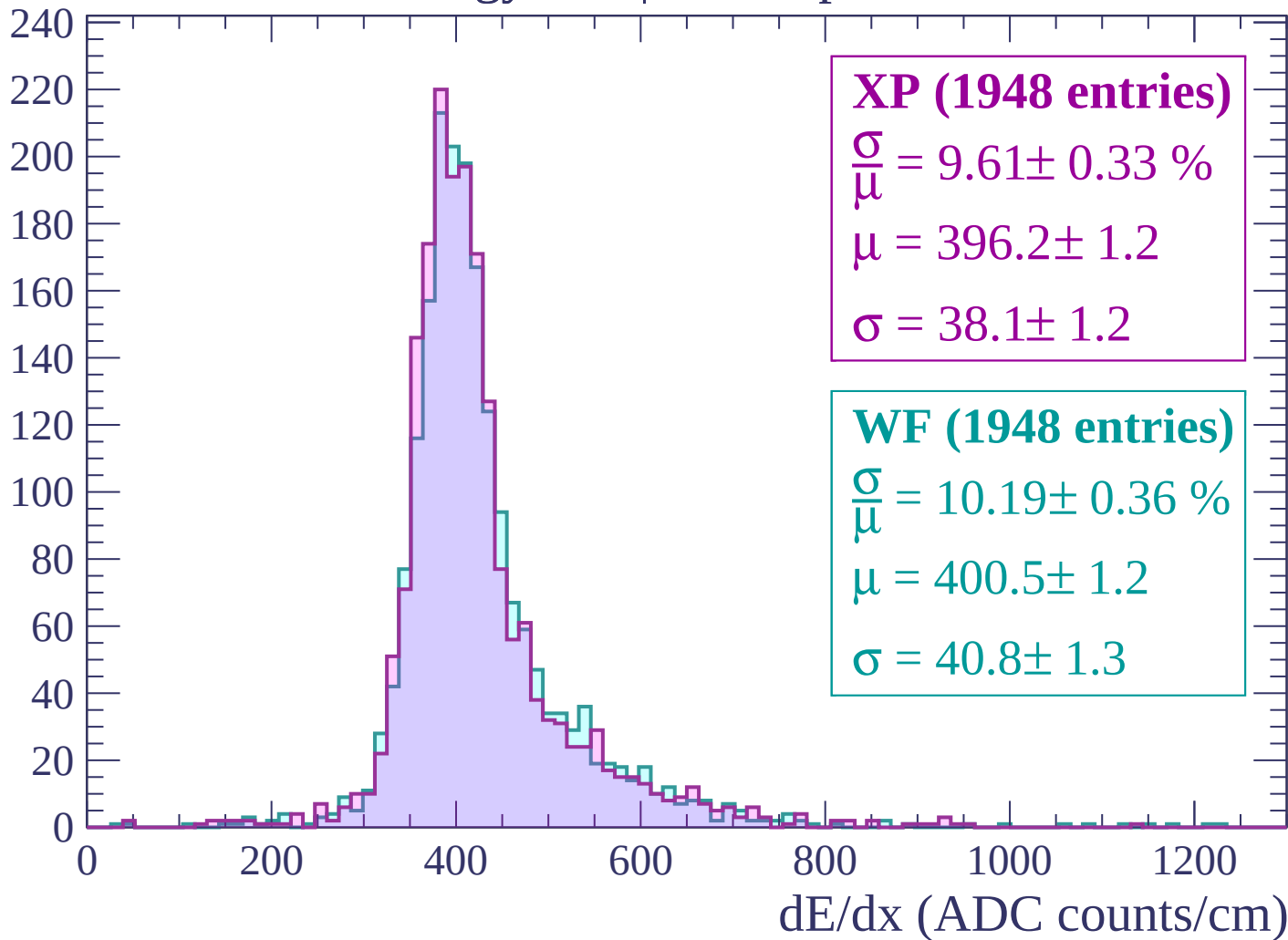
Energy loss | $-420 < p < -360$

Count



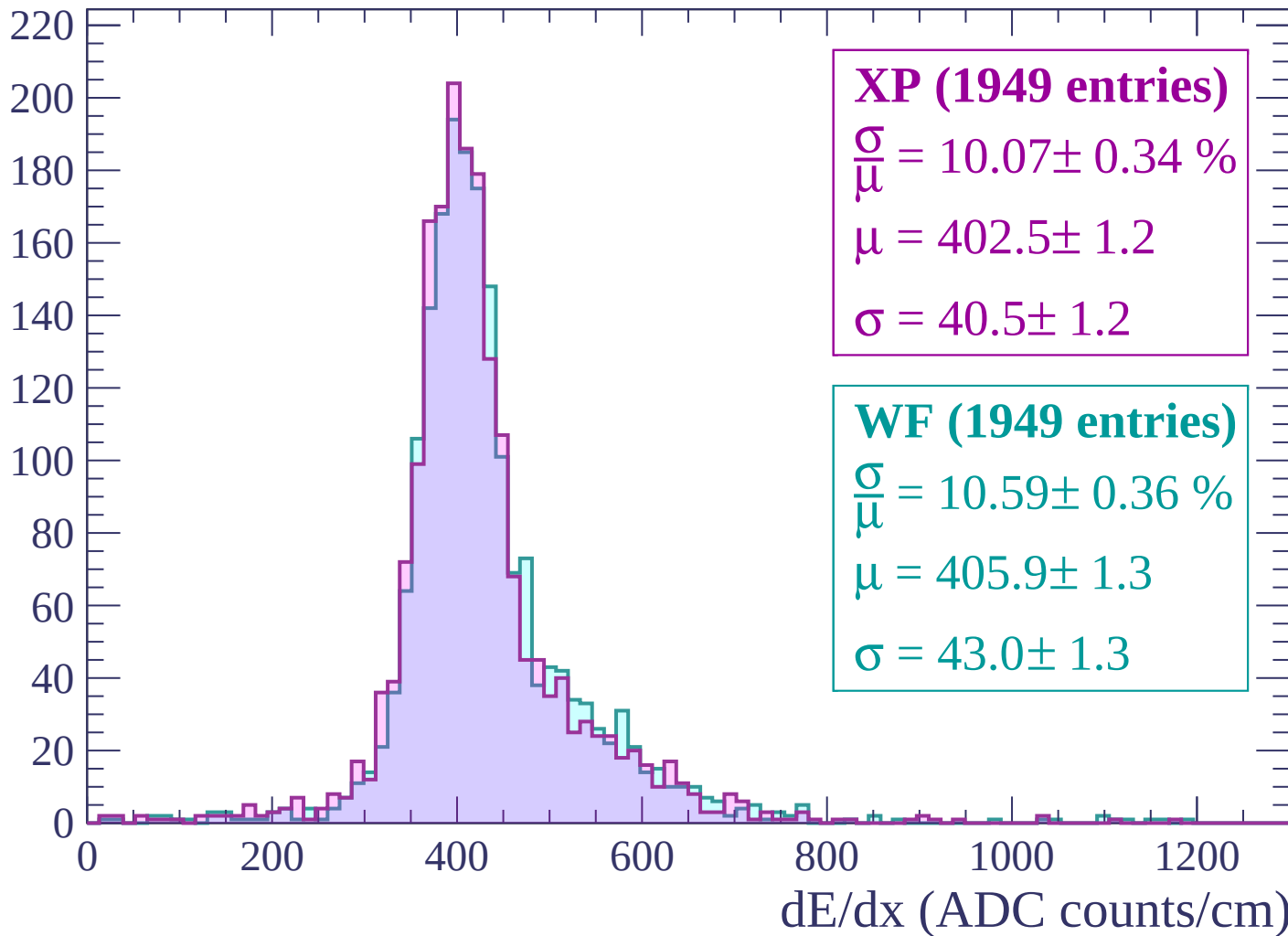
Energy loss | $-360 < p < -300$

Count



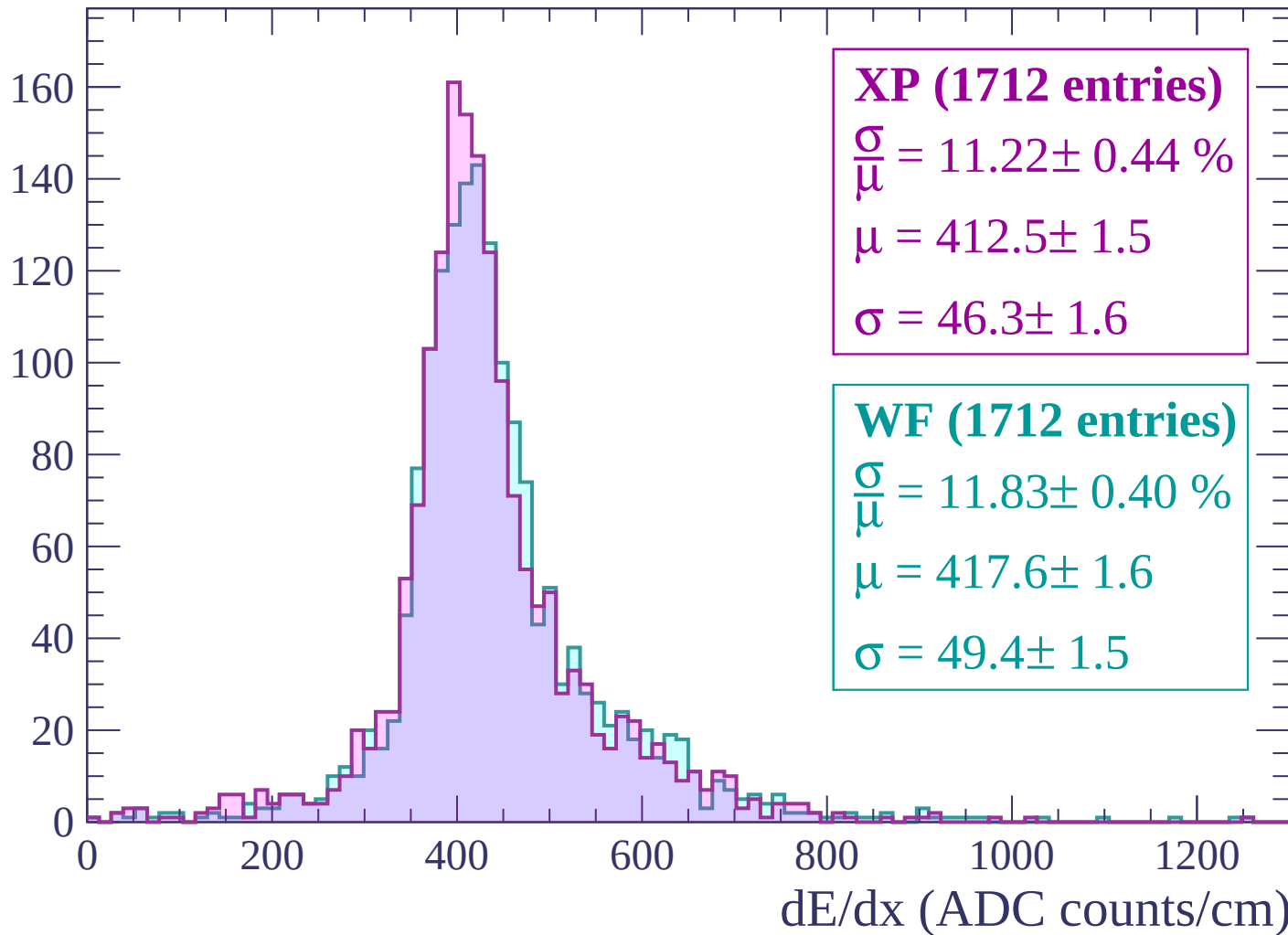
Energy loss | $-300 < p < -240$

Count



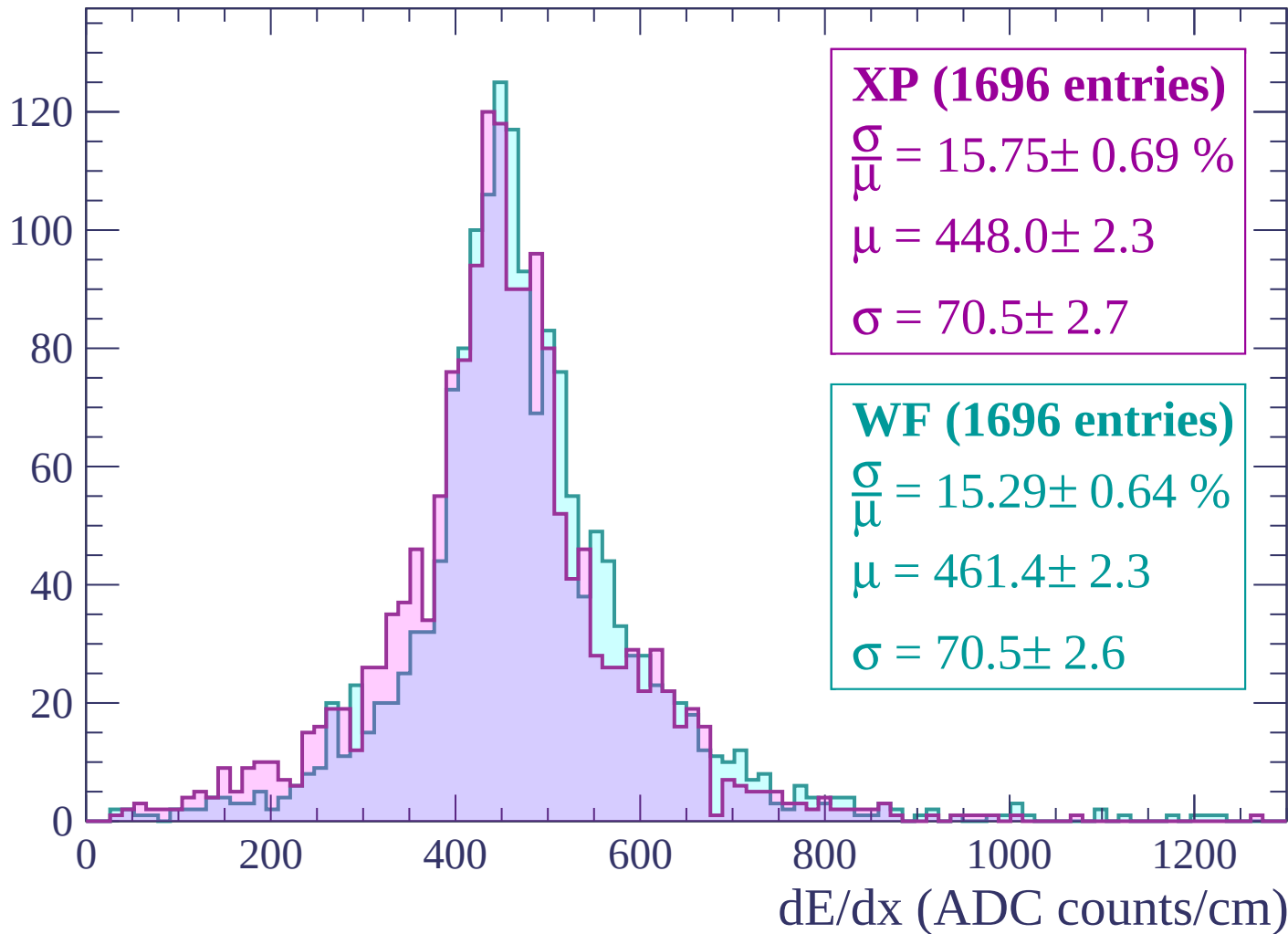
Energy loss | $-240 < p < -180$

Count



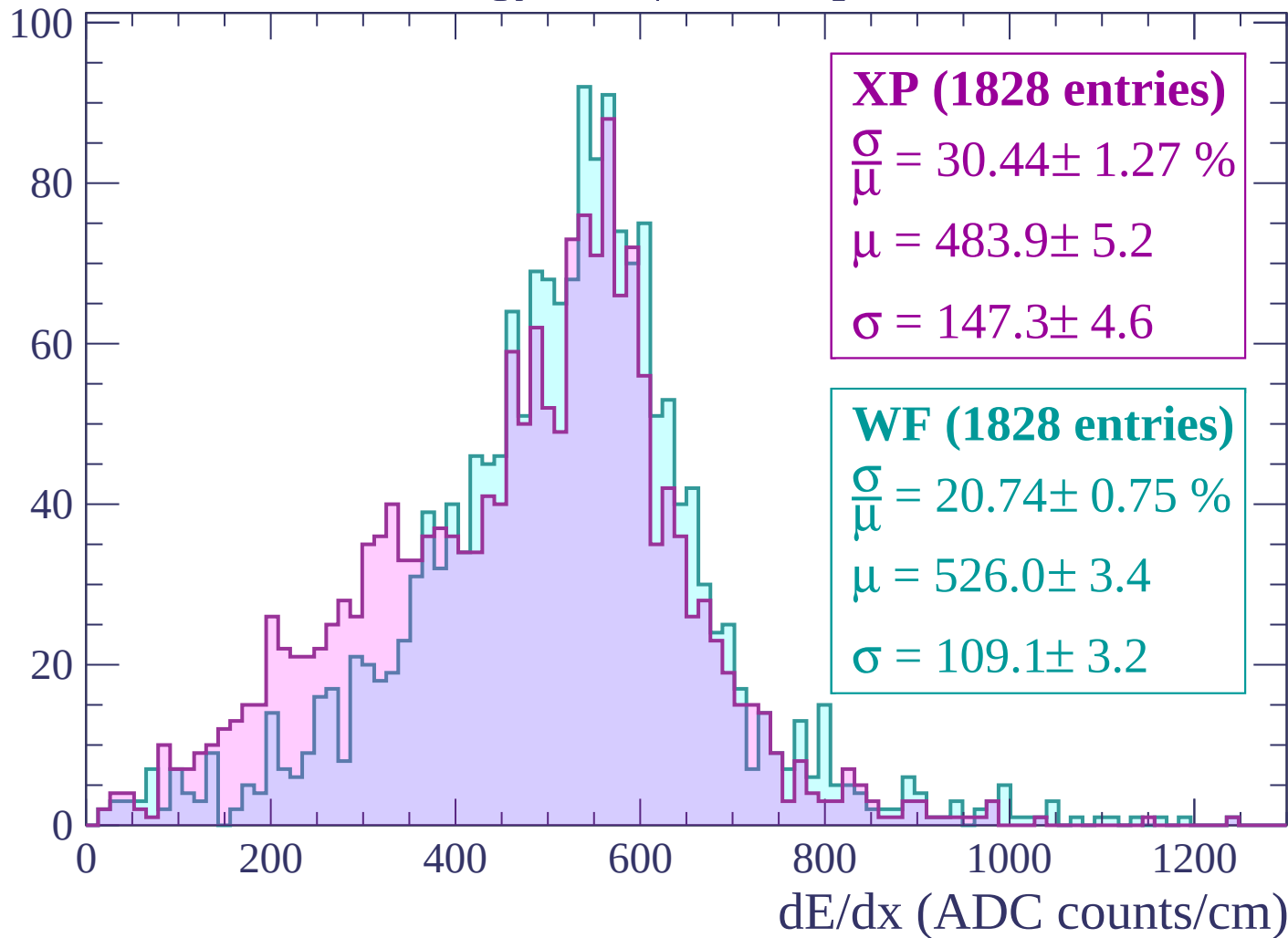
Energy loss | -180 < p < -120

Count



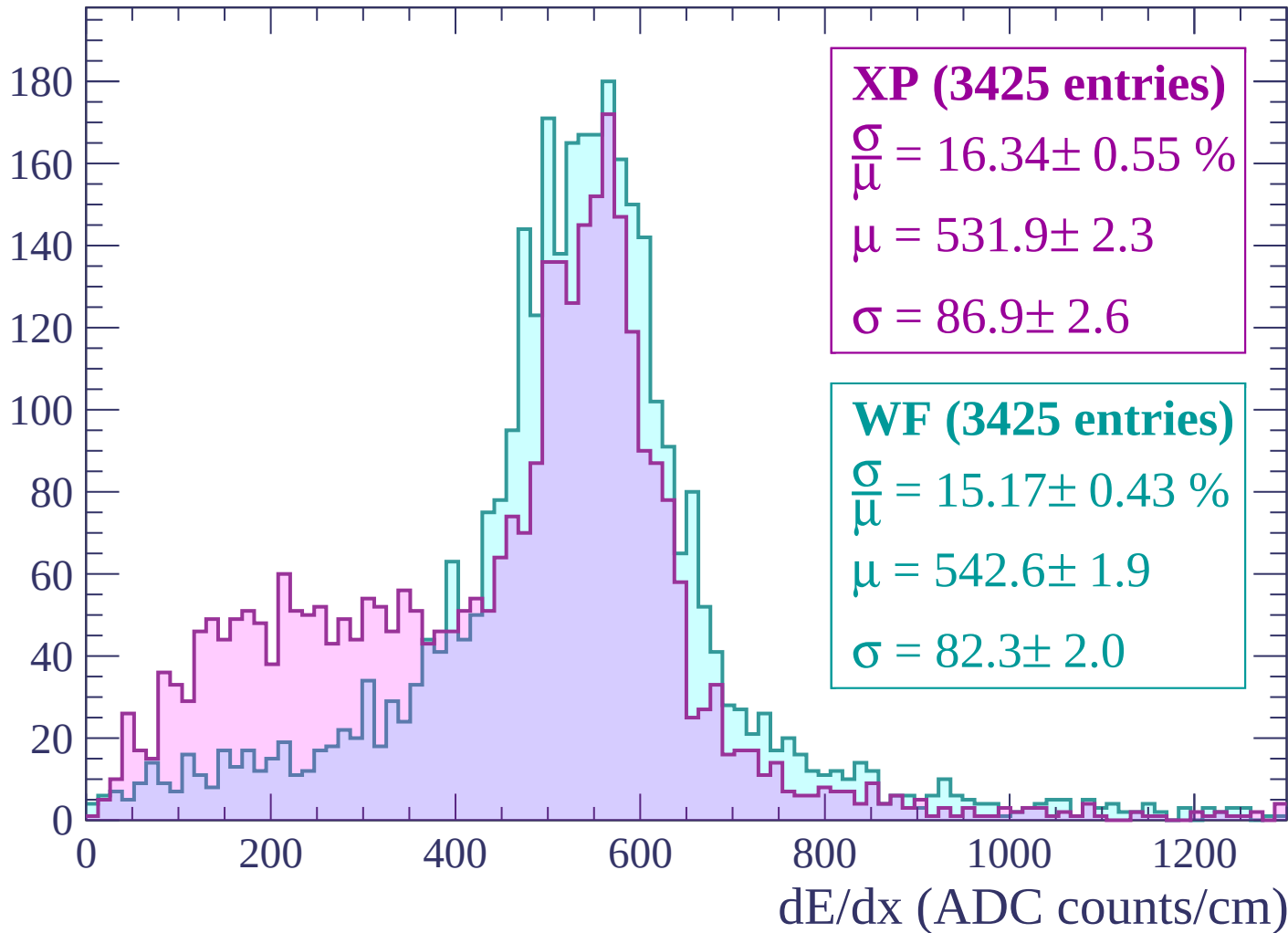
Energy loss | $-120 < p < -60$

Count



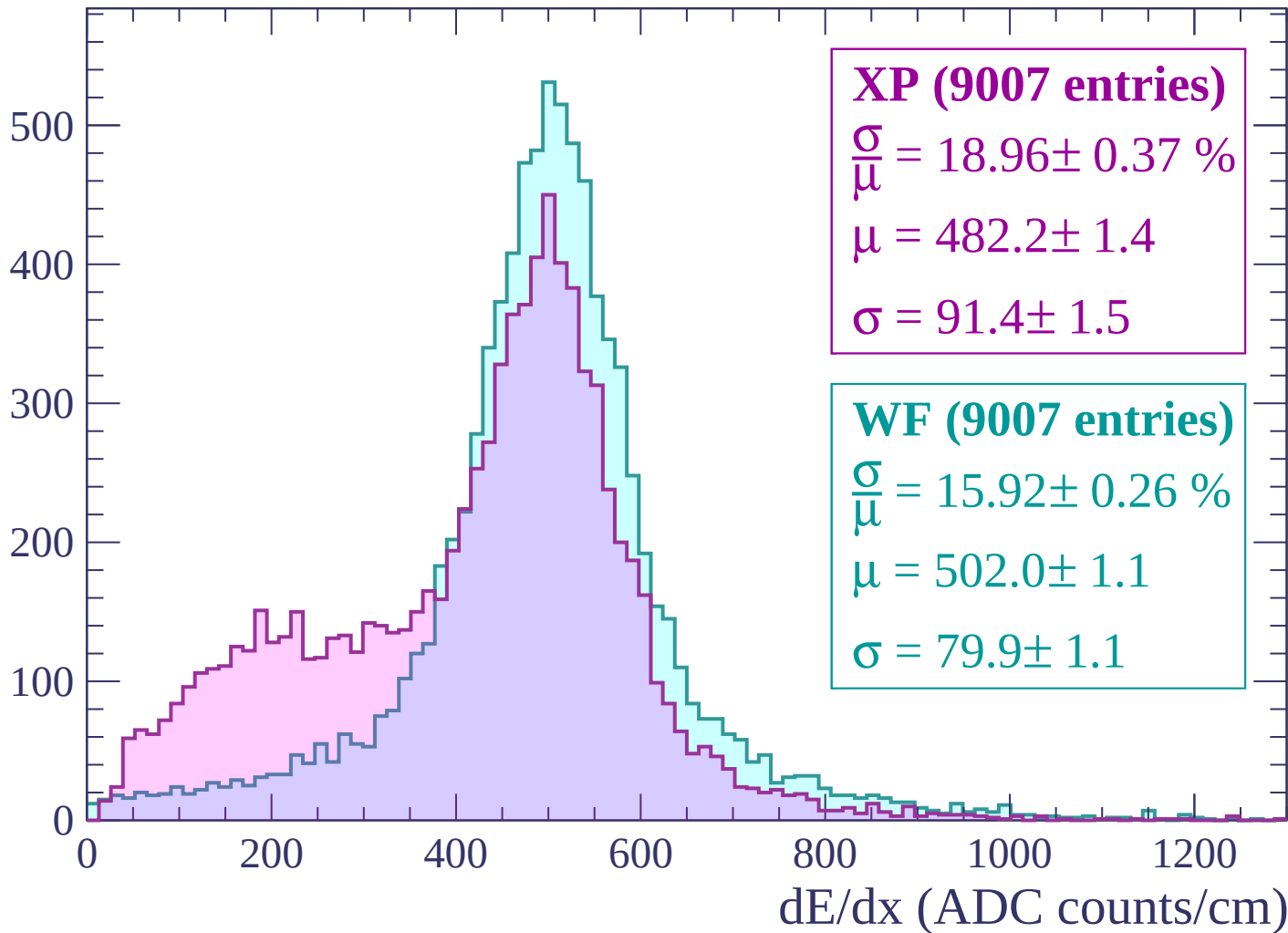
Energy loss | $-60 < p < 0$

Count

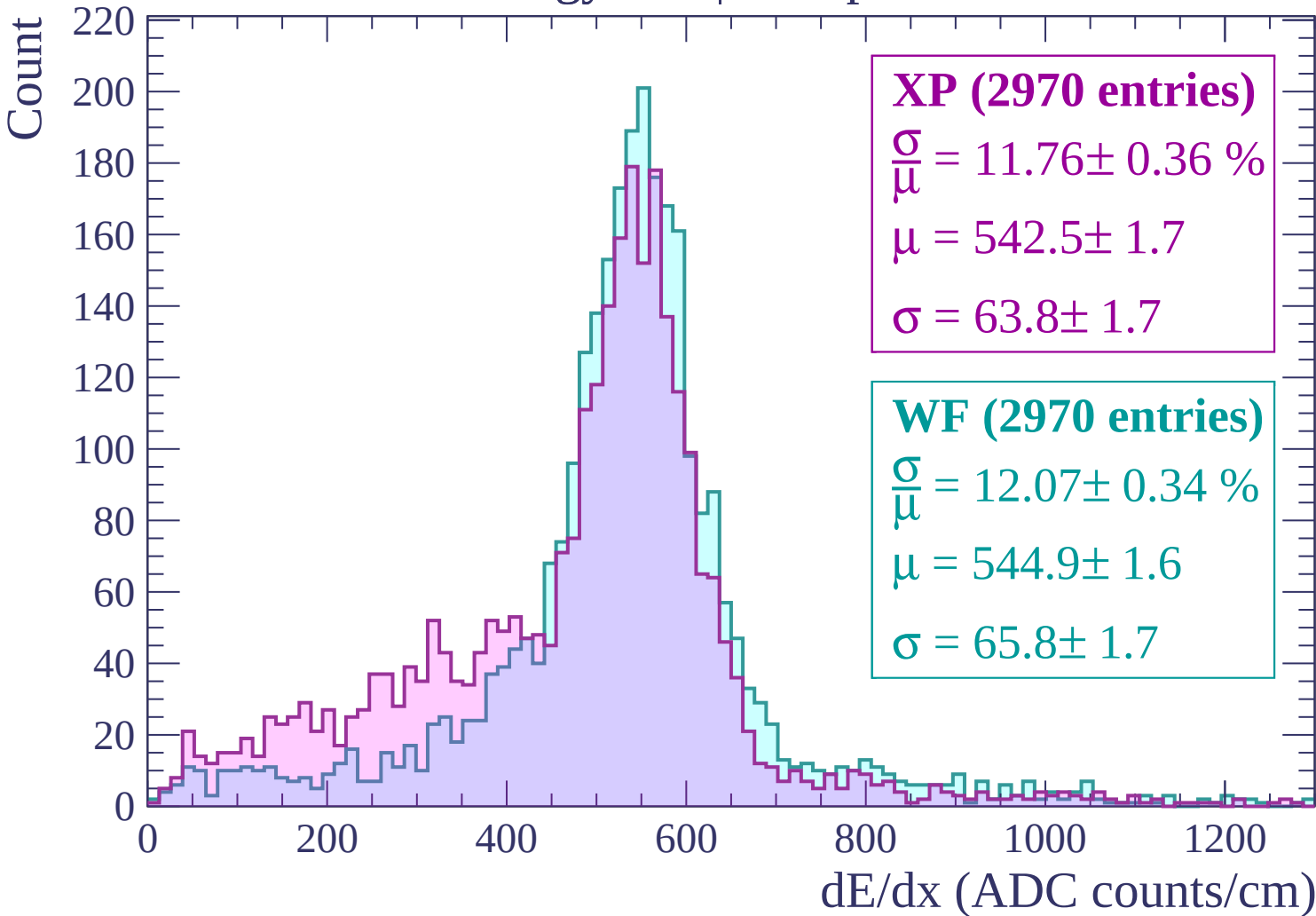


Energy loss | $0 < p < 60$

Count

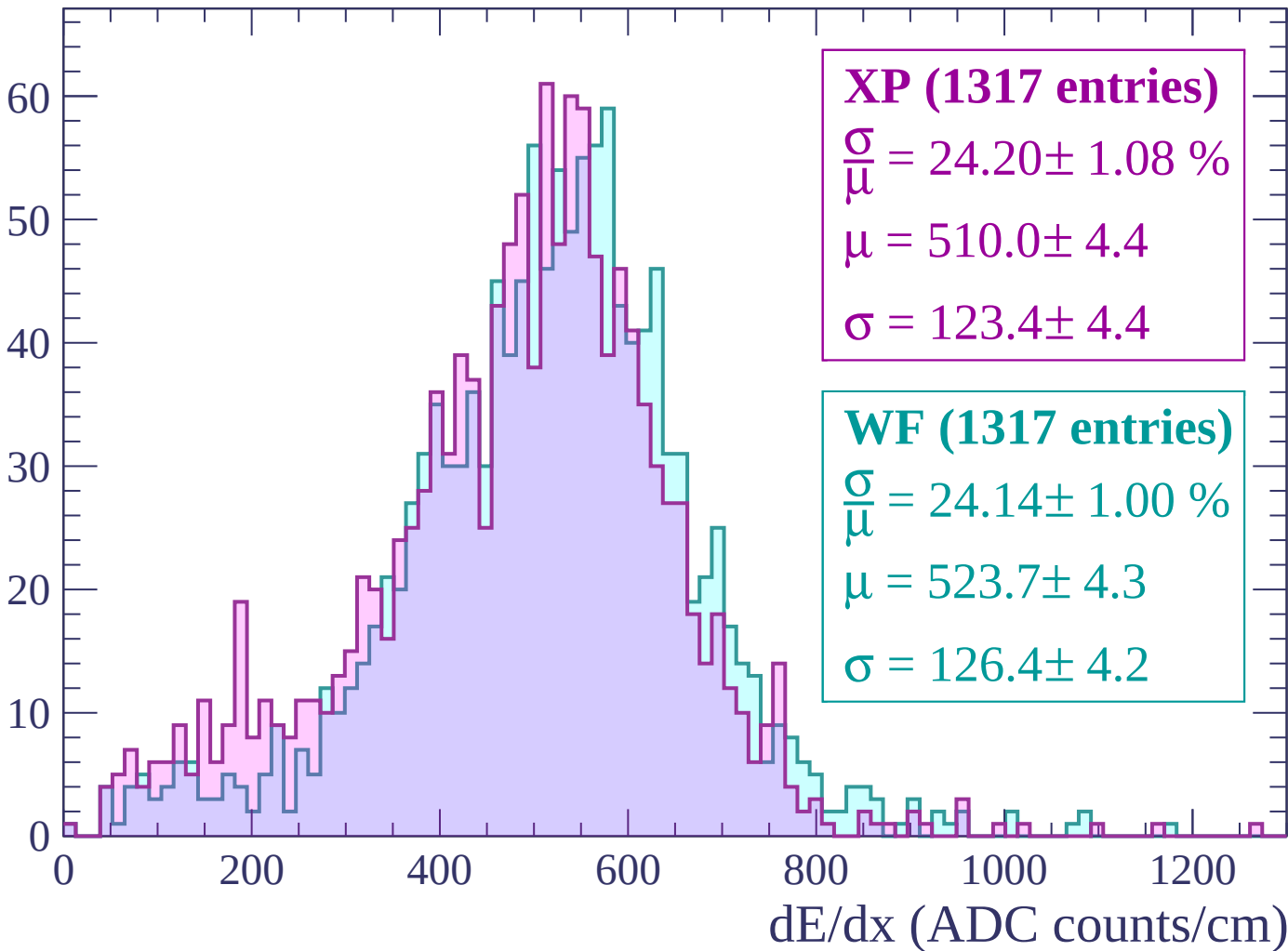


Energy loss | $60 < p < 120$



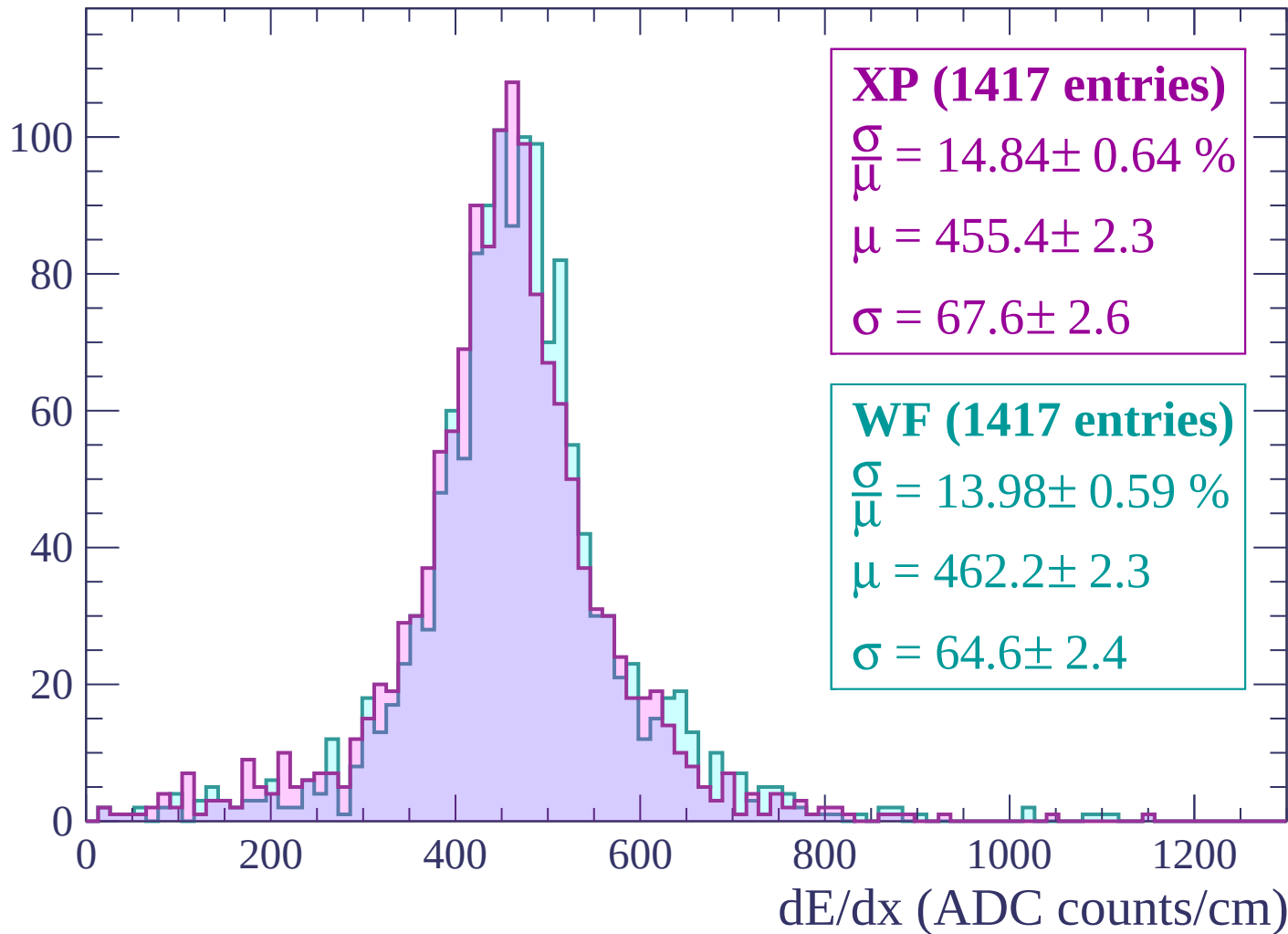
Energy loss | $120 < p < 180$

Count



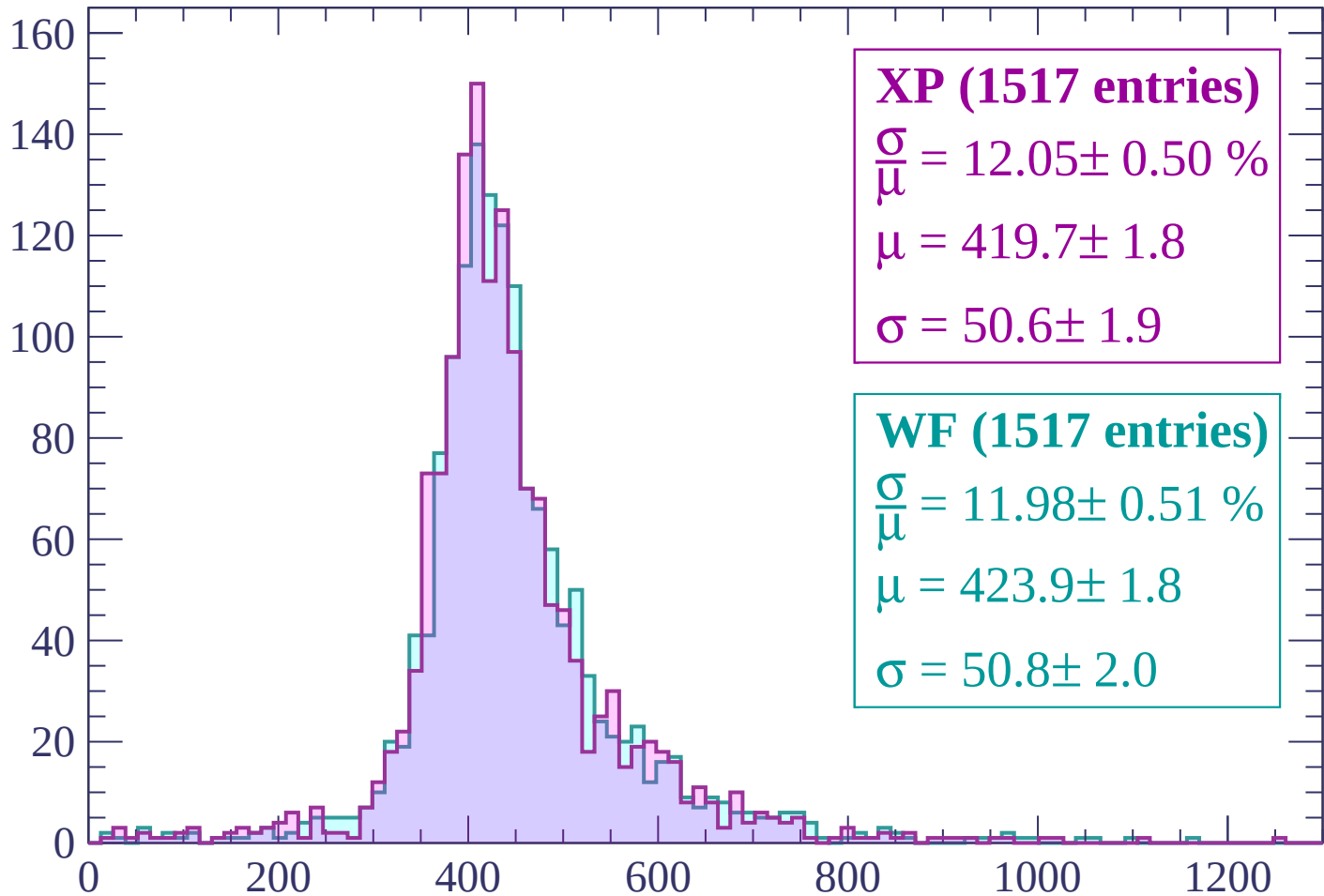
Energy loss | $180 < p < 240$

Count



Energy loss | $240 < p < 300$

Count



XP (1517 entries)

$\frac{\sigma}{\mu} = 12.05 \pm 0.50 \%$

$\mu = 419.7 \pm 1.8$

$\sigma = 50.6 \pm 1.9$

WF (1517 entries)

$\frac{\sigma}{\mu} = 11.98 \pm 0.51 \%$

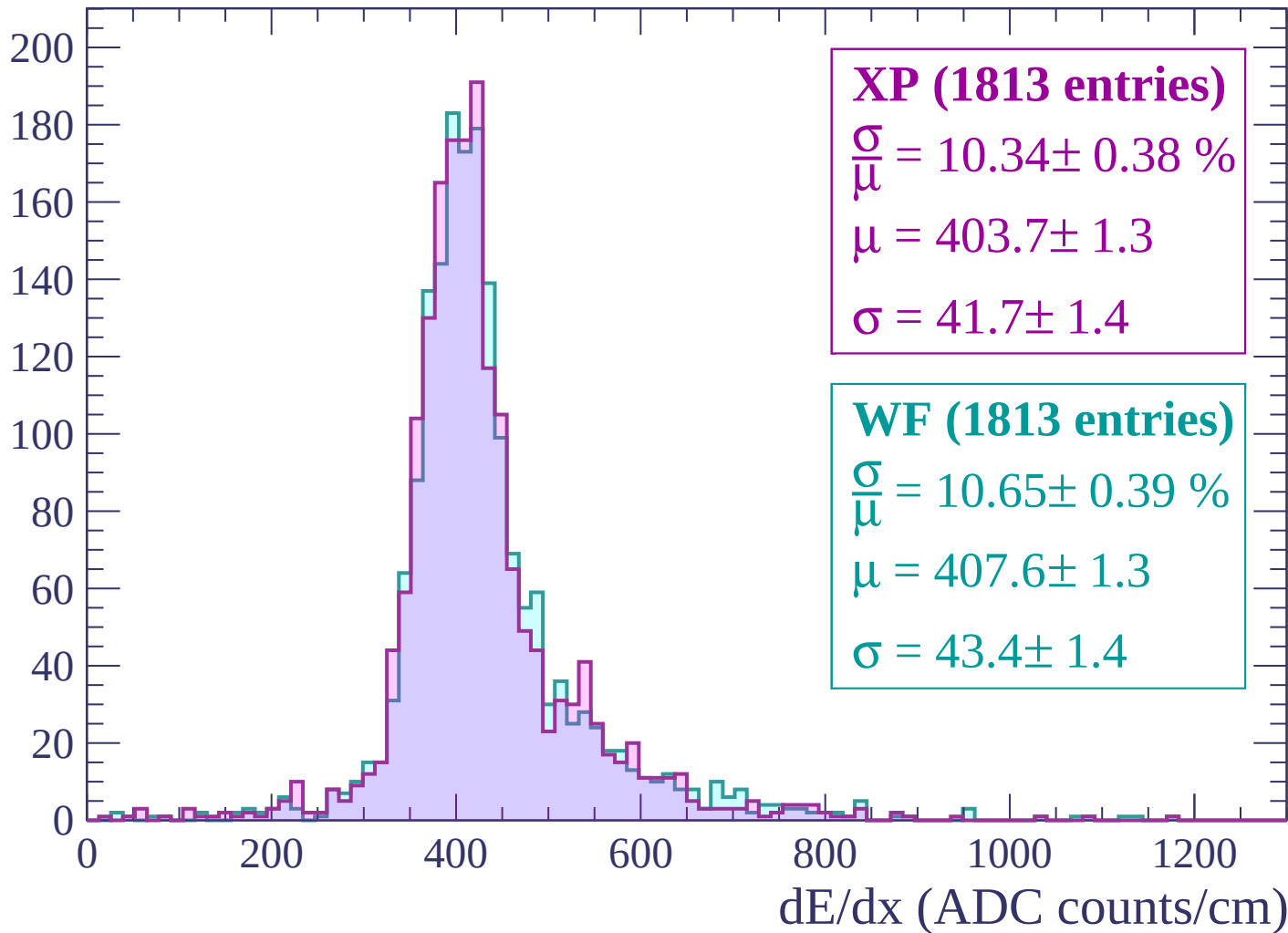
$\mu = 423.9 \pm 1.8$

$\sigma = 50.8 \pm 2.0$

dE/dx (ADC counts/cm)

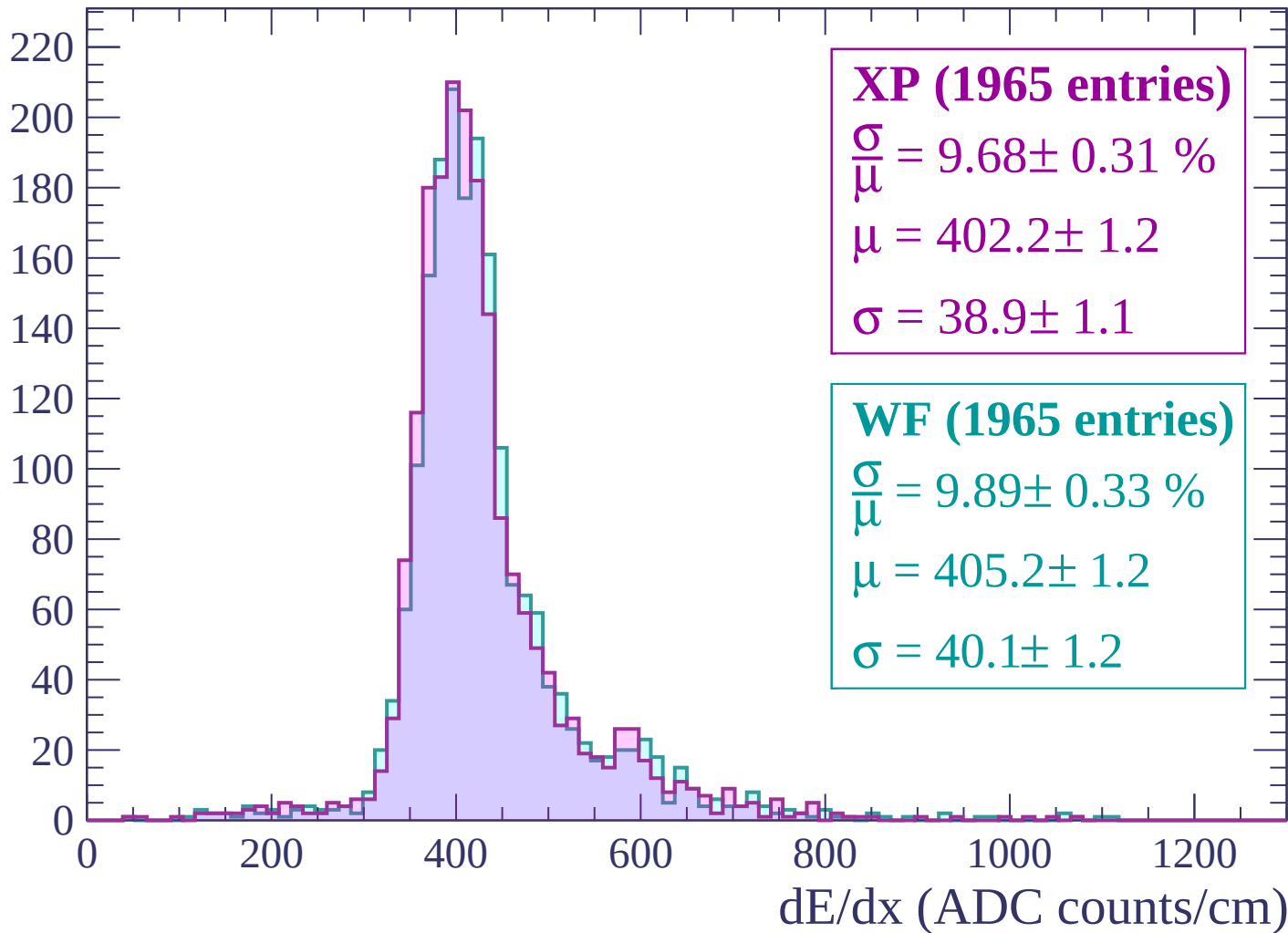
Energy loss | $300 < p < 360$

Count



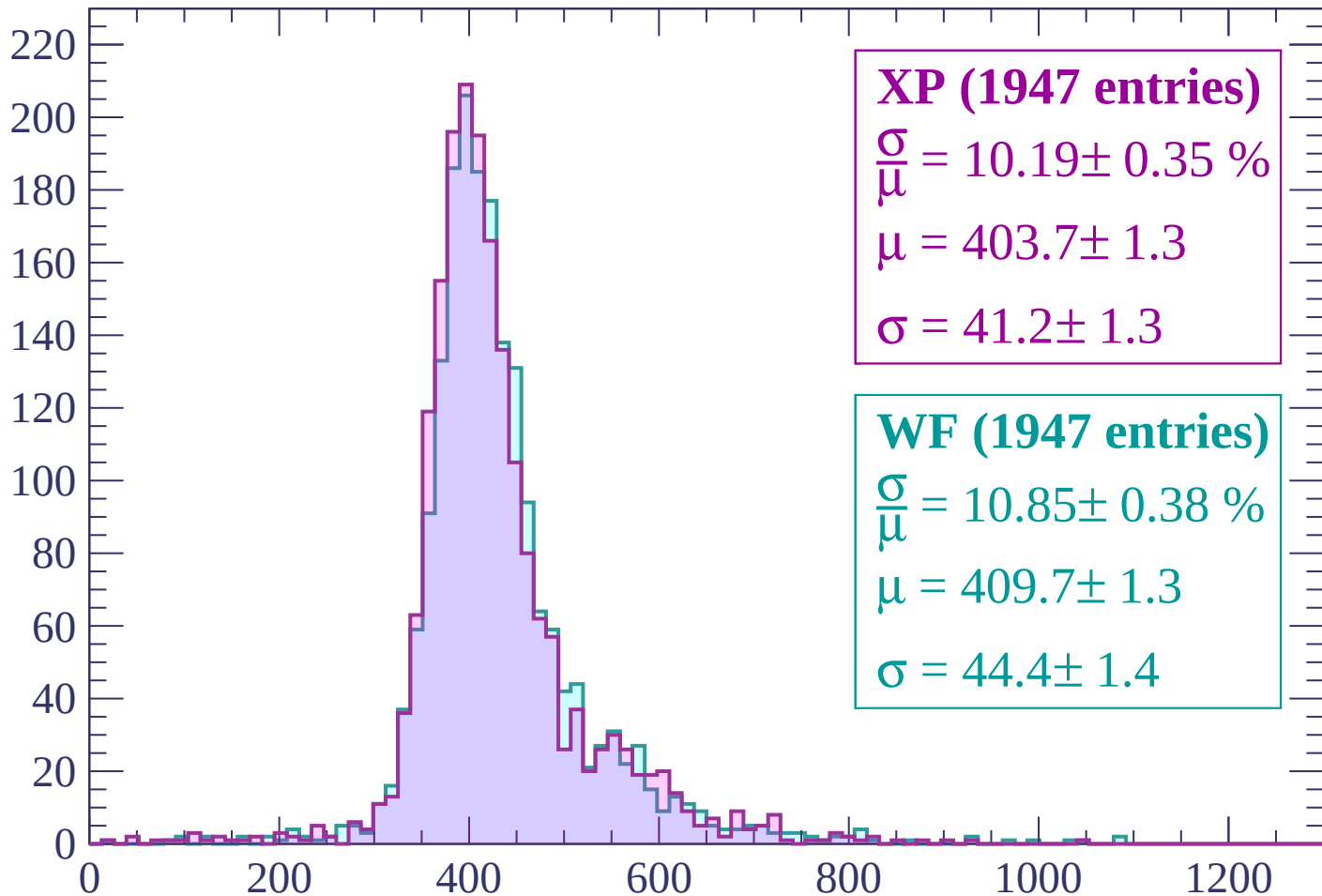
Energy loss | $360 < p < 420$

Count



Energy loss | $420 < p < 480$

Count



XP (1947 entries)

$$\frac{\sigma}{\mu} = 10.19 \pm 0.35 \%$$

$$\mu = 403.7 \pm 1.3$$

$$\sigma = 41.2 \pm 1.3$$

WF (1947 entries)

$$\frac{\sigma}{\mu} = 10.85 \pm 0.38 \%$$

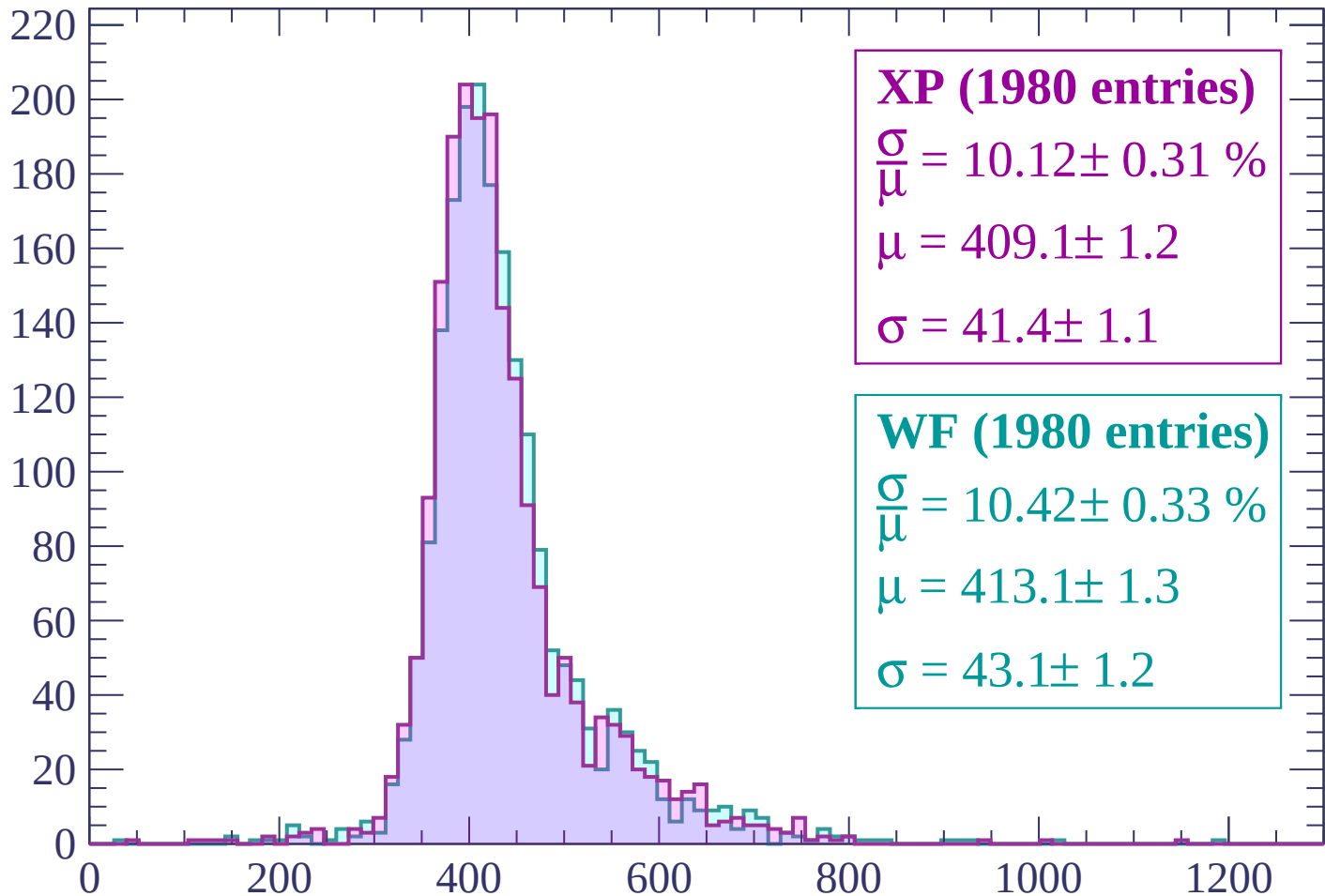
$$\mu = 409.7 \pm 1.3$$

$$\sigma = 44.4 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count



XP (1980 entries)

$$\frac{\sigma}{\mu} = 10.12 \pm 0.31 \%$$

$$\mu = 409.1 \pm 1.2$$

$$\sigma = 41.4 \pm 1.1$$

WF (1980 entries)

$$\frac{\sigma}{\mu} = 10.42 \pm 0.33 \%$$

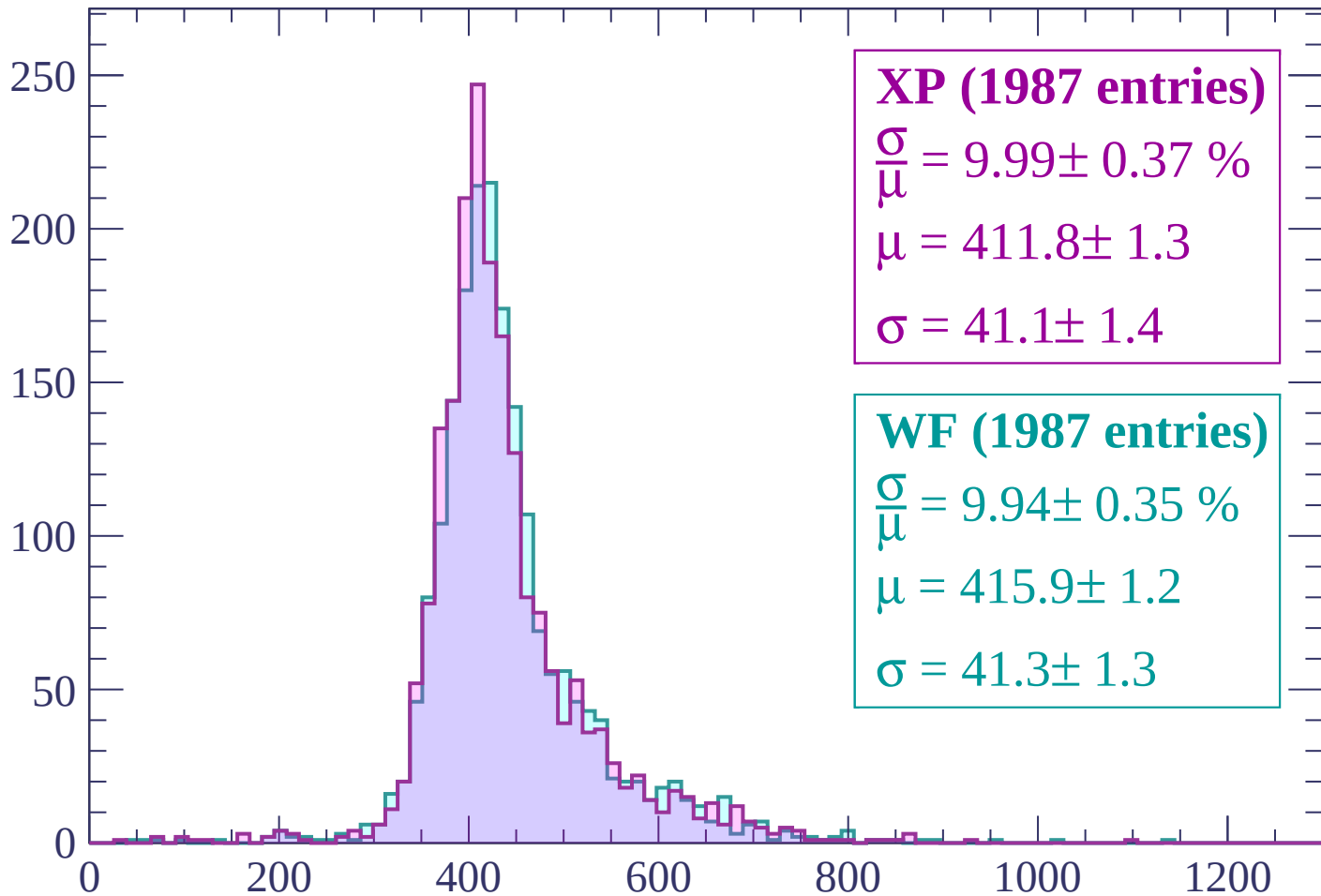
$$\mu = 413.1 \pm 1.3$$

$$\sigma = 43.1 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP (1987 entries)

$$\frac{\sigma}{\mu} = 9.99 \pm 0.37 \%$$

$$\mu = 411.8 \pm 1.3$$

$$\sigma = 41.1 \pm 1.4$$

WF (1987 entries)

$$\frac{\sigma}{\mu} = 9.94 \pm 0.35 \%$$

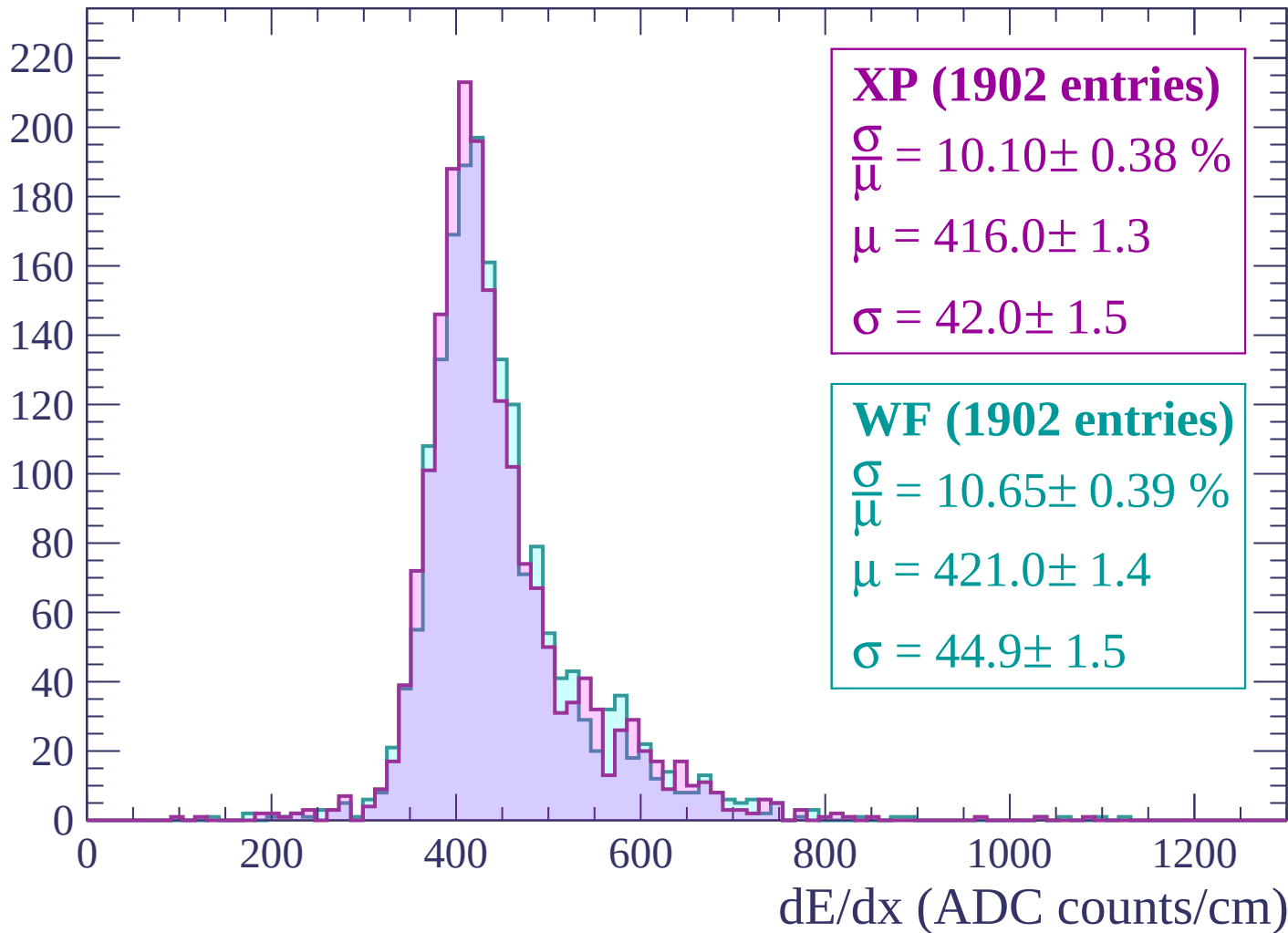
$$\mu = 415.9 \pm 1.2$$

$$\sigma = 41.3 \pm 1.3$$

dE/dx (ADC counts/cm)

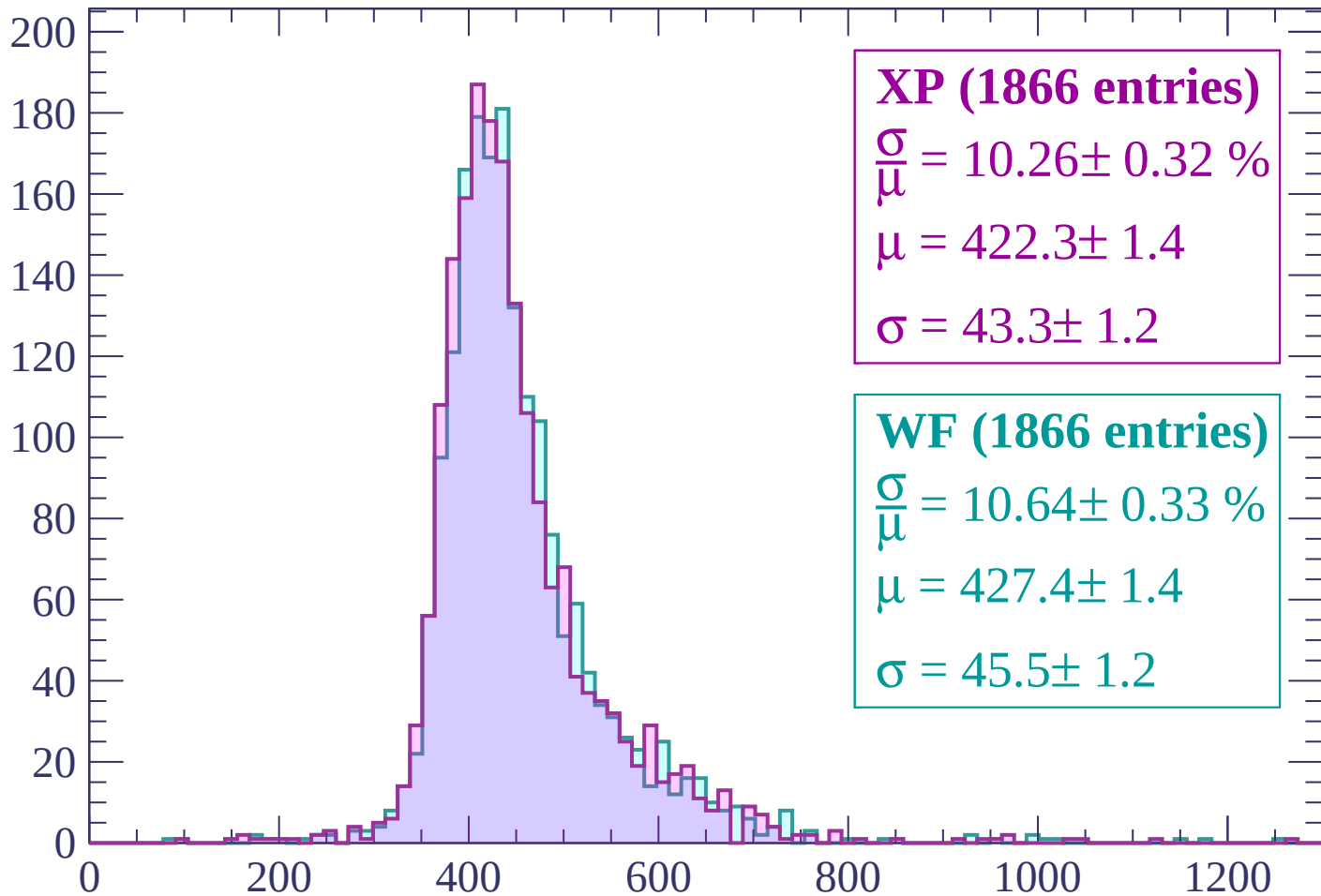
Energy loss | $600 < p < 660$

Count



Energy loss | $660 < p < 720$

Count



XP (1866 entries)

$$\frac{\sigma}{\mu} = 10.26 \pm 0.32 \%$$

$$\mu = 422.3 \pm 1.4$$

$$\sigma = 43.3 \pm 1.2$$

WF (1866 entries)

$$\frac{\sigma}{\mu} = 10.64 \pm 0.33 \%$$

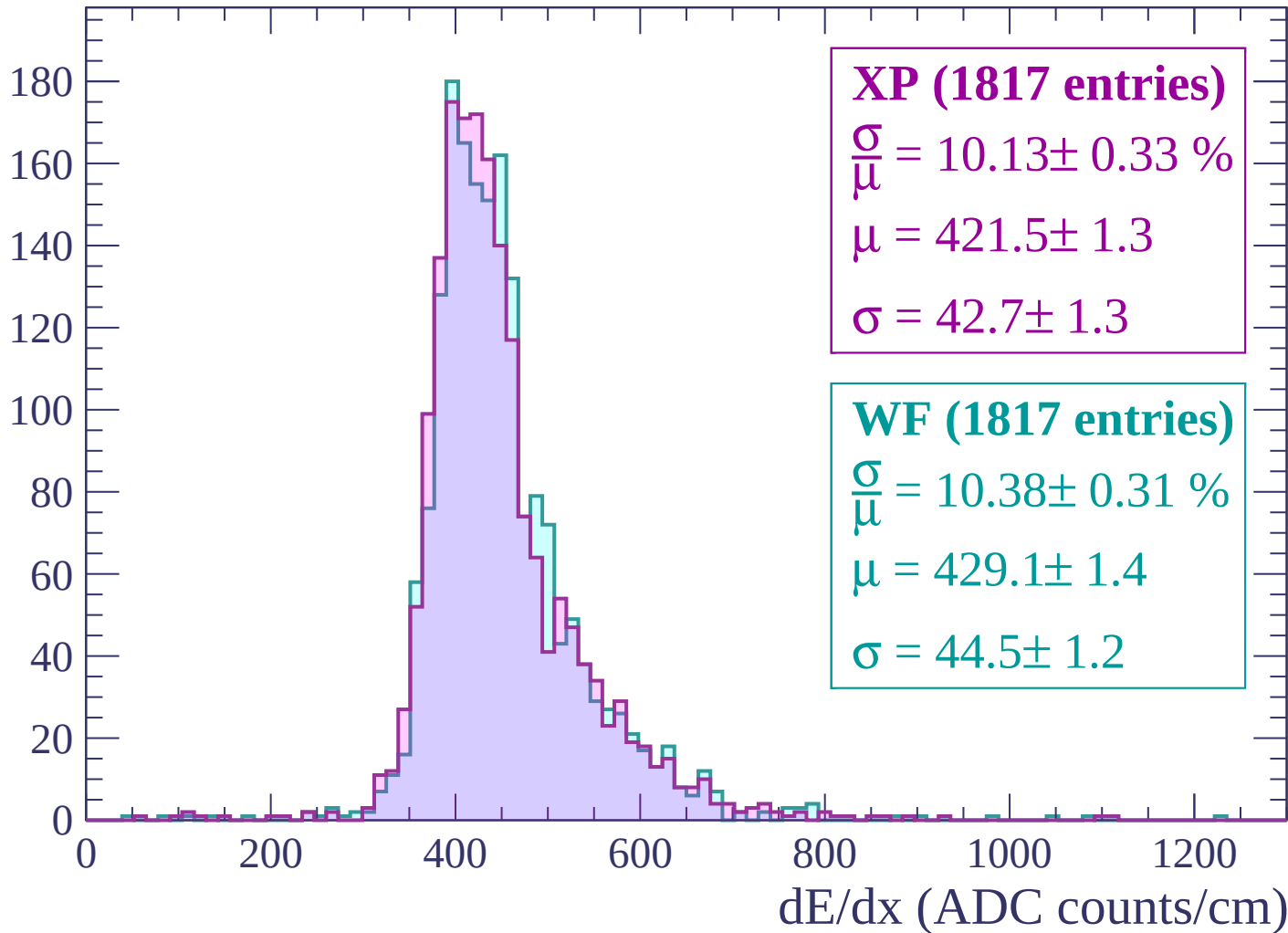
$$\mu = 427.4 \pm 1.4$$

$$\sigma = 45.5 \pm 1.2$$

dE/dx (ADC counts/cm)

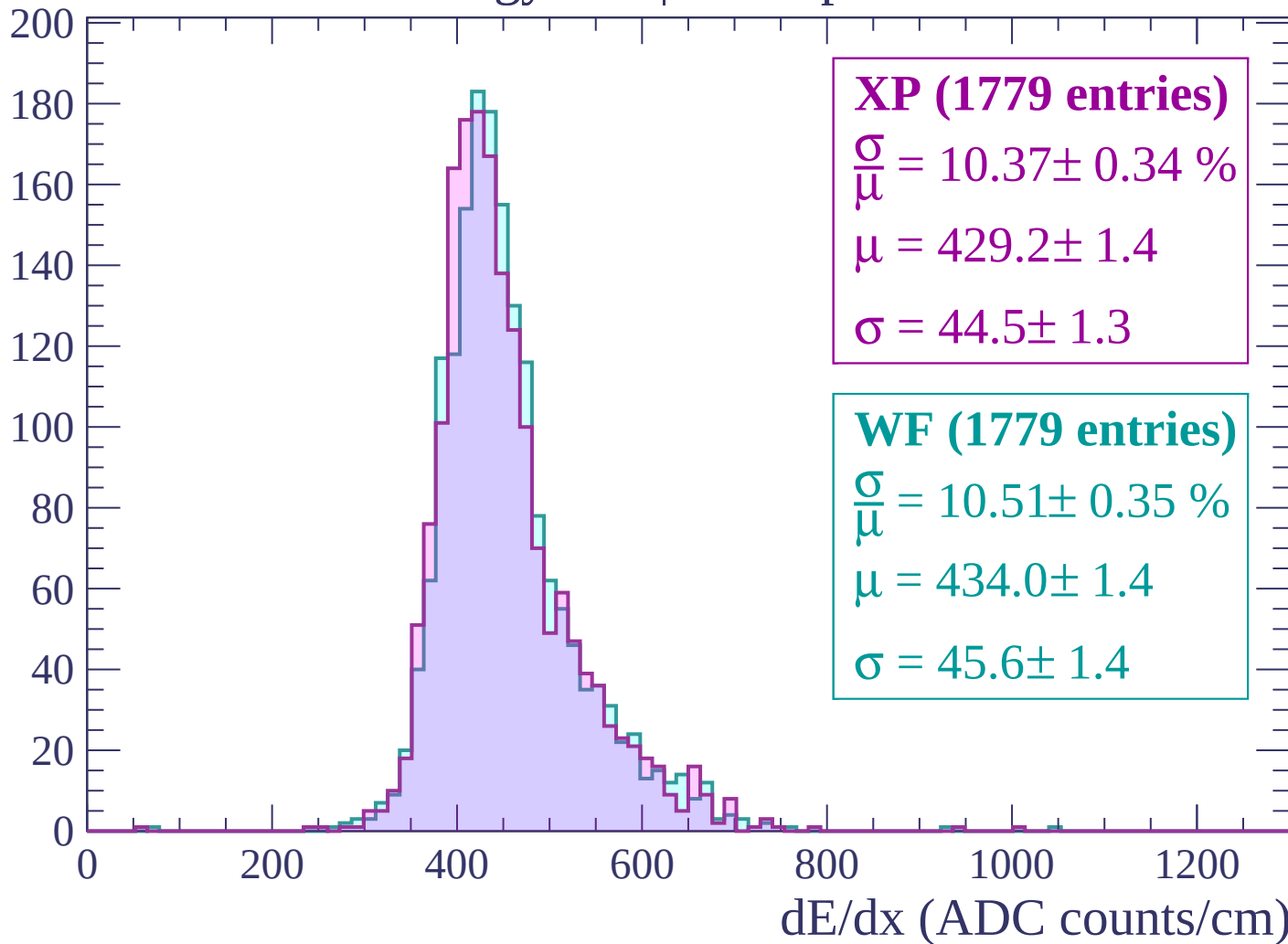
Energy loss | $720 < p < 780$

Count



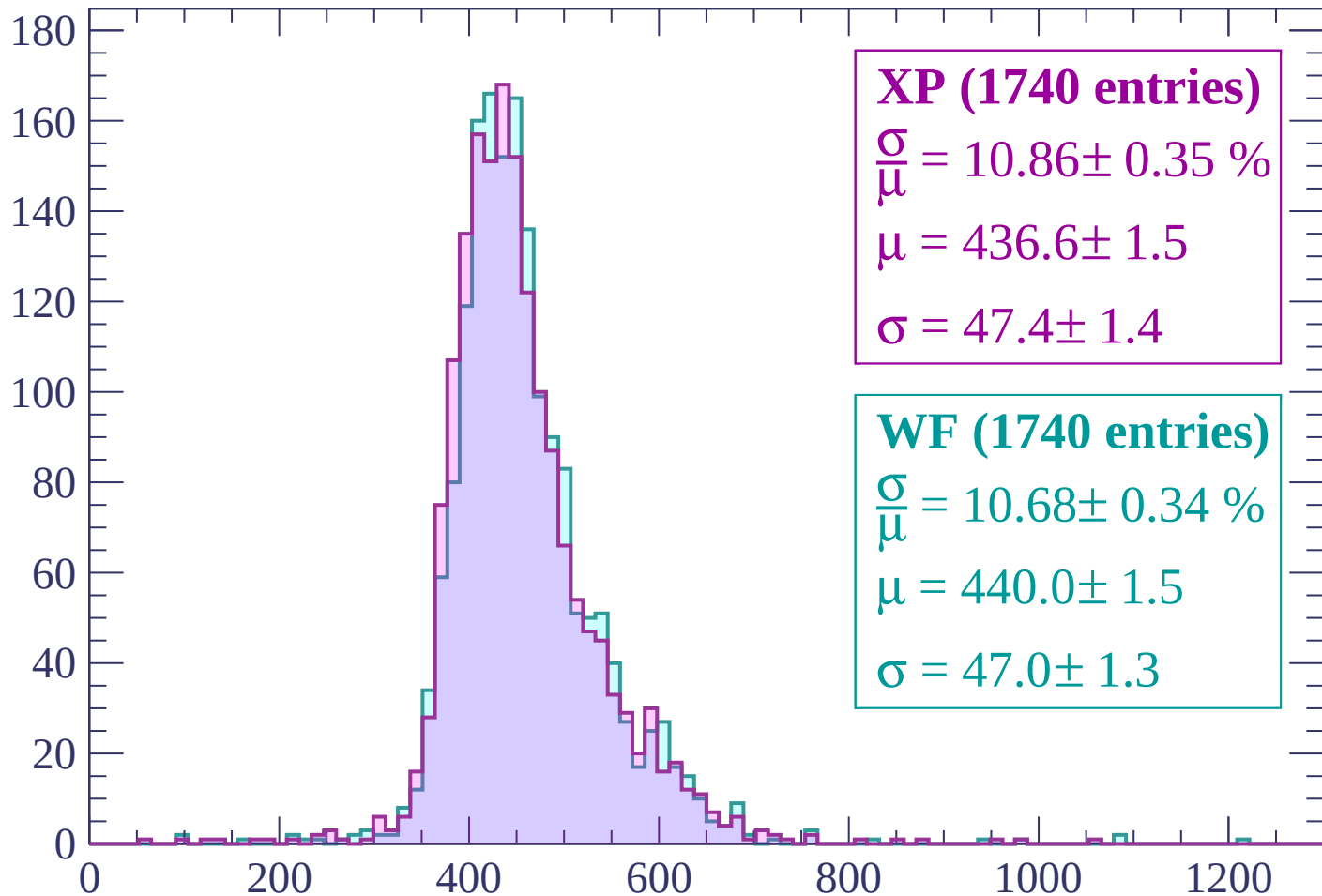
Energy loss | $780 < p < 840$

Count



Energy loss | $840 < p < 900$

Count



XP (1740 entries)

$$\frac{\sigma}{\mu} = 10.86 \pm 0.35 \%$$

$$\mu = 436.6 \pm 1.5$$

$$\sigma = 47.4 \pm 1.4$$

WF (1740 entries)

$$\frac{\sigma}{\mu} = 10.68 \pm 0.34 \%$$

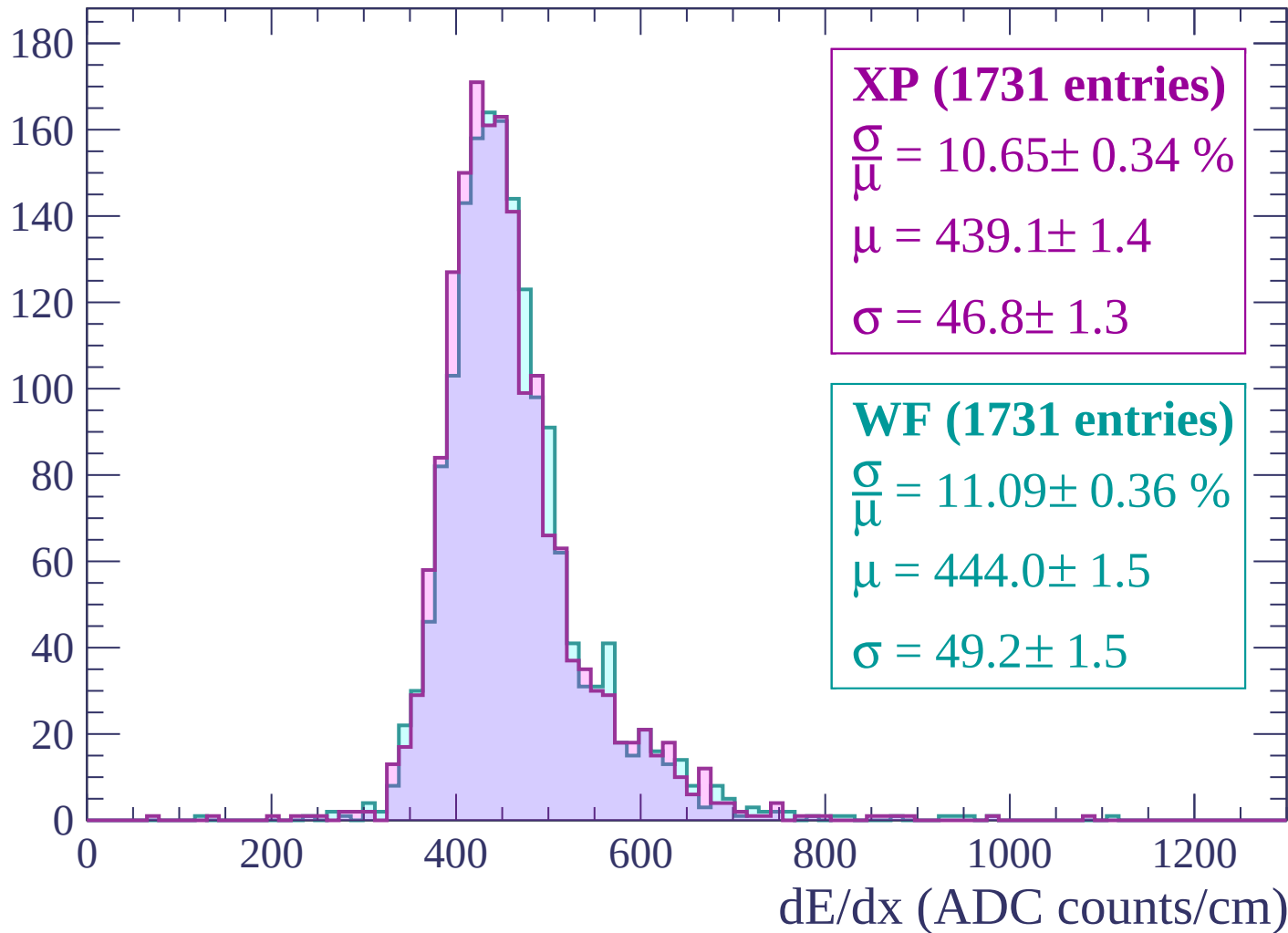
$$\mu = 440.0 \pm 1.5$$

$$\sigma = 47.0 \pm 1.3$$

dE/dx (ADC counts/cm)

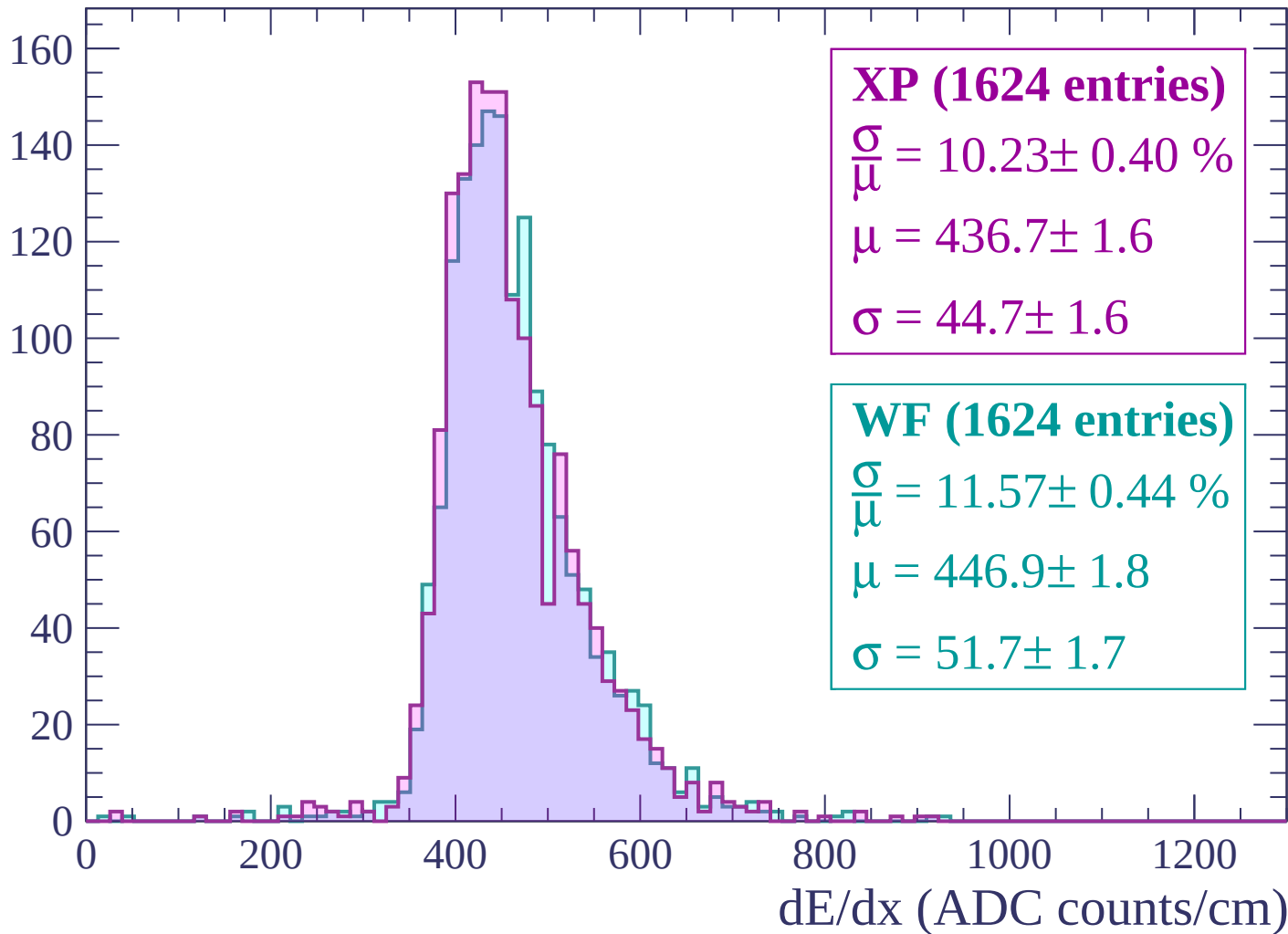
Energy loss | $900 < p < 960$

Count



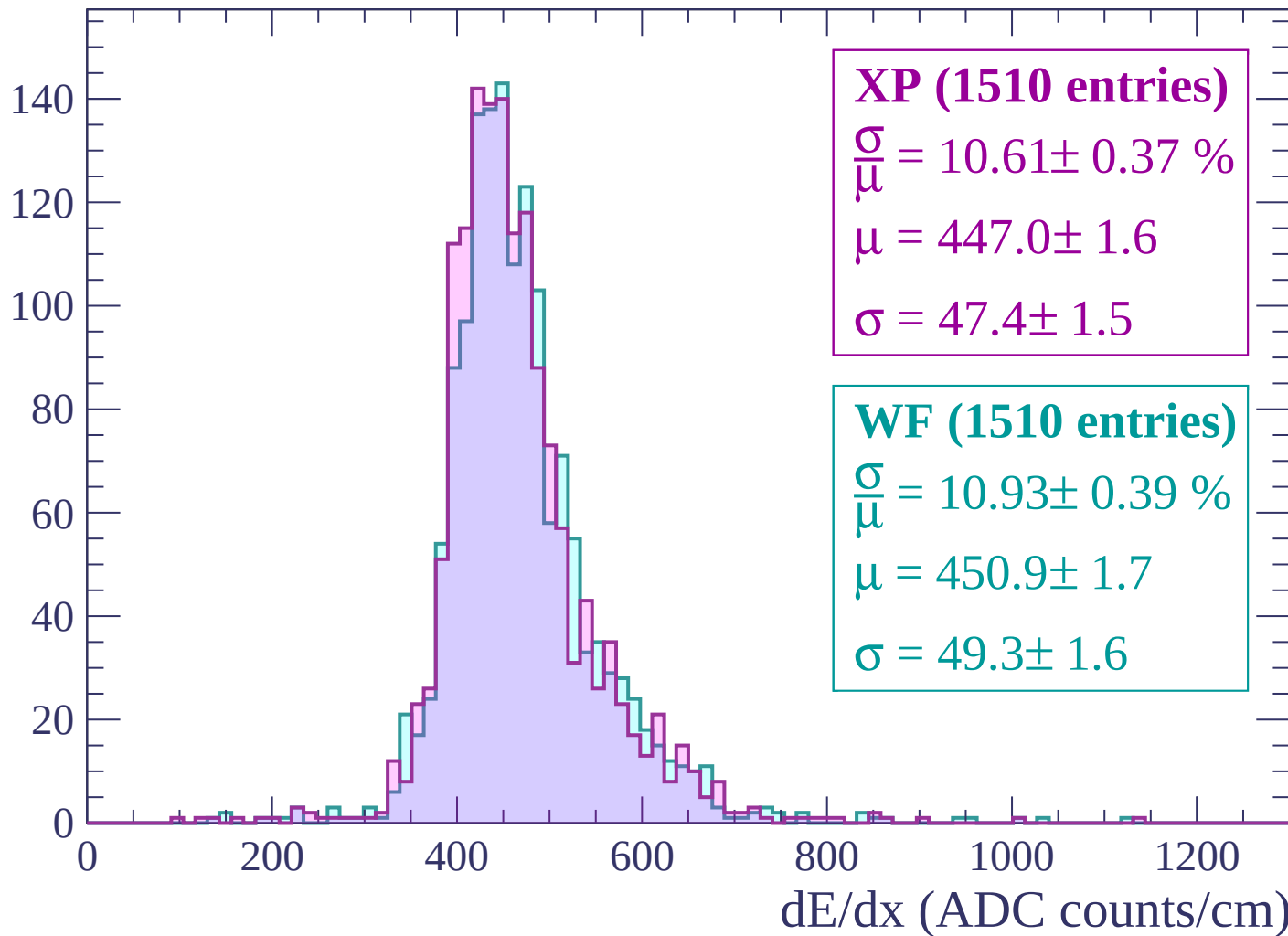
Energy loss | $960 < p < 1020$

Count



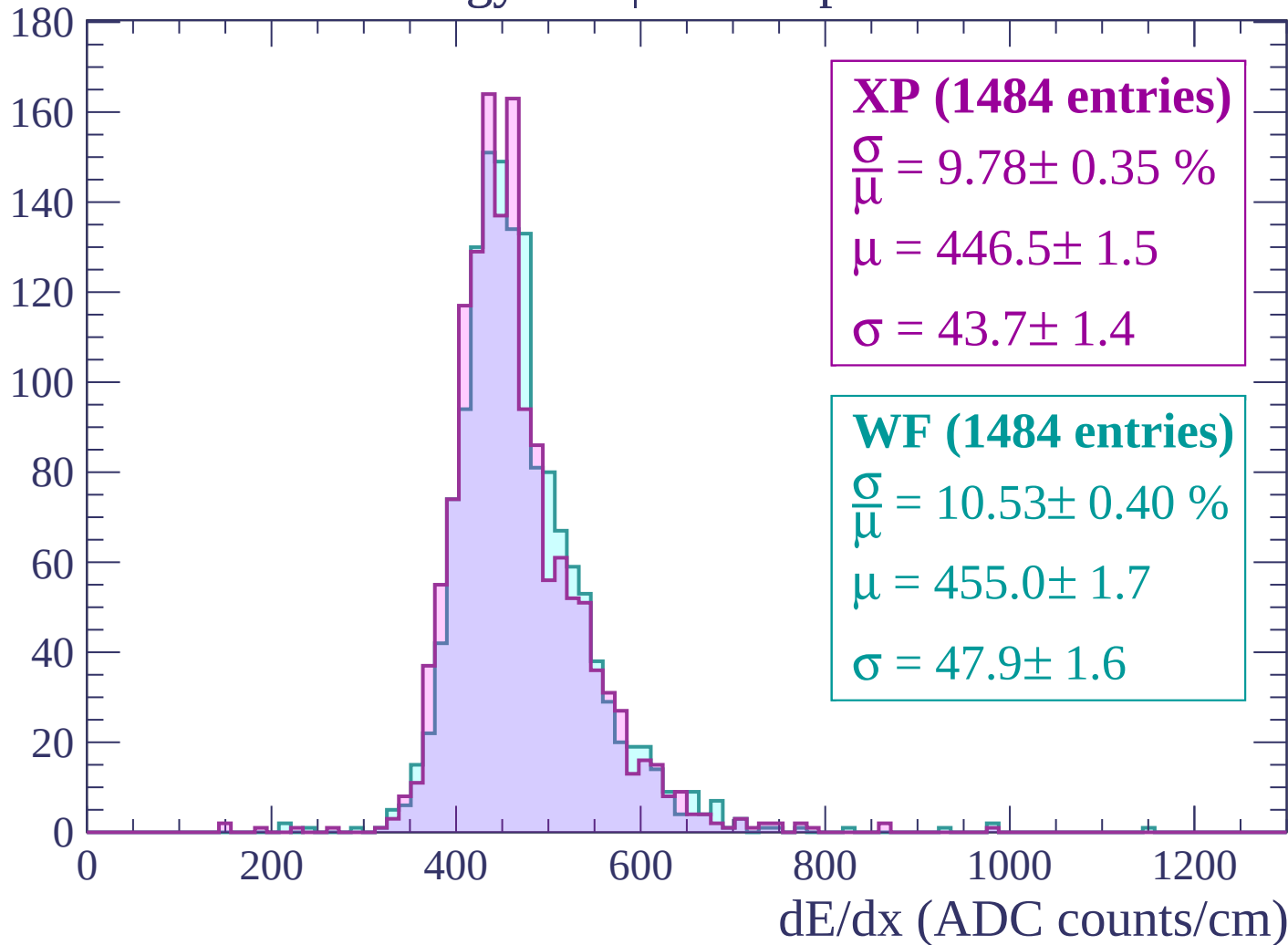
Energy loss | $1020 < p < 1080$

Count



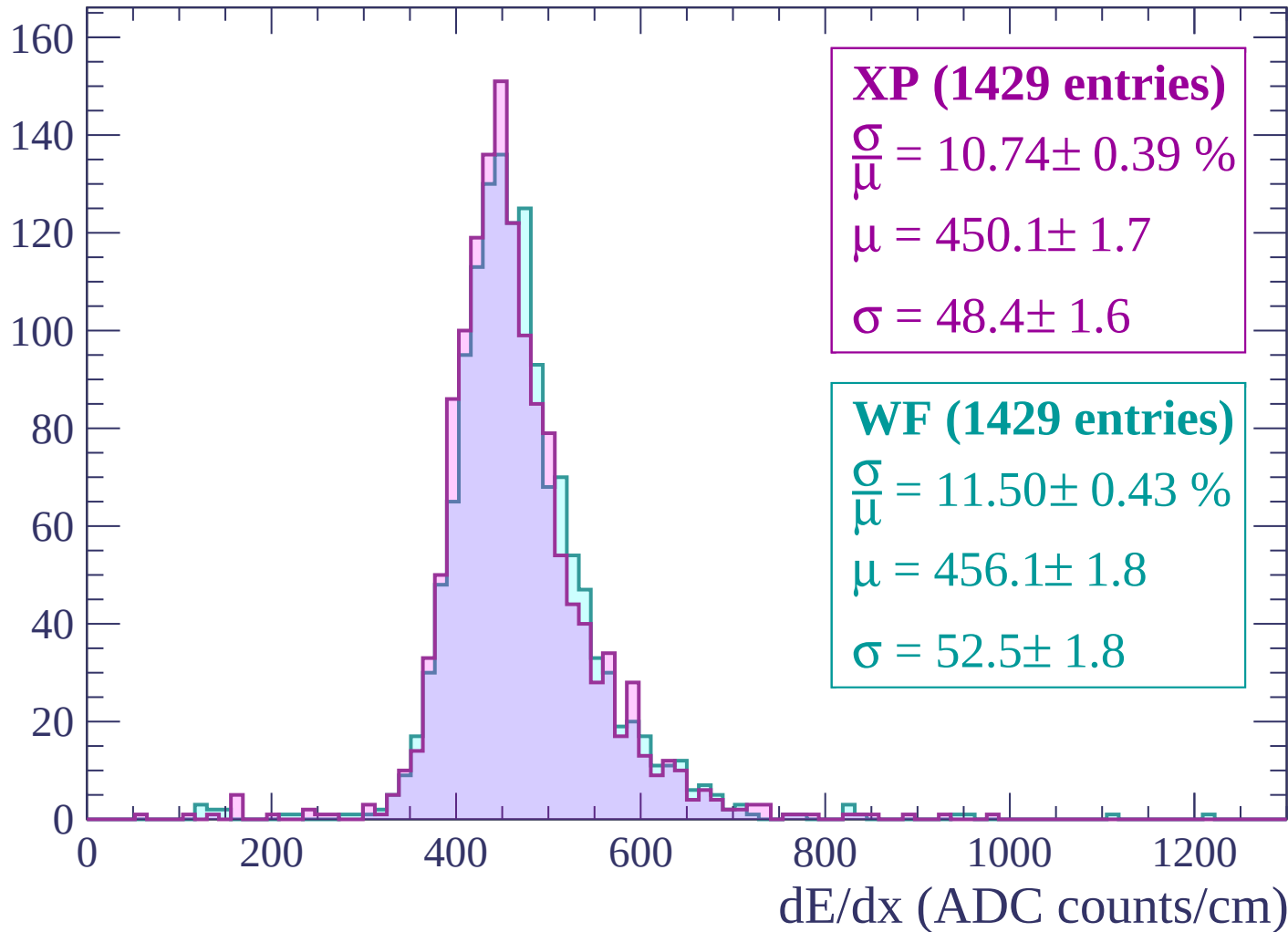
Energy loss | $1080 < p < 1140$

Count



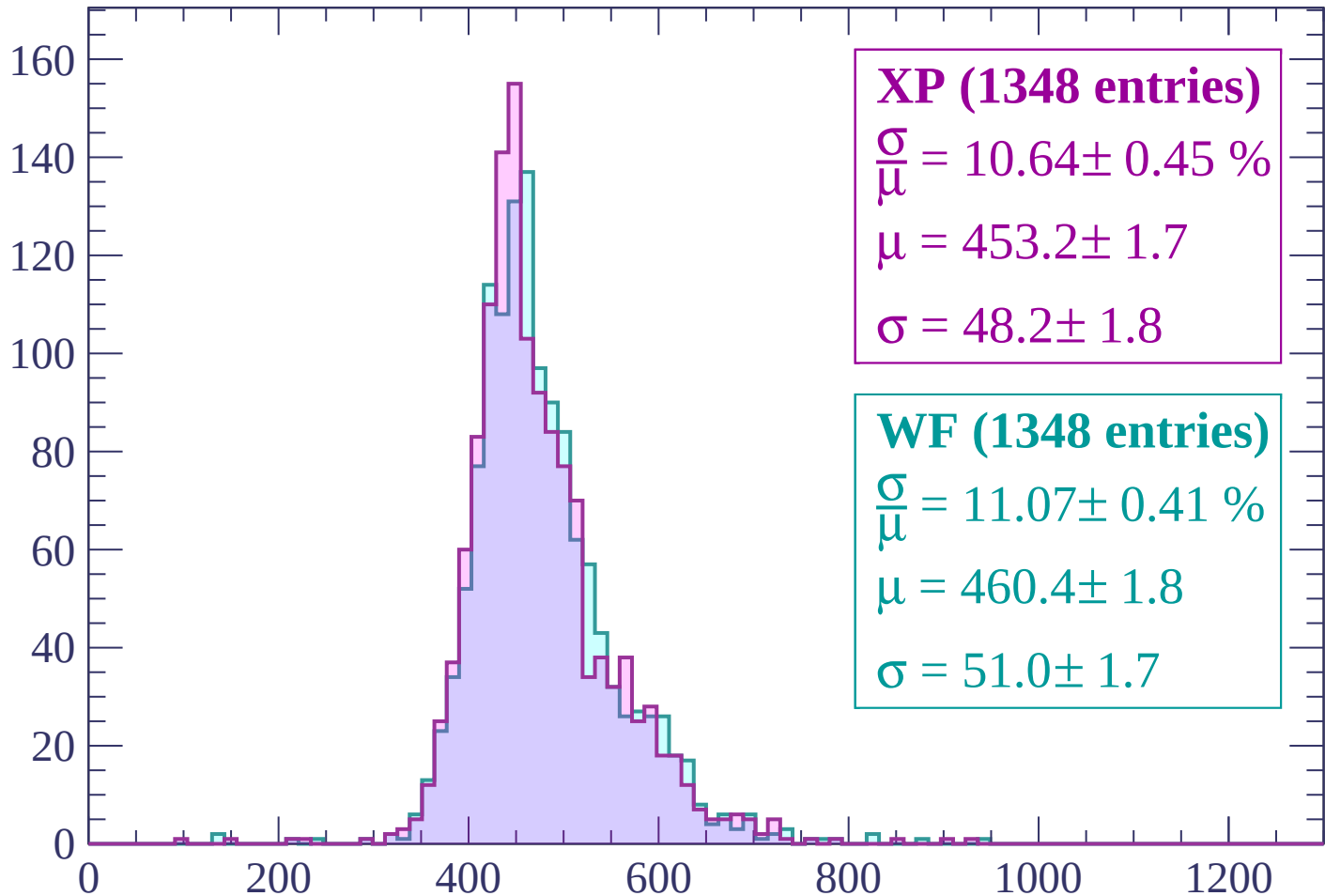
Energy loss | $1140 < p < 1200$

Count



Energy loss | $1200 < p < 1260$

Count



XP (1348 entries)

$\frac{\sigma}{\mu} = 10.64 \pm 0.45 \%$

$\mu = 453.2 \pm 1.7$

$\sigma = 48.2 \pm 1.8$

WF (1348 entries)

$\frac{\sigma}{\mu} = 11.07 \pm 0.41 \%$

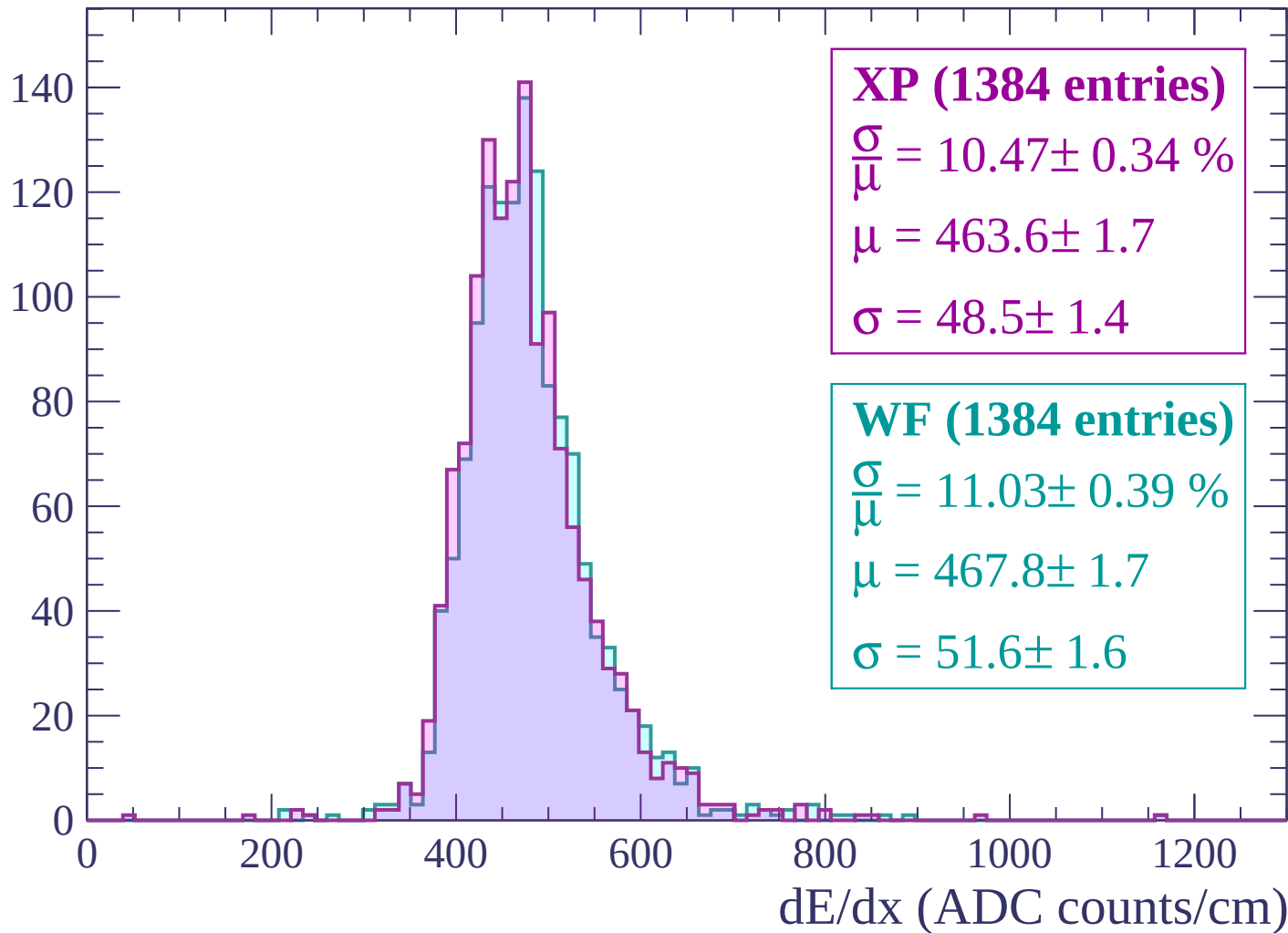
$\mu = 460.4 \pm 1.8$

$\sigma = 51.0 \pm 1.7$

dE/dx (ADC counts/cm)

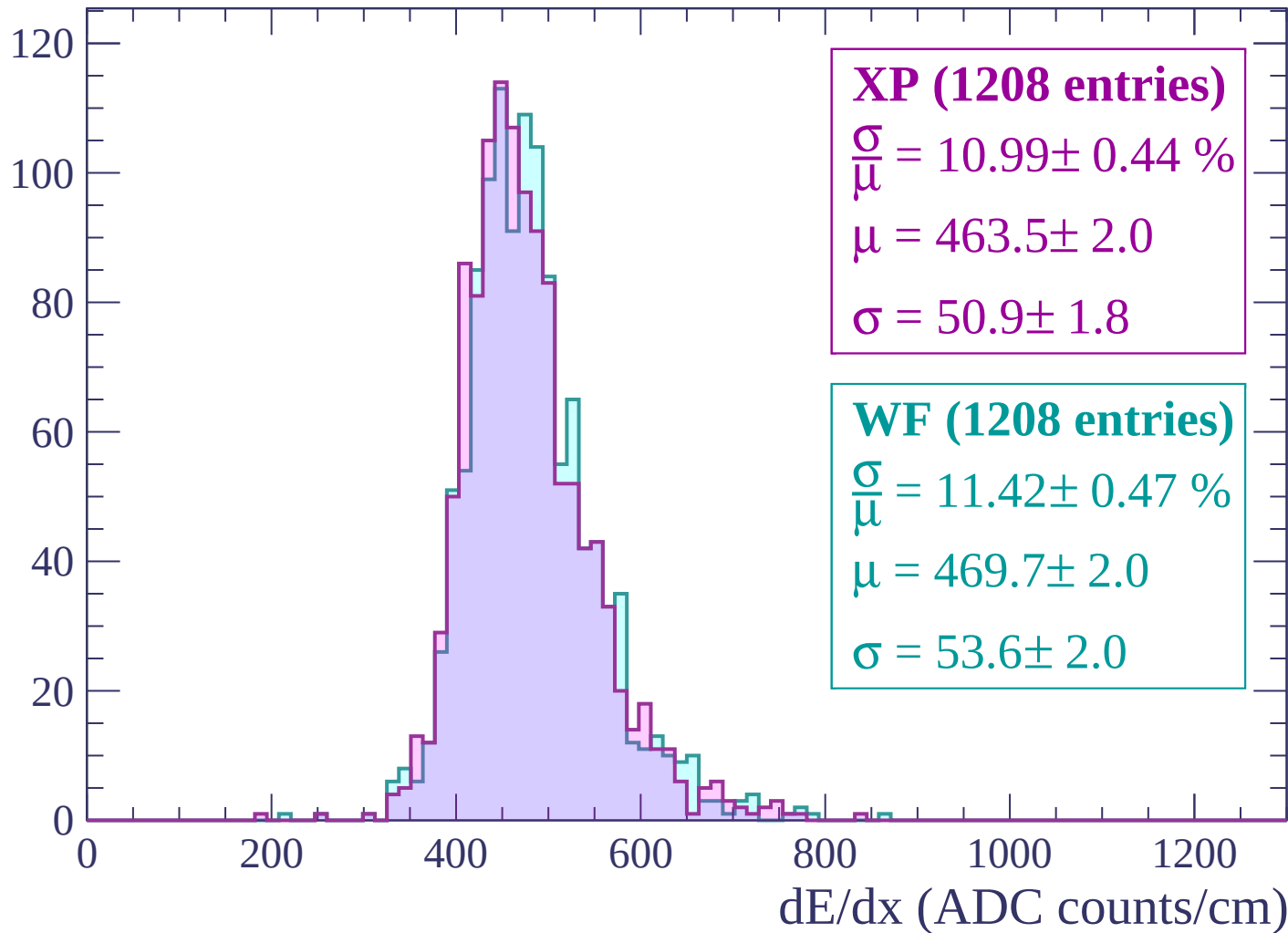
Energy loss | $1260 < p < 1320$

Count



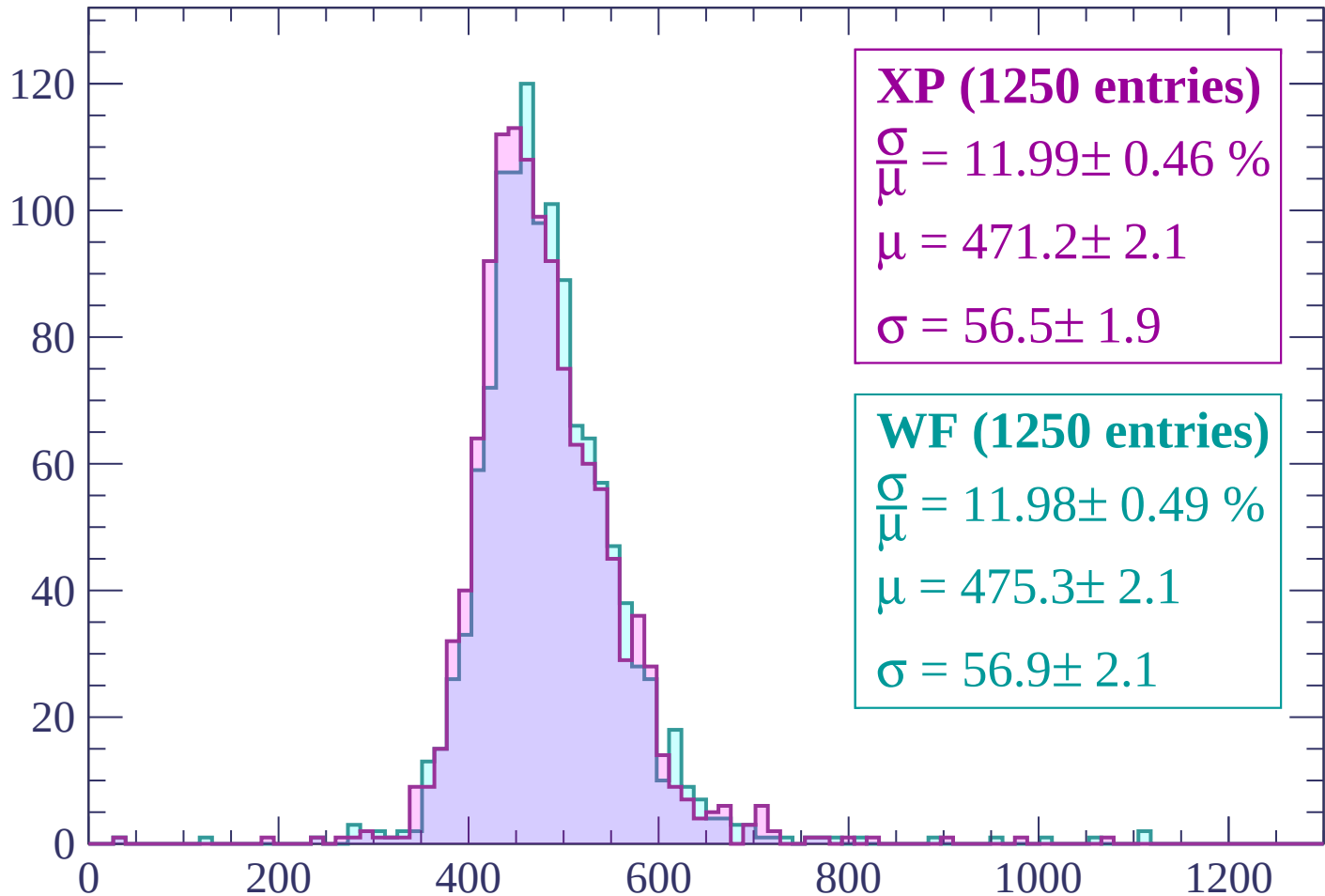
Energy loss | $1320 < p < 1380$

Count



Energy loss | $1380 < p < 1440$

Count



XP (1250 entries)

$$\frac{\sigma}{\mu} = 11.99 \pm 0.46 \%$$

$$\mu = 471.2 \pm 2.1$$

$$\sigma = 56.5 \pm 1.9$$

WF (1250 entries)

$$\frac{\sigma}{\mu} = 11.98 \pm 0.49 \%$$

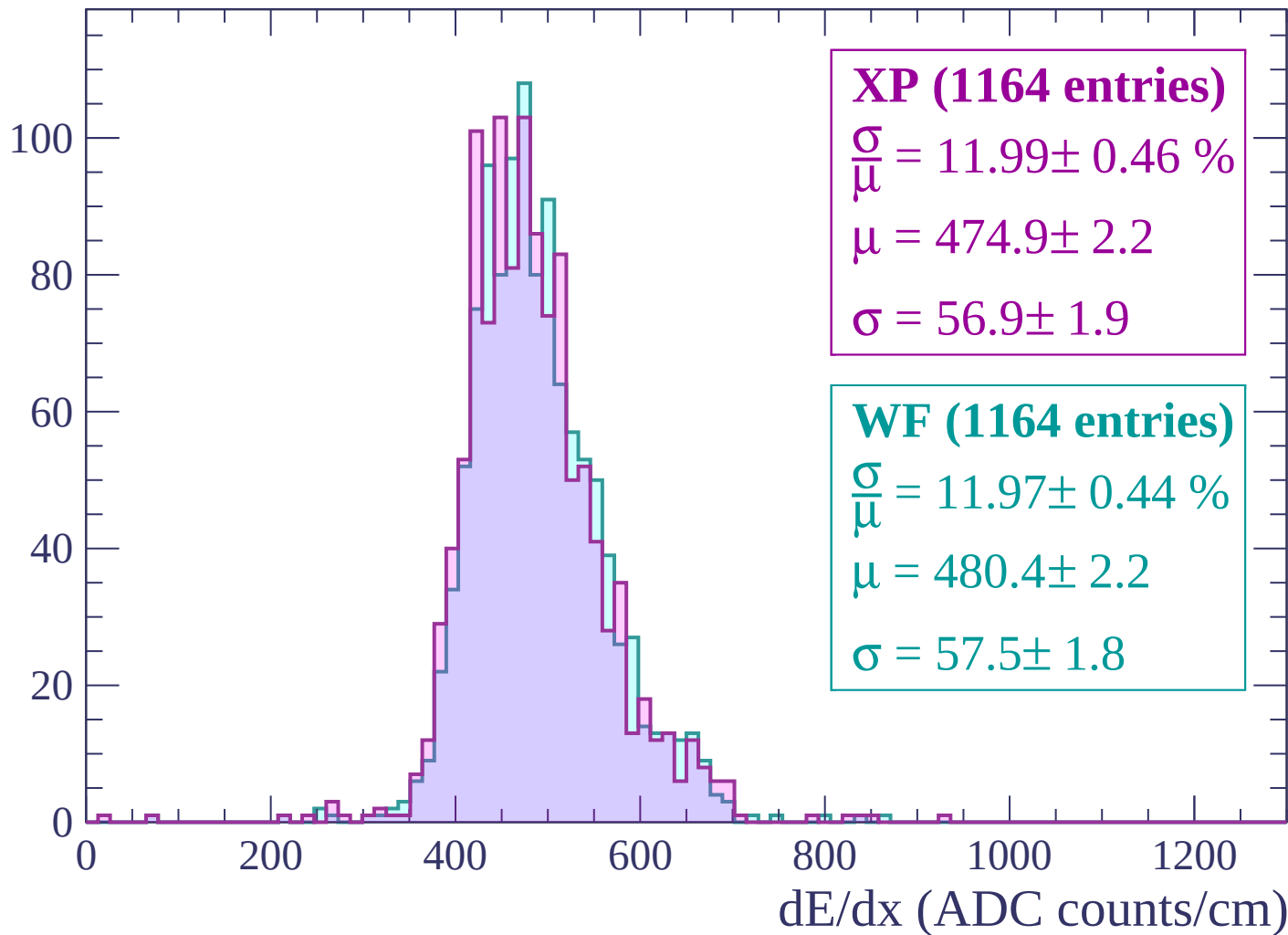
$$\mu = 475.3 \pm 2.1$$

$$\sigma = 56.9 \pm 2.1$$

dE/dx (ADC counts/cm)

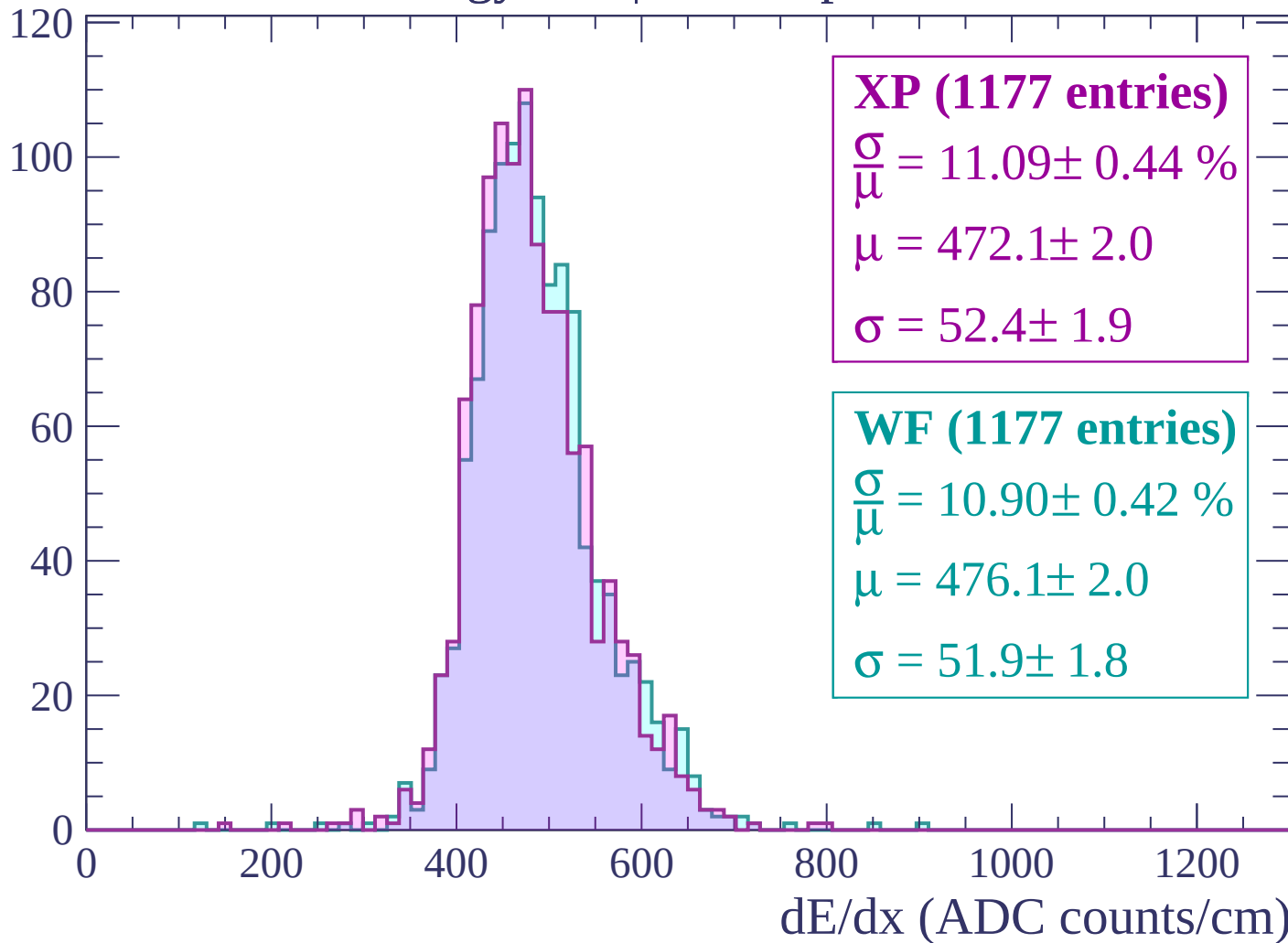
Energy loss | $1440 < p < 1500$

Count



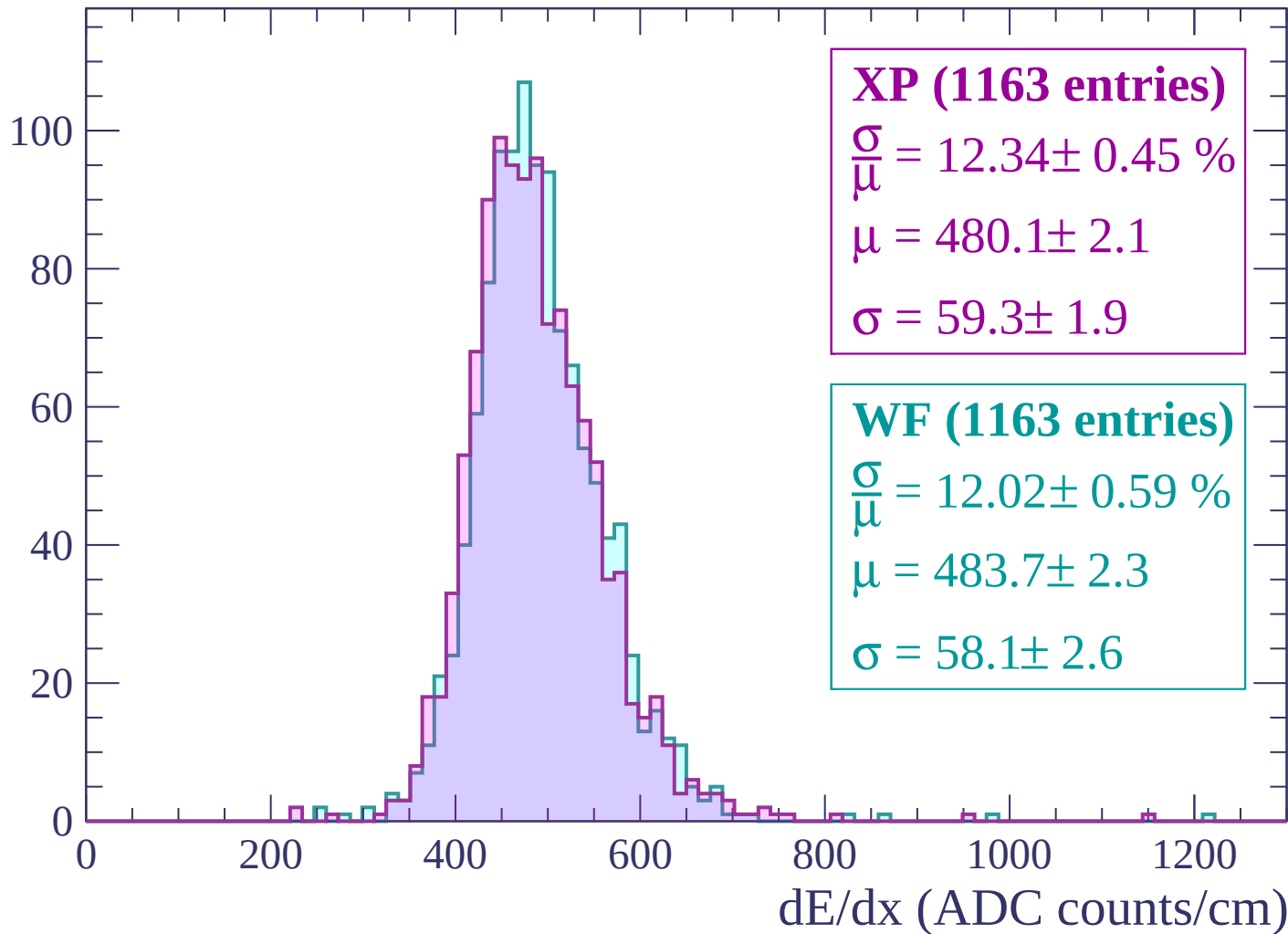
Energy loss | $1500 < p < 1560$

Count



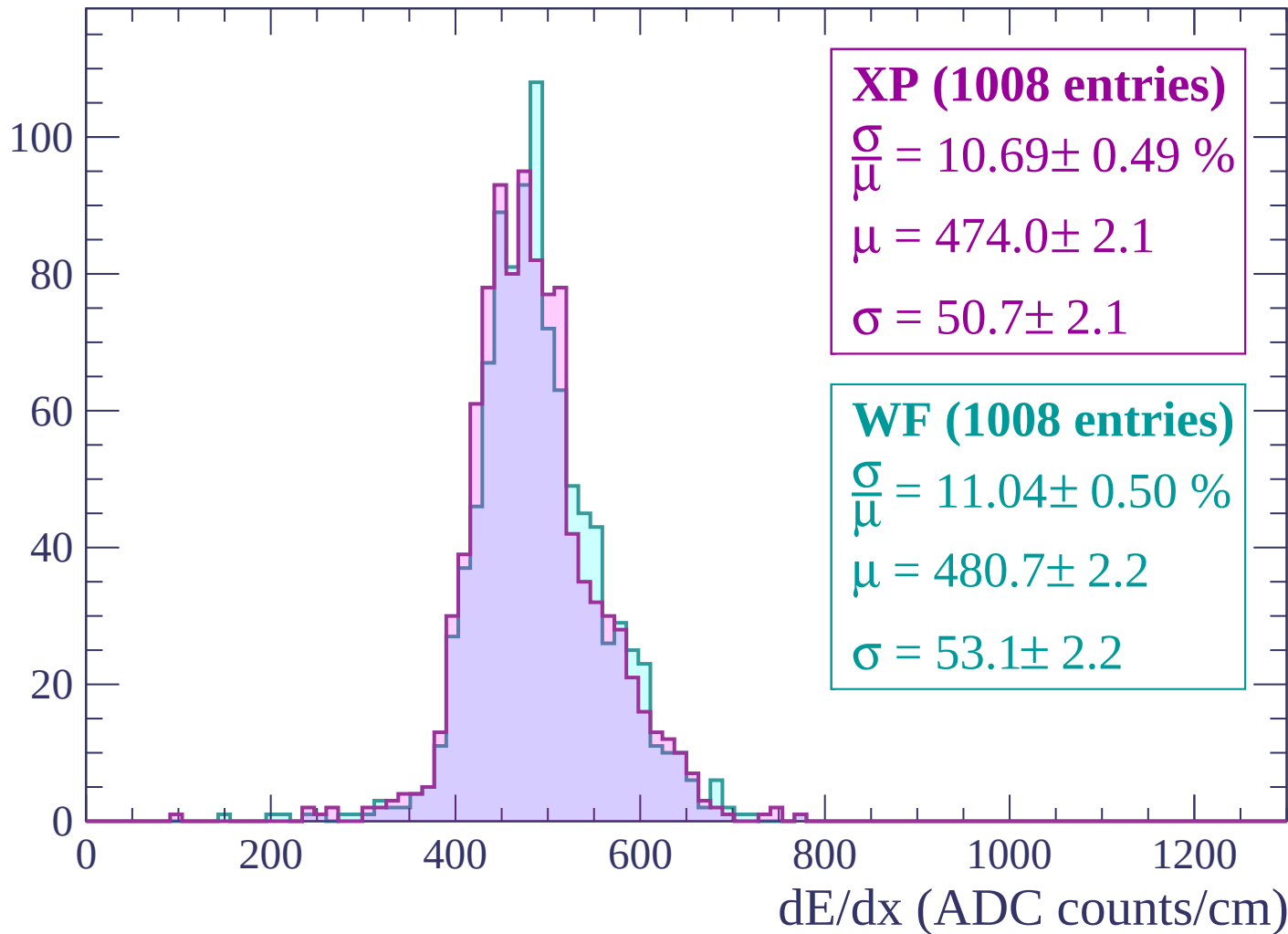
Energy loss | $1560 < p < 1620$

Count



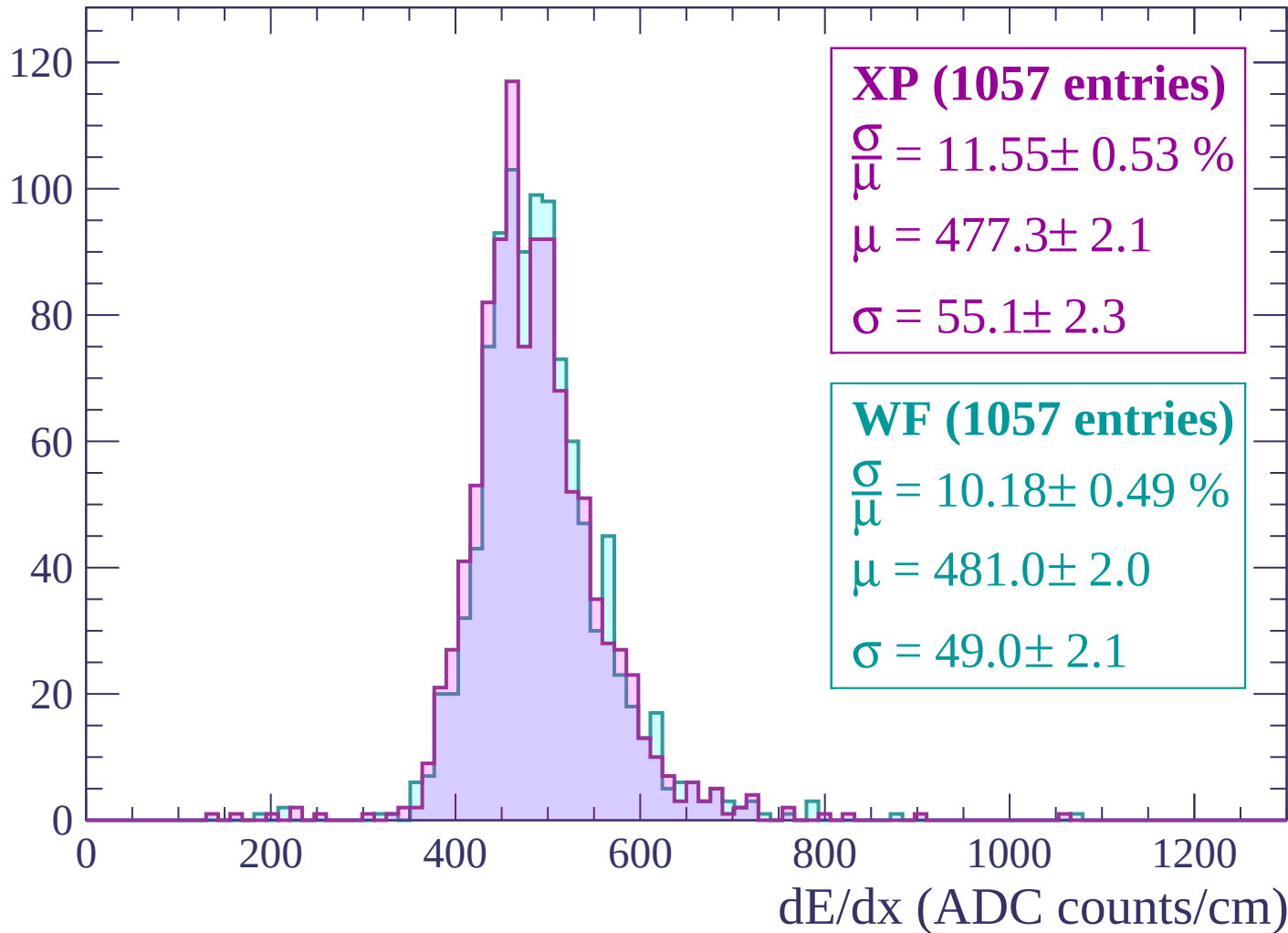
Energy loss | $1620 < p < 1680$

Count



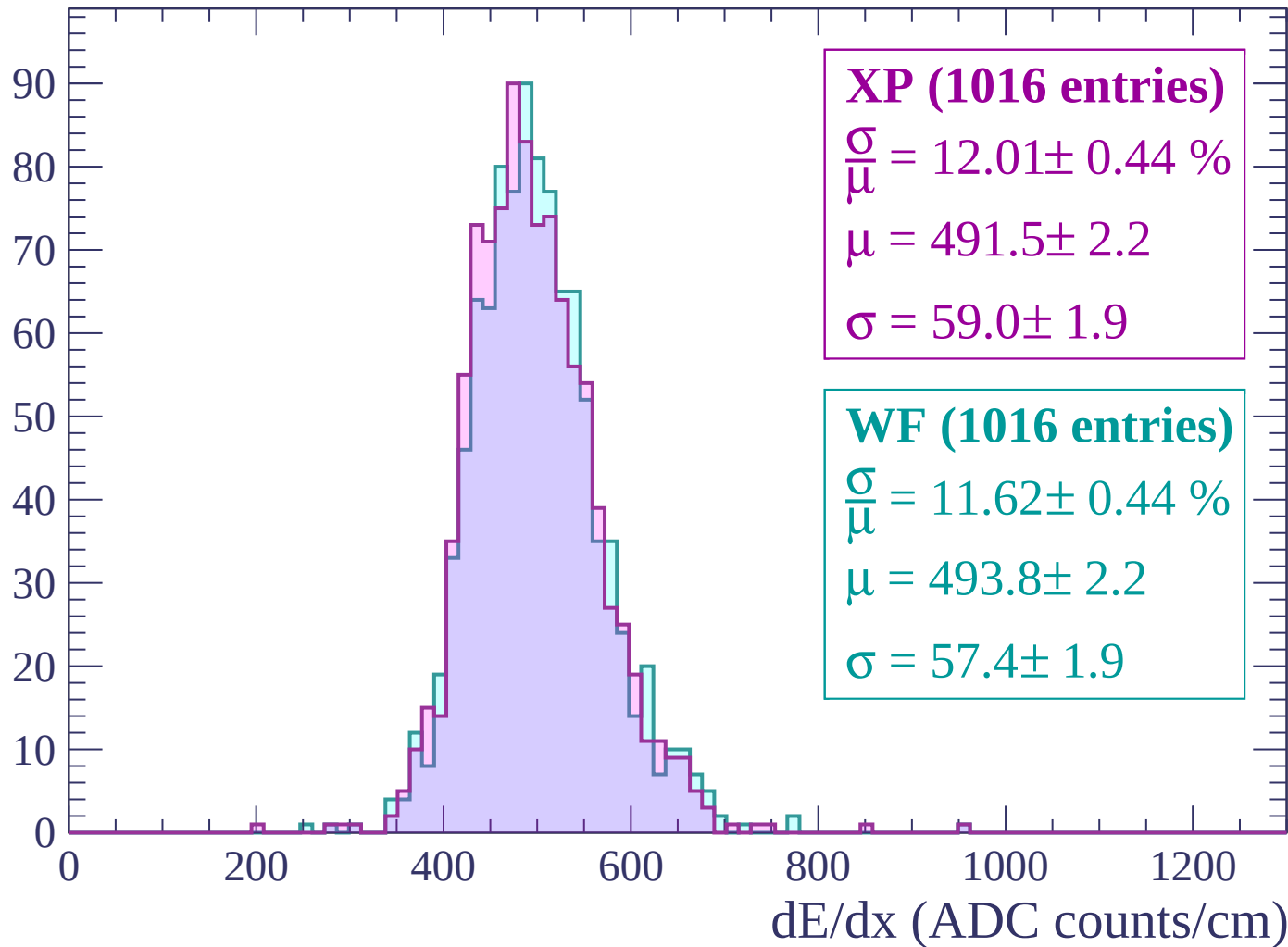
Energy loss | $1680 < p < 1740$

Count



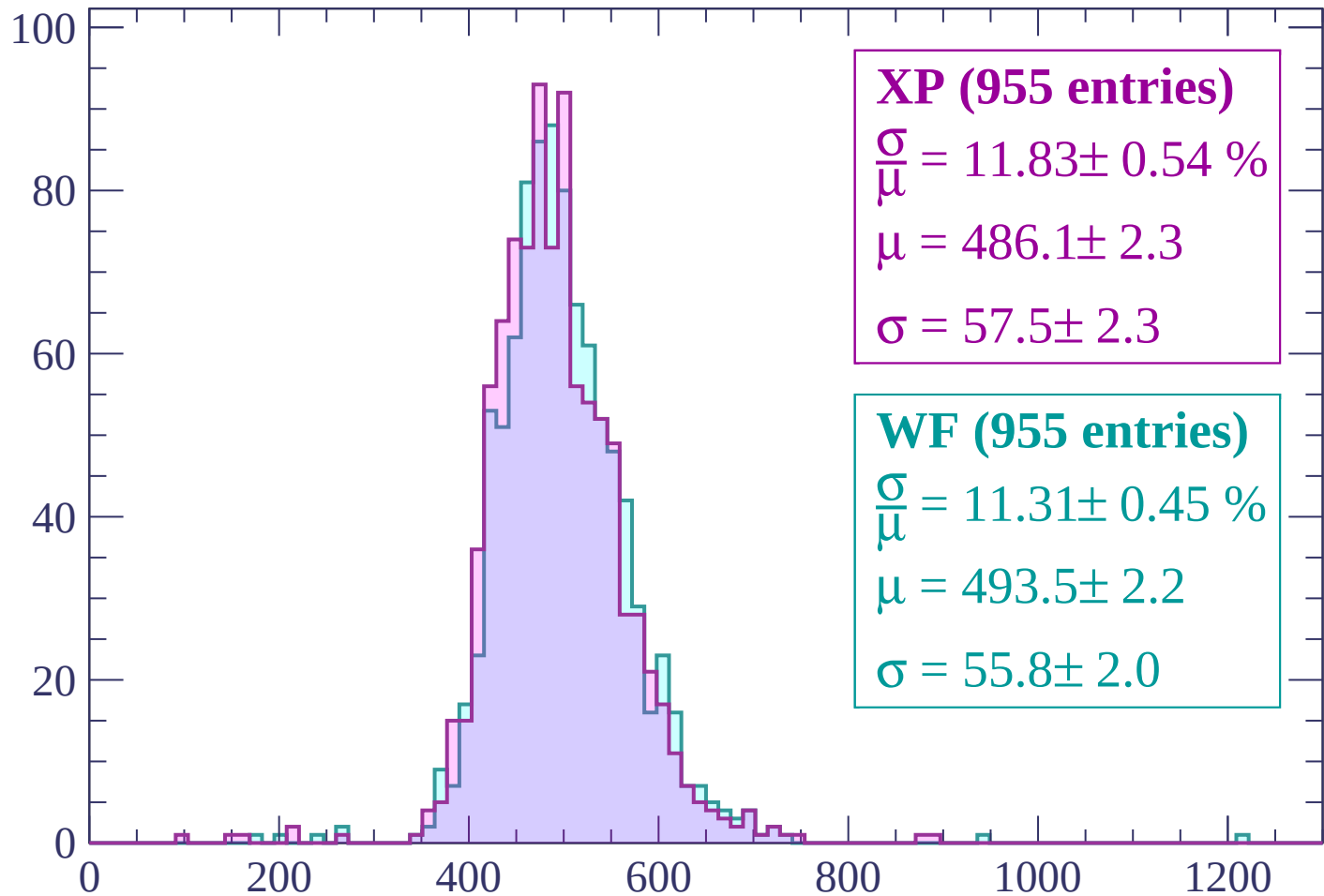
Energy loss | $1740 < p < 1800$

Count



Energy loss | $1800 < p < 1860$

Count



XP (955 entries)

$$\frac{\sigma}{\mu} = 11.83 \pm 0.54 \%$$

$$\mu = 486.1 \pm 2.3$$

$$\sigma = 57.5 \pm 2.3$$

WF (955 entries)

$$\frac{\sigma}{\mu} = 11.31 \pm 0.45 \%$$

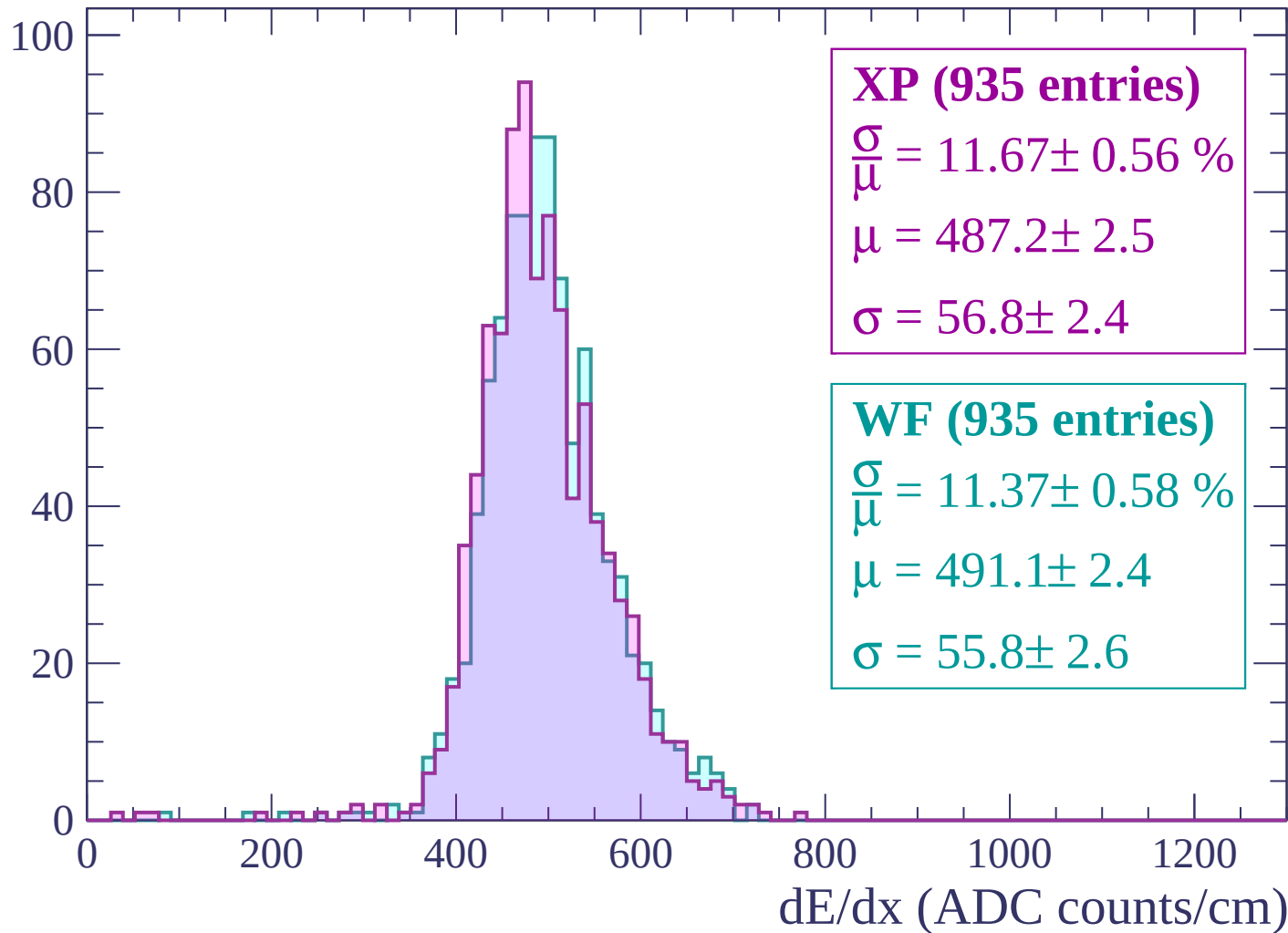
$$\mu = 493.5 \pm 2.2$$

$$\sigma = 55.8 \pm 2.0$$

dE/dx (ADC counts/cm)

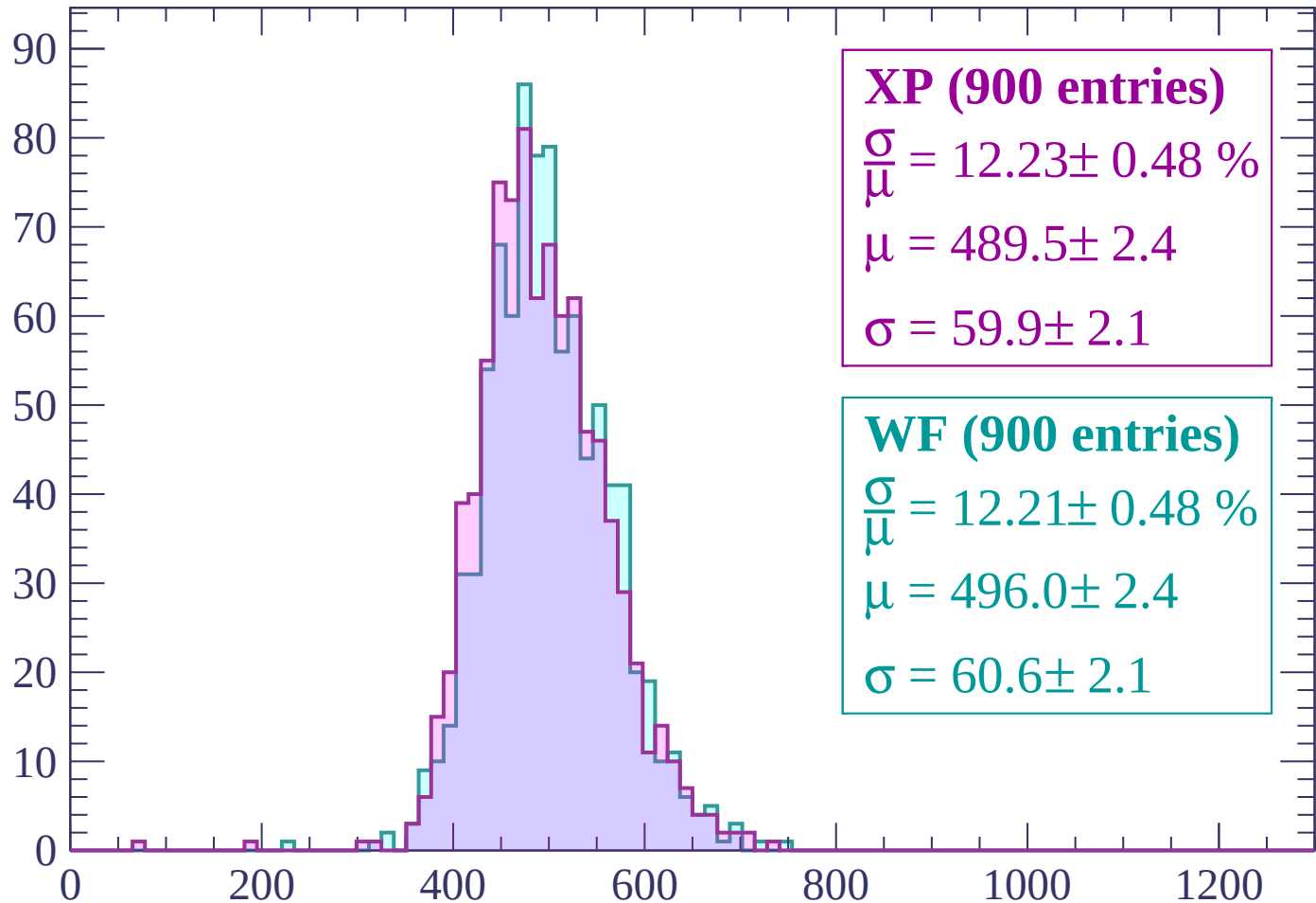
Energy loss | $1860 < p < 1920$

Count



Energy loss | $1920 < p < 1980$

Count



XP (900 entries)

$$\frac{\sigma}{\mu} = 12.23 \pm 0.48 \%$$

$$\mu = 489.5 \pm 2.4$$

$$\sigma = 59.9 \pm 2.1$$

WF (900 entries)

$$\frac{\sigma}{\mu} = 12.21 \pm 0.48 \%$$

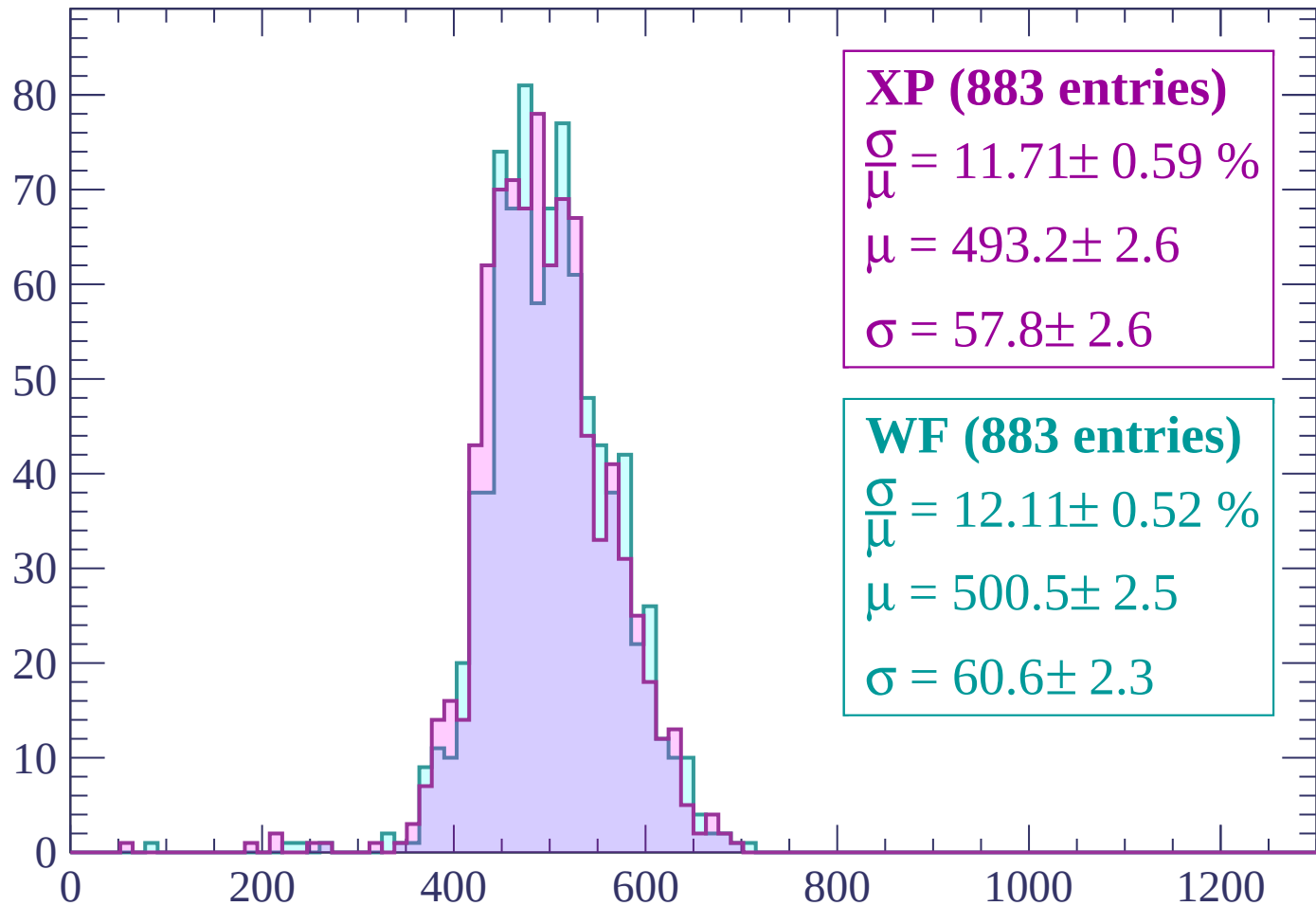
$$\mu = 496.0 \pm 2.4$$

$$\sigma = 60.6 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | 1980 < p < 2040

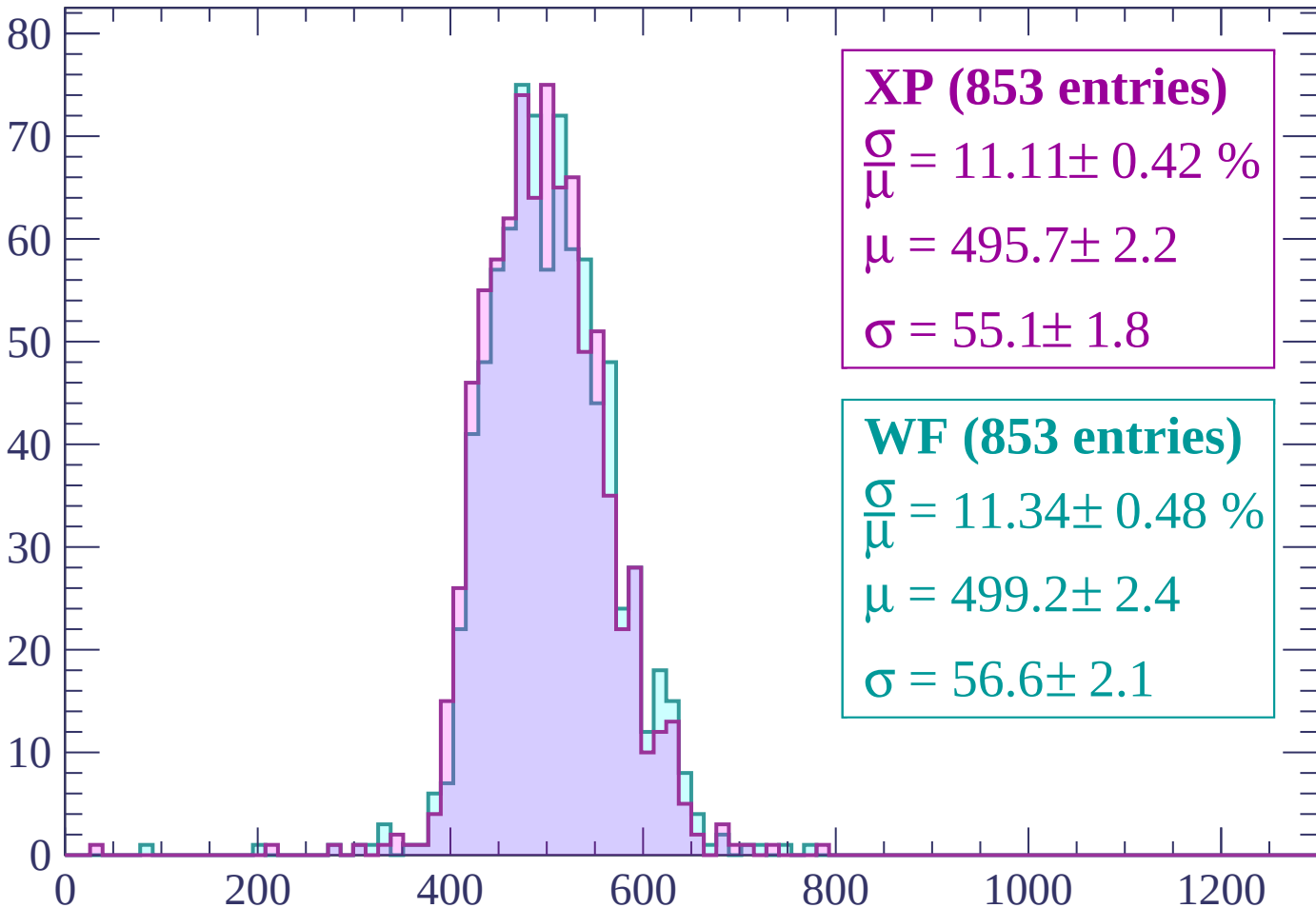
Count



dE/dx (ADC counts/cm)

Energy loss | $2040 < p < 2100$

Count



XP (853 entries)

$\frac{\sigma}{\mu} = 11.11 \pm 0.42 \%$

$\mu = 495.7 \pm 2.2$

$\sigma = 55.1 \pm 1.8$

WF (853 entries)

$\frac{\sigma}{\mu} = 11.34 \pm 0.48 \%$

$\mu = 499.2 \pm 2.4$

$\sigma = 56.6 \pm 2.1$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count

XP (749 entries)

$$\frac{\sigma}{\mu} = 11.27 \pm 0.52 \%$$

$$\mu = 492.8 \pm 2.6$$

$$\sigma = 55.5 \pm 2.3$$

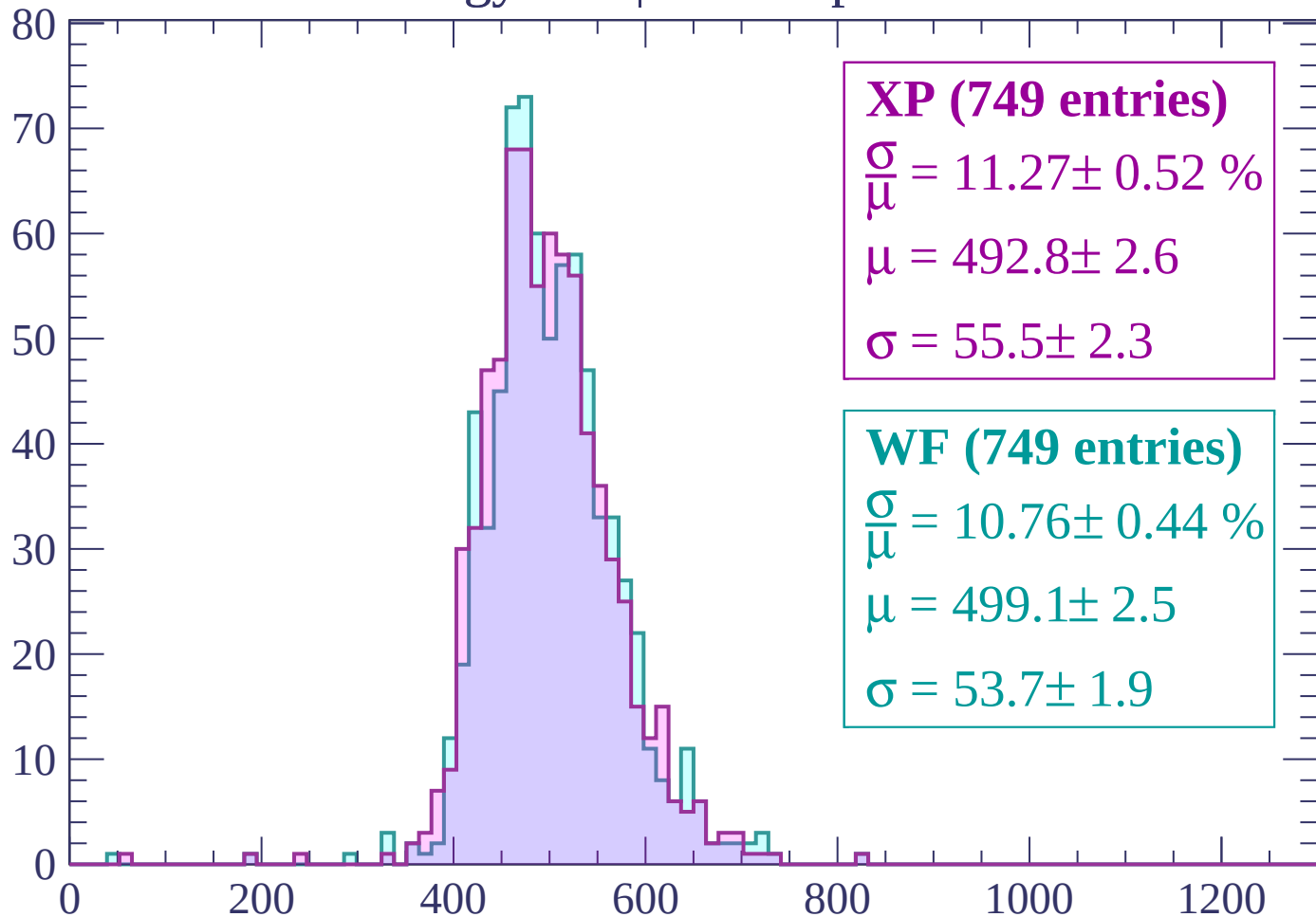
WF (749 entries)

$$\frac{\sigma}{\mu} = 10.76 \pm 0.44 \%$$

$$\mu = 499.1 \pm 2.5$$

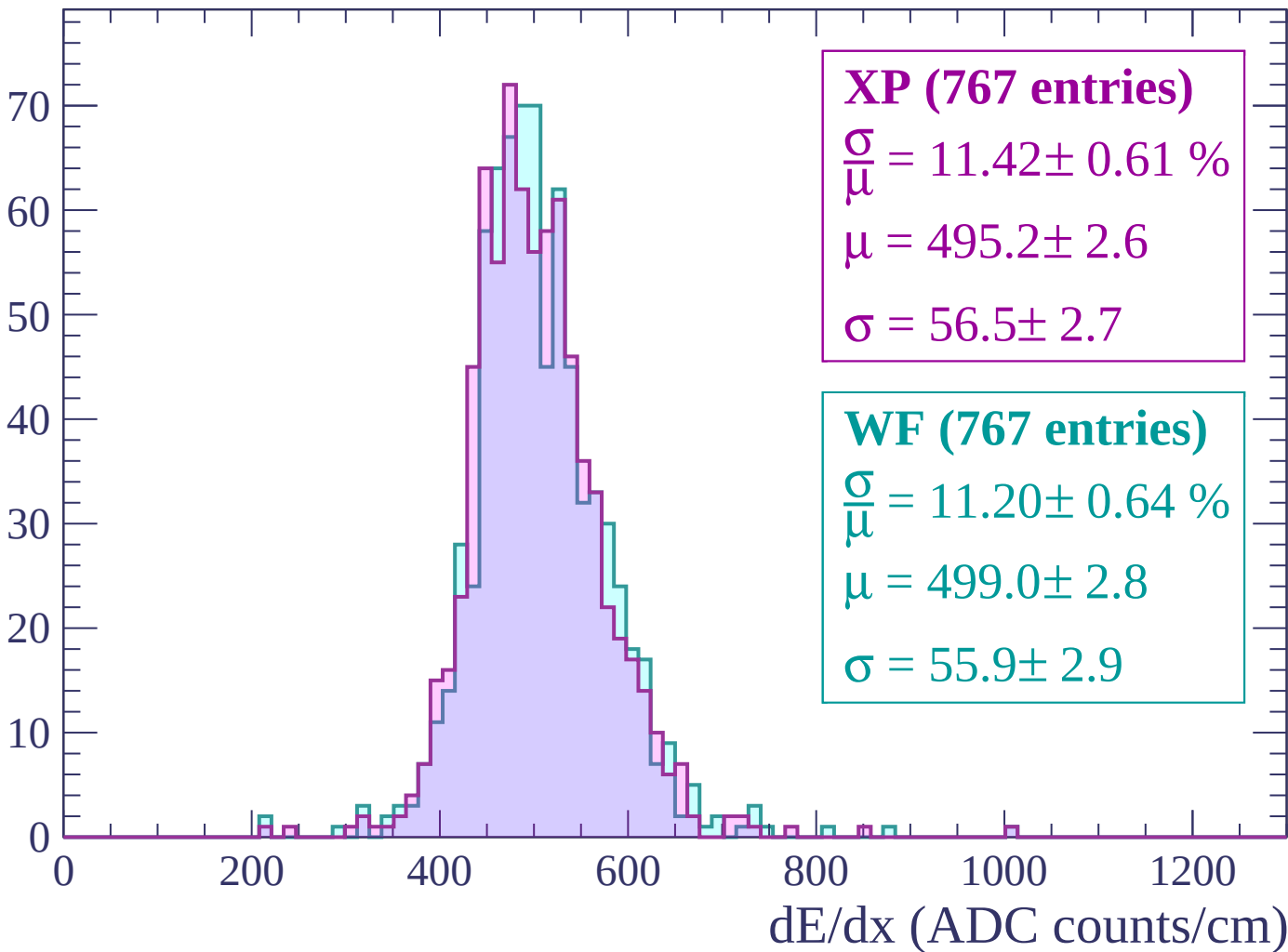
$$\sigma = 53.7 \pm 1.9$$

dE/dx (ADC counts/cm)



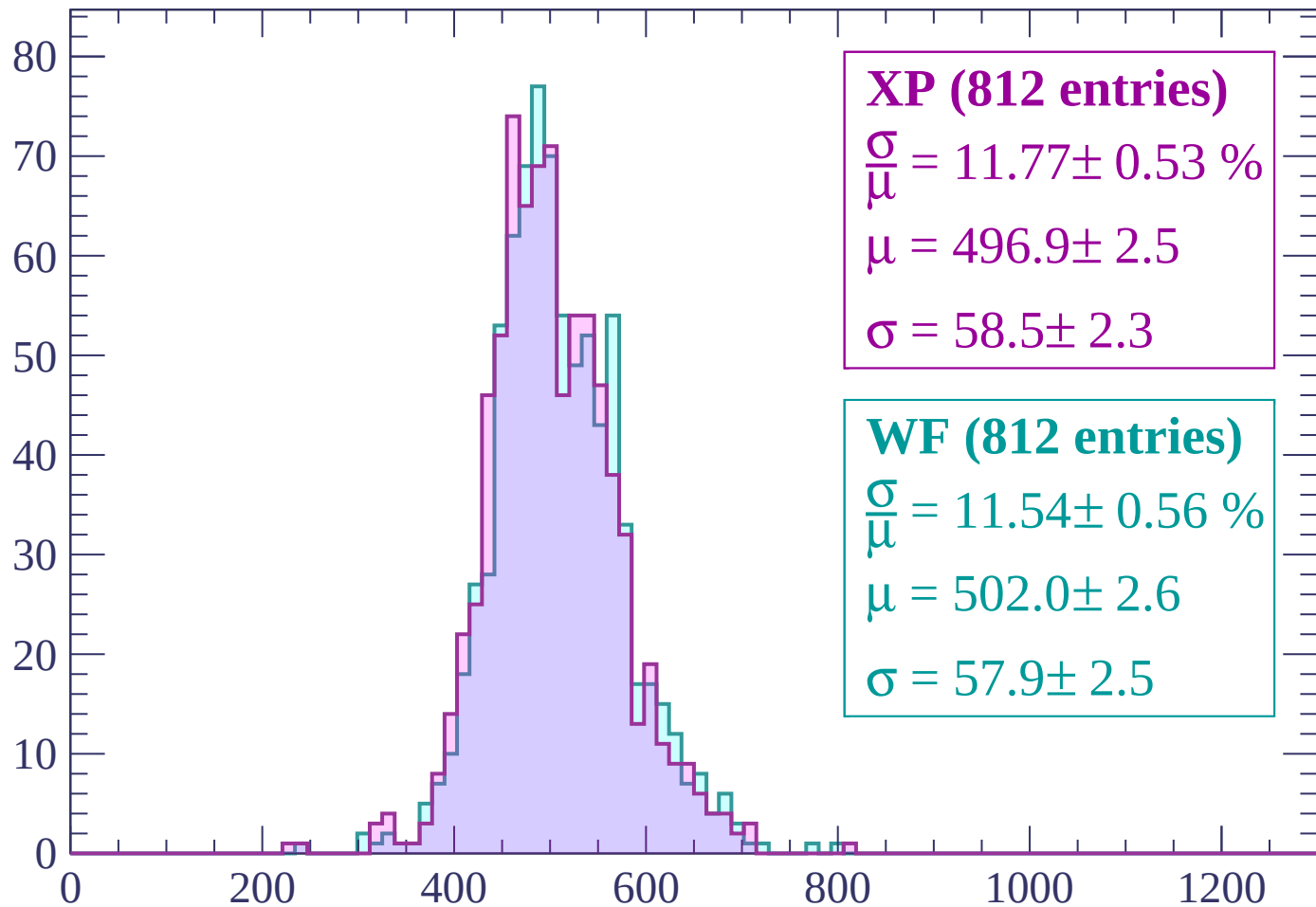
Energy loss | $2160 < p < 2220$

Count



Energy loss | $2220 < p < 2280$

Count



XP (812 entries)

$$\frac{\sigma}{\mu} = 11.77 \pm 0.53 \%$$

$$\mu = 496.9 \pm 2.5$$

$$\sigma = 58.5 \pm 2.3$$

WF (812 entries)

$$\frac{\sigma}{\mu} = 11.54 \pm 0.56 \%$$

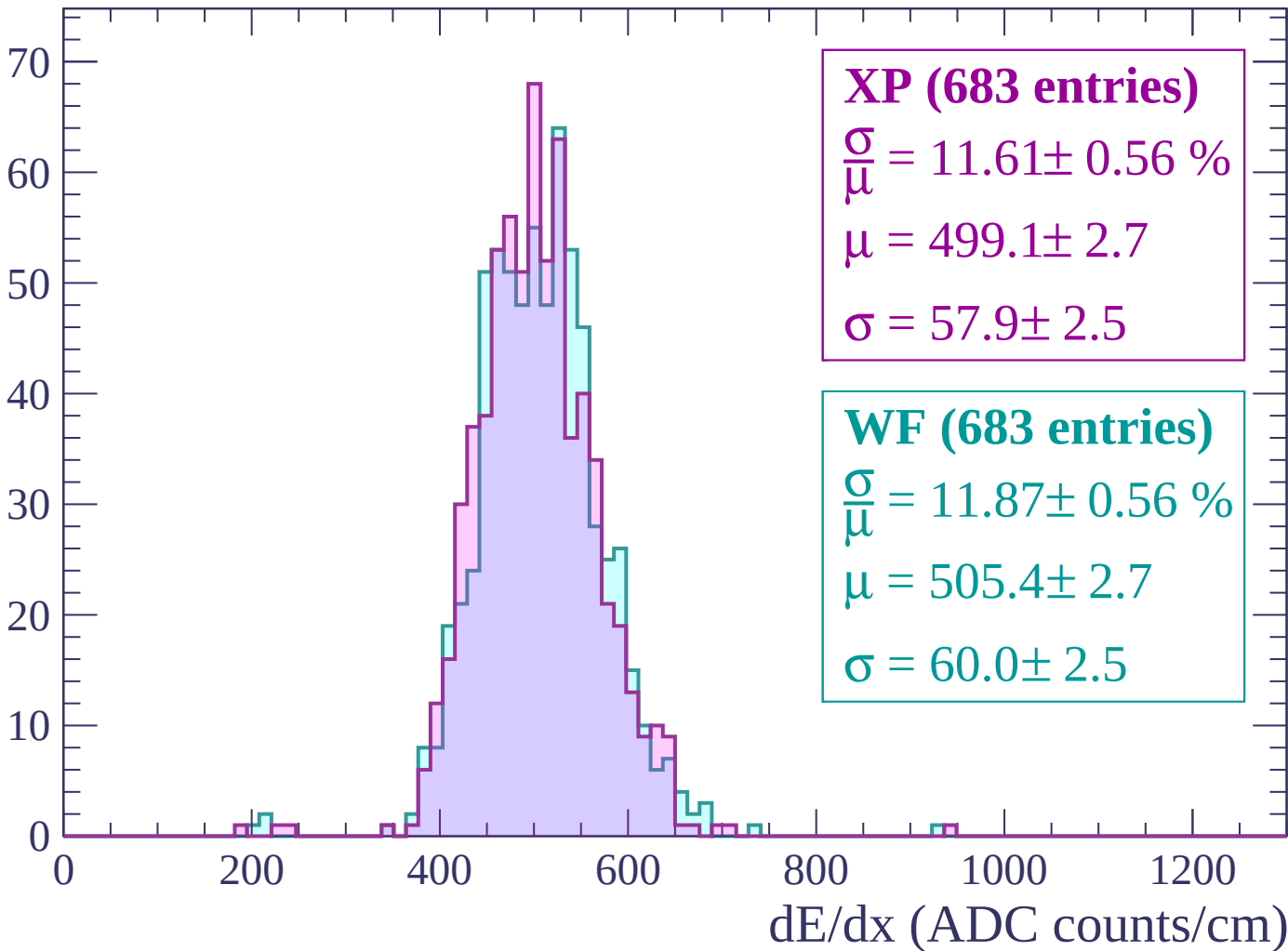
$$\mu = 502.0 \pm 2.6$$

$$\sigma = 57.9 \pm 2.5$$

dE/dx (ADC counts/cm)

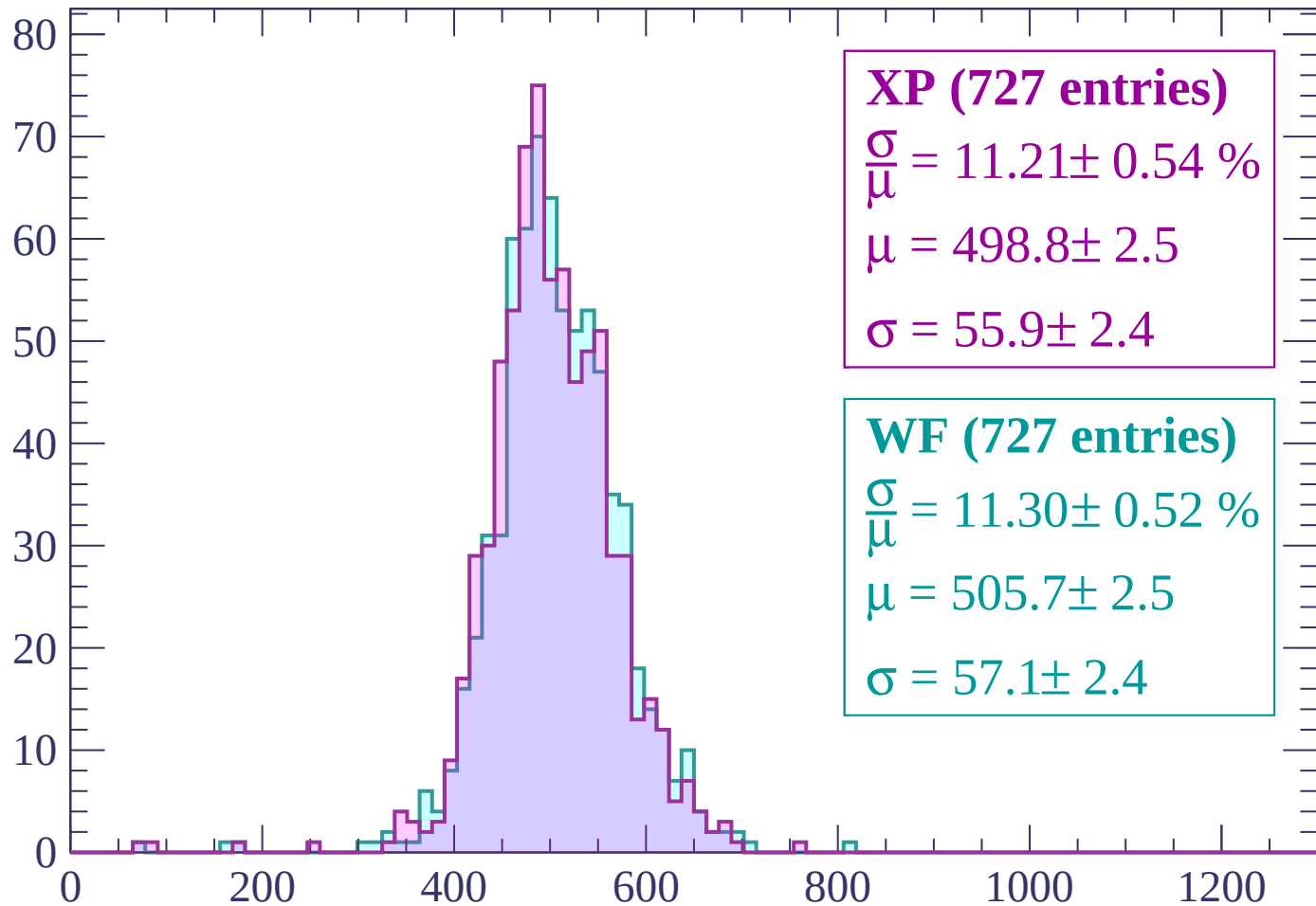
Energy loss | $2280 < p < 2340$

Count



Energy loss | $2340 < p < 2400$

Count



XP (727 entries)

$$\frac{\sigma}{\mu} = 11.21 \pm 0.54 \%$$

$$\mu = 498.8 \pm 2.5$$

$$\sigma = 55.9 \pm 2.4$$

WF (727 entries)

$$\frac{\sigma}{\mu} = 11.30 \pm 0.52 \%$$

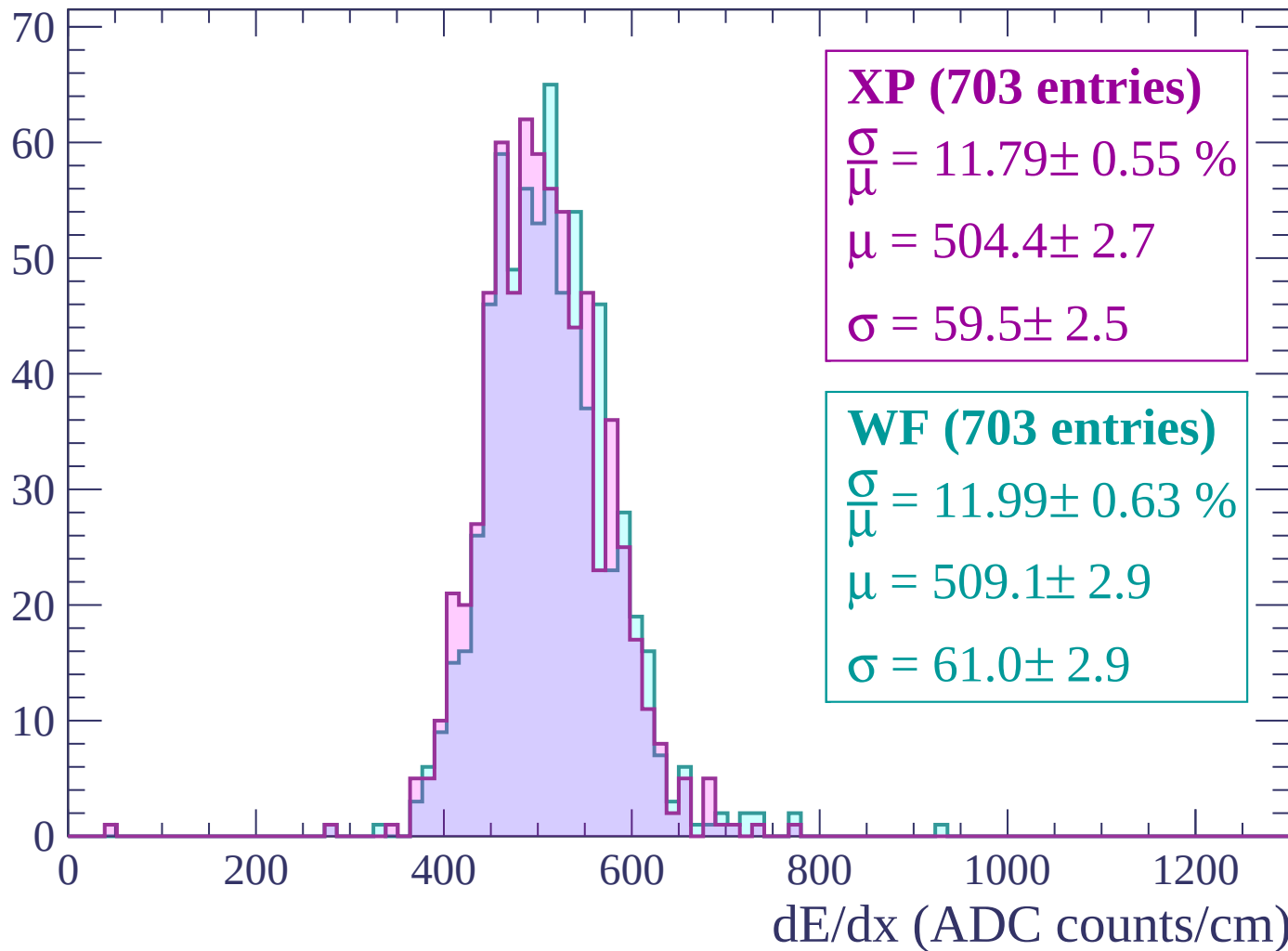
$$\mu = 505.7 \pm 2.5$$

$$\sigma = 57.1 \pm 2.4$$

dE/dx (ADC counts/cm)

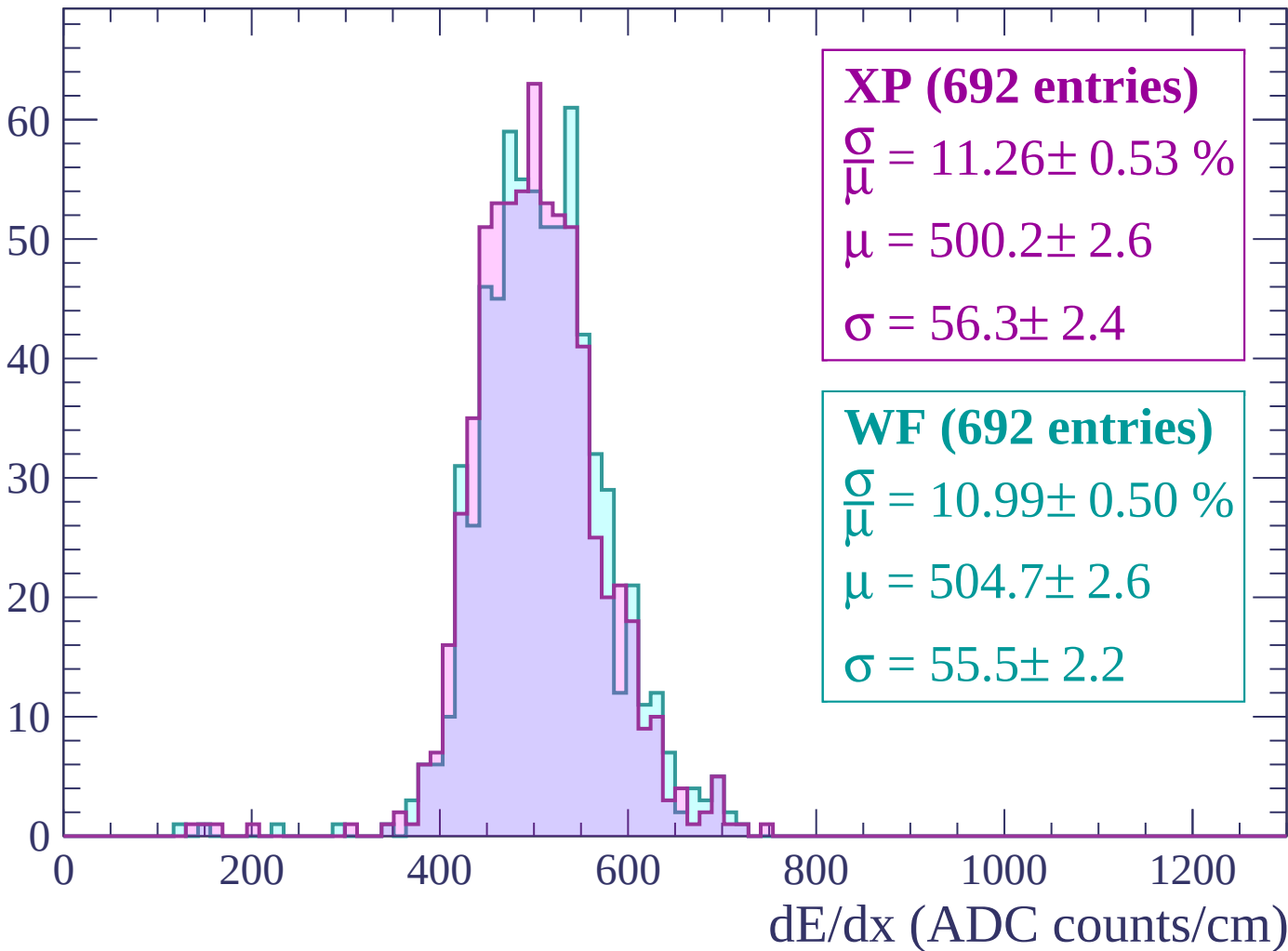
Energy loss | $2400 < p < 2460$

Count



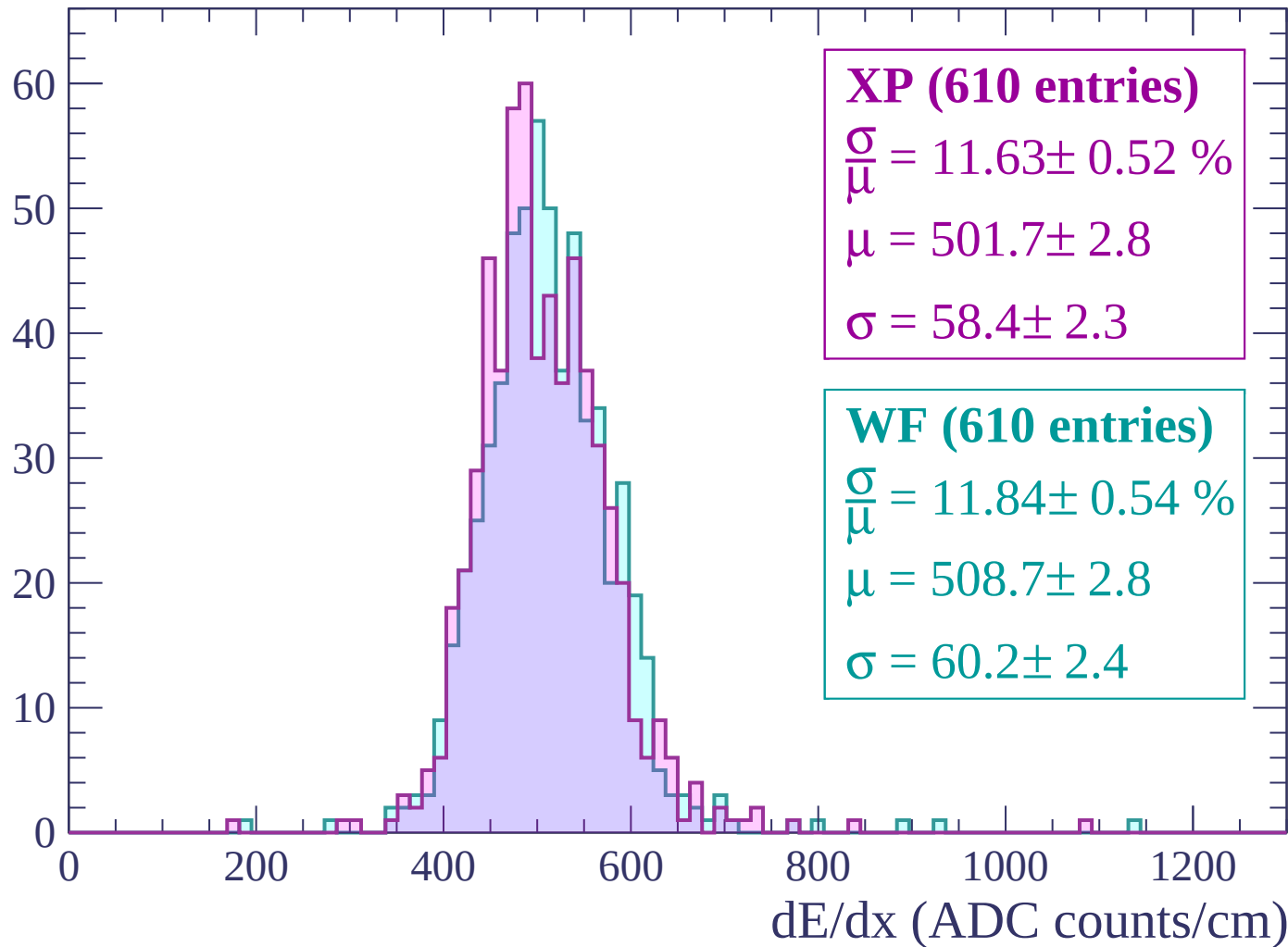
Energy loss | $2460 < p < 2520$

Count



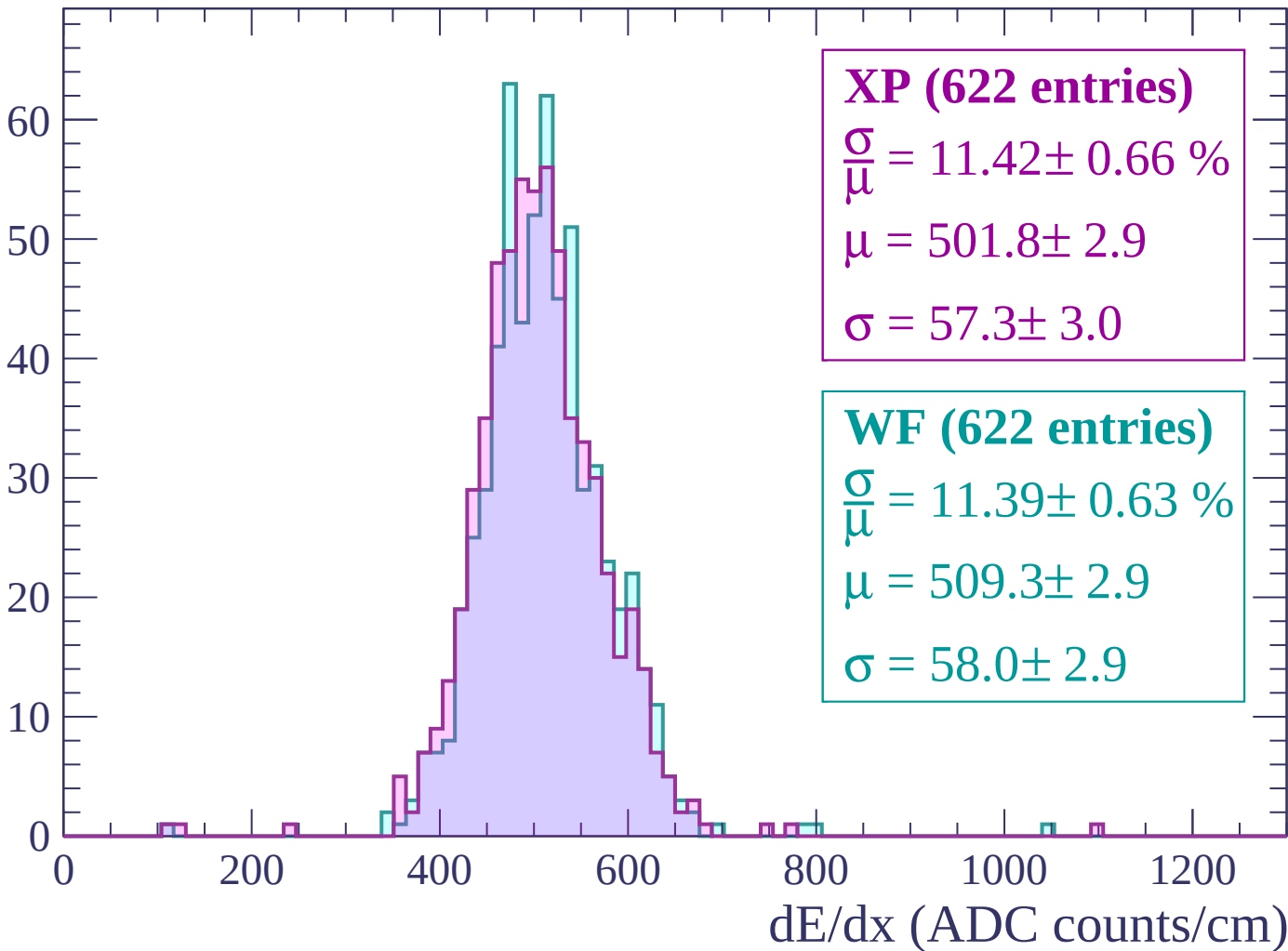
Energy loss | $2520 < p < 2580$

Count



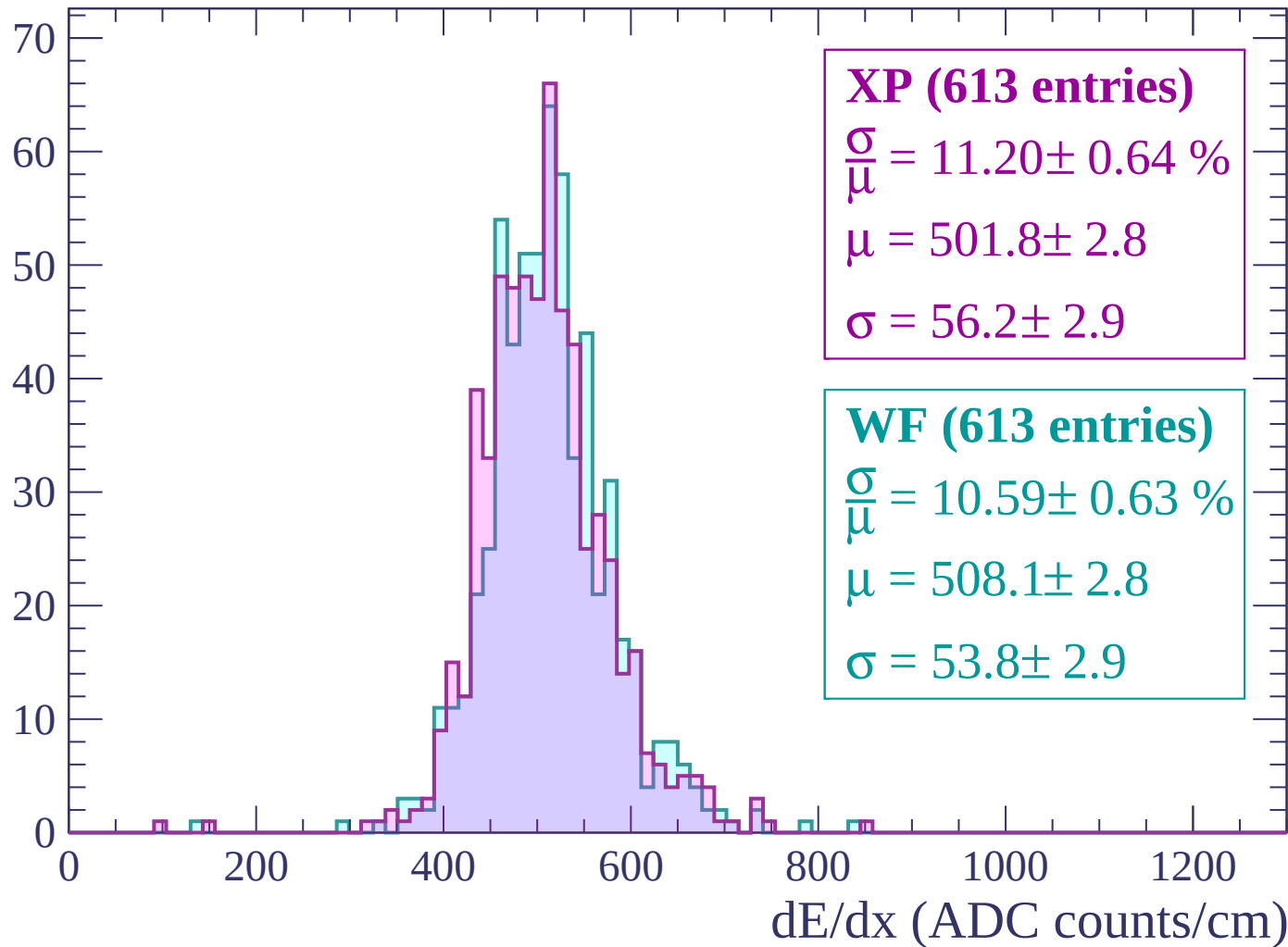
Energy loss | $2580 < p < 2640$

Count



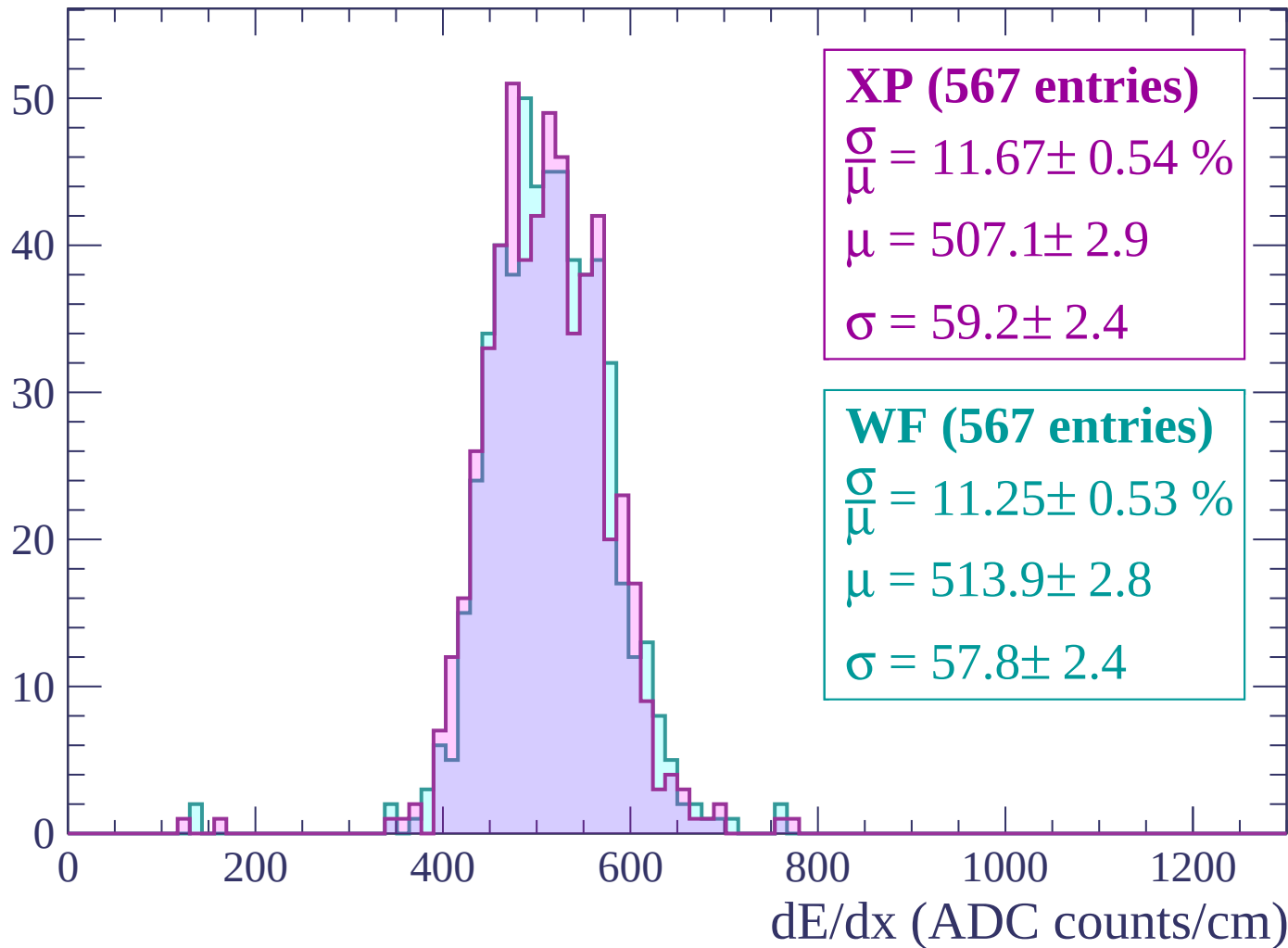
Energy loss | $2640 < p < 2700$

Count



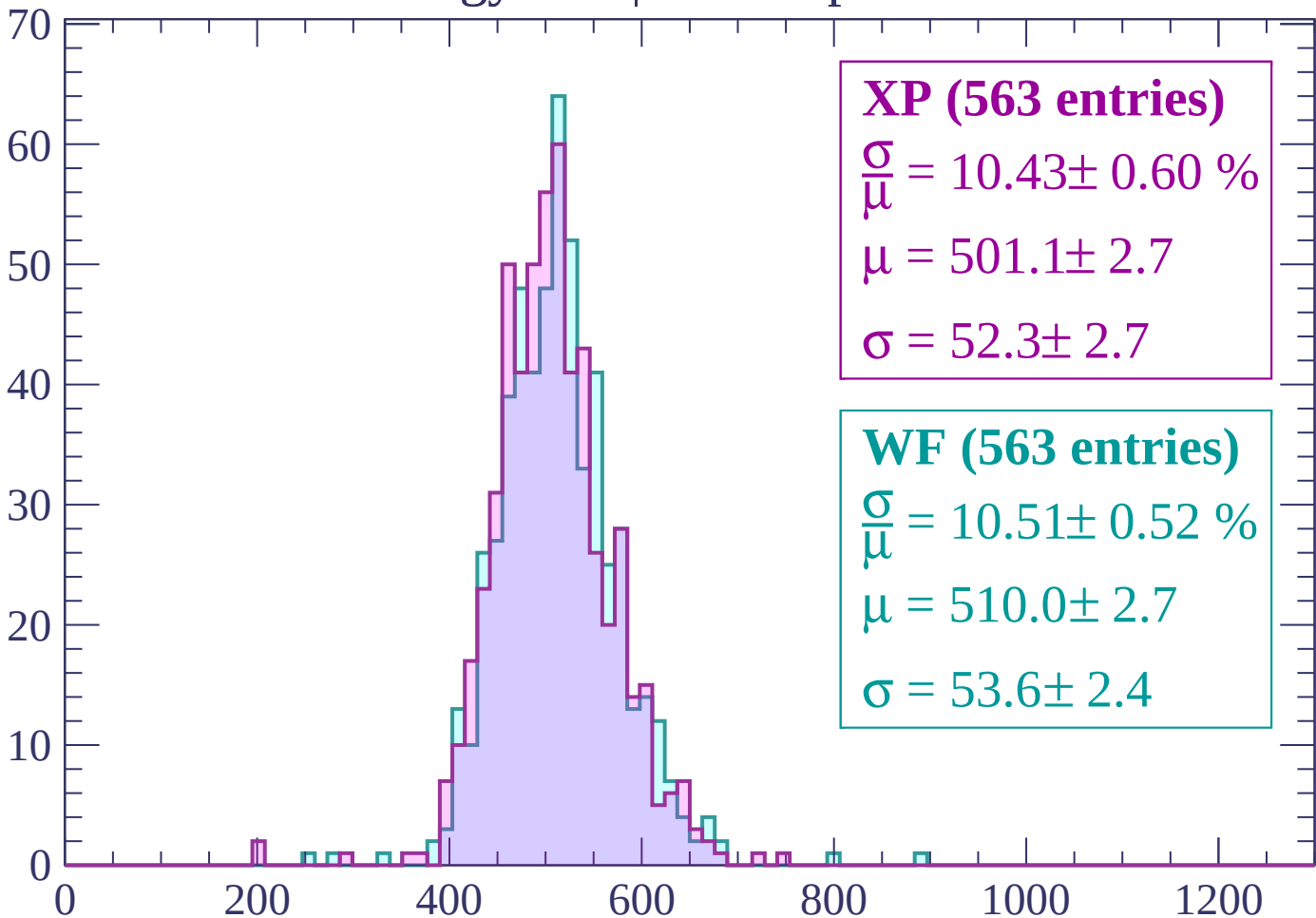
Energy loss | $2700 < p < 2760$

Count



Energy loss | $2760 < p < 2820$

Count



XP (563 entries)

$$\frac{\sigma}{\mu} = 10.43 \pm 0.60 \%$$

$$\mu = 501.1 \pm 2.7$$

$$\sigma = 52.3 \pm 2.7$$

WF (563 entries)

$$\frac{\sigma}{\mu} = 10.51 \pm 0.52 \%$$

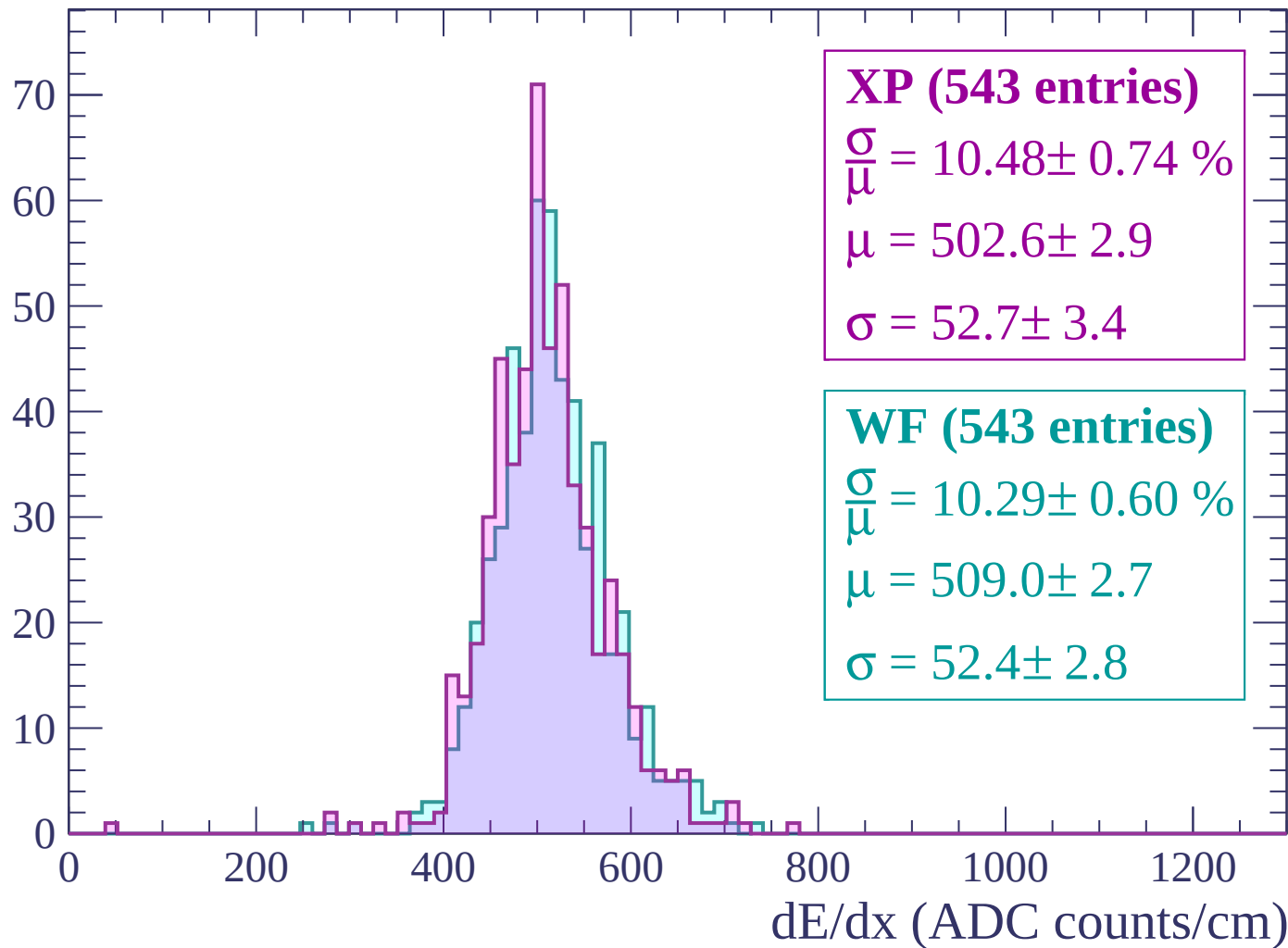
$$\mu = 510.0 \pm 2.7$$

$$\sigma = 53.6 \pm 2.4$$

dE/dx (ADC counts/cm)

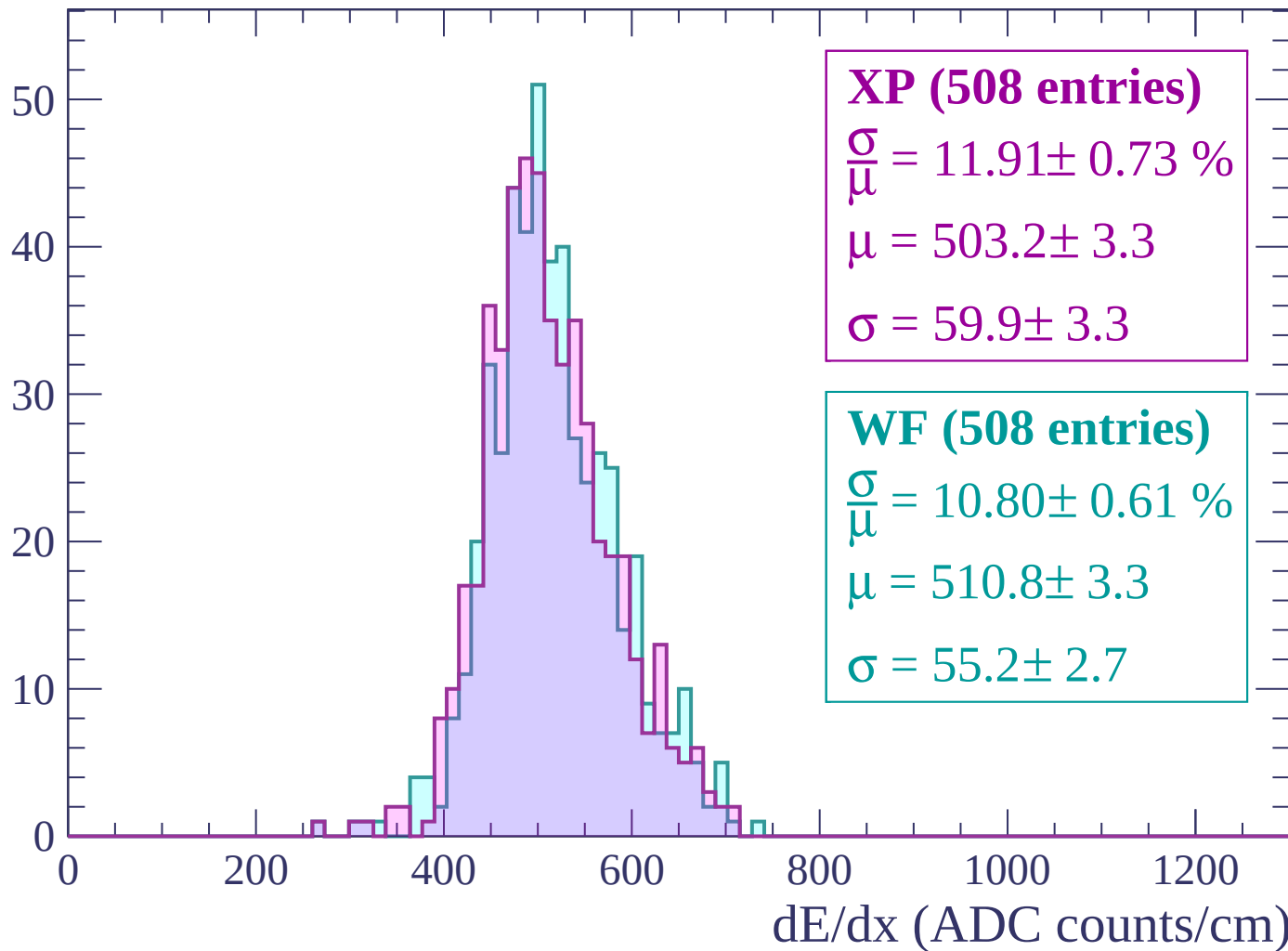
Energy loss | $2820 < p < 2880$

Count



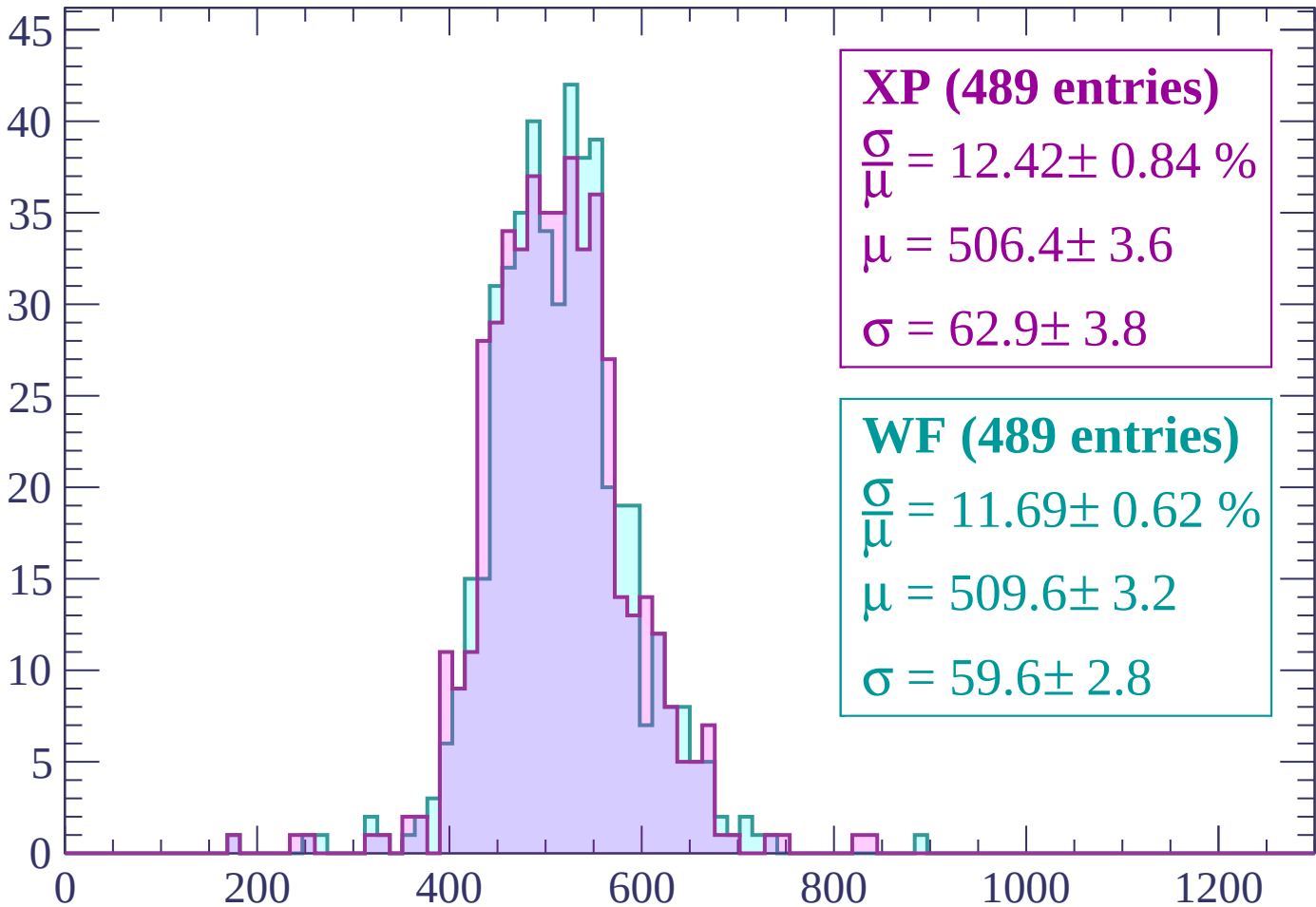
Energy loss | $2880 < p < 2940$

Count



Energy loss | $2940 < p < 3000$

Count



XP (489 entries)

$$\frac{\sigma}{\mu} = 12.42 \pm 0.84 \%$$

$$\mu = 506.4 \pm 3.6$$

$$\sigma = 62.9 \pm 3.8$$

WF (489 entries)

$$\frac{\sigma}{\mu} = 11.69 \pm 0.62 \%$$

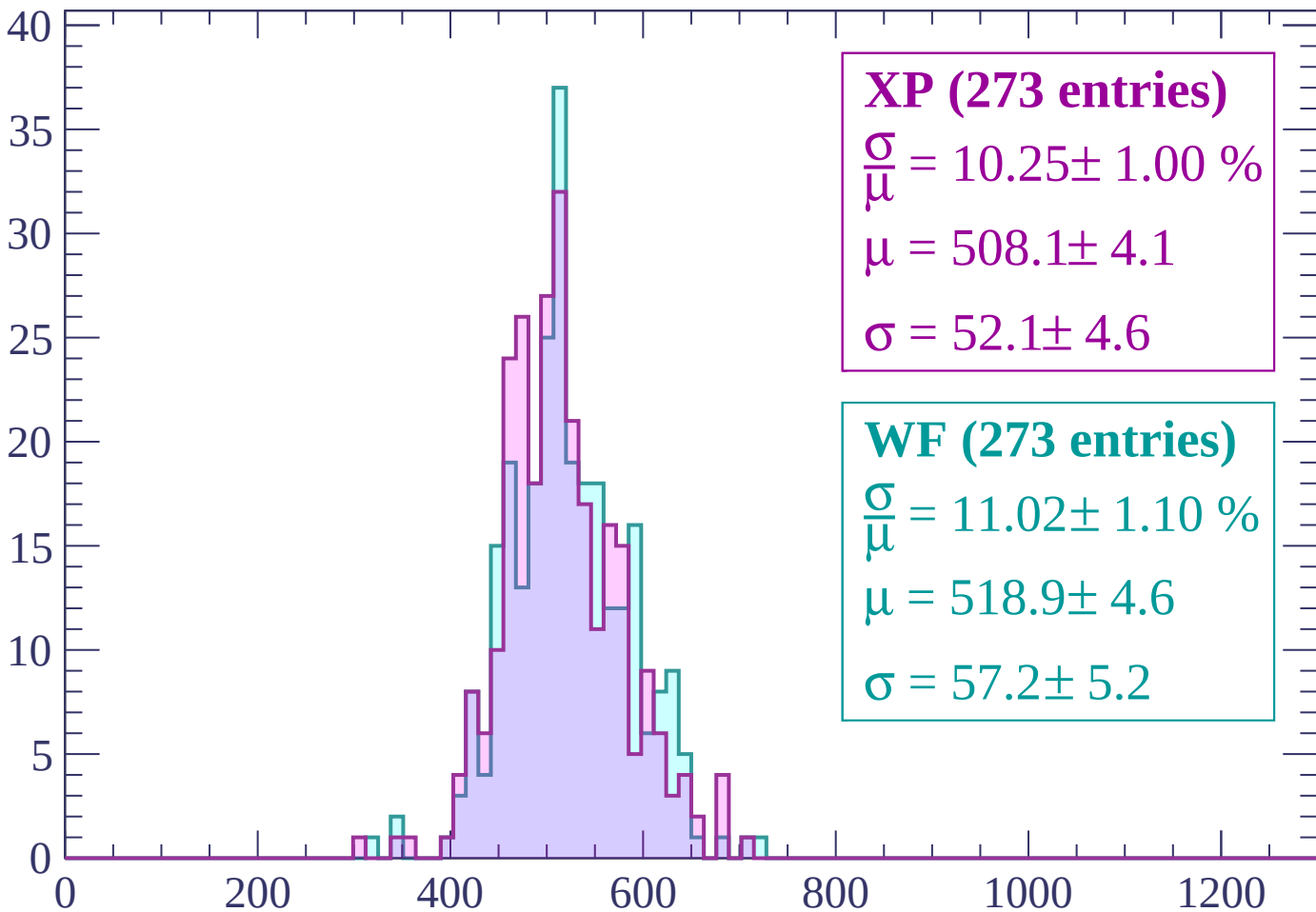
$$\mu = 509.6 \pm 3.2$$

$$\sigma = 59.6 \pm 2.8$$

dE/dx (ADC counts/cm)

Energy loss | $3000 < p < 3060$

Count



XP (273 entries)

$$\frac{\sigma}{\mu} = 10.25 \pm 1.00 \%$$

$$\mu = 508.1 \pm 4.1$$

$$\sigma = 52.1 \pm 4.6$$

WF (273 entries)

$$\frac{\sigma}{\mu} = 11.02 \pm 1.10 \%$$

$$\mu = 518.9 \pm 4.6$$

$$\sigma = 57.2 \pm 5.2$$

dE/dx (ADC counts/cm)

